

lyche-numerical-linear-algebra: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- EQ0706 [conf 0.001]	253	0.001	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} + \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D} \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S}^2 = \boldsymbol{I} / (2 \boldsymbol{h}) \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{T} \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} \\ \boldsymbol{S}^2 = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{I} / (2 \boldsymbol{h}) \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D} \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S}^2 = \boldsymbol{I} / (2 \boldsymbol{h}) \end{array}$
lyche-numerical-linear-algebra- EQ0729 [conf 0.002]	258	0.002	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q} = \omega_{2m}^{j(2k)} = \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k} = \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = \omega_{2m}^j \omega_m^{jk} \end{array}$	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q} = \omega_{2m}^{j(2k)} = \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k} = \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = \omega_{2m}^j \omega_m^{jk} \end{array}$	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q} = \omega_{2m}^{j(2k)} = \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k} = \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = \omega_{2m}^j \omega_m^{jk} \end{array}$

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lyche-numerical-linear-algebra-EQ0326 (5.18)	133	0.118	$\begin{aligned} & \boldsymbol{Q}_1 := \left[\boldsymbol{q}_1, \dots, \boldsymbol{q}_n \right], \\ & \boldsymbol{R}_1 := \\ & \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and} \\ & \left[\begin{array}{ccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & & \ddots & \vdots \\ & & & & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right]. \end{aligned}$	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \left[\begin{array}{cccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right].$	
lyche-numerical-linear-algebra-EQ0029	31	0.138	$\begin{array}{l} \begin{array}{c} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array} \\ \begin{array}{c} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array} \end{array}$	$\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array}$	$\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array}$

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lyche-numerical-linear-algebra- EQ0492 (8.20)	193	■ 0.232	$\begin{aligned} & \text{\texttt{\textbackslash begin{aligned} \& \text{\texttt{\textbackslash boldsymbol{A}\textbackslash _{2}=\sigma_{1} \text\texttt{\textbackslash text} \\ \& \texttt{\textbackslash left\textbackslash \boldsymbol{A}^{\textbackslash -1}\textbackslash right\textbackslash _{2}=\frac{1}{\sigma_{n}}}}}} \texttt{\textbackslash sigma_{n}}}} \texttt{\textbackslash \& \textbackslash \boldsymbol{A}\textbackslash _{2}=\lambda_{1}} \\ \texttt{\textbackslash text} \{ \& \texttt{\textbackslash left\textbackslash \boldsymbol{A}^{\textbackslash -1}\textbackslash right\textbackslash _{2}=\frac{1}{\lambda_{n}}}} \texttt{\textbackslash text} \{ \& \texttt{\textbackslash if} \} \texttt{\textbackslash boldsymbol{A}} \\ \texttt{\textbackslash text} \{ \& \texttt{\textbackslash is positive definite,} \} \texttt{\textbackslash \& \textbackslash \boldsymbol{A}} \\ \texttt{\textbackslash _{2}=\left \lambda_{1}\right \textbackslash right } \texttt{\textbackslash text} \{ \& \texttt{\textbackslash left} \\ \texttt{\textbackslash \boldsymbol{A}^{\textbackslash -1}\textbackslash right\textbackslash _{2}=\frac{1}{\left \lambda_{n}\right }} \texttt{\textbackslash left } \\ \texttt{\textbackslash lambda_{n}\textbackslash right }, \texttt{\textbackslash text} \{ \& \texttt{\textbackslash if} \} \texttt{\textbackslash boldsymbol{A}} \texttt{\textbackslash text} \{ \\ \& \texttt{\textbackslash is normal.} \} \texttt{\textbackslash end{aligned}} \end{aligned}$	$\ \mathbf{A}\ _2 = \sigma_1 \text{ and } \ \mathbf{A}^{-1}\ _2 = \frac{1}{\sigma_n}$ $\ \mathbf{A}\ _2 = \lambda_1 \text{ and } \ \mathbf{A}^{-1}\ _2 = \frac{1}{\lambda_n}, \text{ if } \mathbf{A} \text{ is positive definite,}$ $\ \mathbf{A}\ _2 = \lambda_1 \text{ and } \ \mathbf{A}^{-1}\ _2 = \frac{1}{ \lambda_n }, \text{ if } \mathbf{A} \text{ is normal.}$	$\ \mathbf{A}\ _2 = \sigma_1 \text{ and } \ \mathbf{A}^{-1}\ _2 = \frac{1}{\sigma_n},$ $\ \mathbf{A}\ _2 = \lambda_1 \text{ and } \ \mathbf{A}^{-1}\ _2 = \frac{1}{\lambda_n}, \text{ if } \mathbf{A} \text{ is positive definite,}$ $\ \mathbf{A}\ _2 = \lambda_1 \text{ and } \ \mathbf{A}^{-1}\ _2 = \frac{1}{ \lambda_n }, \text{ if } \mathbf{A} \text{ is normal.}$
lyche-numerical-linear-algebra- EQ0010	24	■ 0.244	$\begin{gathered} \operatorname{diag}(\mathbf{d}_1, \dots, \mathbf{d}_n) := \left[\begin{array}{cccc} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{array} \right] \\ \mathbf{B} = \operatorname{tridiag}(\mathbf{a}_i, \mathbf{d}_i, \mathbf{c}_i) = \operatorname{tridiag}(\mathbf{a}, \mathbf{d}, \mathbf{c}) := \left[\begin{array}{cccc} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{array} \right] \end{gathered}$	$\operatorname{diag}(d_i) = \operatorname{diag}(d_1, \dots, d_n) := \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} = \begin{bmatrix} d_1 & & & \\ & \ddots & & \\ & & d_n & \end{bmatrix},$ $\mathbf{B} = \operatorname{tridiag}(a_i, d_i, c_i) = \operatorname{tridiag}(\mathbf{a}, \mathbf{d}, \mathbf{c}) := \begin{bmatrix} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix}.$	$\operatorname{diag}(d_i) = \operatorname{diag}(d_1, \dots, d_n) := \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} = \begin{bmatrix} d_1 & & & \\ & \ddots & & \\ & & d_n & \end{bmatrix},$ $\mathbf{B} = \operatorname{tridiag}(a_i, d_i, c_i) = \operatorname{tridiag}(\mathbf{a}, \mathbf{d}, \mathbf{c}) := \begin{bmatrix} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0924 (13.53)	314	■ 0.288	$\begin{array}{l} \left(L_h v \right)_{j,k} := \frac{(dv)_{j,k}}{h^2} = f_{j,k}, \quad (j,k) \in I_m, \\ v_{j,k} = 0, \quad (j,k) \in \partial I_m, \end{array}$	$\begin{array}{l} (L_h v)_{j,k} := \frac{(dv)_{j,k}}{h^2} = f_{j,k}, \quad (j,k) \in I_m, \\ v_{j,k} = 0, \quad (j,k) \in \partial I_m, \end{array}$	$(L_h v)_{j,k} := \frac{(dv)_{j,k}}{h^2} = f_{j,k}, \quad (j,k) \in I_m,$ $v_{j,k} = 0, \quad (j,k) \in \partial I_m,$
lyche-numerical-linear-algebra- EQ0369 (6.7)	150	■ 0.298	$\mathbf{J} := \operatorname{diag}(\mathbf{U}_1, \mathbf{U}_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & 1 & & & & \\ & 0 & 1 & & & & & \\ & 0 & 2 & & & & & \\ & & & 0 & & & & \\ & & & 0 & 1 & & & \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$	$\mathbf{J} := \operatorname{diag}(\mathbf{U}_1, \mathbf{U}_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & 1 & & & & \\ & 0 & 1 & & & & & \\ & 0 & 2 & & & & & \\ & & & 0 & & & & \\ & & & 0 & 1 & & & \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$	$\mathbf{J} := \operatorname{diag}(\mathbf{U}_1, \mathbf{U}_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & 1 & & & & \\ & 0 & 1 & & & & & \\ & 0 & 2 & & & & & \\ & & & 0 & & & & \\ & & & 0 & 1 & & & \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$

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lyche-numerical-linear-algebra- EQ0375	152	■0.305	$\begin{aligned} & \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{W}^* (\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V}) \boldsymbol{W} = \\ & \left[\begin{array}{l l} 1 & \mathbf{0}^* \\ \hline \mathbf{0} & \boldsymbol{W}_1^* \end{array} \right] \left[\begin{array}{l l} \lambda_1 & \boldsymbol{x}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right] \left[\begin{array}{l l} 1 & \mathbf{0}^* \\ \hline \mathbf{0} & \boldsymbol{W}_1 \end{array} \right] \\ & = \left[\begin{array}{l l} \lambda_1 & \boldsymbol{x}^* \boldsymbol{W}_1 \\ \hline \mathbf{0} & \boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1 \end{array} \right] \end{aligned}$	$U^*AU = W^*(V^*AV)W = \begin{bmatrix} 1 & \mathbf{0}^* \\ \mathbf{0} & W_1^* \end{bmatrix} \begin{bmatrix} \lambda_1 & \boldsymbol{x}^* \\ \mathbf{0} & M \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^* \\ \mathbf{0} & W_1 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \boldsymbol{x}^* W_1 \\ \mathbf{0} & W_1^* M W_1 \end{bmatrix}$	
lyche-numerical-linear-algebra- EQ0366 (6.4)	149	■0.315	$\boldsymbol{J}_m(\lambda) := \left[\begin{array}{cccccc} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array} \right] = \lambda \boldsymbol{I}_m + \boldsymbol{E}_m, \quad \boldsymbol{E}_m := \left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array} \right].$	$\boldsymbol{J}_m(\lambda) := \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix} = \lambda \boldsymbol{I}_m + \boldsymbol{E}_m, \quad \boldsymbol{E}_m := \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}.$	
lyche-numerical-linear-algebra- EQ0579	215	■0.322	$\begin{aligned} & \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \\ & = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = \boldsymbol{A}^* \boldsymbol{b}, \end{aligned}$	$\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$ $= \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = \boldsymbol{A}^* \boldsymbol{b},$	

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lyche-numerical-linear-algebra- EQ0335 (5.19)	135	■ 0.329	$\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right] = \left[\begin{array}{cc} c & s \\ -sc & c \end{array}\right] \left[\begin{array}{c} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right], [C(:,i) \, C(:,j)] = [\boldsymbol{A}(:,i) \, \boldsymbol{A}(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array}\right].$	$\left[\begin{array}{c} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right] = \left[\begin{array}{cc} c & s \\ -sc & c \end{array}\right] \left[\begin{array}{c} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right], [C(:,i) \, C(:,j)] = [\boldsymbol{A}(:,i) \, \boldsymbol{A}(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array}\right].$	$\begin{bmatrix} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{bmatrix}, [C(:,i) \, C(:,j)] = [\boldsymbol{A}(:,i) \, \boldsymbol{A}(:,j)] \begin{bmatrix} c & s \\ -s & c \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0917	312	■ 0.371	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \quad \text{for } k \geq 0$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \quad \text{for } k \geq 0$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$
lyche-numerical-linear-algebra- EQ0857	207	■ 0.373	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}}\right) & \text{otherwise.} \end{cases}$	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}}\right) & \text{otherwise.} \end{cases}$	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}}\right) & \text{otherwise.} \end{cases}$
lyche-numerical-linear-algebra- EQ0855	296	■ 0.386	$\begin{array}{l} \boldsymbol{t}_0 = \left[\begin{array}{c} 3 \\ -3/2 \end{array}\right], \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \left[\begin{array}{c} -1/4 \\ -1/2 \end{array}\right], \quad \boldsymbol{r}_1 = \left[\begin{array}{c} 0 \\ 3/4 \end{array}\right], \quad \beta_0 = 1/4, \\ \boldsymbol{p}_1 = \left[\begin{array}{c} 3/8 \\ 3/4 \end{array}\right], \quad \boldsymbol{t}_1 = \left[\begin{array}{c} 0 \\ 9/8 \end{array}\right], \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}. \end{array}$	$\begin{array}{l} \boldsymbol{t}_0 = \left[\begin{array}{c} 3 \\ -3/2 \end{array}\right], \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \left[\begin{array}{c} -1/4 \\ -1/2 \end{array}\right], \quad \boldsymbol{r}_1 = \left[\begin{array}{c} 0 \\ 3/4 \end{array}\right], \quad \beta_0 = 1/4, \\ \boldsymbol{p}_1 = \left[\begin{array}{c} 3/8 \\ 3/4 \end{array}\right], \quad \boldsymbol{t}_1 = \left[\begin{array}{c} 0 \\ 9/8 \end{array}\right], \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}. \end{array}$	$\begin{array}{l} \boldsymbol{t}_0 = \left[\begin{array}{c} 3 \\ -3/2 \end{array}\right], \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \left[\begin{array}{c} -1/4 \\ -1/2 \end{array}\right], \quad \boldsymbol{r}_1 = \left[\begin{array}{c} 0 \\ 3/4 \end{array}\right], \quad \beta_0 = 1/4, \\ \boldsymbol{p}_1 = \left[\begin{array}{c} 3/8 \\ 3/4 \end{array}\right], \quad \boldsymbol{t}_1 = \left[\begin{array}{c} 0 \\ 9/8 \end{array}\right], \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}. \end{array}$
lyche-numerical-linear-algebra- EQ0670 (10.10)	241	■ 0.390	$\begin{aligned} a_{ii} &= 2d, i=1, \dots, n, \\ a_{i,i+1} &= a_{i+1,i} = a, \quad i=1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ a_{i,i+m} &= a_{i+m,i} = a, \quad i=1, \dots, n-m, \\ a_{ij} &= 0, \text{ otherwise} \end{aligned}$	$\begin{aligned} a_{ii} &= 2d, i=1, \dots, n, \\ a_{i,i+1} &= a_{i+1,i} = a, \quad i=1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ a_{i,i+m} &= a_{i+m,i} = a, \quad i=1, \dots, n-m, \\ a_{ij} &= 0, \text{ otherwise} \end{aligned}$	$\begin{aligned} a_{ii} &= 2d, i=1, \dots, n, \\ a_{i,i+1} &= a_{i+1,i} = a, \quad i=1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ a_{i,i+m} &= a_{i+m,i} = a, \quad i=1, \dots, n-m, \\ a_{ij} &= 0, \text{ otherwise,} \end{aligned}$
lyche-numerical-linear-algebra- EQ0763 (12.14)	271	■ 0.402	$\boldsymbol{G}_{\omega} = \left[\begin{array}{cc} l & 0 \\ 0 & 1 \end{array}\right] - \left[\begin{array}{cc} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array}\right] \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] = \left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array}\right].$	$\boldsymbol{G}_{\omega} = \left[\begin{array}{cc} l & 0 \\ 0 & 1 \end{array}\right] - \left[\begin{array}{cc} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array}\right] \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] = \left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array}\right].$	$\boldsymbol{G}_{\omega} = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right] - \left[\begin{array}{cc} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array}\right] \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] = \left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array}\right].$
lyche-numerical-linear-algebra- EQ0498	194	■ 0.420	$\begin{aligned} & x_1 + \left(1 - 10^{-16}\right) x_2 = 20 \\ & x_1 + \left(1 - 10^{-15}\right) x_2 = 20 - 10^{-15} \end{aligned}$	$\begin{aligned} & x_1 + \left(1 - 10^{-16}\right) x_2 = 20 \\ & x_1 + \left(1 - 10^{-15}\right) x_2 = 20 - 10^{-15} \end{aligned}$	$\begin{aligned} & x_1 + x_2 = 20 \\ & x_1 + (1 - 10^{-16})x_2 = 20 - 10^{-15} \end{aligned}$

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lyche-numerical-linear-algebra- EQ0262	107	■ 0.421	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{z} = \boldsymbol{z}^* \operatorname{diag}(\lambda_1, \dots, \lambda_n) \boldsymbol{z} = \sum_{j=1}^n \lambda_j \boldsymbol{z}_j ^2 \geq 0$.		
lyche-numerical-linear-algebra- EQ0404	161	■ 0.448	$\left[\begin{array}{llllllll} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \\ \end{array} \right] \in \mathbb{R}^{8,8}?$		$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & \end{bmatrix} \in \mathbb{R}^{8,8}?$
lyche-numerical-linear-algebra- EQ0914 (13.48)	311	■ 0.453	$\begin{aligned} & \boldsymbol{y}_{k+1} = \boldsymbol{y}_k + \alpha_k \boldsymbol{q}_k, \quad \alpha_k = \boldsymbol{z}_k^T \boldsymbol{z}_k / \boldsymbol{q}_k^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{q}_k, \\ & \boldsymbol{z}_{k+1} = \boldsymbol{z}_k - \alpha_k (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{q}_k, \\ & \boldsymbol{q}_{k+1} = \boldsymbol{z}_{k+1} + \beta_k \boldsymbol{q}_k, \quad \beta_k = \boldsymbol{z}_{k+1}^T \boldsymbol{z}_{k+1} / \boldsymbol{z}_k^T \boldsymbol{z}_k. \end{aligned}$		
lyche-numerical-linear-algebra- EQ0245	100	■ 0.458	$\left[\begin{array}{ll} a & \bar{b} \\ b & d \end{array} \right] = \left[\begin{array}{ll} 1 & 0 \\ l_1 & 1 \end{array} \right] \left[\begin{array}{ll} d_1 & 0 \\ 0 & d_2 \end{array} \right] \left[\begin{array}{ll} 1 & \bar{l}_1 \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} d_1 & d_1 \bar{l}_1 \\ d_1 l_1 & d_1 l_1 ^2 + d_2 \end{array} \right]$		
lyche-numerical-linear-algebra- EQ0370	150	■ 0.466	$\begin{aligned} & \boldsymbol{s}_1 = 2\boldsymbol{s}_1, \quad \boldsymbol{s}_2 = \boldsymbol{s}_1 + 2\boldsymbol{s}_2, \quad \boldsymbol{s}_3 = \boldsymbol{s}_2 + 2\boldsymbol{s}_3, \\ & \boldsymbol{s}_4 = 2\boldsymbol{s}_4, \quad \boldsymbol{s}_5 = \boldsymbol{s}_4 + 2\boldsymbol{s}_5, \\ & \boldsymbol{s}_6 = 2\boldsymbol{s}_6, \\ & \boldsymbol{s}_7 = 3\boldsymbol{s}_7, \quad \boldsymbol{s}_8 = \boldsymbol{s}_7 + 3\boldsymbol{s}_8. \end{aligned}$		

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lyche-numerical-linear-algebra- EQ0396	158	■ 0.467	$\begin{aligned} \beta_k &\leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{B}}(\mathbf{x}) \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) \\ &\leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) = \alpha_k + \varepsilon_1, \end{aligned}$	$\beta_k \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{B}}(\mathbf{x}) \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) \\ \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) = \alpha_k + \varepsilon_1,$	$\beta_k \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{B}}(\mathbf{x}) \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) \\ \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) = \alpha_k + \varepsilon_1,$
lyche-numerical-linear-algebra- EQ0308	125	■ 0.469	$\mathbf{P} \mathbf{x} = \mathbf{x} - \frac{\mathbf{v}^* \mathbf{x}}{\mathbf{v}^* \mathbf{v}} \mathbf{v} \stackrel{(5.13)}{=} \mathbf{x} - \frac{1}{2} \mathbf{v} = \frac{1}{2} (\mathbf{x} + \mathbf{y}).$	$\mathbf{P} \mathbf{x} = \mathbf{x} - \frac{\mathbf{v}^* \mathbf{x}}{\mathbf{v}^* \mathbf{v}} \mathbf{v} \stackrel{(5.13)}{=} \mathbf{x} - \frac{1}{2} \mathbf{v} = \frac{1}{2} (\mathbf{x} + \mathbf{y}).$	$\mathbf{P} \mathbf{x} = \mathbf{x} - \frac{\mathbf{v}^* \mathbf{x}}{\mathbf{v}^* \mathbf{v}} \mathbf{v} \stackrel{(5.13)}{=} \mathbf{x} - \frac{1}{2} \mathbf{v} = \frac{1}{2} (\mathbf{x} + \mathbf{y}).$
lyche-numerical-linear-algebra- EQ0600 (0)	221	■ 0.474	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{llll} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array} \right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array} \right],$	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{llll} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array} \right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array} \right],$	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{llll} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array} \right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array} \right],$
lyche-numerical-linear-algebra- EQ0028 (1.11)	31	■ 0.479	$\ \mathbf{x} \ _2 := \left(\sum_{j=1}^n x_j ^2 \right)^{1/2} = \sqrt{\mathbf{x}^* \mathbf{x}}.$	$\ \mathbf{x} \ _2 := \left(\sum_{j=1}^n x_j ^2 \right)^{1/2} = \sqrt{\mathbf{x}^* \mathbf{x}}.$	$\ \mathbf{x} \ _2 := \left(\sum_{j=1}^n x_j ^2 \right)^{1/2} = \sqrt{\mathbf{x}^* \mathbf{x}}.$
lyche-numerical-linear-algebra- EQ0241 (3.33)	97	■ 0.492	$\mathbf{B} := \left[\begin{array}{cc} (1 + \mathbf{u}_1^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\mathbf{u}_1^T \mathbf{B}_{2,2} / u_{1,1} \\ -\mathbf{u}_{2,2}^T \mathbf{B}_{2,2} / u_{1,1} & \mathbf{B}_{2,2} \end{array} \right].$	$\mathbf{B} := \left[\begin{array}{cc} (1 + \mathbf{u}_1^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\mathbf{u}_1^T \mathbf{B}_{2,2} / u_{1,1} \\ -\mathbf{u}_{2,2}^T \mathbf{B}_{2,2} / u_{1,1} & \mathbf{B}_{2,2} \end{array} \right].$	$\mathbf{B} := \left[\begin{array}{cc} (1 + \mathbf{u}_1^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\mathbf{u}_1^T \mathbf{B}_{2,2} / u_{1,1} \\ -\mathbf{u}_{2,2}^T \mathbf{B}_{2,2} / u_{1,1} & \mathbf{B}_{2,2} \end{array} \right].$
lyche-numerical-linear-algebra- EQ0757 (12.11)	270	■ 0.498	$\mathbf{A}_{\mathbf{L}} := \left[\begin{array}{ccc} 0 & -a_{2,1} & 0 \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} \end{array} \right], \quad \mathbf{A}_{\mathbf{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ 0 & -a_{n-1,n} & 0 \end{array} \right].$	$\mathbf{A}_{\mathbf{L}} := \left[\begin{array}{ccc} 0 & -a_{2,1} & 0 \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} \end{array} \right], \quad \mathbf{A}_{\mathbf{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ 0 & -a_{n-1,n} & 0 \end{array} \right].$	$\mathbf{A}_{\mathbf{L}} := \left[\begin{array}{ccc} 0 & -a_{2,1} & 0 \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} \end{array} \right], \quad \mathbf{A}_{\mathbf{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ 0 & -a_{n-1,n} & 0 \end{array} \right].$

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lyche-numerical-linear-algebra-EQ0138	64	0.503	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right] \left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right] = \left[\begin{array}{ll} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right],$		
lyche-numerical-linear-algebra-EQ0465 (8.1)	186	0.507	$\begin{aligned} \ \mathbf{x}\ _p &:= \left(\sum_{j=1}^n x_j ^p\right)^{1/p}, \\ \ \mathbf{x}\ _\infty &:= \max_{1 \leq j \leq n} x_j . \end{aligned}$		$\ \mathbf{x}\ _p := \left(\sum_{j=1}^n x_j ^p\right)^{1/p},$ $\ \mathbf{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$
lyche-numerical-linear-algebra-EQ0486	191	0.508	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p\right)^{1/p}, \quad p \geq 1, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$		$\ \mathbf{x}\ _p := \left(\sum_{j=1}^n x_j ^p\right)^{1/p}, \quad p \geq 1, \quad \ \mathbf{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$
lyche-numerical-linear-algebra-EQ0766	272	0.512	$x_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} x_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} x_k(2) + \frac{3}{4}.$		$\mathbf{x}_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} \mathbf{x}_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} \mathbf{x}_k(2) + \frac{3}{4}.$
lyche-numerical-linear-algebra-EQ1017	345	0.514	$\mathbf{z}_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$		$\mathbf{z}_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$
lyche-numerical-linear-algebra-EQ0798	280	0.516	$\begin{aligned} \ \mathbf{x}_k - \mathbf{x}\ &= \ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x})\ \leq \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}\ \\ &= \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x}\ \leq \ \mathbf{G}\ (\ \mathbf{x}_{k-1} - \mathbf{x}_k\ + \ \mathbf{x}_k - \mathbf{x}\) \end{aligned}$		$\ \mathbf{x}_k - \mathbf{x}\ = \ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x})\ \leq \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}\ $ $= \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x}\ \leq \ \mathbf{G}\ (\ \mathbf{x}_{k-1} - \mathbf{x}_k\ + \ \mathbf{x}_k - \mathbf{x}\).$
lyche-numerical-linear-algebra-EQ0837 (13.3)	291	0.516	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$		$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$

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lyche-numerical-linear-algebra-E09854 (13.18)	296	0.521	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \boldsymbol{\alpha}_k &= (\boldsymbol{r}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= (\boldsymbol{r}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= (\boldsymbol{r}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$
lyche-numerical-linear-algebra-E09329	134	0.523	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$
lyche-numerical-linear-algebra-E09862	300	0.523	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$
lyche-numerical-linear-algebra-E09594 (8.26)	197	0.531	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1/\lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1/\sigma_n, & \text{in general.} \end{cases}$	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1/\lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1/\sigma_n, & \text{in general.} \end{cases}$	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1/\lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1/\sigma_n, & \text{in general.} \end{cases}$
lyche-numerical-linear-algebra-E09830 (12.46)	287	0.531	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L}\boldsymbol{L}^T.$	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L}\boldsymbol{L}^T.$	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L}\boldsymbol{L}^T.$
lyche-numerical-linear-algebra-E09098	54	0.532	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1/u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2/u_2, \quad u_3 = d_3 - l_2 c_2, \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2, \\ x_3 &= z_3/u_3, \quad x_2 = (z_2 - c_2 x_3)/u_2, \quad x_1 = (z_1 - c_1 x_2)/u_1. \end{aligned}$	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1/u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2/u_2, \quad u_3 = d_3 - l_2 c_2, \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2, \\ x_3 &= z_3/u_3, \quad x_2 = (z_2 - c_2 x_3)/u_2, \quad x_1 = (z_1 - c_1 x_2)/u_1 \end{aligned}$	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1/u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2/u_2, \quad u_3 = d_3 - l_2 c_2, \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2, \\ x_3 &= z_3/u_3, \quad x_2 = (z_2 - c_2 x_3)/u_2, \quad x_1 = (z_1 - c_1 x_2)/u_1. \end{aligned}$

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lyche-numerical-linear-algebra-EQ0836 (13.2)	291	0.546	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}\right ^2 \leq \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}\right)\left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$ $\leq \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}\right)\left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$		
lyche-numerical-linear-algebra-EQ0868	301	0.546	$\left \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right _A=\left \boldsymbol{x}-\boldsymbol{x}_k+\boldsymbol{u}\right _A \geq\left \boldsymbol{x}-\boldsymbol{x}_k\right _A,$ $\left \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right _A=\left \boldsymbol{x}-\boldsymbol{x}_k+\boldsymbol{u}\right _A \geq\left \boldsymbol{x}-\boldsymbol{x}_k\right _A,$		
lyche-numerical-linear-algebra-EQ0918	312	0.547	$\frac{\left \boldsymbol{y}-\boldsymbol{y}_k\right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}}{\left \boldsymbol{y}-\boldsymbol{y}_0\right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k, \quad \text { for } k \geq 0,$ $\frac{\left \boldsymbol{y}-\boldsymbol{y}_k\right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}}{\left \boldsymbol{y}-\boldsymbol{y}_0\right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k, \quad \text { for } k \geq 0,$		
lyche-numerical-linear-algebra-EQ0386	156	0.548	$\boldsymbol{R}:=\left[\begin{array}{ccc} D_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \mathbf{0} & D_2 & \boldsymbol{R}_{2,3} \\ \mathbf{0} & \mathbf{0} & D_3 \end{array}\right], \boldsymbol{D}_1:=\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array}\right], \boldsymbol{D}_2:=[1], \boldsymbol{D}_3:=\left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array}\right].$ $\boldsymbol{R}:=\left[\begin{array}{ccc} D_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \mathbf{0} & D_2 & \boldsymbol{R}_{2,3} \\ \mathbf{0} & \mathbf{0} & D_3 \end{array}\right], \boldsymbol{D}_1:=\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array}\right], \boldsymbol{D}_2:=[1], \boldsymbol{D}_3:=\left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array}\right].$		
lyche-numerical-linear-algebra-EQ0185	79	0.548	$l_{i k}^{(k)} a_{k j}^{(k)}=a_{i j}^{(k)}-a_{i j}^{(k+1)} \text { for } k<\min (i, j), \text { and } l_{i j}^{(k)} a_{j j}^{(j)}=a_{i j}^{(j)} \text { for } i>j.$ $l_{i k}^{(k)} a_{k j}^{(k)}=a_{i j}^{(k)}-a_{i j}^{(k+1)} \text { for } k<\min (i, j), \text { and } l_{i j}^{(k)} a_{j j}^{(j)}=a_{i j}^{(j)} \text { for } i>j.$		
lyche-numerical-linear-algebra-EQ0313	126	0.549	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^* \boldsymbol{x}\right) \boldsymbol{u}=\rho\ \boldsymbol{x}\ _2\left(\boldsymbol{z}-\left(\boldsymbol{u}^* \boldsymbol{z}\right) \boldsymbol{u}\right)=\rho\ \boldsymbol{x}\ _2\left(\boldsymbol{z}-\frac{\left(\boldsymbol{z}^*+\boldsymbol{e}_1^*\right) \boldsymbol{z}}{1+z_1}\left(\boldsymbol{z}+\boldsymbol{e}_1\right)\right)$ $\boldsymbol{H} \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^* \boldsymbol{x}\right) \boldsymbol{u}=\rho\ \boldsymbol{x}\ _2\left(\boldsymbol{z}-\left(\boldsymbol{u}^* \boldsymbol{z}\right) \boldsymbol{u}\right)=\rho\ \boldsymbol{x}\ _2\left(\boldsymbol{z}-\frac{\left(\boldsymbol{z}^*+\boldsymbol{e}_1^*\right) \boldsymbol{z}}{1+z_1}\left(\boldsymbol{z}+\boldsymbol{e}_1\right)\right)$ $=\rho\ \boldsymbol{x}\ _2\left(\boldsymbol{z}-\left(\boldsymbol{z}+\boldsymbol{e}_1\right)\right)=-\rho\ \boldsymbol{x}\ _2 \boldsymbol{e}_1=\boldsymbol{a} \boldsymbol{e}_1.$		
lyche-numerical-linear-algebra-EQ0129 (89)	63	0.550	$\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3], \quad \boldsymbol{A}_{21}=\left[\begin{array}{c} 4 \\ 7 \end{array}\right], \quad \text { and } \boldsymbol{A}_{22}=\left[\begin{array}{cc} 5 & 6 \\ 8 & 9 \end{array}\right],$ $\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3], \quad \boldsymbol{A}_{21}=\left[\begin{array}{c} 4 \\ 7 \end{array}\right], \quad \text { and } \boldsymbol{A}_{22}=\left[\begin{array}{cc} 5 & 6 \\ 8 & 9 \end{array}\right],$		

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lyche-numerical-linear-algebra-E00132	63	0.564	$\boldsymbol{A} \boldsymbol{B}=\left[\begin{array}{c} a_{1:}^T \\ a_{2:}^T \\ \vdots \\ a_{m:}^T \end{array}\right] \boldsymbol{B}=\left[\begin{array}{c} a_{1:}^T \boldsymbol{B} \\ a_{2:}^T \boldsymbol{B} \\ \vdots \\ a_{m:}^T \boldsymbol{B} \end{array}\right],$		
lyche-numerical-linear-algebra-E00425 (7.4)	171	0.564	$\begin{aligned} & \boldsymbol{U}=\left[\begin{array}{c} \boldsymbol{U}_1 \\ \boldsymbol{U}_2 \end{array}\right] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_1:=\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\right], \quad \boldsymbol{U}_2:=\left[\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\right], \\ & \boldsymbol{V}=\left[\boldsymbol{V}_1, \boldsymbol{V}_2\right] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_1:=\left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r\right], \quad \boldsymbol{V}_2:=\left[\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\right], \\ & \boldsymbol{\Sigma}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0}_{r, n-r} \\ \boldsymbol{0}_{m-r, r} & \boldsymbol{0}_{m-r, n-r} \end{array}\right] \in \mathbb{R}^{m \times n}, \text { where } \boldsymbol{\Sigma}_1:=\operatorname{diag}\left(\sigma_1, \ldots, \sigma_r\right) . \end{aligned}$		
lyche-numerical-linear-algebra-E01016	345	0.571	$z_{\{1\}}=\boldsymbol{A} z_{\{0\}}=\left[\begin{array}{c} 2 \\ -1 \end{array}\right], \quad z_{\{2\}}=\boldsymbol{A} z_{\{1\}}=\left[\begin{array}{c} 5 \\ -4 \end{array}\right], \quad \cdots, \quad z_{\{k\}}=\frac{1}{2}\left[\begin{array}{c} 1+3^k \\ 1-3^k \end{array}\right], \quad \cdots .$		
lyche-numerical-linear-algebra-E00034	289	0.578	$\boldsymbol{A}:=\left[\begin{array}{ccccc} \lambda & 0 & \cdots & 0 & 0 \\ 0 & \lambda & a & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \\ 0 & 0 & 0 & \cdots & 0 \end{array}\right] \in \mathbb{R}^{n \times n} .$		
lyche-numerical-linear-algebra-E00831 (12.47)	287	0.581	$\lambda=\frac{\boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}+\boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}} .$		
lyche-numerical-linear-algebra-E00420	169	0.585	$\lambda^m \pi_{\boldsymbol{A}^* \boldsymbol{A}}(\lambda)=\lambda^n \pi_{\boldsymbol{A} \boldsymbol{A}^*}(\lambda), \quad \lambda \in \mathbb{C},$		

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lyche-numerical-linear-algebra- EQ0709	254	0.587	$\begin{aligned} & \text{\textbf{X}} = h^4 \text{\textbf{G}} / \text{\textbf{M}}, \text{ where } \text{\textbf{M}} := \begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_m \end{bmatrix} [1, \dots, 1] + \begin{bmatrix} 1 \\ \vdots \\ i \end{bmatrix} [\sigma_1 \dots \sigma_m], \\ & \text{\textbf{S}} = \sin \left(\pi h \begin{bmatrix} \frac{1}{2} \\ \vdots \\ m \end{bmatrix} [1 \ 2 \dots m] \right), \quad \boldsymbol{\sigma} = \sin \left(\frac{\pi h}{2} \begin{bmatrix} \frac{1}{2} \\ \vdots \\ m \end{bmatrix} \right) \wedge 2. \end{aligned}$	$\text{\textbf{X}} = h^4 \text{\textbf{G}} / \text{\textbf{M}}, \text{ where } \text{\textbf{M}} := \begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_m \end{bmatrix} [1, \dots, 1] + \begin{bmatrix} 1 \\ \vdots \\ i \end{bmatrix} [\sigma_1 \dots \sigma_m],$ $\text{\textbf{S}} = \sin \left(\pi h \begin{bmatrix} \frac{1}{2} \\ \vdots \\ m \end{bmatrix} [1 \ 2 \dots m] \right), \quad \boldsymbol{\sigma} = \sin \left(\frac{\pi h}{2} \begin{bmatrix} \frac{1}{2} \\ \vdots \\ m \end{bmatrix} \right) \wedge 2.$	$\text{\textbf{X}} = h^4 \text{\textbf{G}} / \text{\textbf{M}}, \text{ where } \text{\textbf{M}} := \begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_m \end{bmatrix} [1, \dots, 1] + \begin{bmatrix} 1 \\ \vdots \\ i \end{bmatrix} [\sigma_1 \dots \sigma_m],$ $\text{\textbf{S}} = \sin \left(\pi h \begin{bmatrix} \frac{1}{2} \\ \vdots \\ m \end{bmatrix} [1 \ 2 \dots m] \right), \quad \boldsymbol{\sigma} = \sin \left(\frac{\pi h}{2} \begin{bmatrix} \frac{1}{2} \\ \vdots \\ m \end{bmatrix} \right) \wedge 2.$
lyche-numerical-linear-algebra- EQ0644	232	0.587	$\begin{array}{l} \text{\textbf{A B A}} = \text{\textbf{A}} \quad (1) \quad \text{\textbf{A C A}} = \text{\textbf{A}}, \\ \text{\textbf{B A B}} = \text{\textbf{B}} \quad (2) \quad \text{\textbf{C A C}} = \text{\textbf{C}}, \\ (\text{\textbf{A B}})^* = \text{\textbf{A B}} \quad (3) \quad (\text{\textbf{A C}})^* = \text{\textbf{A C}}, \\ (\text{\textbf{B A}})^* = \text{\textbf{B A}} \quad (4) \quad (\text{\textbf{C A}})^* = \text{\textbf{C A}}. \end{array}$	$\begin{array}{l} \text{\textbf{A B A}} = \text{\textbf{A}} \quad (1) \quad \text{\textbf{A C A}} = \text{\textbf{A}}, \\ \text{\textbf{B A B}} = \text{\textbf{B}} \quad (2) \quad \text{\textbf{C A C}} = \text{\textbf{C}}, \\ (\text{\textbf{A B}})^* = \text{\textbf{A B}} \quad (3) \quad (\text{\textbf{A C}})^* = \text{\textbf{A C}}, \\ (\text{\textbf{B A}})^* = \text{\textbf{B A}} \quad (4) \quad (\text{\textbf{C A}})^* = \text{\textbf{C A}}. \end{array}$	$\begin{array}{l} \text{\textbf{A B A}} = \text{\textbf{A}} \quad (1) \quad \text{\textbf{A C A}} = \text{\textbf{A}}, \\ \text{\textbf{B A B}} = \text{\textbf{B}} \quad (2) \quad \text{\textbf{C A C}} = \text{\textbf{C}}, \\ (\text{\textbf{A B}})^* = \text{\textbf{A B}} \quad (3) \quad (\text{\textbf{A C}})^* = \text{\textbf{A C}}, \\ (\text{\textbf{B A}})^* = \text{\textbf{B A}} \quad (4) \quad (\text{\textbf{C A}})^* = \text{\textbf{C A}}. \end{array}$
lyche-numerical-linear-algebra- EQ0171	76	0.588	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \quad \text{II}' \\ & a_{32}^{(2)} x_2 + a_{33}^{(2)} x_3 = b_3^{(2)}, \quad \text{III}', \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \text{II}' \\ & a_{32}^{(2)} x_2 + a_{33}^{(2)} x_3 = b_3^{(2)}, \text{III}', \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \quad \text{II}' \\ & a_{32}^{(2)} x_2 + a_{33}^{(2)} x_3 = b_3^{(2)}, \quad \text{III}', \end{aligned}$
lyche-numerical-linear-algebra- EQ0467 (8.4)	186	0.589	$\lim_{p \rightarrow \infty} \ \text{\textbf{x}}\ _p = \ \text{\textbf{x}}\ _\infty \text{ for all } \text{\textbf{x}} \in \mathbb{C}^n.$	$\lim_{p \rightarrow \infty} \ \text{\textbf{x}}\ _p = \ \text{\textbf{x}}\ _\infty \text{ for all } \text{\textbf{x}} \in \mathbb{C}^n.$	$\lim_{p \rightarrow \infty} \ \text{\textbf{x}}\ _p = \ \text{\textbf{x}}\ _\infty \text{ for all } \text{\textbf{x}} \in \mathbb{C}^n.$
lyche-numerical-linear-algebra- EQ0793	279	0.592	$k = -\frac{s \log(10)}{\log(1-\eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}.$	$k = -\frac{s \log(10)}{\log(1-\eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}.$	$k = -\frac{s \log(10)}{\log(1-\eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}.$
lyche-numerical-linear-algebra- EQ1023 (15.6)	346	0.600	$\begin{array}{l} \text{(i)} \quad \text{\textbf{y}}_k = \text{\textbf{A}} \text{\textbf{x}}_{k-1} \\ \text{(ii)} \quad \text{\textbf{x}}_k = \text{\textbf{y}}_k / \ \text{\textbf{y}}_k\ . \end{array}$	$\begin{array}{l} \text{(i)} \quad \text{\textbf{y}}_k = \text{\textbf{A}} \text{\textbf{x}}_{k-1} \\ \text{(ii)} \quad \text{\textbf{x}}_k = \text{\textbf{y}}_k / \ \text{\textbf{y}}_k\ . \end{array}$	$\begin{array}{l} \text{(i)} \quad \text{\textbf{y}}_k = \text{\textbf{A}} \text{\textbf{x}}_{k-1} \\ \text{(ii)} \quad \text{\textbf{x}}_k = \text{\textbf{y}}_k / \ \text{\textbf{y}}_k\ . \end{array}$
lyche-numerical-linear-algebra- EQ0935	317	0.600	$\lambda = \frac{x^T \text{\textbf{A}} \text{\textbf{x}}}{x^T \text{\textbf{A}}_p \text{\textbf{x}}}.$	$\lambda = \frac{x^T \text{\textbf{A}} \text{\textbf{x}}}{x^T \text{\textbf{A}}_p \text{\textbf{x}}}.$	$\lambda = \frac{x^T \text{\textbf{A}} \text{\textbf{x}}}{x^T \text{\textbf{A}}_p \text{\textbf{x}}}.$

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lyche-numerical-linear-algebra-EQ0901	308	0.602	$\begin{aligned} \left \left \boldsymbol{\epsilon}_k \right \right _{\boldsymbol{A}}^2 &= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k, \\ \left \left \boldsymbol{\epsilon}_{k+1} \right \right _{\boldsymbol{A}}^2 &= (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k) \\ &= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2\alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}. \end{aligned}$	$\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k,$ $\ \boldsymbol{\epsilon}_{k+1}\ _{\boldsymbol{A}}^2 = (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)$ $= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2\alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k,$ $\ \boldsymbol{\epsilon}_{k+1}\ _{\boldsymbol{A}}^2 = (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)$ $= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2\alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}.$
lyche-numerical-linear-algebra-EQ0892 (13.37)	307	0.606	$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$	$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$	$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$
lyche-numerical-linear-algebra-EQ1029	348	0.608	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0379	154	0.611	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^* &= (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and } \\ \boldsymbol{A}^* \boldsymbol{A} &= (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*. \end{aligned}$	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and }$ $\boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and }$ $\boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$
lyche-numerical-linear-algebra-EQ1031 (15.7)	349	0.611	$\begin{aligned} \text{(i)} \quad & (\boldsymbol{A} - s \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1} \\ \text{(ii)} \quad & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{aligned}$	$\text{(i)} \quad (\boldsymbol{A} - s \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}$ $\text{(ii)} \quad \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ .$	$\text{(i)} \quad (\boldsymbol{A} - s \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}$ $\text{(ii)} \quad \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ .$

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lyche-numerical-linear-algebra-EQ1011	341	0.614	$\begin{aligned} \boldsymbol{A} &:= \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{E} := \left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array} \right]. \\ \boldsymbol{A} &:= \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{E} := \left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$\boldsymbol{A} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{E} := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}.$	
lyche-numerical-linear-algebra-EQ0011	24	0.618	$\begin{aligned} \boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) &:= \left[\begin{array}{cccc} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{array} \right]. \\ \boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) &:= \left[\begin{array}{cccc} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{array} \right]. \end{aligned}$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{pmatrix} = \begin{bmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}.$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{pmatrix} = \begin{bmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0249 (4.4)	102	0.629	$\begin{aligned} \boldsymbol{a}_n &= \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}. \\ \boldsymbol{a}_n &= \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}. \end{aligned}$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$
lyche-numerical-linear-algebra-EQ0849	295	0.636	$\begin{aligned} \boldsymbol{r}_{k+1}^T \boldsymbol{r}_j &\stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j \\ &\stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right) \\ \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i &\stackrel{=0}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k. \end{aligned}$	$\begin{aligned} \boldsymbol{r}_{k+1}^T \boldsymbol{r}_j &\stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j \\ &\stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right) \\ \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i &\stackrel{=0}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k. \end{aligned}$	$\begin{aligned} \boldsymbol{r}_{k+1}^T \boldsymbol{r}_j &\stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j \\ &\stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right) \\ \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i &\stackrel{=0}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k. \end{aligned}$
lyche-numerical-linear-algebra-EQ0592 (9.4)	218	0.637	$\begin{aligned} \mathbb{C}^m &= \mathcal{R}(\boldsymbol{A}) \oplus \mathcal{N}(\boldsymbol{A}^*). \\ \mathbb{C}^m &= \mathcal{R}(\boldsymbol{A}) \oplus \mathcal{N}(\boldsymbol{A}^*). \end{aligned}$	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \oplus \mathcal{N}(\boldsymbol{A}^*).$	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \oplus \mathcal{N}(\boldsymbol{A}^*).$
lyche-numerical-linear-algebra-EQ0553 (8.42)	208	0.638	$\begin{aligned} \operatorname{cond}_1(\boldsymbol{T}) &= \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases} \\ \operatorname{cond}_1(\boldsymbol{T}) &= \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases} \end{aligned}$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases}$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases}.$

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lyche-numerical-linear-algebra- EQ1032	349	0.642	<pre> \begin{aligned} \text{ (i) } & \& \left(\right. \\ & \boldsymbol{A}-\boldsymbol{s}_{k-1} \boldsymbol{I} \right) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}, \\ & \boldsymbol{y}_k = \boldsymbol{x}_k / \ \boldsymbol{y}_k\ , \\ & \text{ (ii) } & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ , \\ & \text{ (iii) } & \boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k, \\ & \text{ (iv) } & \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - \boldsymbol{s}_k \boldsymbol{x}_k. \\ \end{aligned} </pre>	(i) $(\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}$, (ii) $\boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\$, (iii) $\boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k$, (iv) $\boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k$.	(i) $(\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}$, (ii) $\boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\$, (iii) $\boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k$, (iv) $\boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k$.
lyche-numerical-linear-algebra- EQ0698	250	0.645	<pre> \begin{aligned} \boldsymbol{A} &= \left[\begin{array}{rrr rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right] \\ &= \left[\begin{array}{ccc} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{array} \right], \end{aligned} </pre>	$\boldsymbol{A} = \left[\begin{array}{ccc ccc ccc} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right]$ $= \left[\begin{array}{ccc} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{array} \right],$	$\boldsymbol{A} = \left[\begin{array}{ccc ccc ccc} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right]$ $= \left[\begin{array}{ccc} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{array} \right],$
lyche-numerical-linear-algebra- EQ0641	232	0.645	<pre> \boldsymbol{r} = \boldsymbol{r}(\boldsymbol{x}) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}. </pre>	$\boldsymbol{r} = \boldsymbol{r}(\boldsymbol{x}) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$	$\boldsymbol{r} = \boldsymbol{r}(\boldsymbol{x}) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$
lyche-numerical-linear-algebra- EQ0601 (9.7)	222	0.650	<pre> \boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{array} \right] = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r), </pre>	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{array} \right] = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{array} \right] = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$
lyche-numerical-linear-algebra- EQ0201 (3.11)	85	0.651	<pre> \boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1. </pre>	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$

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lyche-numerical-linear-algebra- EQ0947	320	0.654	$\begin{aligned} & \left\{ \begin{array}{l} \boldsymbol{x}_0, \\ \boldsymbol{x}_1, \boldsymbol{x}_2, \\ \boldsymbol{x}_3 \end{array} \right\} = \left\{ \begin{array}{l} 0 \\ 2 \text{ \& } 8 / 3 \text{ \& } 3 \\ 0 \text{ \& } 0 \text{ \& } 4 / 3 \text{ \& } 2 \end{array} \right\} \\ & \left\{ \begin{array}{l} \boldsymbol{r}_0, \\ \boldsymbol{r}_1, \boldsymbol{r}_2, \\ \boldsymbol{r}_3 \end{array} \right\} = \left\{ \begin{array}{l} 4 \text{ \& } 0 \\ 0 \text{ \& } 2 \\ 0 \text{ \& } 0 \text{ \& } 4 / 3 \end{array} \right\} \end{aligned}$	$\begin{aligned} [\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3] &= \begin{bmatrix} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ [\boldsymbol{r}_0, \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3] &= \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{bmatrix}, \end{aligned}$	$[\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3] = \begin{bmatrix} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix},$ $[\boldsymbol{r}_0, \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3] = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0415	166	0.666	$R(\boldsymbol{x} - t\boldsymbol{y}) = R(\boldsymbol{x}) - 2t(\boldsymbol{Ax} - R(\boldsymbol{x})\boldsymbol{x})^T \boldsymbol{y} + \mathcal{O}(t^2),$	$R(\boldsymbol{x} - t\boldsymbol{y}) = R(\boldsymbol{x}) - 2t(\boldsymbol{Ax} - R(\boldsymbol{x})\boldsymbol{x})^T \boldsymbol{y} + \mathcal{O}(t^2),$	$R(\boldsymbol{x} - t\boldsymbol{y}) = R(\boldsymbol{x}) - 2t(\boldsymbol{Ax} - R(\boldsymbol{x})\boldsymbol{x})^T \boldsymbol{y} + \mathcal{O}(t^2),$
lyche-numerical-linear-algebra- EQ0602	222	0.667	$\begin{aligned} & \left\{ \begin{array}{l} \boldsymbol{U}_1, \dots, \boldsymbol{U}_r \\ \boldsymbol{V}_1, \dots, \boldsymbol{V}_r \end{array} \right\} = \left\{ \begin{array}{l} \boldsymbol{u}_1, \dots, \boldsymbol{u}_r \\ \boldsymbol{v}_1, \dots, \boldsymbol{v}_r \end{array} \right\} \\ & \left\{ \begin{array}{l} \boldsymbol{U}_{r+1}, \dots, \boldsymbol{U}_m \\ \boldsymbol{V}_{r+1}, \dots, \boldsymbol{V}_n \end{array} \right\} = \left\{ \begin{array}{l} \boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m \\ \boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n \end{array} \right\} \end{aligned}$	$\begin{aligned} \boldsymbol{U}_1 &= [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], & \boldsymbol{U}_2 &= [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], & \boldsymbol{V}_1^* \boldsymbol{V}_1 &= \boldsymbol{I}, & \boldsymbol{V}_2^* \boldsymbol{V}_2 &= \boldsymbol{I}, \\ \boldsymbol{V}_1 &= [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], & \boldsymbol{V}_2 &= [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], & \boldsymbol{V}_1^* \boldsymbol{V}_1 &= \boldsymbol{I}, & \boldsymbol{V}_2^* \boldsymbol{V}_2 &= \boldsymbol{I} \end{aligned}$	$\begin{aligned} \boldsymbol{U}_1 &= [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], & \boldsymbol{U}_2 &= [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], & \boldsymbol{U}_1^* \boldsymbol{U}_1 &= \boldsymbol{I}, & \boldsymbol{U}_2^* \boldsymbol{U}_2 &= \boldsymbol{I}, \\ \boldsymbol{V}_1 &= [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], & \boldsymbol{V}_2 &= [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], & \boldsymbol{V}_1^* \boldsymbol{V}_1 &= \boldsymbol{I}, & \boldsymbol{V}_2^* \boldsymbol{V}_2 &= \boldsymbol{I} \end{aligned}$
lyche-numerical-linear-algebra- EQ0751	267	0.670	$\begin{aligned} & \boldsymbol{J} : \boldsymbol{v}_{k+1}(i, j) = \left(\boldsymbol{v}_k(i-1, j) + \boldsymbol{v}_k(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) \right. \\ & \quad \left. + \boldsymbol{e}(i, j) \right) / 4 \\ & \boldsymbol{GS} : \boldsymbol{v}_{k+1}(i, j) = \left(\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_{k+1}(i+1, j) + \boldsymbol{v}_{k+1}(i, j+1) \right. \\ & \quad \left. + \boldsymbol{e}(i, j) \right) / 4 \end{aligned}$	$\begin{aligned} \boldsymbol{J} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_k(i-1, j) + \boldsymbol{v}_k(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) \\ & \quad + \boldsymbol{e}(i, j)) / 4 \\ \boldsymbol{GS} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_{k+1}(i+1, j) + \boldsymbol{v}_{k+1}(i, j+1) \\ & \quad + \boldsymbol{e}(i, j)) / 4 \end{aligned}$	$\begin{aligned} \boldsymbol{J} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_k(i-1, j) + \boldsymbol{v}_k(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) \\ & \quad + \boldsymbol{e}(i, j)) / 4 \\ \boldsymbol{GS} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_{k+1}(i+1, j) + \boldsymbol{v}_{k+1}(i, j+1) \\ & \quad + \boldsymbol{e}(i, j)) / 4 \end{aligned}$
lyche-numerical-linear-algebra- EQ0190 (3.8)	80	0.672	$\boldsymbol{x}_k = \left(\boldsymbol{b}_k - \sum_{j=l_k}^{k-1} \boldsymbol{a}_{k,j} \boldsymbol{x}_j \right) / \boldsymbol{a}_{kk}, \quad k = 1, 2, \dots, n.$	$\boldsymbol{x}_k = \left(\boldsymbol{b}_k - \sum_{j=l_k}^{k-1} \boldsymbol{a}_{k,j} \boldsymbol{x}_j \right) / \boldsymbol{a}_{kk}, \quad k = 1, 2, \dots, n.$	$\boldsymbol{x}_k = (\boldsymbol{b}_k - \sum_{j=l_k}^{k-1} \boldsymbol{a}_{k,j} \boldsymbol{x}_j) / \boldsymbol{a}_{kk}, \quad k = 1, 2, \dots, n.$

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lyche-numerical-linear-algebra-EQ0719	257	0.674	$\begin{aligned} \left(\boldsymbol{F}_{2m+2}\boldsymbol{z}\right)_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$	$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$	$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$
lyche-numerical-linear-algebra-EQ0599	221	0.676	$\boldsymbol{A} = \left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array} \right] \text{ and } \boldsymbol{b} = \left[\begin{array}{c} 1 \\ 1 \\ 1 \\ 1 \end{array} \right].$	$\boldsymbol{A} = \left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array} \right] \text{ and } \boldsymbol{b} = \left[\begin{array}{c} 1 \\ 1 \\ 1 \\ 1 \end{array} \right].$	$\boldsymbol{A} = \left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array} \right] \text{ and } \boldsymbol{b} = \left[\begin{array}{c} 1 \\ 1 \\ 1 \\ 1 \end{array} \right].$
lyche-numerical-linear-algebra-EQ0726 (11.7)	258	0.678	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} = \left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array} \right],$	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} = \left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array} \right],$	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} = \left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array} \right],$
lyche-numerical-linear-algebra-EQ0513	200	0.679	$\begin{aligned} f(x) &= \frac{x_2 - x}{x_2 - x_1} f(x_1) + \frac{x - x_1}{x_2 - x_1} f(x_2) + (x - x_1)(x - x_2) f''(c)/2 \\ &= (1 - \lambda) f(x_1) + \lambda f(x_2) + (x_2 - x_1)^2 \lambda (\lambda - 1) f''(c)/2, \quad \lambda := \frac{x - x_1}{x_2 - x_1}. \end{aligned}$	$\begin{aligned} f(x) &= \frac{x_2 - x}{x_2 - x_1} f(x_1) + \frac{x - x_1}{x_2 - x_1} f(x_2) + (x - x_1)(x - x_2) f''(c)/2 \\ &= (1 - \lambda) f(x_1) + \lambda f(x_2) + (x_2 - x_1)^2 \lambda (\lambda - 1) f''(c)/2, \quad \lambda := \frac{x - x_1}{x_2 - x_1} \end{aligned}$	$\begin{aligned} f(x) &= \frac{x_2 - x}{x_2 - x_1} f(x_1) + \frac{x - x_1}{x_2 - x_1} f(x_2) + (x - x_1)(x - x_2) f''(c)/2 \\ &= (1 - \lambda) f(x_1) + \lambda f(x_2) + (x_2 - x_1)^2 \lambda (\lambda - 1) f''(c)/2, \quad \lambda := \frac{x - x_1}{x_2 - x_1}. \end{aligned}$
lyche-numerical-linear-algebra-EQ0607	222	0.679	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^\dagger. \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^\dagger \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^\dagger. \end{aligned}$
lyche-numerical-linear-algebra-EQ0638	35	0.682	$\begin{aligned} &\left \begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \right = a_{11} a_{22} a_{33} - a_{11} a_{32} a_{23} - a_{21} a_{12} a_{33} \\ &\quad + a_{21} a_{32} a_{13} + a_{31} a_{12} a_{23} - a_{31} a_{22} a_{13}. \end{aligned}$	$\begin{aligned} &\left \begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \right = a_{11} a_{22} a_{33} - a_{11} a_{32} a_{23} - a_{21} a_{12} a_{33} \\ &\quad + a_{21} a_{32} a_{13} + a_{31} a_{12} a_{23} - a_{31} a_{22} a_{13}. \end{aligned}$	$\begin{aligned} &\left \begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \right = a_{11} a_{22} a_{33} - a_{11} a_{32} a_{23} - a_{21} a_{12} a_{33} \\ &\quad + a_{21} a_{32} a_{13} + a_{31} a_{12} a_{23} - a_{31} a_{22} a_{13}. \end{aligned}$

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lyche-numerical-linear-algebra- EQ8383	155	0.685	$R(x)=R_{\{A\}}(x):=\frac{x^{*}Ax}{x^{*}x}$	$R(x)=R_A(x):=\frac{x^*Ax}{x^*x}$	$R(\boldsymbol{x})=R_A(\boldsymbol{x}):=\frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$
lyche-numerical-linear-algebra- EQ0578	214	0.686	$c_{\{1\}}\left[\begin{array}{c} c_1 \\ \vdots \\ c_m \end{array}\right]+c_{\{2\}}\left[\begin{array}{c} t_1 \\ \vdots \\ t_m \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right]\Rightarrow\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array}\right]\left[\begin{array}{c} c_1 \\ \vdots \\ c_m \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right]\Rightarrow c_1=c_2=0.$	$c_1\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array}\right]+c_2\left[\begin{array}{c} t_1 \\ \vdots \\ t_m \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right]\Rightarrow\left[\begin{array}{cc} 1 & t_i \\ 1 & t_j \end{array}\right]\left[\begin{array}{c} c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right]\Rightarrow c_1=c_2=0.$	$c_1\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array}\right]+c_2\left[\begin{array}{c} t_1 \\ \vdots \\ t_m \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right]\Rightarrow\left[\begin{array}{cc} 1 & t_i \\ 1 & t_j \end{array}\right]\left[\begin{array}{c} c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right]\Rightarrow c_1=c_2=0.$
lyche-numerical-linear-algebra- EQ0535	203	0.693	$a_{\{n\}}\angle\boldsymbol{x},\boldsymbol{y}=\left[\begin{array}{c} \boldsymbol{x} \\ \boldsymbol{y} \end{array}\right]\angle\boldsymbol{a}_{\{n\}}=\frac{1}{4}\left(\left\ \boldsymbol{a}_{\{n\}}\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{a}_{\{n\}}\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).$	$a_n\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle a_n\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\left\ a_n\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ a_n\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).$	$a_n\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle a_n\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\left\ a_n\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ a_n\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).$
lyche-numerical-linear-algebra- EQ1026	347	0.696	$\left[\begin{array}{c} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{array}\right](\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{array}\right].$	$\left[\begin{array}{c} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{array}\right](\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{array}\right].$	$\left[\begin{array}{c} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{array}\right](\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{array}\right].$
lyche-numerical-linear-algebra- EQ0410 (6.19)	164	0.702	$\mu_{\{\boldsymbol{A}\}}(\lambda):=\prod_{i=1}^k(\lambda-\lambda_i)^{m_i}\text{ where }m_i:=\max_{1\leq j\leq g_i}m_{i,j},$	$\mu_{\boldsymbol{A}}(\lambda):=\prod_{i=1}^k(\lambda_i-\lambda)^{m_i}\text{ where }m_i:=\max_{1\leq j\leq g_i}m_{i,j},$	$\mu_{\boldsymbol{A}}(\lambda):=\prod_{i=1}^k(\lambda_i-\lambda)^{m_i}\text{ where }m_i:=\max_{1\leq j\leq g_i}m_{i,j},$
lyche-numerical-linear-algebra- EQ0699 (11.1)	251	0.706	$\left[\begin{array}{cccc} \boldsymbol{D}_{\{1\}} & \boldsymbol{C}_{\{1\}} & \boldsymbol{D}_{\{2\}} & \boldsymbol{C}_{\{2\}} \\ \boldsymbol{A}_{\{1\}} & \boldsymbol{D}_{\{2\}} & \boldsymbol{C}_{\{2\}} & \boldsymbol{D}_{\{3\}} \\ \vdots & \vdots & \vdots & \vdots \\ \boldsymbol{A}_{\{m-1\}} & \boldsymbol{D}_{\{m-1\}} & \boldsymbol{C}_{\{m-1\}} & \boldsymbol{D}_{\{m\}} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{I} \\ \boldsymbol{L}_1 & \vdots \\ \vdots & \ddots \\ \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} & \vdots & \vdots \\ \boldsymbol{U}_m & \vdots & \vdots & \vdots \end{array}\right].$	$\left[\begin{array}{cccc} \boldsymbol{D}_1 & \boldsymbol{C}_1 & & \\ \boldsymbol{A}_1 & \boldsymbol{D}_2 & \boldsymbol{C}_2 & \\ & \ddots & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_m \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{I} \\ \boldsymbol{L}_1 & \vdots \\ \vdots & \ddots \\ \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} & \vdots & \vdots \\ \boldsymbol{U}_m & \vdots & \vdots & \vdots \end{array}\right].$	$\left[\begin{array}{cccc} \boldsymbol{D}_1 & \boldsymbol{C}_1 & & \\ \boldsymbol{A}_1 & \boldsymbol{D}_2 & \boldsymbol{C}_2 & \\ & \ddots & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_m \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{I} \\ \boldsymbol{L}_1 & \vdots \\ \vdots & \ddots \\ \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} & \vdots & \vdots \\ \boldsymbol{U}_m & \vdots & \vdots & \vdots \end{array}\right].$
lyche-numerical-linear-algebra- EQ0717	256	0.709	$(\boldsymbol{S}\boldsymbol{x})_k=\frac{i}{2}(\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1},\quad k=1,\ldots,m.$	$(\boldsymbol{S}\boldsymbol{x})_k=\frac{i}{2}(\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1},\quad k=1,\ldots,m.$	$(\boldsymbol{S}\boldsymbol{x})_k=\frac{i}{2}(\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1},\quad k=1,\ldots,m.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra E00593	218	■ 0.712	$\ A \mathbf{x} - \mathbf{b}\ _2^2 = \left\ \left(A \mathbf{x} - \mathbf{b}_1 \right) - \mathbf{b}_2 \right\ _2^2 = \left\ A \mathbf{x} - \mathbf{b}_1 \right\ _2^2 + \left\ \mathbf{b}_2 \right\ _2^2 \geq \left\ \mathbf{b}_2 \right\ _2^2$	$\ A\mathbf{x} - \mathbf{b}\ _2^2 = \ (A\mathbf{x} - \mathbf{b}_1) - \mathbf{b}_2\ _2^2 = \ A\mathbf{x} - \mathbf{b}_1\ _2^2 + \ \mathbf{b}_2\ _2^2 \geq \ \mathbf{b}_2\ _2^2$	$\ A\mathbf{x} - \mathbf{b}\ _2^2 = \ (A\mathbf{x} - \mathbf{b}_1) - \mathbf{b}_2\ _2^2 = \ A\mathbf{x} - \mathbf{b}_1\ _2^2 + \ \mathbf{b}_2\ _2^2 \geq \ \mathbf{b}_2\ _2^2$
lyche-numerical-linear-algebra E00530 (8.38)	203	■ 0.713	$\begin{aligned} & 4\langle \mathbf{x}, \mathbf{z} \rangle + 4\langle \mathbf{y}, \mathbf{z} \rangle \\ &= \left\ \left(\mathbf{z} + \frac{\mathbf{x} + \mathbf{y}}{2} \right) + \frac{\mathbf{x} - \mathbf{y}}{2} \right\ ^2 - \left\ \left(\mathbf{z} - \frac{\mathbf{x} + \mathbf{y}}{2} \right) + \frac{\mathbf{y} - \mathbf{x}}{2} \right\ ^2 \\ &+ \left\ \left(\mathbf{z} + \frac{\mathbf{x} + \mathbf{y}}{2} \right) - \frac{\mathbf{x} - \mathbf{y}}{2} \right\ ^2 - \left\ \left(\mathbf{z} - \frac{\mathbf{x} + \mathbf{y}}{2} \right) - \frac{\mathbf{y} - \mathbf{x}}{2} \right\ ^2 \\ &\stackrel{(8.35)}{=} 2 \left\ \mathbf{z} + \frac{\mathbf{x} + \mathbf{y}}{2} \right\ ^2 + 2 \left\ \frac{\mathbf{x} - \mathbf{y}}{2} \right\ ^2 - 2 \left\ \mathbf{z} - \frac{\mathbf{x} + \mathbf{y}}{2} \right\ ^2 - 2 \left\ \frac{\mathbf{y} - \mathbf{x}}{2} \right\ ^2 \\ &\stackrel{(8.36)}{=} 8 \left\langle \frac{\mathbf{x} + \mathbf{y}}{2}, \mathbf{z} \right\rangle \end{aligned}$	$4\langle x, z \rangle + 4\langle y, z \rangle \stackrel{(8.36)}{=} \ x + z\ ^2 - \ x - z\ ^2 + \ y + z\ ^2 - \ y - z\ ^2$ $= \left\ \left(z + \frac{x+y}{2} \right) + \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) + \frac{y-x}{2} \right\ ^2$ $+ \left\ \left(z + \frac{x+y}{2} \right) - \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) - \frac{y-x}{2} \right\ ^2$ $\stackrel{(8.35)}{=} 2 \left\ z + \frac{x+y}{2} \right\ ^2 + 2 \left\ \frac{x-y}{2} \right\ ^2 - 2 \left\ z - \frac{x+y}{2} \right\ ^2 - 2 \left\ \frac{y-x}{2} \right\ ^2$ $\stackrel{(8.36)}{=} 8 \left\langle \frac{x+y}{2}, z \right\rangle,$	$4\langle \mathbf{x}, \mathbf{z} \rangle + 4\langle \mathbf{y}, \mathbf{z} \rangle \stackrel{(8.36)}{=} \ \mathbf{x} + \mathbf{z}\ ^2 - \ \mathbf{x} - \mathbf{z}\ ^2 + \ \mathbf{y} + \mathbf{z}\ ^2 - \ \mathbf{y} - \mathbf{z}\ ^2$ $= \left\ \left(\mathbf{z} + \frac{\mathbf{x} + \mathbf{y}}{2} \right) + \frac{\mathbf{x} - \mathbf{y}}{2} \right\ ^2 - \left\ \left(\mathbf{z} - \frac{\mathbf{x} + \mathbf{y}}{2} \right) + \frac{\mathbf{y} - \mathbf{x}}{2} \right\ ^2$ $+ \left\ \left(\mathbf{z} + \frac{\mathbf{x} + \mathbf{y}}{2} \right) - \frac{\mathbf{x} - \mathbf{y}}{2} \right\ ^2 - \left\ \left(\mathbf{z} - \frac{\mathbf{x} + \mathbf{y}}{2} \right) - \frac{\mathbf{y} - \mathbf{x}}{2} \right\ ^2$ $\stackrel{(8.35)}{=} 2\ \mathbf{z} + \frac{\mathbf{x} + \mathbf{y}}{2}\ ^2 + 2\ \frac{\mathbf{x} - \mathbf{y}}{2}\ ^2 - 2\ \mathbf{z} - \frac{\mathbf{x} + \mathbf{y}}{2}\ ^2 - 2\ \frac{\mathbf{y} - \mathbf{x}}{2}\ ^2$ $\stackrel{(8.36)}{=} 8\langle \frac{\mathbf{x} + \mathbf{y}}{2}, \mathbf{z} \rangle,$
lyche-numerical-linear-algebra E00615	226	■ 0.721	$\begin{aligned} & \mathbf{A}^* \mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\mathbf{A}^* \mathbf{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \\ & \mathbf{A}^* \mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\mathbf{A}^* \mathbf{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \end{aligned}$	$\mathbf{A}^* \mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\mathbf{A}^* \mathbf{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$	$\mathbf{A}^* \mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\mathbf{A}^* \mathbf{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$
lyche-numerical-linear-algebra E00460 (2)	182	■ 0.722	$\begin{aligned} & \mathbf{A} := \begin{bmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix} \\ & \mathbf{A} := \begin{bmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix} \end{aligned}$	$\mathbf{A} := \begin{bmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$	$\mathbf{A} := \begin{bmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$
lyche-numerical-linear-algebra E00736	261	■ 0.724	$\mathbf{T} \mathbf{X} + \mathbf{X} \mathbf{D} = \mathbf{C}, \text{ where } \mathbf{X} = \mathbf{V} \mathbf{S}, \text{ and } \mathbf{C} = h^2 \mathbf{F} \mathbf{S}.$	$\mathbf{T}\mathbf{X} + \mathbf{X}\mathbf{D} = \mathbf{C}, \text{ where } \mathbf{X} = \mathbf{V}\mathbf{S}, \text{ and } \mathbf{C} = h^2\mathbf{F}\mathbf{S}.$	$\mathbf{T}\mathbf{X} + \mathbf{X}\mathbf{D} = \mathbf{C}, \text{ where } \mathbf{X} = \mathbf{V}\mathbf{S}, \text{ and } \mathbf{C} = h^2\mathbf{F}\mathbf{S}.$

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lyche-numerical-linear-algebra- EQ0668	241	0.727	$\begin{array}{cccccccc} 4 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array}$	$\mathbf{A} = \begin{bmatrix} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{bmatrix}.$	$\mathbf{A} = \begin{bmatrix} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0976 (14.1)	330	0.729	$ \mu - \lambda \leq K \ E\ _2^{1/n}, \quad K = (\ A\ _2 + \ A + E\ _2)^{1-1/n}.$	$ \mu - \lambda \leq K \ E\ _2^{1/n}, \quad K = (\ A\ _2 + \ A + E\ _2)^{1-1/n}.$	$ \mu - \lambda \leq K \ E\ _2^{1/n}, \quad K = (\ A\ _2 + \ A + E\ _2)^{1-1/n}.$
lyche-numerical-linear-algebra- EQ0676 (10.13)	243	0.737	$\begin{aligned} & (\lambda \mathbf{A}) \otimes (\mu \mathbf{B}) = \lambda \mu (\mathbf{A} \otimes \mathbf{B}), \\ & (\mathbf{A}_1 + \mathbf{A}_2) \otimes \mathbf{B} = \mathbf{A}_1 \otimes \mathbf{B} + \mathbf{A}_2 \otimes \mathbf{B}, \\ & \mathbf{A} \otimes (\mathbf{B}_1 + \mathbf{B}_2) = \mathbf{A} \otimes \mathbf{B}_1 + \mathbf{A} \otimes \mathbf{B}_2, \\ & (\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C}). \end{aligned}$	$\begin{aligned} & (\lambda \mathbf{A}) \otimes (\mu \mathbf{B}) = \lambda \mu (\mathbf{A} \otimes \mathbf{B}) \\ & (\mathbf{A}_1 + \mathbf{A}_2) \otimes \mathbf{B} = \mathbf{A}_1 \otimes \mathbf{B} + \mathbf{A}_2 \otimes \mathbf{B}, \\ & \mathbf{A} \otimes (\mathbf{B}_1 + \mathbf{B}_2) = \mathbf{A} \otimes \mathbf{B}_1 + \mathbf{A} \otimes \mathbf{B}_2 \\ & (\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C}). \end{aligned}$	$\begin{aligned} & (\lambda \mathbf{A}) \otimes (\mu \mathbf{B}) = \lambda \mu (\mathbf{A} \otimes \mathbf{B}), \\ & (\mathbf{A}_1 + \mathbf{A}_2) \otimes \mathbf{B} = \mathbf{A}_1 \otimes \mathbf{B} + \mathbf{A}_2 \otimes \mathbf{B}, \\ & \mathbf{A} \otimes (\mathbf{B}_1 + \mathbf{B}_2) = \mathbf{A} \otimes \mathbf{B}_1 + \mathbf{A} \otimes \mathbf{B}_2, \\ & (\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C}). \end{aligned}$
lyche-numerical-linear-algebra- EQ0466 (8.3)	186	0.737	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \mathbf{x}, \mathbf{y} \in \mathbb{C}^n.$	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \mathbf{x}, \mathbf{y} \in \mathbb{C}^n.$	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \mathbf{x}, \mathbf{y} \in \mathbb{C}^n.$
lyche-numerical-linear-algebra- EQ0897	308	0.743	$\frac{(\mathbf{y}^T \mathbf{A} \mathbf{y}) (\mathbf{y}^T \mathbf{A}^{-1} \mathbf{y})}{(\mathbf{y}^T \mathbf{y})^2} = ab \leq \frac{(M + m)^2}{4 M m},$	$\frac{(\mathbf{y}^T \mathbf{A} \mathbf{y}) (\mathbf{y}^T \mathbf{A}^{-1} \mathbf{y})}{(\mathbf{y}^T \mathbf{y})^2} = ab \leq \frac{(M + m)^2}{4 M m},$	$\frac{(\mathbf{y}^T \mathbf{A} \mathbf{y}) (\mathbf{y}^T \mathbf{A}^{-1} \mathbf{y})}{(\mathbf{y}^T \mathbf{y})^2} = ab \leq \frac{(M + m)^2}{4 M m},$
lyche-numerical-linear-algebra- EQ0582	216	0.744	$\begin{array}{ccccc} \mathbf{u} & \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ \hline & y_1 & y_2 & \cdots & y_m \end{array}$	$\frac{\mathbf{u} \mid \mathbf{u}_1 \mid \mathbf{u}_2 \mid \cdots \mid \mathbf{u}_m}{y \mid y_1 \mid y_2 \mid \cdots \mid y_m}.$	$\frac{\mathbf{u} \mid \mathbf{u}_1 \mid \mathbf{u}_2 \mid \cdots \mid \mathbf{u}_m}{y \mid y_1 \mid y_2 \mid \cdots \mid y_m}.$

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lyche-numerical-linear-algebra- E00772 (12.19)	273	0.772	$\begin{gathered} \rho(\boldsymbol{I}-\alpha \boldsymbol{A})>\rho\left(\boldsymbol{I}-\alpha_{\mathrm{o}}\boldsymbol{A}\right)=\frac{\kappa-1}{\kappa+1}, \\ \boldsymbol{A}\right)=\frac{\kappa-1}{\kappa+1}, \\ \quad \alpha \in \mathbb{R} \backslash \left\{\alpha_{\mathrm{o}}\right\}, \\ \quad \backslash \backslash \quad \kappa:=\frac{\lambda_1}{\lambda_n}, \quad \alpha_o:=\frac{2}{\lambda_1+\lambda_n}. \\ \quad \quad \alpha_{\mathrm{o}}:=\frac{2}{\lambda_1+\lambda_n}. \\ \end{gathered}$	$\rho(\boldsymbol{I}-\alpha \boldsymbol{A})>\rho\left(\boldsymbol{I}-\alpha_{\mathrm{o}}\boldsymbol{A}\right)=\frac{\kappa-1}{\kappa+1}, \quad \alpha \in \mathbb{R} \backslash \left\{\alpha_{\mathrm{o}}\right\},$ $\kappa:=\frac{\lambda_1}{\lambda_n}, \quad \alpha_o:=\frac{2}{\lambda_1+\lambda_n}.$	$\rho(\boldsymbol{I}-\alpha \boldsymbol{A})>\rho\left(\boldsymbol{I}-\alpha_{\mathrm{o}}\boldsymbol{A}\right)=\frac{\kappa-1}{\kappa+1}, \quad \alpha \in \mathbb{R} \backslash \left\{\alpha_{\mathrm{o}}\right\},$ $\kappa:=\frac{\lambda_1}{\lambda_n}, \quad \alpha_o:=\frac{2}{\lambda_1+\lambda_n}.$
lyche-numerical-linear-algebra- E00915 (13.47)	311	0.779	$\begin{aligned} \boldsymbol{x}_{\mathrm{k}} &:= \boldsymbol{C}^T \boldsymbol{y}_{\mathrm{k}}, \quad \boldsymbol{p}_{\mathrm{k}} := \boldsymbol{C}^T \boldsymbol{q}_{\mathrm{k}}, \quad \boldsymbol{s}_{\mathrm{k}} := \boldsymbol{C}^T \boldsymbol{z}_{\mathrm{k}}, \quad \boldsymbol{r}_{\mathrm{k}} := \boldsymbol{C}^{-1} \boldsymbol{z}_{\mathrm{k}} \\ \boldsymbol{y}_{\mathrm{k}}, \quad \boldsymbol{p}_{\mathrm{k}} &:= \boldsymbol{C}^T \boldsymbol{q}_{\mathrm{k}}, \quad \boldsymbol{s}_{\mathrm{k}} := \boldsymbol{C}^T \boldsymbol{z}_{\mathrm{k}}, \quad \boldsymbol{r}_{\mathrm{k}} := \boldsymbol{C}^{-1} \boldsymbol{z}_{\mathrm{k}} \\ \boldsymbol{C}^T \boldsymbol{y}_{\mathrm{k}}, \quad \boldsymbol{p}_{\mathrm{k}} &:= \boldsymbol{C}^T \boldsymbol{q}_{\mathrm{k}}, \quad \boldsymbol{s}_{\mathrm{k}} := \boldsymbol{C}^T \boldsymbol{z}_{\mathrm{k}}, \quad \boldsymbol{r}_{\mathrm{k}} := \boldsymbol{C}^{-1} \boldsymbol{z}_{\mathrm{k}} \\ \boldsymbol{C}^T \boldsymbol{y}_{\mathrm{k}}, \quad \boldsymbol{p}_{\mathrm{k}} &:= \boldsymbol{C}^T \boldsymbol{q}_{\mathrm{k}}, \quad \boldsymbol{s}_{\mathrm{k}} := \boldsymbol{C}^T \boldsymbol{z}_{\mathrm{k}}, \quad \boldsymbol{r}_{\mathrm{k}} := \boldsymbol{C}^{-1} \boldsymbol{z}_{\mathrm{k}} \\ \boldsymbol{C}^T \boldsymbol{y}_{\mathrm{k}}, \quad \boldsymbol{p}_{\mathrm{k}} &:= \boldsymbol{C}^T \boldsymbol{q}_{\mathrm{k}}, \quad \boldsymbol{s}_{\mathrm{k}} := \boldsymbol{C}^T \boldsymbol{z}_{\mathrm{k}}, \quad \boldsymbol{r}_{\mathrm{k}} := \boldsymbol{C}^{-1} \boldsymbol{z}_{\mathrm{k}} \end{aligned}$	$\boldsymbol{x}_k := \boldsymbol{C}^T \boldsymbol{y}_k, \quad \boldsymbol{p}_k := \boldsymbol{C}^T \boldsymbol{q}_k, \quad \boldsymbol{s}_k := \boldsymbol{C}^T \boldsymbol{z}_k, \quad \boldsymbol{r}_k := \boldsymbol{C}^{-1} \boldsymbol{z}_k$	$\boldsymbol{x}_k := \boldsymbol{C}^T \boldsymbol{y}_k, \quad \boldsymbol{p}_k := \boldsymbol{C}^T \boldsymbol{q}_k, \quad \boldsymbol{s}_k := \boldsymbol{C}^T \boldsymbol{z}_k, \quad \boldsymbol{r}_k := \boldsymbol{C}^{-1} \boldsymbol{z}_k$
lyche-numerical-linear-algebra- E00610	223	0.780	$\begin{aligned} \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \\ \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \\ \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \\ \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \\ \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \end{aligned}$	$\boldsymbol{v} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t},$	$\boldsymbol{v} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \begin{bmatrix} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{bmatrix} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t},$
lyche-numerical-linear-algebra- E00203 (3.13)	85	0.781	$\begin{aligned} \boldsymbol{p} &= [1, \dots, n]^T; \\ \text{for } k &= 1 : n - 1 \\ \text{choose } r_k &\geq k \text{ so that } a_{p_{r_k}, k}^{(k)} \neq 0. \\ \boldsymbol{p} &= I_{r_k, k} \boldsymbol{p} \end{aligned}$	$\boldsymbol{p} = [1, \dots, n]^T;$ for $k = 1 : n - 1$ choose $r_k \geq k$ so that $a_{p_{r_k}, k}^{(k)} \neq 0$. $\boldsymbol{p} = I_{r_k, k} \boldsymbol{p}$	$\boldsymbol{p} = [1, \dots, n]^T;$ for $k = 1 : n - 1$ choose $r_k \geq k$ so that $a_{p_{r_k}, k}^{(k)} \neq 0$. $\boldsymbol{p} = I_{r_k, k} \boldsymbol{p}$
lyche-numerical-linear-algebra- E00852 (13.15)	295	0.782	$\begin{aligned} \text{For } k &= 0, 1, 2, \dots \\ \boldsymbol{x}_{k+1} &:= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ \boldsymbol{r}_{k+1} &:= \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \end{aligned}$	For $k = 0, 1, 2, \dots$ $\boldsymbol{x}_{k+1} := \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k},$ $\boldsymbol{r}_{k+1} := \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k,$	For $k = 0, 1, 2, \dots$ $\boldsymbol{x}_{k+1} := \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k},$ $\boldsymbol{r}_{k+1} := \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k,$
lyche-numerical-linear-algebra- E00531	203	0.783	$\begin{aligned} \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle &= 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}. \\ \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle &= 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}. \end{aligned}$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$
lyche-numerical-linear-algebra- E00009 (1.1)	23	0.784	$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$	$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$	$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases}$

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lyche-numerical-linear-algebra-E00108 (2.21)	57	0.788	$\begin{array}{l} \text{\textbf{T}} \text{\textbf{v}}=\left[\begin{array}{rrrrr} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & & -1 & 2 \\ & & & 0 & -1 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{array}\right] = h^2 \left[\begin{array}{c} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{array}\right] =: \text{\textbf{b}}. \end{array}$	$T\mathbf{v} = \begin{bmatrix} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & & -1 & 2 \\ & & & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \mathbf{b}.$	$T\mathbf{v} = \begin{bmatrix} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & & -1 & 2 \\ & & & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \mathbf{b}.$
lyche-numerical-linear-algebra-E00401	160	0.791	$\text{\textbf{V}}^*\text{\textbf{A}}\text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}=\left[\begin{array}{c c} \lambda & \mathbf{z}^* \\ \hline \mathbf{0} & \mathbf{M} \end{array}\right] \text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}=\left[\begin{array}{c c} \lambda & \mathbf{z}^* \\ \hline \mathbf{0} & \mathbf{M} \end{array}\right] \text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}=\left[\begin{array}{c c} \lambda & \mathbf{z}^* \\ \hline \mathbf{0} & \mathbf{M} \end{array}\right] \text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}^* \text{\textbf{A}} \text{\textbf{V}}=\left[\begin{array}{c c} \lambda & \mathbf{z}^* \\ \hline \mathbf{0} & \mathbf{M} \end{array}\right]$	$\mathbf{V}^* \mathbf{A} \mathbf{V} = \left[\begin{array}{c c} \lambda & \mathbf{z}^* \\ \hline \mathbf{0} & \mathbf{M} \end{array} \right],$	$\mathbf{V}^* \mathbf{A} \mathbf{V} = \left[\begin{array}{c c} \lambda & \mathbf{z}^* \\ \hline \mathbf{0} & \mathbf{M} \end{array} \right],$
lyche-numerical-linear-algebra-E00590	218	0.792	$\begin{array}{l} \frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}. \end{array}$	$\frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}$	$\frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}.$
lyche-numerical-linear-algebra-E00096 (0)	54	0.793	$\left[\begin{array}{lll} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{array}\right] = \left[\begin{array}{lll} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array}\right] \left[\begin{array}{lll} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array}\right] = \left[\begin{array}{lll} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{array}\right],$	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$
lyche-numerical-linear-algebra-E00752 (12.7)	268	0.800	$\begin{array}{l} \text{\textbf{S}} \text{\textbf{O}} \text{\textbf{R}}: \text{\textbf{v}}_{k+1}(i, j) = \omega (\text{\textbf{v}}_{k+1}(i-1, j) + \text{\textbf{v}}_{k+1}(i, j-1) + \text{\textbf{v}}_k(i+1, j) + \text{\textbf{v}}_k(i, j+1) + \text{\textbf{e}}(i, j)) / 4 + (1-\omega) \text{\textbf{v}}_k(i, j). \end{array}$	$SOR: \mathbf{v}_{k+1}(i, j) = \omega (\mathbf{v}_{k+1}(i-1, j) + \mathbf{v}_{k+1}(i, j-1) + \mathbf{v}_k(i+1, j) + \mathbf{v}_k(i, j+1) + \mathbf{e}(i, j)) / 4 + (1-\omega) \mathbf{v}_k(i, j).$	$SOR: \mathbf{v}_{k+1}(i, j) = \omega (\mathbf{v}_{k+1}(i-1, j) + \mathbf{v}_{k+1}(i, j-1) + \mathbf{v}_k(i+1, j) + \mathbf{v}_k(i, j+1) + \mathbf{e}(i, j)) / 4 + (1-\omega) \mathbf{v}_k(i, j).$
lyche-numerical-linear-algebra-E00403	161	0.801	$\text{\textbf{V}}^*\text{\textbf{A}}^*\text{\textbf{V}}\text{\textbf{u}} = \left[\begin{array}{c c} \bar{\lambda} & \mathbf{0}^* \\ \hline \mathbf{z} & \mathbf{M}^* \end{array}\right] \left[\begin{array}{c} 0 \\ \mathbf{v} \end{array}\right] = \left[\begin{array}{c} 0 \\ \mathbf{M}^* \mathbf{v} \end{array}\right] = \bar{\lambda} \left[\begin{array}{c} 0 \\ \mathbf{v} \end{array}\right]$	$\mathbf{V}^* \mathbf{A}^* \mathbf{V} \mathbf{u} = \left[\begin{array}{c c} \bar{\lambda} & \mathbf{0}^* \\ \hline \mathbf{z} & \mathbf{M}^* \end{array} \right] \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{M}^* \mathbf{v} \end{bmatrix} = \bar{\lambda} \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix}$	$\mathbf{V}^* \mathbf{A}^* \mathbf{V} \mathbf{u} = \left[\begin{array}{c c} \bar{\lambda} & \mathbf{0}^* \\ \hline \mathbf{z} & \mathbf{M}^* \end{array} \right] \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{M}^* \mathbf{v} \end{bmatrix} = \bar{\lambda} \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix}$

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lyche-numerical-linear-algebra- EQ0588	217	0.812	$\begin{array}{l} \texttt{\boldsymbol{B}_{\{3\}} \boldsymbol{x}:=\left[\right.} \\ \texttt{\begin{array}{ccc} m & \sum t_{\{k\}} & \sum t_{\{k\}}^2 \backslash} \\ \texttt{\backslash \sum t_{\{k\}} & \sum t_{\{k\}}^2 & \sum t_{\{k\}}^3 \backslash \backslash \sum} \\ \texttt{t_{\{k\}}^2 & \sum t_{\{k\}}^3 & \sum t_{\{k\}}^4 \end{array}} \\ \texttt{\left. \right] \left[\begin{array}{l} x_{\{1\}} \backslash \backslash x_{\{2\}} \backslash \backslash x_{\{3\}} \end{array} \right.} \\ \texttt{\end{array} \right] = \left[\begin{array}{c} \sum y_{\{k\}} \\ \sum t_{\{k\}} y_{\{k\}} \\ \sum t_{\{k\}}^2 y_{\{k\}} \end{array} \right.} \\ \texttt{\backslash \sum t_{\{k\}} y_{\{k\}} \backslash \backslash \sum t_{\{k\}}^2 y_{\{k\}} \end{array} \right.} \\ \texttt{\left. \right] .} \end{array}$	$\boldsymbol{B}_3 \boldsymbol{x} := \begin{bmatrix} m & \sum t_k & \sum t_k^2 \\ \sum t_k & \sum t_k^2 & \sum t_k^3 \\ \sum t_k^2 & \sum t_k^3 & \sum t_k^4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \\ \sum t_k^2 y_k \end{bmatrix}.$	$\boldsymbol{B}_3 \boldsymbol{x} := \begin{bmatrix} m & \sum t_k & \sum t_k^2 \\ \sum t_k & \sum t_k^2 & \sum t_k^3 \\ \sum t_k^2 & \sum t_k^3 & \sum t_k^4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \\ \sum t_k^2 y_k \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0529 (8.37)	203	0.818	$\begin{array}{l} \texttt{\begin{gathered} \backslash \texttt{\langle} \boldsymbol{x}, \boldsymbol{z} \rangle, \boldsymbol{z} \in \mathcal{V} \\ \texttt{\langle} \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle = \texttt{\langle} \boldsymbol{x}, \boldsymbol{z} \rangle + \texttt{\langle} \boldsymbol{y}, \boldsymbol{z} \rangle, \\ \texttt{\langle} \boldsymbol{a} \boldsymbol{x}, \boldsymbol{y} \rangle = a \texttt{\langle} \boldsymbol{x}, \boldsymbol{y} \rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}. \end{gathered}}$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ $\langle a \boldsymbol{x}, \boldsymbol{y} \rangle = a \langle \boldsymbol{x}, \boldsymbol{y} \rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V},$ $\langle a \boldsymbol{x}, \boldsymbol{y} \rangle = a \langle \boldsymbol{x}, \boldsymbol{y} \rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$
lyche-numerical-linear-algebra- EQ0689 (2.11)	51	0.822	$\begin{array}{l} \texttt{\left[\begin{array}{ccccc} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array} \right] \left[\begin{array}{c} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{array} \right] = \frac{6}{h^2} \left[\begin{array}{c} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{array} \right], \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}. \\ \texttt{\left[\begin{array}{ccccc} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array} \right] \left[\begin{array}{c} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{array} \right] = \frac{6}{h^2} \left[\begin{array}{c} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{array} \right], \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}.} \end{array}$	$\begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{bmatrix}, \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}.$	$\begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{bmatrix}, \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}.$
lyche-numerical-linear-algebra- EQ0563 (8.44)	210	0.822	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$
lyche-numerical-linear-algebra- EQ0354	142	0.825	$\begin{array}{l} \texttt{\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right] = \boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],} \\ \texttt{\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right] = \boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],} \end{array}$	$\begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \boldsymbol{Q} \begin{bmatrix} \boldsymbol{R} \\ 0 \end{bmatrix},$	$\begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \boldsymbol{Q} \begin{bmatrix} \boldsymbol{R} \\ 0 \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0040 (1.14)	36	0.826	$\begin{array}{l} \texttt{\begin{aligned} \operatorname{det}(\boldsymbol{A}) &= \sum_{j=1}^n (-1)^{i+j} a_{ij} \operatorname{det}(\boldsymbol{A}_{ij}) \text{ for } i = 1, \dots, n, \text{ row} \\ \operatorname{det}(\boldsymbol{A}) &= \sum_{i=1}^n (-1)^{i+j} a_{ij} \operatorname{det}(\boldsymbol{A}_{ij}) \text{ for } j = 1, \dots, n, \text{ column.} \end{aligned}}$	$\det(\boldsymbol{A}) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } i = 1, \dots, n, \text{ row}$ $\det(\boldsymbol{A}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } j = 1, \dots, n, \text{ column.}$	$\det(\boldsymbol{A}) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } i = 1, \dots, n, \text{ row}$ $\det(\boldsymbol{A}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } j = 1, \dots, n, \text{ column.}$
lyche-numerical-linear-algebra- EQ0338	136	0.827	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} \right ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$	$ \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$	$ \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$

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lyche-numerical-linear-algebra- EQ0624	228	0.842	$\begin{aligned} \alpha_j &\leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (B + (A - B))\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ B\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} + \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (A - B)\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \\ &\leq \beta_j + \ A - B\ _2. \end{aligned}$	$\alpha_j \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (B + (A - B))\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ B\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} + \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (A - B)\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \\ \leq \beta_j + \ A - B\ _2.$	$\alpha_j \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (B + (A - B))\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ B\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} + \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (A - B)\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \\ \leq \beta_j + \ A - B\ _2.$
lyche-numerical-linear-algebra- EQ0649	38	0.844	$\begin{aligned} \operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) &= \left(a_{11} - \lambda \right) \begin{vmatrix} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} - \lambda \end{vmatrix} \\ &\quad + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} - \lambda & a_{23} \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) + r(\lambda) \end{aligned}$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda) \begin{vmatrix} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} - \lambda \end{vmatrix} \\ + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} - \lambda & a_{23} \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) + r(\lambda)$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda) \begin{vmatrix} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} - \lambda \end{vmatrix} \\ + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} - \lambda & a_{23} \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) + r(\lambda)$
lyche-numerical-linear-algebra- EQ0851 (13.14)	295	0.846	$\beta_k := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{A} \boldsymbol{p}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.13)}{=} \frac{\boldsymbol{r}_{k+1}^T (\boldsymbol{r}_{k+1} - \boldsymbol{r}_k)}{\alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.12)}{=} \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$	$\beta_k := -\frac{\boldsymbol{r}_{k+1}^T \boldsymbol{A} \boldsymbol{p}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.13)}{=} \frac{\boldsymbol{r}_{k+1}^T (\boldsymbol{r}_{k+1} - \boldsymbol{r}_k)}{\alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.12)}{=} \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$	$\beta_k := -\frac{\boldsymbol{r}_{k+1}^T \boldsymbol{A} \boldsymbol{p}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.13)}{=} \frac{\boldsymbol{r}_{k+1}^T (\boldsymbol{r}_{k+1} - \boldsymbol{r}_k)}{\alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.12)}{=} \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$
lyche-numerical-linear-algebra- EQ0387	156	0.846	$\pi_{D_1}(\lambda) = \pi_{D_3}(\lambda) = \lambda^2 - 4\lambda + 5, \quad \pi_{D_2}(\lambda) = \lambda - 1,$	$\pi_{D_1}(\lambda) = \pi_{D_3}(\lambda) = \lambda^2 - 4\lambda + 5, \quad \pi_{D_2}(\lambda) = \lambda - 1,$	$\pi_{D_1}(\lambda) = \pi_{D_3}(\lambda) = \lambda^2 - 4\lambda + 5, \quad \pi_{D_2}(\lambda) = \lambda - 1,$
lyche-numerical-linear-algebra- EQ0650	234	0.847	$\left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & 0 \end{array} \right] y = c.$	$\left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & 0 \end{array} \right] y = c.$	$\left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & 0 \end{array} \right] y = c.$
lyche-numerical-linear-algebra- EQ0034	33	0.848	$C=C \quad I=C(A \ B)=(C \ A) \quad B=I \quad B=B$	$C = CI = C(AB) = (CA)B = IB = B$	$\boldsymbol{C} = \boldsymbol{CI} = \boldsymbol{C}(\boldsymbol{AB}) = (\boldsymbol{CA})\boldsymbol{B} = \boldsymbol{IB} = \boldsymbol{B}$
lyche-numerical-linear-algebra- EQ0084 (2.6)	50	0.850	$p_i(x_i) = y_i, \quad p_i(x_{i+1}) = y_{i+1}, \quad p_i'(x_i) = \mu_i, \quad p_i'(x_{i+1}) = \mu_{i+1}.$	$p_i(x_i) = y_i, \quad p_i(x_{i+1}) = y_{i+1}, \quad p_i''(x_i) = \mu_i, \quad p_i''(x_{i+1}) = \mu_{i+1}.$	$p_i(x_i) = y_i, \quad p_i(x_{i+1}) = y_{i+1}, \quad p_i''(x_i) = \mu_i, \quad p_i''(x_{i+1}) = \mu_{i+1}.$
lyche-numerical-linear-algebra- EQ0409	163	0.850	$\boldsymbol{A} = \left[\begin{array}{cc} a & 1 \\ 0 & a \end{array} \right]$	$\boldsymbol{A} = \left[\begin{array}{cc} a & 1 \\ 0 & a \end{array} \right]$	$\boldsymbol{A} = \left[\begin{array}{cc} a & 1 \\ 0 & a \end{array} \right]$

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lyche-numerical-linear-algebra-EQ0144	65	0.853	$\boldsymbol{B} \boldsymbol{A} = \left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array} \right] \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array} \right] = \left[\begin{array}{ll} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array} \right] = \boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{A} = \begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{bmatrix} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} = \boldsymbol{I}$	
lyche-numerical-linear-algebra-EQ0264 (4.8)	109	0.854	$\boldsymbol{A} = \left[\begin{array}{ll} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right], \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$	$\boldsymbol{A} = \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix}, \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$	
lyche-numerical-linear-algebra-EQ0392	157	0.857	$\max_{\boldsymbol{x} \in \mathcal{S}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$	$\max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \boldsymbol{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$	
lyche-numerical-linear-algebra-EQ0032	32	0.859	$\boldsymbol{B} \boldsymbol{z} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{b} \end{bmatrix} \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix} = \boldsymbol{A} \tilde{\boldsymbol{z}} + z_{n+1} \boldsymbol{b} = \boldsymbol{0}.$	$\boldsymbol{B} \boldsymbol{z} = [\boldsymbol{A} \ \boldsymbol{b}] \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix} = \boldsymbol{A} \tilde{\boldsymbol{z}} + z_{n+1} \boldsymbol{b} = \boldsymbol{0}.$	
lyche-numerical-linear-algebra-EQ0891 (13.36)	307	0.859	$1 \leq \frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} \leq \frac{(M+m)^2}{4Mm} \quad \boldsymbol{y} \neq \boldsymbol{0}, \boldsymbol{y} \in \mathbb{R}^n,$	$1 \leq \frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} \leq \frac{(M+m)^2}{4Mm} \quad \boldsymbol{y} \neq \boldsymbol{0}, \boldsymbol{y} \in \mathbb{R}^n,$	
lyche-numerical-linear-algebra-EQ0033	33	0.860	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} (-z_{n+1} \boldsymbol{b}) = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} (-z_{n+1} \boldsymbol{b}) = \boldsymbol{b},$	
lyche-numerical-linear-algebra-EQ0451	180	0.869	$\boldsymbol{p}_i = \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{q}_i = \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{r}_j = \begin{bmatrix} \boldsymbol{u}_j \\ \boldsymbol{0} \end{bmatrix}, \text{ for } i = 1, \dots, n, j = n+1, \dots, m.$	$\boldsymbol{p}_i = \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{q}_i = \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{r}_j = \begin{bmatrix} \boldsymbol{u}_j \\ \boldsymbol{0} \end{bmatrix}, \text{ for } i = 1, \dots, n, j = n+1, \dots, m.$	
lyche-numerical-linear-algebra-EQ0835 (13.1)	290	0.869	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_A := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _A := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_A := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _A := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	

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lyche-numerical-linear-algebra-EQ0910 (13.46)	310	0.869	$\begin{aligned} & \left \left \boldsymbol{\epsilon}_k \right _2^2 - \left \boldsymbol{\epsilon}_{k+1} \right _2^2 \right = \alpha_k \left(2 \boldsymbol{p}_k^T \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k^T \boldsymbol{p}_k \right) \\ & \stackrel{(13.41)}{=} \alpha_k \left(2 \sum_{i=k+1}^m \alpha_i \boldsymbol{p}_i^T \boldsymbol{p}_k + \alpha_k \boldsymbol{p}_k^T \boldsymbol{p}_k \right) = \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \alpha_k \alpha_i \boldsymbol{p}_i^T \boldsymbol{p}_k \\ & \stackrel{(13.43)}{=} \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k \\ & \stackrel{(13.42)}{=} \frac{\left\ \boldsymbol{p}_k \right\ _2^2}{\left\ \boldsymbol{p}_k \right\ _A^2} \left(\left\ \boldsymbol{\epsilon}_k \right\ _A^2 + \left\ \boldsymbol{\epsilon}_{k+1} \right\ _A^2 \right). \end{aligned}$	$\left\ \boldsymbol{\epsilon}_k \right\ _2^2 - \left\ \boldsymbol{\epsilon}_{k+1} \right\ _2^2 = \alpha_k \left(2 \boldsymbol{p}_k^T \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k^T \boldsymbol{p}_k \right)$ $\stackrel{(13.41)}{=} \alpha_k \left(2 \sum_{i=k+1}^m \alpha_i \boldsymbol{p}_i^T \boldsymbol{p}_k + \alpha_k \boldsymbol{p}_k^T \boldsymbol{p}_k \right) = \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \alpha_k \alpha_i \boldsymbol{p}_i^T \boldsymbol{p}_k$ $\stackrel{(13.43)}{=} \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k$ $\stackrel{(13.42)}{=} \frac{\left\ \boldsymbol{p}_k \right\ _2^2}{\left\ \boldsymbol{p}_k \right\ _A^2} \left(\left\ \boldsymbol{\epsilon}_k \right\ _A^2 + \left\ \boldsymbol{\epsilon}_{k+1} \right\ _A^2 \right).$	$\left\ \boldsymbol{\epsilon}_k \right\ _2^2 - \left\ \boldsymbol{\epsilon}_{k+1} \right\ _2^2 = \alpha_k \left(2 \boldsymbol{p}_k^T \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k^T \boldsymbol{p}_k \right)$ $\stackrel{(13.41)}{=} \alpha_k \left(2 \sum_{i=k+1}^m \alpha_i \boldsymbol{p}_i^T \boldsymbol{p}_k + \alpha_k \boldsymbol{p}_k^T \boldsymbol{p}_k \right) = \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \alpha_k \alpha_i \boldsymbol{p}_i^T \boldsymbol{p}_k$ $\stackrel{(13.43)}{=} \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k$ $\stackrel{(13.42)}{=} \frac{\left\ \boldsymbol{p}_k \right\ _2^2}{\left\ \boldsymbol{p}_k \right\ _A^2} \left(\left\ \boldsymbol{\epsilon}_k \right\ _A^2 + \left\ \boldsymbol{\epsilon}_{k+1} \right\ _A^2 \right).$
lyche-numerical-linear-algebra-EQ1014 (15.1)	344	0.870	$\begin{aligned} \boldsymbol{z}_k &:= \boldsymbol{A}^k \boldsymbol{z}_0 = \boldsymbol{A} \boldsymbol{z}_{k-1}, \quad k = 1, 2, \dots \\ \boldsymbol{z}_0 &:= \boldsymbol{A} \boldsymbol{z}_{-1}, \quad \text{for } k=1, 2, \dots \end{aligned}$	$\boldsymbol{z}_k := \boldsymbol{A}^k \boldsymbol{z}_0 = \boldsymbol{A} \boldsymbol{z}_{k-1}, \quad k = 1, 2, \dots$	$\boldsymbol{z}_k := \boldsymbol{A}^k \boldsymbol{z}_0 = \boldsymbol{A} \boldsymbol{z}_{k-1}, \quad k = 1, 2, \dots$
lyche-numerical-linear-algebra-EQ0591	218	0.877	$\begin{aligned} E(\boldsymbol{x}) &:= \left\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b} \right\ _2^2 = (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b})^* (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} - 2 \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{b}. \end{aligned}$	$E(\boldsymbol{x}) := \left\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b} \right\ _2^2 = (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b})^* (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} - 2 \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{b}.$	$E(\boldsymbol{x}) := \left\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b} \right\ _2^2 = (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b})^* (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} - 2 \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{b}.$
lyche-numerical-linear-algebra-EQ0181	78	0.880	$\begin{aligned} & [1], [5], [9], \left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array} \right], \left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array} \right], \boldsymbol{A}. \end{aligned}$	$[1], [5], [9], \left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array} \right], \left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array} \right], \boldsymbol{A}.$	$[1], [5], [9], \left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array} \right], \left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array} \right], \boldsymbol{A}.$

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lyche-numerical-linear-algebra-EQ0323	132	0.881	$\boldsymbol{A}=\left[\begin{array}{rrr} 1 & 3 & 1 \\ 3 & 7 & 1 \\ 1 & -1 & -4 \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{array}\right]\times\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}\boldsymbol{R}.$	$\boldsymbol{A}=\left[\begin{array}{rrr} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{array}\right]\times\left[\begin{array}{rrr} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{array}\right]=\boldsymbol{Q}\boldsymbol{R}.$	
lyche-numerical-linear-algebra-EQ0314	127	0.884	$\boldsymbol{H}\boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{y} \\ a\boldsymbol{e}_1 \end{array}\right], \text{ where } \boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]-\left[\begin{array}{r} \boldsymbol{0} \\ \hat{\boldsymbol{u}} \end{array}\right]\left[\begin{array}{rr} \boldsymbol{0} & \hat{\boldsymbol{u}}^* \end{array}\right]=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{H}} \end{array}\right],$	$\boldsymbol{H}\boldsymbol{x}=\left[\begin{array}{r} \boldsymbol{y} \\ a\boldsymbol{e}_1 \end{array}\right], \text{ where } \boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]-\left[\begin{array}{r} \boldsymbol{0} \\ \hat{\boldsymbol{u}} \end{array}\right]\left[\begin{array}{rr} \boldsymbol{0} & \hat{\boldsymbol{u}}^* \end{array}\right]=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{H}} \end{array}\right],$	
lyche-numerical-linear-algebra-EQ0367 (6.5)	150	0.884	$\boldsymbol{J}:=\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}=\text{diag}(\boldsymbol{U}_1,\dots,\boldsymbol{U}_k), \text{ with } \boldsymbol{U}_i\in\mathbb{C}^{a_i\times a_i},$	$\boldsymbol{J}:=\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}=\text{diag}(\boldsymbol{U}_1,\dots,\boldsymbol{U}_k), \text{ with } \boldsymbol{U}_i\in\mathbb{C}^{a_i\times a_i},$	
lyche-numerical-linear-algebra-EQ0594 (9.5)	219	0.885	$\left\{\boldsymbol{x}\in\mathbb{C}^n:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1\right\}.$	$\left\{\boldsymbol{x}\in\mathbb{C}^n:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1\right\}.$	
lyche-numerical-linear-algebra-EQ0064	42	0.887	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -6 & 3 \\ 3 & -2 & -6 \\ 6 & 3 & 2 \end{array}\right],$	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -6 & 3 \\ 3 & -2 & -6 \\ 6 & 3 & 2 \end{array}\right],$	
lyche-numerical-linear-algebra-EQ0977	330	0.892	$\begin{gathered} \left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_1\right\ _2=\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\boldsymbol{u}_1-\boldsymbol{A}\boldsymbol{u}_1\right\ _2=\left\ \boldsymbol{E}\boldsymbol{u}_1\right\ _2\leq\left\ \boldsymbol{E}\right\ _2, \\ \prod_{j=2}^n\left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_j\right\ _2\leq\prod_{j=2}^n\left(\left \mu\right +\left\ \boldsymbol{A}\boldsymbol{u}_j\right\ _2\right)\leq\left(\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1}. \end{gathered}$	$\begin{gathered} \left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_1\right\ _2=\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\boldsymbol{u}_1-\boldsymbol{A}\boldsymbol{u}_1\right\ _2=\left\ \boldsymbol{E}\boldsymbol{u}_1\right\ _2\leq\left\ \boldsymbol{E}\right\ _2, \\ \prod_{j=2}^n\left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_j\right\ _2\leq\prod_{j=2}^n\left(\left \mu\right +\left\ \boldsymbol{A}\boldsymbol{u}_j\right\ _2\right)\leq\left(\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1}. \end{gathered}$	
lyche-numerical-linear-algebra-EQ0147	66	0.896	$\left[\begin{array}{rr} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right]\left[\begin{array}{rr} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]=\boldsymbol{I}$	$\left[\begin{array}{rr} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right]\left[\begin{array}{rr} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]=\boldsymbol{I}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-EQ0825	286	■ 0.896	$\rho\left(\boldsymbol{G}_J\right)=\sqrt{\left a_{21}a_{12}/a_{11}a_{22}\right }.$ $a_{12} / a_{11} a_{22}\right } .$	$\rho(\boldsymbol{G}_J) = \sqrt{ a_{21}a_{12}/a_{11}a_{22} }.$	$\rho(\boldsymbol{G}_J) = \sqrt{ a_{21}a_{12}/a_{11}a_{22} }.$
lyche-numerical-linear-algebra-EQ0989	333	■ 0.899	$\boldsymbol{A}_{\boldsymbol{k}}=\left[\begin{array}{ll}\boldsymbol{B}_{\boldsymbol{k}} & \boldsymbol{C}_{\boldsymbol{k}} \\ \boldsymbol{D}_{\boldsymbol{k}} & \boldsymbol{E}_{\boldsymbol{k}}\end{array}\right].$	$\boldsymbol{A}_k = \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array} \right].$	$\boldsymbol{A}_k = \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array} \right].$
lyche-numerical-linear-algebra-EQ0824 (1.9)	29	■ 0.902	$\boldsymbol{s}_{\boldsymbol{j}}=\sum_{\boldsymbol{i}=1}^{\boldsymbol{m}}\boldsymbol{a}_{\boldsymbol{i}\boldsymbol{j}}$ $\boldsymbol{v}_{\boldsymbol{i}}\text{ for } \boldsymbol{j}=1,\ldots,\boldsymbol{n} .$	$\boldsymbol{s}_j = \sum_{i=1}^m a_{ij} \boldsymbol{v}_i \text{ for } j = 1, \dots, n.$	$\boldsymbol{s}_j = \sum_{i=1}^m a_{ij} \boldsymbol{v}_i \text{ for } j = 1, \dots, n.$
lyche-numerical-linear-algebra-EQ0614	225	■ 0.902	$\boldsymbol{A}=\left[\begin{array}{ll}1&1\\0&0\end{array}\right],\quad\boldsymbol{b}=\left[\begin{array}{l}10^{-4}\\0\\1\end{array}\right],\quad\boldsymbol{e}=\left[\begin{array}{l}10^{-6}\\0\\0\end{array}\right].$	$\boldsymbol{A} = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array} \right], \quad \boldsymbol{b} = \left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array} \right], \quad \boldsymbol{e} = \left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array} \right].$	$\boldsymbol{A} = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array} \right], \quad \boldsymbol{b} = \left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array} \right], \quad \boldsymbol{e} = \left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array} \right].$
lyche-numerical-linear-algebra-EQ0168	74	■ 0.904	$\boldsymbol{A}:=\left[\begin{array}{l}\lambda&\boldsymbol{a}^T\\0&\boldsymbol{A}_1\end{array}\right],\quad\boldsymbol{B}:=\left[\begin{array}{l}1&\boldsymbol{0}^T\\0&\boldsymbol{B}_1\end{array}\right],\quad\boldsymbol{C}:=\left[\begin{array}{l}1&\boldsymbol{0}^T\\0&\boldsymbol{C}_1\end{array}\right],$	$\boldsymbol{A} := \left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \\ 0 & \boldsymbol{A}_1 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ 0 & \boldsymbol{B}_1 \end{array} \right], \quad \boldsymbol{C} := \left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ 0 & \boldsymbol{C}_1 \end{array} \right],$	$\boldsymbol{A} := \left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \\ 0 & \boldsymbol{A}_1 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ 0 & \boldsymbol{B}_1 \end{array} \right], \quad \boldsymbol{C} := \left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ 0 & \boldsymbol{C}_1 \end{array} \right],$
lyche-numerical-linear-algebra-EQ0925 (13.55)	314	■ 0.907	$\begin{aligned} &\left(d_1\boldsymbol{v}\right)_{\boldsymbol{j},\boldsymbol{k}}:=c_{\boldsymbol{j}-\frac{1}{2},\boldsymbol{k}}\left(\boldsymbol{v}_{\boldsymbol{j},\boldsymbol{k}}-\boldsymbol{v}_{\boldsymbol{j}-1,\boldsymbol{k}}\right)-c_{\boldsymbol{j}+\frac{1}{2},\boldsymbol{k}}\left(\boldsymbol{v}_{\boldsymbol{j}+1,\boldsymbol{k}}-\boldsymbol{v}_{\boldsymbol{j},\boldsymbol{k}}\right)\approx-\boldsymbol{h}^2\frac{\partial}{\partial\boldsymbol{x}}\left(c\frac{\partial\boldsymbol{u}}{\partial\boldsymbol{x}}\right)_{\boldsymbol{j},\boldsymbol{k}}, \\ &\left(d_2\boldsymbol{v}\right)_{\boldsymbol{j},\boldsymbol{k}}:=c_{\boldsymbol{j},\boldsymbol{k}-\frac{1}{2}}\left(\boldsymbol{v}_{\boldsymbol{j},\boldsymbol{k}}-\boldsymbol{v}_{\boldsymbol{j},\boldsymbol{k}-1}\right)-c_{\boldsymbol{j},\boldsymbol{k}+\frac{1}{2}}\left(\boldsymbol{v}_{\boldsymbol{j},\boldsymbol{k}+1}-\boldsymbol{v}_{\boldsymbol{j},\boldsymbol{k}}\right)\approx-\boldsymbol{h}^2\frac{\partial}{\partial\boldsymbol{y}}\left(c\frac{\partial\boldsymbol{u}}{\partial\boldsymbol{y}}\right)_{\boldsymbol{j},\boldsymbol{k}}, \end{aligned}$	$\begin{aligned} (d_1 \boldsymbol{v})_{j,k} &:= c_{j-\frac{1}{2},k} (v_{j,k} - v_{j-1,k}) - c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) \approx -h^2 \frac{\partial}{\partial x} \left(c \frac{\partial u}{\partial x} \right)_{j,k}, \\ (d_2 \boldsymbol{v})_{j,k} &:= c_{j,k-\frac{1}{2}} (v_{j,k} - v_{j,k-1}) - c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) \approx -h^2 \frac{\partial}{\partial y} \left(c \frac{\partial u}{\partial y} \right)_{j,k}, \end{aligned}$	$\begin{aligned} (d_1 \boldsymbol{v})_{j,k} &:= c_{j-\frac{1}{2},k} (v_{j,k} - v_{j-1,k}) - c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) \approx -h^2 \frac{\partial}{\partial x} \left(c \frac{\partial u}{\partial x} \right)_{j,k}, \\ (d_2 \boldsymbol{v})_{j,k} &:= c_{j,k-\frac{1}{2}} (v_{j,k} - v_{j,k-1}) - c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) \approx -h^2 \frac{\partial}{\partial y} \left(c \frac{\partial u}{\partial y} \right)_{j,k}, \end{aligned}$
lyche-numerical-linear-algebra-EQ0714 (11.6)	256	■ 0.908	$\boldsymbol{F}_{\boldsymbol{4}}=\left[\begin{array}{cccc}1&1&1&1\\1&\omega_4&\omega_4^2&\omega_4^3\\1&\omega_4^2&\omega_4^4&\omega_4^6\\1&\omega_4^3&\omega_4^6&\omega_4^9\end{array}\right]=\left[\begin{array}{cccc}1&1&1&1\\1&-i&-1&i\\1&-1&1&-1\\1&i&-1&-i\end{array}\right].$	$\boldsymbol{F}_4 = \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & \omega_4 & \omega_4^2 & \omega_4^3 \\ 1 & \omega_4^2 & \omega_4^4 & \omega_4^6 \\ 1 & \omega_4^3 & \omega_4^6 & \omega_4^9 \end{array} \right] = \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{array} \right].$	$\boldsymbol{F}_4 = \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & \omega_4 & \omega_4^2 & \omega_4^3 \\ 1 & \omega_4^2 & \omega_4^4 & \omega_4^6 \\ 1 & \omega_4^3 & \omega_4^6 & \omega_4^9 \end{array} \right] = \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{array} \right].$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-EQ0488 (8.18)	191	■ 0.909	$\begin{aligned} & \& \ \boldsymbol{A}\ _1 := \max_{1 \leq j \leq n} \sum_{k=1}^m a_{k,j} , \quad (\text{max column sum}) \\ & \ \boldsymbol{A}\ _2 := \sigma_1, \quad (\text{largest singular value of } \boldsymbol{A}) \\ & \ \boldsymbol{A}\ _\infty = \max_{1 \leq k \leq m} \sum_{j=1}^n a_{k,j} . \quad (\text{max row sum}) \end{aligned}$	$\ \boldsymbol{A}\ _1 := \max_{1 \leq j \leq n} \ \boldsymbol{A} \boldsymbol{e}_j\ _1 = \max_{1 \leq j \leq n} \sum_{k=1}^m a_{k,j} , \quad (\text{max column sum})$ $\ \boldsymbol{A}\ _2 := \sigma_1, \quad (\text{largest singular value of } \boldsymbol{A})$ $\ \boldsymbol{A}\ _\infty = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \sum_{j=1}^n a_{k,j} . \quad (\text{max row sum})$	$\ \boldsymbol{A}\ _1 := \max_{1 \leq j \leq n} \ \boldsymbol{A} \boldsymbol{e}_j\ _1 = \max_{1 \leq j \leq n} \sum_{k=1}^m a_{k,j} , \quad (\text{max column sum})$ $\ \boldsymbol{A}\ _2 := \sigma_1, \quad (\text{largest singular value of } \boldsymbol{A})$ $\ \boldsymbol{A}\ _\infty = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \sum_{j=1}^n a_{k,j} . \quad (\text{max row sum})$
lyche-numerical-linear-algebra-EQ0303	124	■ 0.910	$\boldsymbol{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 1 - u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1 - u_2^2 \end{bmatrix}.$	$\boldsymbol{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 1 - u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1 - u_2^2 \end{bmatrix}.$	$\boldsymbol{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 1 - u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1 - u_2^2 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0252	103	■ 0.911	$z^* \boldsymbol{A} z = (\boldsymbol{x} - i \boldsymbol{y})^T \boldsymbol{A} (\boldsymbol{x} + i \boldsymbol{y}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} - i \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} + i \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} - i^2 \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y},$	$z^* \boldsymbol{A} z = (\boldsymbol{x} - i \boldsymbol{y})^T \boldsymbol{A} (\boldsymbol{x} + i \boldsymbol{y}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} - i \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} + i \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} - i^2 \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y},$	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} = (\boldsymbol{x} - i \boldsymbol{y})^T \boldsymbol{A} (\boldsymbol{x} + i \boldsymbol{y}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} - i \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} + i \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} - i^2 \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y},$
lyche-numerical-linear-algebra-EQ0170	75	■ 0.914	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{21}^{(1)} x_1 + a_{22}^{(1)} x_2 + a_{23}^{(1)} x_3 = b_2^{(1)}, \quad \text{II} \\ & a_{31}^{(1)} x_1 + a_{32}^{(1)} x_2 + a_{33}^{(1)} x_3 = b_3^{(1)}. \quad \text{III.} \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{21}^{(1)} x_1 + a_{22}^{(1)} x_2 + a_{23}^{(1)} x_3 = b_2^{(1)}, \quad \text{II} \\ & a_{31}^{(1)} x_1 + a_{32}^{(1)} x_2 + a_{33}^{(1)} x_3 = b_3^{(1)}. \quad \text{III.} \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{21}^{(1)} x_1 + a_{22}^{(1)} x_2 + a_{23}^{(1)} x_3 = b_2^{(1)}, \quad \text{II} \\ & a_{31}^{(1)} x_1 + a_{32}^{(1)} x_2 + a_{33}^{(1)} x_3 = b_3^{(1)}. \quad \text{III.} \end{aligned}$
lyche-numerical-linear-algebra-EQ0494	194	■ 0.915	$\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F \text{ and } \ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2.$	$\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F \text{ and } \ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2.$	$\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F \text{ and } \ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2.$
lyche-numerical-linear-algebra-EQ0485	191	■ 0.917	$\begin{aligned} & \ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ & \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ . \end{aligned}$	$\ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ .$	$\ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ .$

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lyche-numerical-linear-algebra- EQ1822 (15.5)	246	■ 0.918	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 \mathbf{v}_1,$	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 \mathbf{v}_1,$	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 \mathbf{v}_1,$
lyche-numerical-linear-algebra- EQ8580	215	■ 0.924	$\begin{array}{c c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$	$\begin{array}{c c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$	$\begin{array}{c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$
lyche-numerical-linear-algebra- EQ8912 (13.45)	310	■ 0.931	$\begin{aligned} A \mathbf{x} &= \mathbf{b} \\ B A \mathbf{x} &= B \mathbf{b} \end{aligned} \quad \left(C A C^T \right) \mathbf{y} = C \mathbf{b}, \quad \mathbf{x} = C^T \mathbf{y}.$	$A \mathbf{x} = \mathbf{b}$ $B A \mathbf{x} = B \mathbf{b}$ $(C A C^T) \mathbf{y} = C \mathbf{b}, \mathbf{x} = C^T \mathbf{y}.$	$A \mathbf{x} = \mathbf{b}$ $B A \mathbf{x} = B \mathbf{b}$ $(C A C^T) \mathbf{y} = C \mathbf{b}, \text{ \& } \mathbf{x} = C^T \mathbf{y}.$
lyche-numerical-linear-algebra- EQ8229 (3.20)	92	■ 0.933	$\mathbf{A} = \mathbf{L} \mathbf{U}$	$A = LU = \begin{bmatrix} \mathbf{I} & & & \\ \mathbf{L}_{21} & \mathbf{I} & & \\ \vdots & & \ddots & \\ \mathbf{L}_{m1} & \cdots & \mathbf{L}_{m,m-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{11} & & \cdots & \mathbf{U}_{1m} \\ & \mathbf{U}_{22} & \cdots & \mathbf{U}_{2m} \\ & & \ddots & \vdots \\ & & & \mathbf{U}_{mm} \end{bmatrix}$	$A = LU = \begin{bmatrix} \mathbf{I} & & & \\ \mathbf{L}_{21} & \mathbf{I} & & \\ \vdots & & \ddots & \\ \mathbf{L}_{m1} & \cdots & \mathbf{L}_{m,m-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{11} & \cdots & \mathbf{U}_{1m} \\ & \mathbf{U}_{22} & \cdots & \mathbf{U}_{2m} \\ & & \ddots & \vdots \\ & & & \mathbf{U}_{mm} \end{bmatrix}$
lyche-numerical-linear-algebra- EQ8253	103	■ 0.934	$\nabla f(\mathbf{x}) = \left[\frac{\partial f(\mathbf{x})}{\partial x_1}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right] \in \mathbb{R}^n$	$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n, \quad Hf(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_n} \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n, \quad Hf(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_n} \end{bmatrix} \in \mathbb{R}^{n \times n}.$
lyche-numerical-linear-algebra- EQ8659 (10.3)	239	■ 0.934	$-\Delta_h v_{j,k} = f_{j,k}, \quad (jh, kh) \in \Omega_h,$	$-\Delta_h v_{j,k} = f_{j,k}, \quad (jh, kh) \in \Omega_h,$	$-\Delta_h v_{j,k} = f_{j,k}, \quad (jh, kh) \in \Omega_h,$
lyche-numerical-linear-algebra- EQ8873	303	■ 0.936	$\left\ \mathbf{x} - \mathbf{x}_k \right\ _A^2 \leq \left\ \mathbf{x} - \mathbf{x}_0 - \mathbf{w} \right\ _A^2 \leq \max_{a \leq x \leq b} Q(x) ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} = \max_{a \leq x \leq b} Q(x) ^2 \left\ \mathbf{x} - \mathbf{x}_0 \right\ _A^2,$	$\left\ \mathbf{x} - \mathbf{x}_k \right\ _A^2 \leq \left\ \mathbf{x} - \mathbf{x}_0 - \mathbf{w} \right\ _A^2 \leq \max_{a \leq x \leq b} Q(x) ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} = \max_{a \leq x \leq b} Q(x) ^2 \left\ \mathbf{x} - \mathbf{x}_0 \right\ _A^2,$	$\left\ \mathbf{x} - \mathbf{x}_k \right\ _A^2 \leq \left\ \mathbf{x} - \mathbf{x}_0 - \mathbf{w} \right\ _A^2 \leq \max_{a \leq x \leq b} Q(x) ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} = \max_{a \leq x \leq b} Q(x) ^2 \left\ \mathbf{x} - \mathbf{x}_0 \right\ _A^2,$

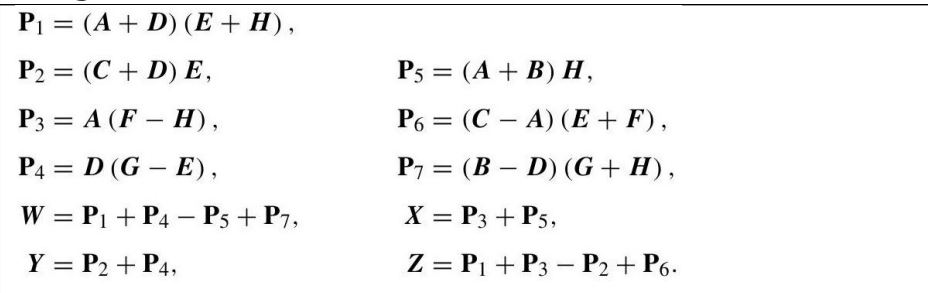
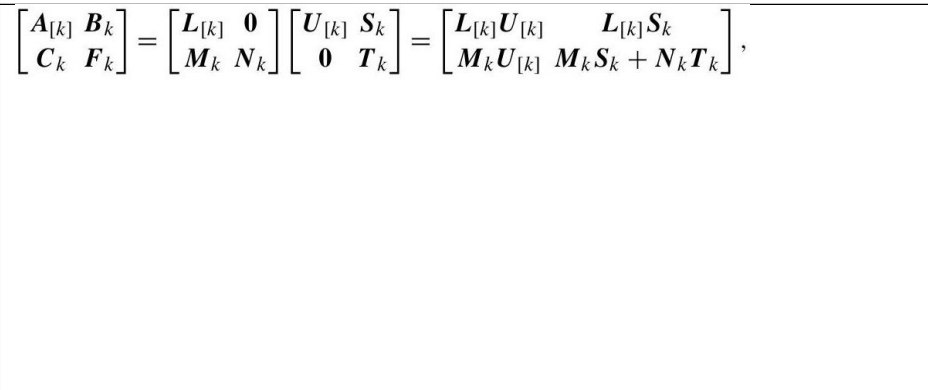
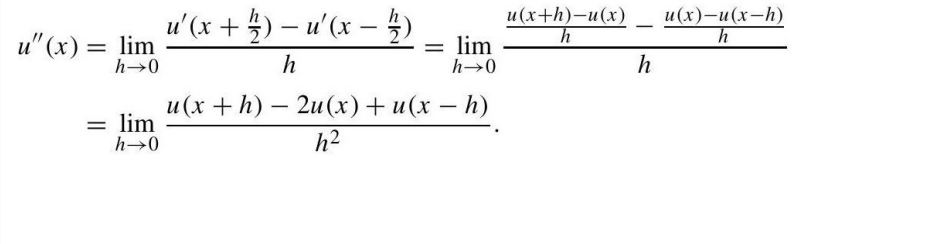
Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- EQ1052	359	■ 0.936	$\begin{aligned} & \nabla f(x, y) = \left[\begin{array}{c} 2x - y \\ -x + 2y \end{array} \right], \quad \nabla^T \boldsymbol{g}(x, y) = \left[\begin{array}{cc} 2x - y & -x + 2y \\ 1 & -1 \end{array} \right], \\ & \boldsymbol{g}(x, y) = \left[\begin{array}{c} 2x - y - x + 2y \\ 1 \end{array} \right], \quad \boldsymbol{H} f(x, y) = \left[\begin{array}{cc} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{array} \right] = \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]. \end{aligned}$	$\nabla f(x, y) = \left[\begin{array}{c} 2x - y \\ -x + 2y \end{array} \right], \quad \nabla^T \boldsymbol{g}(x, y) = \left[\begin{array}{cc} 2x - y & -x + 2y \\ 1 & -1 \end{array} \right],$ $\boldsymbol{H} f(x, y) = \left[\begin{array}{cc} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{array} \right] = \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right].$	$\nabla f(x, y) = \left[\begin{array}{c} 2x - y \\ -x + 2y \end{array} \right], \quad \nabla^T \boldsymbol{g}(x, y) = \left[\begin{array}{cc} 2x - y & -x + 2y \\ 1 & -1 \end{array} \right],$ $\boldsymbol{H} f(x, y) = \left[\begin{array}{cc} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{array} \right] = \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right].$
lyche-numerical-linear-algebra- EQ0782	276	■ 0.938	$\begin{aligned} & (\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y} + \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y} \\ & = \boldsymbol{y}^* (2\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{y} = \omega^{-1} (2 - \omega) 1 - \lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}. \end{aligned}$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y} + \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$ $= \boldsymbol{y}^* (2\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{y} = \omega^{-1} (2 - \omega) 1 - \lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}.$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y} + \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$ $= \boldsymbol{y}^* (2\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{y} = \omega^{-1} (2 - \omega) 1 - \lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}.$
lyche-numerical-linear-algebra- EQ0330	134	■ 0.938	$\boldsymbol{P} := \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right], \text{ where } c^2 + s^2 = 1.$	$\boldsymbol{P} := \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right], \text{ where } c^2 + s^2 = 1.$	$\boldsymbol{P} := \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right], \text{ where } c^2 + s^2 = 1.$
lyche-numerical-linear-algebra- EQ0770 (12.17)	273	■ 0.939	$\rho(\boldsymbol{A}) := \max_{\lambda \in \sigma(\boldsymbol{A})} \lambda .$	$\rho(\boldsymbol{A}) := \max_{\lambda \in \sigma(\boldsymbol{A})} \lambda .$	$\rho(\boldsymbol{A}) := \max_{\lambda \in \sigma(\boldsymbol{A})} \lambda .$
lyche-numerical-linear-algebra- EQ0482	190	■ 0.940	$\max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq 0} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{A} \boldsymbol{y}\ .$	$\max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq 0} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{A} \boldsymbol{y}\ .$	$\max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq 0} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{A} \boldsymbol{y}\ .$
lyche-numerical-linear-algebra- EQ0604 (9.9)	222	■ 0.940	$\boldsymbol{A}^\dagger = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$	$\boldsymbol{A}^\dagger = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$	$\boldsymbol{A}^\dagger = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$
lyche-numerical-linear-algebra- EQ0806 (12.35)	282	■ 0.941	$\ (I - \boldsymbol{B})^{-1}\ \leq \frac{1}{1 - \ \boldsymbol{B}\ }.$	$\ (I - \boldsymbol{B})^{-1}\ \leq \frac{1}{1 - \ \boldsymbol{B}\ }.$	$\ (I - \boldsymbol{B})^{-1}\ \leq \frac{1}{1 - \ \boldsymbol{B}\ }.$
lyche-numerical-linear-algebra- EQ1012	342	■ 0.942	$\boldsymbol{A}_n = \left[\begin{array}{ccccc} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{array} \right] \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A}_n = \left[\begin{array}{ccccc} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{array} \right] \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A}_n = \left[\begin{array}{ccccc} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{array} \right] \in \mathbb{R}^{n \times n}.$

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lyche-numerical-linear-algebra-EQ0174	76	0.942	$\begin{aligned} \boldsymbol{L} \boldsymbol{U} &:= \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array}\right] \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array}\right] \\ &= \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{array}\right] = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array}\right] = \boldsymbol{A}. \end{aligned}$	$\boldsymbol{LU} := \begin{bmatrix} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{bmatrix} \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{bmatrix}$ $= \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{bmatrix} = \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{bmatrix} = \boldsymbol{A}.$	
lyche-numerical-linear-algebra-EQ0628	229	0.943	$\boldsymbol{C} := \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array}\right] \text{ and } \boldsymbol{D} := \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0} \end{array}\right] \in \mathbb{C}^{(m+n) \times (m+n)}.$	$\boldsymbol{C} := \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \text{ and } \boldsymbol{D} := \begin{bmatrix} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0} \end{bmatrix} \in \mathbb{C}^{(m+n) \times (m+n)}.$	
lyche-numerical-linear-algebra-EQ1054	359	0.944	$\begin{aligned} g(0) &= f(\boldsymbol{x}) \quad g(1) = f(\boldsymbol{x} + \boldsymbol{h}), \\ g'(t) &= \sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i} = \boldsymbol{h}^T \nabla f(\boldsymbol{x} + t\boldsymbol{h}), \\ g''(t) &= \sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i \partial x_j} = \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x} + t\boldsymbol{h}) \boldsymbol{h}. \end{aligned}$	$g(0) = f(\boldsymbol{x}) \quad g(1) = f(\boldsymbol{x} + \boldsymbol{h}),$ $g'(t) = \sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i} = \boldsymbol{h}^T \nabla f(\boldsymbol{x} + t\boldsymbol{h}),$ $g''(t) = \sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i \partial x_j} = \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x} + t\boldsymbol{h}) \boldsymbol{h}.$	
lyche-numerical-linear-algebra-EQ0373 (6.8)	152	0.945	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda_1 & \boldsymbol{x}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right], \text{ for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text{ and } \boldsymbol{x} \in \mathbb{C}^k.$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda_1 & \boldsymbol{x}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right], \text{ for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text{ and } \boldsymbol{x} \in \mathbb{C}^k.$	

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lyche-numerical-linear-algebra-E09533	203	■ 0.946	$m^{\left\langle 2\right\rangle \left\lceil \angle \frac{n}{m}\right\rceil \boldsymbol{x},\boldsymbol{y}\right\rangle \stackrel{(8.39)}{=} m\langle n\boldsymbol{x},\boldsymbol{y}\rangle \stackrel{(8.39)}{=} mn\langle \boldsymbol{x},\boldsymbol{y}\rangle ,$	$m^2\left\langle \frac{n}{m}\boldsymbol{x},\boldsymbol{y}\right\rangle \stackrel{(8.39)}{=} m\langle n\boldsymbol{x},\boldsymbol{y}\rangle \stackrel{(8.39)}{=} mn\langle \boldsymbol{x},\boldsymbol{y}\rangle ,$	$m^2\langle \frac{n}{m}\boldsymbol{x},\boldsymbol{y}\rangle \stackrel{(8.39)}{=} m\langle n\boldsymbol{x},\boldsymbol{y}\rangle \stackrel{(8.39)}{=} mn\langle \boldsymbol{x},\boldsymbol{y}\rangle ,$
lyche-numerical-linear-algebra-E09686 (10.19)	246	■ 0.946	$\left(\boldsymbol{s}_j\otimes \boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p\otimes \boldsymbol{s}_q\right)=\frac{1}{4h^2}\delta _{j,p}\delta _{k,q},\quad j,k,p,q=1,\ldots ,m,$	$\left(\boldsymbol{s}_j\otimes \boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p\otimes \boldsymbol{s}_q\right)=\frac{1}{4h^2}\delta _{j,p}\delta _{k,q},\quad j,k,p,q=1,\ldots ,m,$	$\left(\boldsymbol{s}_j\otimes \boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p\otimes \boldsymbol{s}_q\right)=\frac{1}{4h^2}\delta _{j,p}\delta _{k,q},\quad j,k,p,q=1,\ldots ,m,$
lyche-numerical-linear-algebra-E09324	132	■ 0.946	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc}1&1&-1\\1&1&1\\1&-1&-1\\1&-1&1\end{array}\right]\times \left[\begin{array}{ccc}2&2&3\\0&4&5\\0&0&6\end{array}\right]=\boldsymbol{Q}_1\boldsymbol{R}_1.$	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc}1&1&-1\\1&1&1\\1&-1&-1\\1&-1&1\end{array}\right]\times \left[\begin{array}{ccc}2&2&3\\0&4&5\\0&0&6\end{array}\right]=\boldsymbol{Q}_1\boldsymbol{R}_1.$	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc}1&1&-1\\1&1&1\\1&-1&-1\\1&-1&1\end{array}\right]\times \left[\begin{array}{ccc}2&2&3\\0&4&5\\0&0&6\end{array}\right]=\boldsymbol{Q}_1\boldsymbol{R}_1.$
lyche-numerical-linear-algebra-E09331	135	■ 0.947	$\boldsymbol{x}=\left[\begin{array}{l}x_1\\x_2\end{array}\right]\neq \boldsymbol{0},\quad c:=\frac{x_1}{r},\quad s:=\frac{x_2}{r},\quad r:=\ \boldsymbol{x}\ _2.$	$\boldsymbol{x}=\left[\begin{array}{l}x_1\\x_2\end{array}\right]\neq \boldsymbol{0},\quad c:=\frac{x_1}{r},\quad s:=\frac{x_2}{r},\quad r:=\ \boldsymbol{x}\ _2.$	$\boldsymbol{x}=\left[\begin{array}{l}x_1\\x_2\end{array}\right]\neq \boldsymbol{0},\quad c:=\frac{x_1}{r},\quad s:=\frac{x_2}{r},\quad r:=\ \boldsymbol{x}\ _2.$
lyche-numerical-linear-algebra-E01900	337	■ 0.947	$y_0<y_1<z_1<y_2<z_2\\ \cdots <z_{k-1}<y_k<y_{k+1}.$	$y_0<y_1<z_1<y_2<z_2\cdots <z_{k-1}<y_k<y_{k+1}.$	$y_0<y_1<z_1<y_2<z_2\cdots <z_{k-1}<y_k<y_{k+1}.$
lyche-numerical-linear-algebra-E09999	337	■ 0.950	$p_{\{3\}(-M)}<0,\quad p_{\{3\}}\left(y_1\right)>0,\quad p_{\{3\}}\left(y_2\right)<0,\quad p_{\{3\}}(+M)>0.$	$p_3(-M)<0,\quad p_3(y_1)>0,\quad p_3(y_2)<0,\quad p_3(+M)>0.$	$p_3(-M)<0,\quad p_3(y_1)>0,\quad p_3(y_2)<0,\quad p_3(+M)>0.$
lyche-numerical-linear-algebra-E09948 (13.61)	321	■ 0.950	$\begin{aligned}&\left(\boldsymbol{A}\boldsymbol{p}_0,\boldsymbol{A}\boldsymbol{p}_1,\boldsymbol{A}\boldsymbol{p}_2\right)=\left[\begin{array}{ccc}8&0&0\\-4&3&0\\0&-2&16/9\end{array}\right],\\&\left(\boldsymbol{p}_0,\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3\right)=\left[\begin{array}{ccc}4&1&4/9&0\\0&2&8/9&0\\0&0&12/9&0\end{array}\right],\end{aligned}$	$\begin{aligned}&\left[\boldsymbol{A}\boldsymbol{p}_0,\boldsymbol{A}\boldsymbol{p}_1,\boldsymbol{A}\boldsymbol{p}_2\right]=\left[\begin{array}{ccc}8&0&0\\-4&3&0\\0&-2&16/9\end{array}\right],\\&\left[\boldsymbol{p}_0,\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3\right]=\left[\begin{array}{ccc}4&1&4/9&0\\0&2&8/9&0\\0&0&12/9&0\end{array}\right],\end{aligned}$	$\begin{aligned}&\left[\boldsymbol{A}\boldsymbol{p}_0,\boldsymbol{A}\boldsymbol{p}_1,\boldsymbol{A}\boldsymbol{p}_2\right]=\left[\begin{array}{ccc}8&0&0\\-4&3&0\\0&-2&16/9\end{array}\right],\\&\left[\boldsymbol{p}_0,\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3\right]=\left[\begin{array}{ccc}4&1&4/9&0\\0&2&8/9&0\\0&0&12/9&0\end{array}\right],\end{aligned}$
lyche-numerical-linear-algebra-E09058	41	■ 0.951	$\left[\begin{array}{ll}a&b\\c&d\end{array}\right]^{-1}=\alpha\left[\begin{array}{cc}d&-b\\-c&a\end{array}\right],\quad \alpha=\frac{1}{ad-bc},$	$\left[\begin{array}{cc}a&b\\c&d\end{array}\right]^{-1}=\alpha\left[\begin{array}{cc}d&-b\\-c&a\end{array}\right],\quad \alpha=\frac{1}{ad-bc},$	$\left[\begin{array}{cc}a&b\\c&d\end{array}\right]^{-1}=\alpha\left[\begin{array}{cc}d&-b\\-c&a\end{array}\right],\quad \alpha=\frac{1}{ad-bc},$
lyche-numerical-linear-algebra-E09220	89	■ 0.951	$\left[\begin{array}{ll}a&b\\c&d\end{array}\right]=\left[\begin{array}{cc}1&0\\l_1&1\end{array}\right]\left[\begin{array}{cc}u_1&u_2\\0&u_3\end{array}\right]=\left[\begin{array}{cc}u_1&u_2\\u_1l_1&u_2l_1+u_3\end{array}\right]$	$\left[\begin{array}{cc}a&b\\c&d\end{array}\right]=\left[\begin{array}{cc}1&0\\l_1&1\end{array}\right]\left[\begin{array}{cc}u_1&u_2\\0&u_3\end{array}\right]=\left[\begin{array}{cc}u_1&u_2\\u_1l_1&u_2l_1+u_3\end{array}\right]$	$\left[\begin{array}{cc}a&b\\c&d\end{array}\right]=\left[\begin{array}{cc}1&0\\l_1&1\end{array}\right]\left[\begin{array}{cc}u_1&u_2\\0&u_3\end{array}\right]=\left[\begin{array}{cc}u_1&u_2\\u_1l_1&u_2l_1+u_3\end{array}\right]$

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lyche-numerical-linear-algebra- EQ0110	58	■ 0.951	$\boldsymbol{A}_1=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_3=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right].$	$\boldsymbol{A}_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \boldsymbol{A}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{A}_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$	$\boldsymbol{A}_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \boldsymbol{A}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{A}_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0278	117	■ 0.953	$\angle \boldsymbol{x}, \boldsymbol{y} = \angle \boldsymbol{y}, \boldsymbol{x} \quad (\text{symmetry}).$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle \quad (\text{symmetry}).$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle \quad (\text{symmetry}).$
lyche-numerical-linear-algebra- EQ0570	211	■ 0.953	$\frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _\infty}{\ \boldsymbol{s}\ _\infty} \quad \text{and} \quad \frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$	$\frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _\infty}{\ \boldsymbol{s}\ _\infty} \quad \text{and} \quad \frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$	$\frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _\infty}{\ \boldsymbol{s}\ _\infty} \quad \text{and} \quad \frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$
lyche-numerical-linear-algebra- EQ0620 (9.16)	227	■ 0.953	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$
lyche-numerical-linear-algebra- EQ0722	257	■ 0.954	$\boldsymbol{P}_4=\left[\begin{array}{lll} \boldsymbol{e}_1 & \boldsymbol{e}_3 & \boldsymbol{e}_2 \end{array}\right] = \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \quad \boldsymbol{F}_4\boldsymbol{P}_4 = \left[\begin{array}{cc cc} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array}\right],$	$\boldsymbol{P}_4 = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_3 & \boldsymbol{e}_2 \\ \boldsymbol{e}_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{F}_4\boldsymbol{P}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{bmatrix},$	$\boldsymbol{P}_4 = [\boldsymbol{e}_1 \ \boldsymbol{e}_3 \ \boldsymbol{e}_2 \ \boldsymbol{e}_4] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{F}_4\boldsymbol{P}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0609 (9.12)	223	■ 0.955	$\begin{gathered} \boldsymbol{P}_{\mathcal{R}}(\boldsymbol{A}) = \boldsymbol{A}\boldsymbol{A}^\dagger = \boldsymbol{U}_1\boldsymbol{U}_1^* = \sum_{j=1}^r \boldsymbol{u}_j\boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}, \\ \boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)} = \boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^\dagger = \boldsymbol{U}_2\boldsymbol{U}_2^* = \sum_{j=r+1}^m \boldsymbol{u}_j\boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}. \end{gathered}$	$\boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})} = \boldsymbol{A}\boldsymbol{A}^\dagger = \boldsymbol{U}_1\boldsymbol{U}_1^* = \sum_{j=1}^r \boldsymbol{u}_j\boldsymbol{u}_j^* \in \mathbb{C}^{m \times m},$ $\boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)} = \boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^\dagger = \boldsymbol{U}_2\boldsymbol{U}_2^* = \sum_{j=r+1}^m \boldsymbol{u}_j\boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}.$	$\boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})} = \boldsymbol{A}\boldsymbol{A}^\dagger = \boldsymbol{U}_1\boldsymbol{U}_1^* = \sum_{j=1}^r \boldsymbol{u}_j\boldsymbol{u}_j^* \in \mathbb{C}^{m \times m},$ $\boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)} = \boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^\dagger = \boldsymbol{U}_2\boldsymbol{U}_2^* = \sum_{j=r+1}^m \boldsymbol{u}_j\boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}.$
lyche-numerical-linear-algebra- EQ0753 (12.8)	270	■ 0.958	$\boldsymbol{M}\boldsymbol{x} = (\boldsymbol{M} - \boldsymbol{A})\boldsymbol{x} + \boldsymbol{b}.$	$\boldsymbol{M}\boldsymbol{x} = (\boldsymbol{M} - \boldsymbol{A})\boldsymbol{x} + \boldsymbol{b}.$	$\boldsymbol{M}\boldsymbol{x} = (\boldsymbol{M} - \boldsymbol{A})\boldsymbol{x} + \boldsymbol{b}.$

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lyche-numerical-linear-algebra- EQ0654	235	■ 0.959	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 2 \\ 3 \end{array}\right], \quad \epsilon \in \mathbb{R}.$	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 2 \\ 3 \end{array}\right], \quad \epsilon \in \mathbb{R}.$	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 2 \\ 3 \end{array}\right], \quad \epsilon \in \mathbb{R}.$
lyche-numerical-linear-algebra- EQ0443	177	■ 0.959	$\ \boldsymbol{A}\ _F \stackrel{3.}{=} \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V}\ _F = \ \boldsymbol{\Sigma}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}.$	$\ \boldsymbol{A}\ _F \stackrel{3.}{=} \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V}\ _F = \ \boldsymbol{\Sigma}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}.$	$\ \boldsymbol{A}\ _F \stackrel{3.}{=} \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V}\ _F = \ \boldsymbol{\Sigma}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}.$
lyche-numerical-linear-algebra- EQ0469 (8.5)	186	■ 0.960	$\ x\ _\infty \leq \ x\ _p \leq n^{1/p} \ x\ _\infty, \quad p \geq 1.$	$\ x\ _\infty \leq \ x\ _p \leq n^{1/p} \ x\ _\infty, \quad p \geq 1.$	$\ \boldsymbol{x}\ _\infty \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _\infty, \quad p \geq 1.$
lyche-numerical-linear-algebra- EQ0606 (9.10)	222	■ 0.961	$\begin{aligned} & \boldsymbol{A}^{\dagger} \boldsymbol{A} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*, \\ & \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*. \end{aligned}$	$\begin{aligned} \boldsymbol{A}^{\dagger} \boldsymbol{A} &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{V}_1 \boldsymbol{V}_1^* \\ \boldsymbol{A} \boldsymbol{A}^{\dagger} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{U}_1 \boldsymbol{U}_1^* \end{aligned}$	$\begin{aligned} \boldsymbol{A}^{\dagger} \boldsymbol{A} &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*, \\ \boldsymbol{A} \boldsymbol{A}^{\dagger} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*. \end{aligned}$
lyche-numerical-linear-algebra- EQ0350	141	■ 0.962	$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2,m-1} \boldsymbol{P}_{m-1,m} \boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix},$	$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2,m-1} \boldsymbol{P}_{m-1,m} \boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix},$	$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2,m-1} \boldsymbol{P}_{m-1,m} \boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0961 (13.71)	323	■ 0.962	$\boldsymbol{B} \boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{r}_{k-1},$	$\boldsymbol{B} \boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{r}_{k-1},$	$\boldsymbol{B} \boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{r}_{k-1},$
lyche-numerical-linear-algebra- EQ0332	135	■ 0.963	$\boldsymbol{P} \boldsymbol{x} = \frac{1}{r} \begin{bmatrix} x_1 & x_2 \\ -x_2 & x_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} x_1^2 + x_2^2 \\ 0 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$	$\boldsymbol{P} \boldsymbol{x} = \frac{1}{r} \begin{bmatrix} x_1 & x_2 \\ -x_2 & x_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} x_1^2 + x_2^2 \\ 0 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$	$\boldsymbol{P} \boldsymbol{x} = \frac{1}{r} \begin{bmatrix} x_1 & x_2 \\ -x_2 & x_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} x_1^2 + x_2^2 \\ 0 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$

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lyche-numerical-linear-algebra-EQ0057	40	0.969	$\begin{array}{ll} \mathbf{P}_1=(\mathbf{A}+\mathbf{D})(\mathbf{E}+\mathbf{H}), & \\ \mathbf{P}_2=(\mathbf{C}+\mathbf{D})\mathbf{E}, & \mathbf{P}_5=(\mathbf{A}+\mathbf{B})\mathbf{H}, \\ \mathbf{P}_3=\mathbf{A}(\mathbf{F}-\mathbf{H}), & \mathbf{P}_6=(\mathbf{C}-\mathbf{A})(\mathbf{E}+\mathbf{F}), \\ \mathbf{P}_4=\mathbf{D}(\mathbf{G}-\mathbf{E}), & \mathbf{P}_7=(\mathbf{B}-\mathbf{D})(\mathbf{G}+\mathbf{H}), \\ \mathbf{W}=\mathbf{P}_1+\mathbf{P}_4-\mathbf{P}_5+\mathbf{P}_7, & \mathbf{X}=\mathbf{P}_3+\mathbf{P}_5, \\ \mathbf{Y}=\mathbf{P}_2+\mathbf{P}_4, & \mathbf{Z}=\mathbf{P}_1+\mathbf{P}_3-\mathbf{P}_2+\mathbf{P}_6. \end{array}$		
lyche-numerical-linear-algebra-EQ0224 (3.16)	90	0.969	$\left[\begin{array}{cc} \mathbf{A}_{[k]} & \mathbf{B}_k \\ \mathbf{C}_k & \mathbf{F}_k \end{array}\right] = \left[\begin{array}{cc} \mathbf{L}_{[k]} & \mathbf{0} \\ \mathbf{M}_k & \mathbf{N}_k \end{array}\right] \left[\begin{array}{cc} \mathbf{U}_{[k]} & \mathbf{S}_k \\ \mathbf{0} & \mathbf{T}_k \end{array}\right] = \left[\begin{array}{cc} \mathbf{L}_{[k]}\mathbf{U}_{[k]} & \mathbf{L}_{[k]}\mathbf{S}_k \\ \mathbf{M}_k\mathbf{U}_{[k]} & \mathbf{M}_k\mathbf{S}_k + \mathbf{N}_k\mathbf{T}_k \end{array}\right],$		
lyche-numerical-linear-algebra-EQ0106	57	0.970	$\begin{aligned} u^{\prime\prime}(x) &= \lim_{h \rightarrow 0} \frac{u^{\prime}\left(x+\frac{h}{2}\right)-u^{\prime}\left(x-\frac{h}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x+h)-2 u(x)+u(x-h)}{h^2} . \end{aligned}$		

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lyche-numerical-linear-algebra- EQ0680	245	■ 0.970	<pre> \begin{aligned} (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) &= \operatorname{vec}(\boldsymbol{F}) \\ &\Leftrightarrow \begin{bmatrix} \boldsymbol{A}b_{11} & \cdots & \boldsymbol{A}b_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}b_{s1} & \cdots & \boldsymbol{A}b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix} \\ &\Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s] \\ &\Leftrightarrow \boldsymbol{A} [\boldsymbol{V}b_1, \dots, \boldsymbol{V}b_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned} </pre>	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow \begin{bmatrix} \boldsymbol{A}b_{11} & \cdots & \boldsymbol{A}b_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}b_{s1} & \cdots & \boldsymbol{A}b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix}$ $\Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$ $\Leftrightarrow \boldsymbol{A} [\boldsymbol{V}b_1, \dots, \boldsymbol{V}b_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow \begin{bmatrix} \boldsymbol{A}b_{11} & \cdots & \boldsymbol{A}b_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}b_{s1} & \cdots & \boldsymbol{A}b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix}$ $\Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$ $\Leftrightarrow \boldsymbol{A} [\boldsymbol{V}b_1, \dots, \boldsymbol{V}b_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$
lyche-numerical-linear-algebra- EQ0161	71	■ 0.971	<pre> a_{jj}=-1-\frac{h}{2}r\left(x_{jj}\right),\ c_{jj}=-1+\frac{h}{2}r\left(x_{jj}\right),\ d_{jj}=2+h^2q\left(x_{jj}\right)\ \text{where}\ \left\{ \right\} </pre>	$a_j = -1 - \frac{h}{2}r(x_j), c_j = -1 + \frac{h}{2}r(x_j), d_j = 2 + h^2q(x_j),$	$a_j = -1 - \frac{h}{2}r(x_j), c_j = -1 + \frac{h}{2}r(x_j), d_j = 2 + h^2q(x_j),$
lyche-numerical-linear-algebra- EQ0312 (5.14)	126	■ 0.971	<pre> \boldsymbol{u}:=\frac{\boldsymbol{z}+\boldsymbol{e}_1}{\sqrt{1+z_1}},\ \text{where}\ \boldsymbol{z}:=\bar{\rho}\boldsymbol{x}/\ \boldsymbol{x}\ _2. </pre>	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \text{ where } \boldsymbol{z} := \bar{\rho}\boldsymbol{x}/\ \boldsymbol{x}\ _2.$	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \text{ where } \boldsymbol{z} := \bar{\rho}\boldsymbol{x}/\ \boldsymbol{x}\ _2.$
lyche-numerical-linear-algebra- EQ1043	353	■ 0.971	<pre> \begin{figure} \begin{array}{l} \boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{P_{12}^*} \left[\begin{array}{llll} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{P_{23}^*} \left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{P_{34}^*} \left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{array} \right]. \end{array} \caption{Post multiplication in a QR step} \end{figure} </pre>	(not rendered)	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{23}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}.$

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lyche-numerical-linear-algebra-EQ0407	162	■ 0.971	$\backslash\boldsymbol{B}=\backslash\left[\backslash\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_0 \\ & 0 & -q_{\{0\}} \backslash \backslash 1 & 0 & \cdots & 0 & -q_{\{1\}} \backslash \backslash 0 & 1 & \\ & \cdots & 0 & -q_{\{2\}} \backslash \backslash \vdots & \vdots & \ddots & \vdots & \\ & & 0 & 0 & \cdots & 1 & -q_{\{n-1\}} \end{array}\right] .$	$\boldsymbol{B} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -q_0 \\ 1 & 0 & \cdots & 0 & -q_1 \\ 0 & 1 & \cdots & 0 & -q_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{bmatrix} .$	$\boldsymbol{B} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -q_0 \\ 1 & 0 & \cdots & 0 & -q_1 \\ 0 & 1 & \cdots & 0 & -q_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{bmatrix} .$
lyche-numerical-linear-algebra-EQ0289 (5.8)	120	■ 0.971	$\backslash\boldsymbol{v}_{\{1\}}:=\boldsymbol{s}_{\{1\}}, \quad \backslash\boldsymbol{v}_{\{j\}}:=\boldsymbol{s}_{\{j\}}-\backslash\sum_{i=1}^{\{j-1\}}\frac{\backslash\left\langle\backslash\mathrm{angle}\boldsymbol{s}_{\{j\}}, \boldsymbol{v}_{\{i\}}\right\rangle\backslash\left\langle\backslash\mathrm{angle}\boldsymbol{v}_{\{i\}}, \boldsymbol{v}_{\{i\}}\right\rangle}{\backslash\left\langle\backslash\mathrm{angle}\boldsymbol{v}_{\{i\}}, \boldsymbol{v}_{\{i\}}\right\rangle} \boldsymbol{v}_{\{i\}}, \quad \backslash\mathrm{quad} \ j=2, \backslash\mathrm{dots}, \ k .$	$\boldsymbol{v}_1 := \boldsymbol{s}_1, \quad \boldsymbol{v}_j := \boldsymbol{s}_j - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \quad j = 2, \dots, k.$	$\boldsymbol{v}_1 := \boldsymbol{s}_1, \quad \boldsymbol{v}_j := \boldsymbol{s}_j - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \quad j = 2, \dots, k.$
lyche-numerical-linear-algebra-EQ0783	276	■ 0.973	$(\backslash\boldsymbol{A} \ \backslash\boldsymbol{s})^{\{*\}} \ \backslash\boldsymbol{y}+\backslash\boldsymbol{y}^{\{*\}}(\backslash\lambda \ \backslash\boldsymbol{A} \ \backslash\boldsymbol{x})=(1-\backslash\lambda) \ \backslash\boldsymbol{x}+\backslash\lambda(1-\overline{\backslash\lambda}) \ \backslash\boldsymbol{x}=\left(1-\backslash\left \lambda\right ^2\right) \ \backslash\boldsymbol{x},$	$(\boldsymbol{Ax})^* \boldsymbol{y} + \boldsymbol{y}^*(\lambda \boldsymbol{Ax}) = (1 - \lambda) \boldsymbol{x}^* \boldsymbol{Ax} + \lambda(1 - \bar{\lambda}) \boldsymbol{x}^* \boldsymbol{Ax} = (1 - \lambda ^2) \boldsymbol{x}^* \boldsymbol{Ax},$	$(\boldsymbol{Ax})^* \boldsymbol{y} + \boldsymbol{y}^*(\lambda \boldsymbol{Ax}) = (1 - \lambda) \boldsymbol{x}^* \boldsymbol{Ax} + \lambda(1 - \bar{\lambda}) \boldsymbol{x}^* \boldsymbol{Ax} = (1 - \lambda ^2) \boldsymbol{x}^* \boldsymbol{Ax},$
lyche-numerical-linear-algebra-EQ0677 (10.14)	243	■ 0.974	$(\backslash\boldsymbol{A} \ \backslash\boldsymbol{B}) \ \backslash\boldsymbol{C} \ \backslash\boldsymbol{D})=(\backslash\boldsymbol{A} \ \backslash\boldsymbol{C}) \ \backslash\boldsymbol{B} \ \backslash\boldsymbol{D}) .$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = (\boldsymbol{AC}) \otimes (\boldsymbol{BD}).$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = (\boldsymbol{AC}) \otimes (\boldsymbol{BD}).$
lyche-numerical-linear-algebra-EQ0527	202	■ 0.975	$\backslash\backslash\boldsymbol{x}+\boldsymbol{y}\backslash\backslash^{\{2\}}+\backslash\backslash\boldsymbol{x}-\boldsymbol{y}\backslash\backslash^{\{2\}}=\backslash\mathrm{angle}\boldsymbol{x}+\boldsymbol{y}, \ \backslash\boldsymbol{x}+\boldsymbol{y}\backslash\backslash\mathrm{angle}+\backslash\mathrm{angle}\boldsymbol{x}-\boldsymbol{y}, \ \backslash\boldsymbol{x}-\boldsymbol{y}\backslash\backslash\mathrm{angle}=2\backslash\backslash\boldsymbol{x}\backslash\backslash^{\{2\}}+2\backslash\backslash\boldsymbol{y}\backslash\backslash^{\{2\}}$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 + \ \boldsymbol{x} - \boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle + \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle = 2\ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 + \ \boldsymbol{x} - \boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle + \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle = 2\ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{y}\ ^2$
lyche-numerical-linear-algebra-EQ0994 (14.6)	336	■ 0.975	$\backslash\boldsymbol{A}=\backslash\left[\backslash\begin{array}{ccccc} d_{\{1\}} & c_{\{1\}} & & & \\ & d_{\{2\}} & c_{\{2\}} & & \\ & & \ddots & \ddots & \\ & & & d_{\{n-1\}} & c_{\{n-1\}} \\ & & & & d_n \end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} d_1 & c_1 & & & \\ c_1 & d_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{bmatrix} .$	$\boldsymbol{A} = \begin{bmatrix} d_1 & c_1 & & & \\ c_1 & d_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{bmatrix} .$
lyche-numerical-linear-algebra-EQ0081 (2.4)	47	■ 0.975	$g(x):= \backslash\begin{cases} p_{\{1\}}(x), & \backslash\mathrm{if} \ x_{\{1\}} \leq x < x_{\{2\}}, \\ p_{\{2\}}(x), & \backslash\mathrm{if} \ x_{\{2\}} \leq x < x_{\{3\}}, \\ \vdots & \\ p_{\{n-1\}}(x), & \backslash\mathrm{if} \ x_{\{n-1\}} \leq x < x_n, \\ p_{\{n\}}(x), & \backslash\mathrm{if} \ x_{\{n\}} \leq x \leq x_{\{n+1\}}, \end{cases}$	$g(x) := \begin{cases} p_1(x), & \text{if } x_1 \leq x < x_2, \\ p_2(x), & \text{if } x_2 \leq x < x_3, \\ \vdots & \\ p_{n-1}(x), & \text{if } x_{n-1} \leq x < x_n, \\ p_n(x), & \text{if } x_n \leq x \leq x_{n+1}, \end{cases}$	$g(x) := \begin{cases} p_1(x), & \text{if } x_1 \leq x < x_2, \\ p_2(x), & \text{if } x_2 \leq x < x_3, \\ \vdots & \\ p_{n-1}(x), & \text{if } x_{n-1} \leq x < x_n, \\ p_n(x), & \text{if } x_n \leq x \leq x_{n+1}, \end{cases}$
lyche-numerical-linear-algebra-EQ0998	336	■ 0.976	$p_{\{3\}}(x)=\left(x-d_{\{3\}}\right) p_{\{2\}}(x)-c_{\{2\}}^2 \backslash\left(x-d_{\{1\}}\right) p_{\{1\}}(x)=\left(x-d_{\{3\}}\right) \backslash\left(x-y_{\{1\}}\right) \backslash\left(x-y_{\{2\}}\right)-c_{\{2\}}^2 \backslash\left(x-d_{\{1\}}\right) .$	$p_3(x) = (x - d_3) p_2(x) - c_2^2 p_1(x) = (x - d_3) (x - y_1) (x - y_2) - c_2^2 (x - d_1) .$	$p_3(x) = (x - d_3) p_2(x) - c_2^2 p_1(x) = (x - d_3) (x - y_1) (x - y_2) - c_2^2 (x - d_1) .$

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lyche-numerical-linear-algebra- EQ0731	259	■ 0.976	$\begin{aligned} & \boldsymbol{F}_{2m} \boldsymbol{y} = \boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \boldsymbol{P}_{2m}^T \boldsymbol{y} = \boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \boldsymbol{w} \\ & = \left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array} \right] \left[\begin{array}{c} \boldsymbol{w}_1 \\ \boldsymbol{w}_2 \end{array} \right] = \left[\begin{array}{c} \boldsymbol{q}_1 + \boldsymbol{q}_2 \\ \boldsymbol{q}_1 - \boldsymbol{q}_2 \end{array} \right], \end{aligned}$	$\boldsymbol{F}_{2m} \boldsymbol{y} = \boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \boldsymbol{P}_{2m}^T \boldsymbol{y} = \boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \boldsymbol{w}$ $= \left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array} \right] \left[\begin{array}{c} \boldsymbol{w}_1 \\ \boldsymbol{w}_2 \end{array} \right] = \left[\begin{array}{c} \boldsymbol{q}_1 + \boldsymbol{q}_2 \\ \boldsymbol{q}_1 - \boldsymbol{q}_2 \end{array} \right],$	
lyche-numerical-linear-algebra- EQ0046	37	■ 0.976	$\left[\begin{array}{ccc} c_{i_1,j_1} & \cdots & c_{i_1,j_r} \\ \vdots & & \vdots \\ c_{i_r,j_1} & \cdots & c_{i_r,j_r} \end{array} \right] = \sum_{\boldsymbol{k}} \left[\begin{array}{ccc} a_{i_1,k_1} & \cdots & a_{i_1,k_r} \\ \vdots & & \vdots \\ a_{i_r,k_1} & \cdots & a_{i_r,k_r} \end{array} \right] \left[\begin{array}{ccc} b_{k_1,j_1} & \cdots & b_{k_1,j_r} \\ \vdots & & \vdots \\ b_{k_r,j_1} & \cdots & b_{k_r,j_r} \end{array} \right],$	$\left[\begin{array}{ccc} c_{i_1,j_1} & \cdots & c_{i_1,j_r} \\ \vdots & & \vdots \\ c_{i_r,j_1} & \cdots & c_{i_r,j_r} \end{array} \right] = \sum_{\boldsymbol{k}} \left[\begin{array}{ccc} a_{i_1,k_1} & \cdots & a_{i_1,k_r} \\ \vdots & & \vdots \\ a_{i_r,k_1} & \cdots & a_{i_r,k_r} \end{array} \right] \left[\begin{array}{ccc} b_{k_1,j_1} & \cdots & b_{k_1,j_r} \\ \vdots & & \vdots \\ b_{k_r,j_1} & \cdots & b_{k_r,j_r} \end{array} \right],$	
lyche-numerical-linear-algebra- EQ0470 (8.6)	187	■ 0.977	$\ \boldsymbol{x}\ _{p'} \leq \ \boldsymbol{x}\ _p \leq n^{1/p-1/p'} \ \boldsymbol{x}\ _{p'}, \quad \boldsymbol{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty.$	$\ \boldsymbol{x}\ _{p'} \leq \ \boldsymbol{x}\ _p \leq n^{1/p-1/p'} \ \boldsymbol{x}\ _{p'}, \quad \boldsymbol{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty.$	
lyche-numerical-linear-algebra- EQ0969	327	■ 0.977	$\begin{aligned} R_i &= \{z \in \mathbb{C} : z - a_{ii} \leq r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} , \\ C_j &= \{z \in \mathbb{C} : z - a_{jj} \leq c_j\}, \quad c_j := \sum_{\substack{i=1 \\ i \neq j}}^n a_{ij} . \end{aligned}$	$R_i = \{z \in \mathbb{C} : z - a_{ii} \leq r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} ,$ $C_j = \{z \in \mathbb{C} : z - a_{jj} \leq c_j\}, \quad c_j := \sum_{\substack{i=1 \\ i \neq j}}^n a_{ij} .$	
lyche-numerical-linear-algebra- EQ0342	137	■ 0.977	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{Q} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}, \quad \boldsymbol{R} = \begin{bmatrix} 2 & 2 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{Q} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}, \quad \boldsymbol{R} = \begin{bmatrix} 2 & 2 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$	

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lyche-numerical-linear-algebra- EQ0418 (7.2)	168	■ 0.977	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$
lyche-numerical-linear-algebra- EQ0536	204	■ 0.978	$\begin{aligned} (-a)\angle\boldsymbol{x},\boldsymbol{y}\rangle,\boldsymbol{y}\rangle\angle(-a)\boldsymbol{x},\boldsymbol{y}\rangle &= \angle(-a)\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=} \frac{1}{4}(\ \boldsymbol{-ax+y}\ ^2-\ \boldsymbol{-ax-y}\ ^2) = -\langle\boldsymbol{ax},\boldsymbol{y}\rangle, \\ \stackrel{(8.36)}{=} \frac{1}{4}\left(\ \boldsymbol{-ax+y}\ ^2-\ \boldsymbol{-ax-y}\ ^2\right) &= -\langle\boldsymbol{ax},\boldsymbol{y}\rangle, \end{aligned}$	$(-a)\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle(-a)\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ \boldsymbol{-ax+y}\ ^2-\ \boldsymbol{-ax-y}\ ^2\right)=-\langle\boldsymbol{ax},\boldsymbol{y}\rangle,$	$(-a)\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle(-a)\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ \boldsymbol{-ax+y}\ ^2-\ \boldsymbol{-ax-y}\ ^2\right)=-\langle\boldsymbol{ax},\boldsymbol{y}\rangle,$
lyche-numerical-linear-algebra- EQ0767	272	■ 0.979	$\begin{aligned} \boldsymbol{x}_{k+1} &= \left[\begin{array}{c} \boldsymbol{x}_{k+1}(1) \\ \boldsymbol{x}_{k+1}(2) \end{array}\right] = \left[\begin{array}{cc} 0 & 1/2 \\ 0 & 1/4 \end{array}\right] \left[\begin{array}{c} \boldsymbol{x}_k(1) \\ \boldsymbol{x}_k(2) \end{array}\right] + \left[\begin{array}{c} 1/2 \\ 3/4 \end{array}\right] = \boldsymbol{G}_1\boldsymbol{x}_k + \boldsymbol{c}. \end{aligned}$	$\boldsymbol{x}_{k+1}=\left[\begin{array}{c} \boldsymbol{x}_{k+1}(1) \\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0 & 1/2 \\ 0 & 1/4 \end{array}\right]\left[\begin{array}{c} \boldsymbol{x}_k(1) \\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2 \\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$	$\boldsymbol{x}_{k+1}=\left[\begin{array}{c} \boldsymbol{x}_{k+1}(1) \\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0 & 1/2 \\ 0 & 1/4 \end{array}\right]\left[\begin{array}{c} \boldsymbol{x}_k(1) \\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2 \\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$
lyche-numerical-linear-algebra- EQ0125	61	■ 0.979	$\begin{aligned} \begin{aligned} \boldsymbol{s}_j^T\boldsymbol{s}_j &= \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h)) \\ &= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2}, \end{aligned} \end{aligned}$	$\begin{aligned} \boldsymbol{s}_j^T\boldsymbol{s}_j &= \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h)) \\ &= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2}, \end{aligned}$	$\begin{aligned} \boldsymbol{s}_j^T\boldsymbol{s}_j &= \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h)) \\ &= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2}, \end{aligned}$
lyche-numerical-linear-algebra- EQ0172	76	■ 0.980	$\begin{aligned} \begin{aligned} a_{11}^{(1)}x_1+a_{12}^{(1)}x_2+a_{13}^{(1)}x_3 &=b_1^{(1)},\quad \text{I} \\ a_{22}^{(2)}x_2+a_{23}^{(2)}x_3 &=b_2^{(2)},\quad \text{II}' \\ a_{33}^{(3)}x_3 &=b_3^{(3)},\quad \text{III}'', \end{aligned} \end{aligned}$	$\begin{aligned} a_{11}^{(1)}x_1+a_{12}^{(1)}x_2+a_{13}^{(1)}x_3 &=b_1^{(1)},\text{I} \\ a_{22}^{(2)}x_2+a_{23}^{(2)}x_3 &=b_2^{(2)},\text{II}' \\ a_{33}^{(3)}x_3 &=b_3^{(3)},\text{III}'', \end{aligned}$	$\begin{aligned} a_{11}^{(1)}x_1+a_{12}^{(1)}x_2+a_{13}^{(1)}x_3 &=b_1^{(1)},\quad \text{I} \\ a_{22}^{(2)}x_2+a_{23}^{(2)}x_3 &=b_2^{(2)},\quad \text{II}' \\ a_{33}^{(3)}x_3 &=b_3^{(3)},\quad \text{III}'', \end{aligned}$
lyche-numerical-linear-algebra- EQ0703	253	■ 0.981	$\begin{aligned} \begin{aligned} \boldsymbol{T}\boldsymbol{s}_j &= \lambda_j\boldsymbol{s}_j, \quad j=1,\dots,m, \\ \boldsymbol{s}_j &= [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ \lambda_j &= 2-2\cos(j\pi h) = 4\sin^2(j\pi h/2), \quad h=1/(m+1), \\ \boldsymbol{s}_j^T\boldsymbol{s}_k &= \delta_{j,k}/(2h) \text{ for all } j,k. \end{aligned} \end{aligned}$	$\begin{aligned} \boldsymbol{T}\boldsymbol{s}_j &= \lambda_j\boldsymbol{s}_j, \quad j=1,\dots,m, \\ \boldsymbol{s}_j &= [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ \lambda_j &= 2-2\cos(j\pi h) = 4\sin^2(j\pi h/2), \quad h=1/(m+1), \\ \boldsymbol{s}_j^T\boldsymbol{s}_k &= \delta_{j,k}/(2h) \text{ for all } j,k. \end{aligned}$	$\begin{aligned} \boldsymbol{T}\boldsymbol{s}_j &= \lambda_j\boldsymbol{s}_j, \quad j=1,\dots,m, \\ \boldsymbol{s}_j &= [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ \lambda_j &= 2-2\cos(j\pi h) = 4\sin^2(j\pi h/2), \quad h=1/(m+1), \\ \boldsymbol{s}_j^T\boldsymbol{s}_k &= \delta_{j,k}/(2h) \text{ for all } j,k. \end{aligned}$

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lyche-numerical-linear-algebra-EQ0674	243	■ 0.981	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$	
lyche-numerical-linear-algebra-EQ0613 (9.13)	225	■ 0.981	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$
lyche-numerical-linear-algebra-EQ0127	62	■ 0.981	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\mu \boldsymbol{x}^* \boldsymbol{y})^* = \mu \boldsymbol{y}^* \boldsymbol{x},$	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\mu \boldsymbol{x}^* \boldsymbol{y})^* = \mu \boldsymbol{y}^* \boldsymbol{x},$	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\mu \boldsymbol{x}^* \boldsymbol{y})^* = \mu \boldsymbol{y}^* \boldsymbol{x},$
lyche-numerical-linear-algebra-EQ1020 (15.3)	345	■ 0.981	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 + \cdots + c_n \lambda_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 + \cdots + c_n \lambda_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 + \cdots + c_n \lambda_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra-EQ1021 (15.4)	345	■ 0.982	$\frac{\boldsymbol{z}_k}{\lambda_1^k} = c_1 \boldsymbol{v}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\frac{\boldsymbol{z}_k}{\lambda_1^k} = c_1 \boldsymbol{v}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\frac{\boldsymbol{z}_k}{\lambda_1^k} = c_1 \boldsymbol{v}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra-EQ0497	194	■ 0.982	$ x_i \leq y_i , i = 1, \dots, n \implies \ \boldsymbol{x}\ \leq \ \boldsymbol{y}\ , \text{ for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$ x_i \leq y_i , i = 1, \dots, n \implies \ \boldsymbol{x}\ \leq \ \boldsymbol{y}\ , \text{ for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$ x_i \leq y_i , i = 1, \dots, n \implies \ \boldsymbol{x}\ \leq \ \boldsymbol{y}\ , \text{ for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$
lyche-numerical-linear-algebra-EQ0537	204	■ 0.983	$\ \boldsymbol{x}\ _A \leq \ \boldsymbol{y}\ _A \implies \ \boldsymbol{x}\ _2 \leq \sqrt{\frac{\lambda_1}{\lambda_n}} \ \boldsymbol{y}\ _2,$	$\ \boldsymbol{x}\ _A \leq \ \boldsymbol{y}\ _A \implies \ \boldsymbol{x}\ _2 \leq \sqrt{\frac{\lambda_1}{\lambda_n}} \ \boldsymbol{y}\ _2,$	$\ \boldsymbol{x}\ _A \leq \ \boldsymbol{y}\ _A \implies \ \boldsymbol{x}\ _2 \leq \sqrt{\frac{\lambda_1}{\lambda_n}} \ \boldsymbol{y}\ _2,$
lyche-numerical-linear-algebra-EQ0573	212	■ 0.984	$\ \boldsymbol{x}\ _p \leq \ \boldsymbol{x}\ _q \leq n^{1/q-1/p} \ \boldsymbol{x}\ _p, \quad \boldsymbol{x} \in \mathbb{C}^n.$	$\ \boldsymbol{x}\ _p \leq \ \boldsymbol{x}\ _q \leq n^{1/q-1/p} \ \boldsymbol{x}\ _p, \quad \boldsymbol{x} \in \mathbb{C}^n.$	$\ \boldsymbol{x}\ _p \leq \ \boldsymbol{x}\ _q \leq n^{1/q-1/p} \ \boldsymbol{x}\ _p, \quad \boldsymbol{x} \in \mathbb{C}^n.$

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lyche-numerical-linear-algebra-E00184 (3.4)	79	■ 0.984	$\boldsymbol{L}=\left[\begin{array}{cccc} 1 & & & \\ l_{21}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ l_{n1}^{(1)} & l_{n2}^{(2)} & \cdots & 1 \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)} & \cdots & a_{1n}^{(1)} \\ & \ddots & \vdots \\ & & a_{nn}^{(n)} \end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{ccc} 1 & & \\ l_{21}^{(1)} & 1 & \\ \vdots & & \ddots \\ l_{n1}^{(1)} & l_{n2}^{(2)} & \cdots & 1 \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)} & \cdots & a_{1n}^{(1)} \\ & \ddots & \vdots \\ & & a_{nn}^{(n)} \end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{ccc} 1 & & \\ l_{21}^{(1)} & 1 & \\ \vdots & & \ddots \\ l_{n1}^{(1)} & l_{n2}^{(2)} & \cdots & 1 \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)} & \cdots & a_{1n}^{(1)} \\ & \ddots & \vdots \\ & & a_{nn}^{(n)} \end{array}\right],$
lyche-numerical-linear-algebra-E00019 (1.7)	29	■ 0.984	$\operatorname{dim}(\mathcal{S}+\mathcal{T})=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S}\cap\mathcal{T}).$	$\dim(\mathcal{S}+\mathcal{T})=\dim(\mathcal{S})+\dim(\mathcal{T})-\dim(\mathcal{S}\cap\mathcal{T}).$	$\dim(\mathcal{S}+\mathcal{T})=\dim(\mathcal{S})+\dim(\mathcal{T})-\dim(\mathcal{S}\cap\mathcal{T}).$
lyche-numerical-linear-algebra-E00502 (8.23)	196	■ 0.984	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$
lyche-numerical-linear-algebra-E00740	262	■ 0.984	$x_{j,k}=\frac{h^4g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k},\text{ where }\sigma_j=\sin^2((j\pi h)/2)\text{ for }j,k=1,2,\ldots,m.$	$x_{j,k}=\frac{h^4g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k},\text{ where }\sigma_j=\sin^2((j\pi h)/2)\text{ for }j,k=1,2,\ldots,m.$	$x_{j,k}=\frac{h^4g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k},\text{ where }\sigma_j=\sin^2((j\pi h)/2)\text{ for }j,k=1,2,\ldots,m.$
lyche-numerical-linear-algebra-E00114	59	■ 0.985	$u^{(\prime\prime)}(jh)\approx\frac{u((j+1)h)-2u(jh)+u((j-1)h)}{h^2},\quad j=1,\ldots,m,$	$u''(jh)\approx\frac{u((j+1)h)-2u(jh)+u((j-1)h)}{h^2},\quad j=1,\ldots,m,$	$u''(jh)\approx\frac{u((j+1)h)-2u(jh)+u((j-1)h)}{h^2},\quad j=1,\ldots,m,$
lyche-numerical-linear-algebra-E00564	210	■ 0.985	$\boldsymbol{A}:=\left[\begin{array}{cccccc} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{array}\right],$	$\boldsymbol{A}:=\left[\begin{array}{cccccc} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{array}\right],$	$\boldsymbol{A}:=\left[\begin{array}{cccccc} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{array}\right],$
lyche-numerical-linear-algebra-E00701 (11.3)	252	■ 0.985	$\begin{aligned} \boldsymbol{y}_1&=\boldsymbol{b}_1, \quad \boldsymbol{y}_k=\boldsymbol{b}_k-\boldsymbol{L}_{k-1}\boldsymbol{y}_{k-1}, \quad k=2,3,\ldots,m, \\ \boldsymbol{x}_m&=\boldsymbol{U}_m^{-1}\boldsymbol{y}_m, \quad \boldsymbol{x}_k=\boldsymbol{U}_k^{-1}(\boldsymbol{y}_k-\boldsymbol{C}_k\boldsymbol{x}_{k+1}), \quad k=m-1,\ldots,2,1. \end{aligned}$	$\boldsymbol{y}_1=\boldsymbol{b}_1, \quad \boldsymbol{y}_k=\boldsymbol{b}_k-\boldsymbol{L}_{k-1}\boldsymbol{y}_{k-1}, \quad k=2,3,\ldots,m, \\ \boldsymbol{x}_m=\boldsymbol{U}_m^{-1}\boldsymbol{y}_m, \quad \boldsymbol{x}_k=\boldsymbol{U}_k^{-1}(\boldsymbol{y}_k-\boldsymbol{C}_k\boldsymbol{x}_{k+1}), \quad k=m-1,\ldots,2,1.$	$\boldsymbol{y}_1=\boldsymbol{b}_1, \quad \boldsymbol{y}_k=\boldsymbol{b}_k-\boldsymbol{L}_{k-1}\boldsymbol{y}_{k-1}, \quad k=2,3,\ldots,m, \\ \boldsymbol{x}_m=\boldsymbol{U}_m^{-1}\boldsymbol{y}_m, \quad \boldsymbol{x}_k=\boldsymbol{U}_k^{-1}(\boldsymbol{y}_k-\boldsymbol{C}_k\boldsymbol{x}_{k+1}), \quad k=m-1,\ldots,2,1.$

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lyche-numerical-linear-algebra- EQ0848 (13.12)	294	■ 0.985	$\begin{aligned} \boldsymbol{p}_{\boldsymbol{j}} &:= \boldsymbol{r}_{\boldsymbol{j}} - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_{\boldsymbol{j}}^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i, \\ \boldsymbol{x}_{j+1} &:= \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j}, \\ \boldsymbol{r}_{j+1} &= \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j. \end{aligned}$	$\boldsymbol{p}_j := \boldsymbol{r}_j - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i,$ $\boldsymbol{x}_{j+1} := \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j},$ $\boldsymbol{r}_{j+1} = \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j.$	$\boldsymbol{p}_j := \boldsymbol{r}_j - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i,$ $\boldsymbol{x}_{j+1} := \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j},$ $\boldsymbol{r}_{j+1} = \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j.$
lyche-numerical-linear-algebra- EQ0210	87	■ 0.985	$\begin{array}{r} 10^{-4} x_1 + 2 x_2 = 4 \\ x_1 + x_2 = 3 \end{array}$	$10^{-4} x_1 + 2 x_2 = 4$ $x_1 + x_2 = 3$	$10^{-4} x_1 + 2 x_2 = 4$ $x_1 + x_2 = 3$
lyche-numerical-linear-algebra- EQ0429	172	■ 0.985	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0145	65	■ 0.985	$\begin{aligned} \boldsymbol{B}_{11} \boldsymbol{A}_{11} &= \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}, \\ \boldsymbol{B}_{11} \boldsymbol{A}_{21} &= \mathbf{0}, \boldsymbol{B}_{12} \boldsymbol{A}_{11} + \boldsymbol{B}_{22} \boldsymbol{A}_{21} = \mathbf{0}, \boldsymbol{B}_{12} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \mathbf{0}, \\ \boldsymbol{B}_{11} \boldsymbol{A}_{22} &= \mathbf{0}, \boldsymbol{B}_{12} \boldsymbol{A}_{21} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \mathbf{0}, \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}. \end{aligned}$	$\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}.$	$\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}.$
lyche-numerical-linear-algebra- EQ0622 (9.17)	227	■ 0.985	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$
lyche-numerical-linear-algebra- EQ0272	113	■ 0.986	$\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$	$\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$	$\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$

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lyche-numerical-linear-algebra- EQ0268	110	■ 0.986	$\boldsymbol{L} \boldsymbol{L}^* = \left[\begin{array}{l} 0 \\ 0 \end{array} \right] \left[\begin{array}{l} 0 \\ 0 \end{array} \right]^* = \left[\begin{array}{l} 0 \\ 0 \end{array} \right] \left[\begin{array}{l} 0 \\ 0 \end{array} \right]^* = \boldsymbol{A}.$	$LL^* = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & L_1 \end{bmatrix} \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & L_1^* \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & B \end{bmatrix} = A.$	$LL^* = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1 \end{bmatrix} \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1^* \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & B \end{bmatrix} = A.$
lyche-numerical-linear-algebra- EQ0205 (3.14)	86	■ 0.986	$\boldsymbol{L} = \left[\begin{array}{ccc} 1 & & \\ a_{p_2,1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{array} \right], \quad \boldsymbol{U} = \left[\begin{array}{ccc} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{array} \right].$	$L = \begin{bmatrix} 1 & & & \\ a_{p_2,1}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad U = \begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{bmatrix}.$	$L = \begin{bmatrix} 1 & & & \\ a_{p_2,1}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad U = \begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0995	336	■ 0.986	$\boldsymbol{A}_1 = \left[\begin{array}{cccc} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{array} \right] \text{ and } \boldsymbol{A}_2 = \left[\begin{array}{cccc} d_{i+1} & c_{i+1} & & \\ c_{i+1} & d_{i+2} & c_{i+2} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array} \right].$	$A_1 = \begin{bmatrix} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{bmatrix} \text{ and } A_2 = \begin{bmatrix} d_{i+1} & c_{i+1} & & \\ c_{i+1} & d_{i+2} & c_{i+2} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{bmatrix}.$	$A_1 = \begin{bmatrix} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{bmatrix} \text{ and } A_2 = \begin{bmatrix} d_{i+1} & c_{i+1} & & \\ c_{i+1} & d_{i+2} & c_{i+2} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0300	124	■ 0.987	$\langle Ux, Uy \rangle = (Uy)^*(Ux) = y^* U^* Ux = y^* x = \langle x, y \rangle.$	$\langle Ux, Uy \rangle = (Uy)^*(Ux) = y^* U^* Ux = y^* x = \langle x, y \rangle.$	$\langle Ux, Uy \rangle = (Uy)^*(Ux) = y^* U^* Ux = y^* x = \langle x, y \rangle.$
lyche-numerical-linear-algebra- EQ0426 (7.5)	172	■ 0.987	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*.$	$A = U \Sigma V^* = U_1 \Sigma_1 V_1^*.$	$A = U \Sigma V^* = U_1 \Sigma_1 V_1^*.$
lyche-numerical-linear-algebra- EQ0397 (6.15)	159	■ 0.987	$\sum_{j=1}^n \mu_{i_j} - \lambda_j ^2 \leq \sum_{i=1}^n \sum_{j=1}^n a_{ij} - b_{ij} ^2.$	$\sum_{j=1}^n \mu_{i_j} - \lambda_j ^2 \leq \sum_{i=1}^n \sum_{j=1}^n a_{ij} - b_{ij} ^2.$	$\sum_{j=1}^n \mu_{i_j} - \lambda_j ^2 \leq \sum_{i=1}^n \sum_{j=1}^n a_{ij} - b_{ij} ^2.$
lyche-numerical-linear-algebra- EQ0511	199	■ 0.987	$\ \boldsymbol{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \quad \ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ \boldsymbol{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \quad \ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ \boldsymbol{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \quad \ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$

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lyche-numerical-linear-algebra-EQ0139	64	■ 0.987	$\boldsymbol{A} \boldsymbol{B} = \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array} \right] \left[\begin{array}{l} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{array} \right], \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array} \right] \left[\begin{array}{l} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{array} \right]$	$AB = \left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right] \left[\begin{array}{l} B_{11} \\ B_{21} \end{array} \right], \left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right] \left[\begin{array}{l} B_{12} \\ B_{22} \end{array} \right]$	$AB = \left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right] \left[\begin{array}{l} B_{11} \\ B_{21} \end{array} \right], \left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right] \left[\begin{array}{l} B_{12} \\ B_{22} \end{array} \right]$
lyche-numerical-linear-algebra-EQ0164 (2.38)	73	■ 0.987	$\boldsymbol{L} = \left[\begin{array}{cccccc} 1 & 0 & \cdots & \cdots & 0 & \vdots \\ -\frac{1}{2} & 1 & \ddots & & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 & \vdots \end{array} \right], \boldsymbol{U} = \left[\begin{array}{cccccc} 2 & -1 & 0 & \cdots & 0 & \vdots \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \frac{m}{m-1} & -1 & & \vdots \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} & \vdots \end{array} \right]$	$L = \left[\begin{array}{cccccc} 1 & 0 & \cdots & \cdots & 0 & \vdots \\ -\frac{1}{2} & 1 & \ddots & & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 & \vdots \end{array} \right], U = \left[\begin{array}{cccccc} 2 & -1 & 0 & \cdots & 0 & \vdots \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \frac{m}{m-1} & -1 & & \vdots \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} & \vdots \end{array} \right]$	$L = \left[\begin{array}{cccccc} 1 & 0 & \cdots & \cdots & 0 & \vdots \\ -\frac{1}{2} & 1 & \ddots & & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 & \vdots \end{array} \right], U = \left[\begin{array}{cccccc} 2 & -1 & 0 & \cdots & 0 & \vdots \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \frac{m}{m-1} & -1 & & \vdots \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} & \vdots \end{array} \right]$
lyche-numerical-linear-algebra-EQ0572	212	■ 0.988	$S_p = \{ \boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x}\ _p = 1 \}$	$S_p = \{ \boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x}\ _p = 1 \}$	$S_p = \{ \boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x}\ _p = 1 \}$
lyche-numerical-linear-algebra-EQ0949 (13.62)	221	■ 0.988	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$ $\langle \boldsymbol{C}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{C}^T \boldsymbol{x} \rangle \stackrel{(13.61)}{=} \langle \boldsymbol{C}^T \boldsymbol{x}, \boldsymbol{y} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n \times n}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$ $\langle \boldsymbol{C}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{C}^T \boldsymbol{x} \rangle \stackrel{(13.61)}{=} \langle \boldsymbol{C}^T \boldsymbol{x}, \boldsymbol{y} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n \times n}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$ $\langle \boldsymbol{C}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{C}^T \boldsymbol{x} \rangle \stackrel{(13.61)}{=} \langle \boldsymbol{C}^T \boldsymbol{x}, \boldsymbol{y} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n \times n}.$
lyche-numerical-linear-algebra-EQ0166	73	■ 0.988	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{a}_{:1} \boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2} \boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n} \boldsymbol{b}_{:n}^T.$	$AB^T = \boldsymbol{a}_{:1} \boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2} \boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n} \boldsymbol{b}_{:n}^T.$	$AB^T = \boldsymbol{a}_{:1} \boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2} \boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n} \boldsymbol{b}_{:n}^T.$
lyche-numerical-linear-algebra-EQ0589	217	■ 0.989	$\boldsymbol{H}_3 = \left[\begin{array}{ccc} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array} \right].$	$H_3 = \left[\begin{array}{ccc} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array} \right].$	$H_3 = \left[\begin{array}{ccc} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array} \right].$

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lyche-numerical-linear-algebra-EQ0911 (13.44)	310	■ 0.989	$\begin{aligned} &\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b}, \\ &\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b}, \end{aligned}$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b},$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b},$
lyche-numerical-linear-algebra-EQ09500	195	■ 0.989	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and } \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ .$	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and } \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ .$	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and } \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ .$
lyche-numerical-linear-algebra-EQ0137	64	■ 0.989	$\begin{aligned} &\boldsymbol{A}_{ij} = \sum_{k=1}^p \boldsymbol{a}_{ik} \boldsymbol{b}_{kj} = \sum_{k=1}^s \boldsymbol{a}_{ik} \boldsymbol{b}_{kj} + \sum_{k=s+1}^p \boldsymbol{a}_{ik} \boldsymbol{b}_{kj} \\ &= (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}. \end{aligned}$	$\begin{aligned} (\boldsymbol{A} \boldsymbol{B})_{ij} &= \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj} \\ &= (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}. \end{aligned}$	$\begin{aligned} (\boldsymbol{A} \boldsymbol{B})_{ij} &= \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj} \\ &= (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}. \end{aligned}$
lyche-numerical-linear-algebra-EQ0436 (7.7)	174	■ 0.989	$\begin{aligned} &\boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ &\boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m. \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{v}_i &= \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ \boldsymbol{A}^* \boldsymbol{u}_i &= \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m. \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{v}_i &= \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ \boldsymbol{A}^* \boldsymbol{u}_i &= \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m. \end{aligned}$
lyche-numerical-linear-algebra-EQ0001	21	■ 0.990	$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$	$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$	$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0060	41	■ 0.990	$\left(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T \right)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} \left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$	$\left(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T \right)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} \left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$	$\left(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T \right)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} \left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$

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lyche-numerical-linear-algebra-E00231 (3.21)	92	■ 0.990	$\begin{aligned} \boldsymbol{A} &= \left[\begin{array}{cc} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{array} \right] \\ &= \left[\begin{array}{cc} \boldsymbol{L}_{\{m-1\}} & \boldsymbol{0} \\ \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{array} \right] \left[\begin{array}{cc} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{mm} - \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{array} \right], \end{aligned}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{\{m-1\}} & \boldsymbol{0} \\ \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{mm} - \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{bmatrix},$	
lyche-numerical-linear-algebra-E00293 (5.9)	122	■ 0.990	$\begin{aligned} \left\langle \boldsymbol{s}_0, \boldsymbol{s} \right\rangle &= \left\langle \boldsymbol{v}, \boldsymbol{s} \right\rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \left\langle \boldsymbol{t}_0, \boldsymbol{t} \right\rangle = \left\langle \boldsymbol{v}, \boldsymbol{t} \right\rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}. \\ \angle \boldsymbol{v} &= \angle \boldsymbol{v}, \quad \angle \boldsymbol{s} = \angle \boldsymbol{s}, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \\ \angle \boldsymbol{t} &= \angle \boldsymbol{t}, \quad \angle \boldsymbol{v} = \angle \boldsymbol{v}, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}. \end{aligned}$	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \langle \boldsymbol{t}_0, \boldsymbol{t} \rangle = \langle \boldsymbol{v}, \boldsymbol{t} \rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$	
lyche-numerical-linear-algebra-E00543	206	■ 0.990	$\left\ \boldsymbol{A}^{-1} \right\ _2 = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{x} \right\ _2}{\left\ \boldsymbol{A} \boldsymbol{x} \right\ _2}.$	$\left\ \boldsymbol{A}^{-1} \right\ _2 = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{x} \right\ _2}{\left\ \boldsymbol{A} \boldsymbol{x} \right\ _2}.$	
lyche-numerical-linear-algebra-E00920 (13.52)	313	■ 0.990	$\begin{aligned} -\frac{\partial}{\partial x} \left(c(x, y) \frac{\partial u}{\partial x} \right) - \frac{\partial}{\partial y} \left(c(x, y) \frac{\partial u}{\partial y} \right) &= f(x, y), \quad (x, y) \in \Omega = (0, 1)^2, \\ u(x, y) &= 0, \quad (x, y) \in \partial \Omega. \end{aligned}$	$-\frac{\partial}{\partial x} \left(c(x, y) \frac{\partial u}{\partial x} \right) - \frac{\partial}{\partial y} \left(c(x, y) \frac{\partial u}{\partial y} \right) = f(x, y), \quad (x, y) \in \Omega = (0, 1)^2, \\ u(x, y) = 0, \quad (x, y) \in \partial \Omega.$	
lyche-numerical-linear-algebra-E00679	244	■ 0.990	$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i,j} = \boldsymbol{A} \boldsymbol{C} \sum_{k=1}^t b_{i,k} d_{k,j} = (\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i,j} = ((\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}))_{i,j}.$	$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i,j} = \boldsymbol{A} \boldsymbol{C} \sum_{k=1}^t b_{i,k} d_{k,j} = (\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i,j} = ((\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}))_{i,j}.$	
lyche-numerical-linear-algebra-E00042	36	■ 0.991	$\begin{aligned} \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right &= \left \begin{array}{ccc} 1 & x & y \\ 0 & x_1 - x & y_1 - y \\ 0 & x_2 - x_1 & y_2 - y_1 \end{array} \right \\ &= \left \begin{array}{cc} x_1 - x & y_1 - y \\ x_2 - x_1 & y_2 - y_1 \end{array} \right = (x_1 - x)(y_2 - y_1) - (y_1 - y)(x_2 - x_1) \end{aligned}$	$\begin{aligned} \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right &= \left \begin{array}{ccc} 1 & x & y \\ 0 & x_1 - x & y_1 - y \\ 0 & x_2 - x_1 & y_2 - y_1 \end{array} \right \\ &= \left \begin{array}{cc} x_1 - x & y_1 - y \\ x_2 - x_1 & y_2 - y_1 \end{array} \right = (x_1 - x)(y_2 - y_1) - (y_1 - y)(x_2 - x_1). \end{aligned}$	
lyche-numerical-linear-algebra-E00213	87	■ 0.991	$\mathrm{fl}(x_2) = 2, \quad \mathrm{fl}(x_1) = 0.$	$\mathrm{fl}(x_2) = 2, \quad \mathrm{fl}(x_1) = 0.$	

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lyche-numerical-linear-algebra-EQ0234 (3.25)	94	■ 0.991	$\begin{aligned} 1+2+\cdots+m &= \frac{1}{2} m(m+1), \\ 1^2+2^2+\cdots+m^2 &= \frac{1}{3} m\left(m+\frac{1}{2}\right)(m+1), \\ 1+3+5+\cdots+2m-1 &= m^2, \\ 1*2+2*3+3*4+\cdots+(m-1)m &= \frac{1}{3}(m-1)m(m+1). \end{aligned}$	$\begin{aligned} 1+2+\cdots+m &= \frac{1}{2} m(m+1), \\ 1^2+2^2+\cdots+m^2 &= \frac{1}{3} m\left(m+\frac{1}{2}\right)(m+1), \\ 1+3+5+\cdots+2m-1 &= m^2, \\ 1*2+2*3+3*4+\cdots+(m-1)m &= \frac{1}{3}(m-1)m(m+1). \end{aligned}$	$\begin{aligned} 1+2+\cdots+m &= \frac{1}{2} m(m+1), \\ 1^2+2^2+\cdots+m^2 &= \frac{1}{3} m\left(m+\frac{1}{2}\right)(m+1), \\ 1+3+5+\cdots+2m-1 &= m^2, \\ 1*2+2*3+3*4+\cdots+(m-1)m &= \frac{1}{3}(m-1)m(m+1). \end{aligned}$
lyche-numerical-linear-algebra-EQ0468 (8.2)	186	■ 0.991	$\ \boldsymbol{x}\ _p := \ \boldsymbol{x}\ _\infty \left(\sum_{j=1}^n \left(\frac{ x_j }{\ \boldsymbol{x}\ _\infty} \right)^p \right)^{1/p}.$	$\ \boldsymbol{x}\ _p := \ \boldsymbol{x}\ _\infty \left(\sum_{j=1}^n \left(\frac{ x_j }{\ \boldsymbol{x}\ _\infty} \right)^p \right)^{1/p}.$	$\ \boldsymbol{x}\ _p := \ \boldsymbol{x}\ _\infty \left(\sum_{j=1}^n \left(\frac{ x_j }{\ \boldsymbol{x}\ _\infty} \right)^p \right)^{1/p}.$
lyche-numerical-linear-algebra-EQ0584 (9.1)	216	■ 0.991	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} p(t_1) \\ \vdots \\ p(t_m) \end{bmatrix} = \begin{bmatrix} \phi_1(t_1) & \cdots & \phi_n(t_1) \\ \vdots & & \vdots \\ \phi_1(t_m) & \cdots & \phi_n(t_m) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} =: \boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} p(t_1) \\ \vdots \\ p(t_m) \end{bmatrix} = \begin{bmatrix} \phi_1(t_1) & \cdots & \phi_n(t_1) \\ \vdots & & \vdots \\ \phi_1(t_m) & \cdots & \phi_n(t_m) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} =: \boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} p(t_1) \\ \vdots \\ p(t_m) \end{bmatrix} = \begin{bmatrix} \phi_1(t_1) & \cdots & \phi_n(t_1) \\ \vdots & & \vdots \\ \phi_1(t_m) & \cdots & \phi_n(t_m) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} =: \boldsymbol{b}.$
lyche-numerical-linear-algebra-EQ0725	258	■ 0.991	$\boldsymbol{F}_4 \boldsymbol{P}_4 = \begin{bmatrix} \boldsymbol{F}_2 & \boldsymbol{D}_2 \boldsymbol{F}_2 \\ \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{bmatrix}.$	$\boldsymbol{F}_4 \boldsymbol{P}_4 = \begin{bmatrix} \boldsymbol{F}_2 & \boldsymbol{D}_2 \boldsymbol{F}_2 \\ \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{bmatrix}.$	$\boldsymbol{F}_4 \boldsymbol{P}_4 = \begin{bmatrix} \boldsymbol{F}_2 & \boldsymbol{D}_2 \boldsymbol{F}_2 \\ \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0169	74	■ 0.991	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{B} = \begin{bmatrix} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \mathbf{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{bmatrix}.$	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{B} = \begin{bmatrix} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \mathbf{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{bmatrix}.$	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{B} = \begin{bmatrix} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \mathbf{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0047 (1.15)	37	■ 0.991	$\det(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})) = \sum_{\boldsymbol{k}} \det(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \det(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$	$\det(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})) = \sum_{\boldsymbol{k}} \det(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \det(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$	$\det(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})) = \sum_{\boldsymbol{k}} \det(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \det(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$
lyche-numerical-linear-algebra-EQ0422	170	■ 0.992	$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} &= [\sigma_1 \boldsymbol{e}_1, \dots, \sigma_r \boldsymbol{e}_r, 0, \dots, 0] \\ &= [\sigma_1 \boldsymbol{u}_1, \dots, \sigma_r \boldsymbol{u}_r, 0, \dots, 0] \\ &= [\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n]. \end{aligned}$	$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} &= \boldsymbol{U} [\sigma_1 \boldsymbol{e}_1, \dots, \sigma_r \boldsymbol{e}_r, 0, \dots, 0] \\ &= [\sigma_1 \boldsymbol{u}_1, \dots, \sigma_r \boldsymbol{u}_r, 0, \dots, 0] \\ &= [\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n]. \end{aligned}$	$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} &= \boldsymbol{U} [\sigma_1 \boldsymbol{e}_1, \dots, \sigma_r \boldsymbol{e}_r, 0, \dots, 0] \\ &= [\sigma_1 \boldsymbol{u}_1, \dots, \sigma_r \boldsymbol{u}_r, 0, \dots, 0] \\ &= [\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n]. \end{aligned}$

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lyche-numerical-linear-algebra-EQ0702	252	0.992	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}$ with $h = 1/(m+1)$, $\boldsymbol{T} = h^2 \boldsymbol{F}$ with $h = 1/(m+1)$,	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}$ with $h = 1/(m+1)$,	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}$ with $h = 1/(m+1)$,
lyche-numerical-linear-algebra-EQ0857 (13.19)	299	0.992	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k < e^{-\frac{2}{\kappa}k}, \quad k > 0,$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k < e^{-\frac{2}{\kappa}k}, \quad k > 0,$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k < e^{-\frac{2}{\kappa}k}, \quad k > 0,$
lyche-numerical-linear-algebra-EQ0687 (4h)	246	0.992	$\left(\boldsymbol{s}_j \otimes \boldsymbol{s}_k\right)^T \left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right) = \left(\boldsymbol{s}_j^T \otimes \boldsymbol{s}_k^T\right) \left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right) = \left(\boldsymbol{s}_j^T \boldsymbol{s}_p\right) \otimes \left(\boldsymbol{s}_k^T \boldsymbol{s}_q\right) = \frac{1}{4h^2} \delta_{j,p} \delta_{k,q}$	$\left(\boldsymbol{s}_j \otimes \boldsymbol{s}_k\right)^T \left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right) = \left(\boldsymbol{s}_j^T \otimes \boldsymbol{s}_k^T\right) \left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right) = \left(\boldsymbol{s}_j^T \boldsymbol{s}_p\right) \otimes \left(\boldsymbol{s}_k^T \boldsymbol{s}_q\right) = \frac{1}{4h^2} \delta_{j,p} \delta_{k,q}$	$\left(\boldsymbol{s}_j \otimes \boldsymbol{s}_k\right)^T \left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right) = \left(\boldsymbol{s}_j^T \otimes \boldsymbol{s}_k^T\right) \left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right) = \left(\boldsymbol{s}_j^T \boldsymbol{s}_p\right) \otimes \left(\boldsymbol{s}_k^T \boldsymbol{s}_q\right) = \frac{1}{4h^2} \delta_{j,p} \delta_{k,q}$
lyche-numerical-linear-algebra-EQ0775 (12.20)	274	0.992	$\left\ \boldsymbol{x}_k - \boldsymbol{x}\right\ _2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \left\ \boldsymbol{x}_0 - \boldsymbol{x}\right\ _2, \quad k = 0, 1, 2, \dots$	$\left\ \boldsymbol{x}_k - \boldsymbol{x}\right\ _2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \left\ \boldsymbol{x}_0 - \boldsymbol{x}\right\ _2, \quad k = 0, 1, 2, \dots$	$\left\ \boldsymbol{x}_k - \boldsymbol{x}\right\ _2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \left\ \boldsymbol{x}_0 - \boldsymbol{x}\right\ _2, \quad k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra-EQ0904	309	0.992	$\boldsymbol{\epsilon}_j = \boldsymbol{x}_{m+1} - \boldsymbol{x}_j = \boldsymbol{x}_m - \boldsymbol{x}_j + \alpha_m \boldsymbol{p}_m = \boldsymbol{x}_{m-1} - \boldsymbol{x}_j + \alpha_{m-1} \boldsymbol{p}_{m-1} + \alpha_m \boldsymbol{p}_m = \dots$	$\boldsymbol{\epsilon}_j = \boldsymbol{x}_{m+1} - \boldsymbol{x}_j = \boldsymbol{x}_m - \boldsymbol{x}_j + \alpha_m \boldsymbol{p}_m = \boldsymbol{x}_{m-1} - \boldsymbol{x}_j + \alpha_{m-1} \boldsymbol{p}_{m-1} + \alpha_m \boldsymbol{p}_m = \dots$	$\boldsymbol{\epsilon}_j = \boldsymbol{x}_{m+1} - \boldsymbol{x}_j = \boldsymbol{x}_m - \boldsymbol{x}_j + \alpha_m \boldsymbol{p}_m = \boldsymbol{x}_{m-1} - \boldsymbol{x}_j + \alpha_{m-1} \boldsymbol{p}_{m-1} + \alpha_m \boldsymbol{p}_m = \dots$
lyche-numerical-linear-algebra-EQ0462	183	0.992	$\boldsymbol{A} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0015 (1.3)	26	0.993	$\boldsymbol{c}_1 \boldsymbol{x}_1 + \dots + \boldsymbol{c}_n \boldsymbol{x}_n = \mathbf{0} \Longleftrightarrow \boldsymbol{c}_1 = \dots = \boldsymbol{c}_n = 0.$	$c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n = \mathbf{0} \implies c_1 = \dots = c_n = 0.$	$c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n = \mathbf{0} \implies c_1 = \dots = c_n = 0.$
lyche-numerical-linear-algebra-EQ0248 (4.3)	101	0.993	$\begin{aligned} \boldsymbol{A} &= \begin{bmatrix} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^* & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{nn} \end{bmatrix} \begin{bmatrix} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n \\ \mathbf{0}^* & 1 \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n \\ \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn} \end{bmatrix} \end{aligned}$	$\begin{aligned} \boldsymbol{A} &= \begin{bmatrix} \boldsymbol{A}_{[n-1]} & \boldsymbol{a}_n \\ \boldsymbol{a}_n^* & a_{nn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^* & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{nn} \end{bmatrix} \begin{bmatrix} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n \\ \mathbf{0}^* & 1 \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n \\ \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn} \end{bmatrix} \end{aligned}$	$\begin{aligned} \boldsymbol{A} &= \begin{bmatrix} \boldsymbol{A}_{[n-1]} & \boldsymbol{a}_n \\ \boldsymbol{a}_n^* & a_{nn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^* & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{nn} \end{bmatrix} \begin{bmatrix} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n \\ \mathbf{0}^* & 1 \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n \\ \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn} \end{bmatrix} \end{aligned}$

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lyche-numerical-linear-algebra- EQ0583	216	■ 0.993	$\boldsymbol{A} \, \boldsymbol{x} = \left[\begin{array}{c} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{array} \right] \boldsymbol{x} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b} .$	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \boldsymbol{b} .$	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \boldsymbol{b} .$
lyche-numerical-linear-algebra- EQ0432	173	■ 0.993	$\boldsymbol{\Sigma} = \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right] , \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} .$	$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} , \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} .$	$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} , \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} .$
lyche-numerical-linear-algebra- EQ0769	273	■ 0.993	$\left\ \boldsymbol{x}_k - \boldsymbol{x} \right\ = \left\ \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \right\ \leq \left\ \boldsymbol{G}^k \right\ \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \leq \left\ \boldsymbol{G} \right\ ^k \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \rightarrow 0, \quad k \rightarrow \infty .$	$\ \boldsymbol{x}_k - \boldsymbol{x} \ = \ \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \ \leq \ \boldsymbol{G}^k \ \ \boldsymbol{x}_0 - \boldsymbol{x} \ \leq \ \boldsymbol{G} \ ^k \ \boldsymbol{x}_0 - \boldsymbol{x} \ \rightarrow 0, \quad k \rightarrow \infty .$	$\ \boldsymbol{x}_k - \boldsymbol{x} \ = \ \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \ \leq \ \boldsymbol{G}^k \ \ \boldsymbol{x}_0 - \boldsymbol{x} \ \leq \ \boldsymbol{G} \ ^k \ \boldsymbol{x}_0 - \boldsymbol{x} \ \rightarrow 0, \quad k \rightarrow \infty .$
lyche-numerical-linear-algebra- EQ0863	300	■ 0.993	$\mathbb{W}_k = \operatorname{span} \left(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0 \right), \quad k = 1, 2, 3, \dots .$	$\mathbb{W}_k = \operatorname{span} \left(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0 \right), \quad k = 1, 2, 3, \dots .$	$\mathbb{W}_k = \operatorname{span} (\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0), \quad k = 1, 2, 3, \dots .$
lyche-numerical-linear-algebra- EQ0279	117	■ 0.993	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j} .$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j} .$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j} .$
lyche-numerical-linear-algebra- EQ0414	165	■ 0.994	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} .$	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} .$	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} .$
lyche-numerical-linear-algebra- EQ0802	281	■ 0.994	$\begin{aligned} & \left\ \boldsymbol{A} \right\ = \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1 = \left\ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \right\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right \\ & \leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon . \end{aligned}$	$\ \boldsymbol{A} \ = \ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \ _1 = \ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon .$	$\ \boldsymbol{A} \ = \ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \ _1 = \ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon .$

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lyche-numerical-linear-algebra-EQ0175 (3.1)	77	■ 0.994	$\boldsymbol{A}^{(k)}=\left[\begin{array}{cccccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & a_{1,k}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & \vdots & \ddots & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & a_{k,k}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & \vdots & \ddots & \vdots \\ & & & a_{i,k}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & \vdots & \ddots & \vdots \\ & & & a_{n,k}^{(k)} & \cdots & a_{n,n}^{(k)} \end{array}\right].$	$\boldsymbol{A}^{(k)}=\left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & \vdots & \ddots & \vdots & \ddots & \vdots \\ & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & \vdots & \ddots & \vdots & \ddots & \vdots \\ & & & a_{n,k}^{(k)} & \cdots & a_{n,j}^{(k)} & \cdots & a_{n,n}^{(k)} \end{array}\right].$	
lyche-numerical-linear-algebra-EQ0833	288	■ 0.994	$\boldsymbol{A}:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$
lyche-numerical-linear-algebra-EQ0992	335	■ 0.994	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k \boldsymbol{Q}_{k+1} \text{ for } k=n-2, n-3, \ldots, 1.$	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k \boldsymbol{Q}_{k+1} \text{ for } k=n-2, n-3, \ldots, 1.$	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k \boldsymbol{Q}_{k+1} \text{ for } k=n-2, n-3, \ldots, 1.$
lyche-numerical-linear-algebra-EQ1019 (15.2)	345	■ 0.994	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_n , \\ & \text{(ii) } z_0^T v_1 \neq 0 \end{aligned}$	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_n , \\ & \text{(ii) } z_0^T v_1 \neq 0 \end{aligned}$	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_n , \\ & \text{(ii) } z_0^T v_1 \neq 0 \end{aligned}$
lyche-numerical-linear-algebra-EQ0124	61	■ 0.995	$\begin{aligned} & \left(\boldsymbol{T}_1 \boldsymbol{s}_j \right)_k = \sum_{l=1}^m t_{k,l} \sin(lj\pi h) \\ & = a[\sin((k-1)j\pi h) + \sin((k+1)j\pi h)] + d \sin(kj\pi h) \\ & = 2a \cos(j\pi h) \sin(kj\pi h) + d \sin(kj\pi h) = \lambda_j s_{k,j}. \end{aligned}$	$\begin{aligned} (\boldsymbol{T}_1 \boldsymbol{s}_j)_k &= \sum_{l=1}^m t_{k,l} \sin(lj\pi h) \\ &= a[\sin((k-1)j\pi h) + \sin((k+1)j\pi h)] + d \sin(kj\pi h) \\ &= 2a \cos(j\pi h) \sin(kj\pi h) + d \sin(kj\pi h) = \lambda_j s_{k,j}. \end{aligned}$	$\begin{aligned} (\boldsymbol{T}_1 \boldsymbol{s}_j)_k &= \sum_{l=1}^m t_{k,l} \sin(lj\pi h) \\ &= a[\sin((k-1)j\pi h) + \sin((k+1)j\pi h)] + d \sin(kj\pi h) \\ &= 2a \cos(j\pi h) \sin(kj\pi h) + d \sin(kj\pi h) = \lambda_j s_{k,j}. \end{aligned}$
lyche-numerical-linear-algebra-EQ0384 (6.9)	155	■ 0.995	$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^n \lambda_i\left c_i\right ^2}{\sum_{j=1}^n\left c_j\right ^2}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j.$	$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^n \lambda_i\left c_i\right ^2}{\sum_{j=1}^n\left c_j\right ^2}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j.$	$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^n \lambda_i\left c_i\right ^2}{\sum_{j=1}^n\left c_j\right ^2}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j.$

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lyche-numerical-linear-algebra-E00632	229	■ 0.995	$\begin{gathered} \lambda_1 \geq \cdots \geq \lambda_{m+n} = \alpha_1 \geq \cdots \geq \alpha_t \geq 0 = \cdots = 0 \geq -\alpha_t \geq \cdots \geq -\alpha_1, \\ \mu_1 \geq \cdots \geq \mu_{m+n} = \beta_1 \geq \cdots \geq \beta_t \geq 0 = \cdots = 0 \geq -\beta_t \geq \cdots \geq -\beta_1. \end{gathered}$	$\lambda_1 \geq \cdots \geq \lambda_{m+n} = \alpha_1 \geq \cdots \geq \alpha_t \geq 0 = \cdots = 0 \geq -\alpha_t \geq \cdots \geq -\alpha_1, \\ \mu_1 \geq \cdots \geq \mu_{m+n} = \beta_1 \geq \cdots \geq \beta_t \geq 0 = \cdots = 0 \geq -\beta_t \geq \cdots \geq -\beta_1.$	$\lambda_1 \geq \cdots \geq \lambda_{m+n} = \alpha_1 \geq \cdots \geq \alpha_t \geq 0 = \cdots = 0 \geq -\alpha_t \geq \cdots \geq -\alpha_1, \\ \mu_1 \geq \cdots \geq \mu_{m+n} = \beta_1 \geq \cdots \geq \beta_t \geq 0 = \cdots = 0 \geq -\beta_t \geq \cdots \geq -\beta_1.$
lyche-numerical-linear-algebra-E00647	233	■ 0.995	$\boldsymbol{A}^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^*, \quad \alpha = \ \boldsymbol{u}\ _2^2 \ \boldsymbol{v}\ _2^2.$	$\boldsymbol{A}^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^*, \quad \alpha = \ \boldsymbol{u}\ _2^2 \ \boldsymbol{v}\ _2^2.$	$\boldsymbol{A}^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^*, \quad \alpha = \ \boldsymbol{u}\ _2^2 \ \boldsymbol{v}\ _2^2.$
lyche-numerical-linear-algebra-E00881 (13.31)	304	■ 0.995	$x_{n+1} - 2tx_n + x_{n-1} = 0 \text{ for } n \geq 1, \text{ with } x_0 = 1, x_1 = t.$	$x_{n+1} - 2tx_n + x_{n-1} = 0 \text{ for } n \geq 1, \text{ with } x_0 = 1, x_1 = t.$	$x_{n+1} - 2tx_n + x_{n-1} = 0 \text{ for } n \geq 1, \text{ with } x_0 = 1, x_1 = t.$
lyche-numerical-linear-algebra-E00301	124	■ 0.995	$\left(\boldsymbol{U}^* \boldsymbol{U} \right)_{i,j} = \boldsymbol{e}_i^* \boldsymbol{U}^* \boldsymbol{U} \boldsymbol{e}_j = (\boldsymbol{U} \boldsymbol{e}_i)^* (\boldsymbol{U} \boldsymbol{e}_j) = \langle \boldsymbol{U} \boldsymbol{e}_j, \boldsymbol{U} \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \boldsymbol{e}_i^* \boldsymbol{e}_j,$	$(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \boldsymbol{e}_i^* \boldsymbol{U}^* \boldsymbol{U} \boldsymbol{e}_j = (\boldsymbol{U} \boldsymbol{e}_i)^* (\boldsymbol{U} \boldsymbol{e}_j) = \langle \boldsymbol{U} \boldsymbol{e}_j, \boldsymbol{U} \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \boldsymbol{e}_i^* \boldsymbol{e}_j,$	$(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \boldsymbol{e}_i^* \boldsymbol{U}^* \boldsymbol{U} \boldsymbol{e}_j = (\boldsymbol{U} \boldsymbol{e}_i)^* (\boldsymbol{U} \boldsymbol{e}_j) = \langle \boldsymbol{U} \boldsymbol{e}_j, \boldsymbol{U} \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \boldsymbol{e}_i^* \boldsymbol{e}_j,$
lyche-numerical-linear-algebra-E00311	126	■ 0.995	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{a} \boldsymbol{e}_1, \quad \boldsymbol{a} = -\rho \ \boldsymbol{x}\ _2, \quad \rho := \begin{cases} x_1/ x_1 , & \text{if } x_1 \neq 0, \\ 1, & \text{otherwise.} \end{cases}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{a} \boldsymbol{e}_1, \quad \boldsymbol{a} = -\rho \ \boldsymbol{x}\ _2, \quad \rho := \begin{cases} x_1/ x_1 , & \text{if } x_1 \neq 0, \\ 1, & \text{otherwise.} \end{cases}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{a} \boldsymbol{e}_1, \quad \boldsymbol{a} = -\rho \ \boldsymbol{x}\ _2, \quad \rho := \begin{cases} x_1/ x_1 , & \text{if } x_1 \neq 0, \\ 1, & \text{otherwise.} \end{cases}$
lyche-numerical-linear-algebra-E00013	25	■ 0.995	$(\boldsymbol{f} + \boldsymbol{g})(t) := \boldsymbol{f}(t) + \boldsymbol{g}(t), \quad (c\boldsymbol{f})(t) := c\boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$	$(\boldsymbol{f} + \boldsymbol{g})(t) := \boldsymbol{f}(t) + \boldsymbol{g}(t), \quad (c\boldsymbol{f})(t) := c\boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$	$(\boldsymbol{f} + \boldsymbol{g})(t) := \boldsymbol{f}(t) + \boldsymbol{g}(t), \quad (c\boldsymbol{f})(t) := c\boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$
lyche-numerical-linear-algebra-E00061 (1.20)	41	■ 0.995	$\lambda := \boldsymbol{a}^T \boldsymbol{C} \boldsymbol{e}_i \neq 0,$	$\lambda := \boldsymbol{a}^T \boldsymbol{C} \boldsymbol{e}_i \neq 0,$	$\lambda := \boldsymbol{a}^T \boldsymbol{C} \boldsymbol{e}_i \neq 0,$
lyche-numerical-linear-algebra-E00216	88	■ 0.995	$x_2 = \frac{3.9997}{1.9999} \approx 2, \quad x_1 = 3 - x_2 \approx 1.$	$x_2 = \frac{3.9997}{1.9999} \approx 2, \quad x_1 = 3 - x_2 \approx 1.$	$x_2 = \frac{3.9997}{1.9999} \approx 2, \quad x_1 = 3 - x_2 \approx 1.$
lyche-numerical-linear-algebra-E00284	118	■ 0.995	$\begin{aligned} \ \boldsymbol{x}\ ^2 &= \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y}, \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y} \rangle \\ &\stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle \boldsymbol{a} \boldsymbol{y}, \boldsymbol{a} \boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + a ^2 \ \boldsymbol{y}\ ^2 \\ &\geq a ^2 \ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}. \end{aligned}$	$\ \boldsymbol{x}\ ^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y}, \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y} \rangle \\ \stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle \boldsymbol{a} \boldsymbol{y}, \boldsymbol{a} \boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + a ^2 \ \boldsymbol{y}\ ^2 \\ \geq a ^2 \ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}.$	$\ \boldsymbol{x}\ ^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y}, \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y} \rangle \\ \stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle \boldsymbol{a} \boldsymbol{y}, \boldsymbol{a} \boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + a ^2 \ \boldsymbol{y}\ ^2 \\ \geq a ^2 \ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}.$
lyche-numerical-linear-algebra-E00374	152	■ 0.995	$\boldsymbol{W} = \left[\begin{array}{c c} \boldsymbol{1} & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1 \end{array} \right] \text{ and } \boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}.$	$\boldsymbol{W} = \left[\begin{array}{c c} \boldsymbol{1} & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1 \end{array} \right] \text{ and } \boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}.$	$\boldsymbol{W} = \left[\begin{array}{c c} \boldsymbol{1} & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1 \end{array} \right] \text{ and } \boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}.$

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lyche-numerical-linear-algebra- E08858 (13.20)	299	■ 0.995	$\frac{\ \mathbf{x}-\mathbf{x}_k\ _{\mathbf{A}}}{\ \mathbf{x}-\mathbf{x}_0\ _{\mathbf{A}}}\leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k<2e^{-\frac{2}{\sqrt{\kappa}}k},\quad k\geq 0.$		$\frac{\ \mathbf{x}-\mathbf{x}_k\ _A}{\ \mathbf{x}-\mathbf{x}_0\ _A}\leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k<2e^{-\frac{2}{\sqrt{\kappa}}k},\quad k\geq 0.$
lyche-numerical-linear-algebra- E08459	182	■ 0.995	$\begin{aligned} & \mathbf{x}_{\{1\}}-\mathbf{x}_{\{3\}}=4, \quad \& \\ & \mathbf{x}_{\{1\}}+\mathbf{x}_{\{2\}}+\mathbf{x}_{\{3\}}=12 \quad . \end{aligned}$	$\begin{aligned} x_1-x_3&=4, \\ x_1+x_2+x_3&=12. \end{aligned}$	$\begin{aligned} x_1-x_3&=4, \\ x_1+x_2+x_3&=12. \end{aligned}$
lyche-numerical-linear-algebra- E08982	331	■ 0.995	$1=\ \mathbf{D}_1^{-1}\mathbf{X}^{-1}\mathbf{r}\ _p\leq\ \mathbf{D}_1^{-1}\ _pK_p(\mathbf{X})\ \mathbf{r}\ _p=\frac{K_p(\mathbf{X})\ \mathbf{r}\ _p}{\min_j \lambda_j-\mu }.$	$1=\ \mathbf{x}\ _p=\ \mathbf{X}\mathbf{D}_1^{-1}\mathbf{X}^{-1}\mathbf{r}\ _p\leq\ \mathbf{D}_1^{-1}\ _pK_p(\mathbf{X})\ \mathbf{r}\ _p=\frac{K_p(\mathbf{X})\ \mathbf{r}\ _p}{\min_j \lambda_j-\mu }.$	$1=\ \mathbf{x}\ _p=\ \mathbf{X}\mathbf{D}_1^{-1}\mathbf{X}^{-1}\mathbf{r}\ _p\leq\ \mathbf{D}_1^{-1}\ _pK_p(\mathbf{X})\ \mathbf{r}\ _p=\frac{K_p(\mathbf{X})\ \mathbf{r}\ _p}{\min_j \lambda_j-\mu }.$
lyche-numerical-linear-algebra- E08951 (13.63)	321	■ 0.995	$\ \mathbf{A}^{-1}\ _2=\max_{\mathbf{x}\neq\mathbf{0}}\frac{\ \mathbf{x}\ _2}{\ \mathbf{A}\mathbf{x}\ _2},$		$\ \mathbf{A}^{-1}\ _2=\max_{\mathbf{x}\neq\mathbf{0}}\frac{\ \mathbf{x}\ _2}{\ \mathbf{A}\mathbf{x}\ _2},$
lyche-numerical-linear-algebra- E08648	233	■ 0.995	$\lambda_i^\dagger=\begin{cases} 1/\lambda_i, & \lambda_i\neq 0 \\ 0 & \lambda_i=0. \end{cases}$	$\lambda_i^\dagger=\begin{cases} 1/\lambda_i, & \lambda_i\neq 0 \\ 0 & \lambda_i=0. \end{cases}$	$\lambda_i^\dagger=\begin{cases} 1/\lambda_i, & \lambda_i\neq 0 \\ 0 & \lambda_i=0. \end{cases}$
lyche-numerical-linear-algebra- E08532 (8.39)	203	■ 0.995	$\langle n\mathbf{x},\mathbf{y}\rangle=\langle(n-1)\mathbf{x}+\mathbf{x},\mathbf{y}\rangle\stackrel{(8.37)}{=}\langle(n-1)\mathbf{x},\mathbf{y}\rangle+\langle\mathbf{x},\mathbf{y}\rangle=n\langle\mathbf{x},\mathbf{y}\rangle.$	$\langle n\mathbf{x},\mathbf{y}\rangle=\langle(n-1)\mathbf{x}+\mathbf{x},\mathbf{y}\rangle\stackrel{(8.37)}{=}\langle(n-1)\mathbf{x},\mathbf{y}\rangle+\langle\mathbf{x},\mathbf{y}\rangle=n\langle\mathbf{x},\mathbf{y}\rangle.$	$\langle n\mathbf{x},\mathbf{y}\rangle=\langle(n-1)\mathbf{x}+\mathbf{x},\mathbf{y}\rangle\stackrel{(8.37)}{=}\langle(n-1)\mathbf{x},\mathbf{y}\rangle+\langle\mathbf{x},\mathbf{y}\rangle=n\langle\mathbf{x},\mathbf{y}\rangle.$
lyche-numerical-linear-algebra- E08975	330	■ 0.995	$\mathbf{A}_1:=\begin{bmatrix} 0&1&0&\cdots&0&0 \\ 0&0&1&\cdots&0&0 \\ \vdots&\vdots&\vdots&&\vdots&\vdots \\ 0&0&0&\cdots&0&1 \\ 0&0&0&\cdots&0&0 \end{bmatrix},\quad \mathbf{E}:=\begin{bmatrix} 0&0&0&\cdots&0&0 \\ 0&0&0&\cdots&0&0 \\ \vdots&\vdots&\vdots&&\vdots&\vdots \\ 0&0&0&\cdots&0&0 \\ \epsilon&0&0&\cdots&0&0 \end{bmatrix}=\epsilon\mathbf{e}_n\mathbf{e}_1^T.$	$\mathbf{A}_1:=\begin{bmatrix} 0&1&0&\cdots&0&0 \\ 0&0&1&\cdots&0&0 \\ \vdots&\vdots&\vdots&&\vdots&\vdots \\ 0&0&0&\cdots&0&1 \\ 0&0&0&\cdots&0&0 \end{bmatrix},\quad \mathbf{E}:=\begin{bmatrix} 0&0&0&\cdots&0&0 \\ 0&0&0&\cdots&0&0 \\ \vdots&\vdots&\vdots&&\vdots&\vdots \\ 0&0&0&\cdots&0&0 \\ \epsilon&0&0&\cdots&0&0 \end{bmatrix}=\epsilon\mathbf{e}_n\mathbf{e}_1^T.$	$\mathbf{A}_1:=\begin{bmatrix} 0&1&0&\cdots&0&0 \\ 0&0&1&\cdots&0&0 \\ \vdots&\vdots&\vdots&&\vdots&\vdots \\ 0&0&0&\cdots&0&1 \\ 0&0&0&\cdots&0&0 \end{bmatrix},\quad \mathbf{E}:=\begin{bmatrix} 0&0&0&\cdots&0&0 \\ 0&0&0&\cdots&0&0 \\ \vdots&\vdots&\vdots&&\vdots&\vdots \\ 0&0&0&\cdots&0&0 \\ \epsilon&0&0&\cdots&0&0 \end{bmatrix}=\epsilon\mathbf{e}_n\mathbf{e}_1^T.$
lyche-numerical-linear-algebra- E08926 (13.56)	314	■ 0.996	$\mathbf{L}_h\mathbf{v}=\mathbf{F},\quad \mathbf{L}_h\mathbf{v}:=\frac{1}{h^2}\begin{bmatrix} (dv)_{1,1}&\cdots&(dv)_{1,m} \\ \vdots&&\vdots \\ (dv)_{m,1}&\cdots&(dv)_{m,m} \end{bmatrix},\quad \mathbf{F}:=\begin{bmatrix} f_{1,1}&\cdots&f_{1,m} \\ \vdots&&\vdots \\ f_{m,1}&\cdots&f_{m,m} \end{bmatrix}.$	$\mathbf{L}_h\mathbf{v}=\mathbf{F},\quad \mathbf{L}_h\mathbf{v}:=\frac{1}{h^2}\begin{bmatrix} (dv)_{1,1}&\cdots&(dv)_{1,m} \\ \vdots&&\vdots \\ (dv)_{m,1}&\cdots&(dv)_{m,m} \end{bmatrix},\quad \mathbf{F}:=\begin{bmatrix} f_{1,1}&\cdots&f_{1,m} \\ \vdots&&\vdots \\ f_{m,1}&\cdots&f_{m,m} \end{bmatrix}.$	$\mathbf{L}_h\mathbf{v}=\mathbf{F},\quad \mathbf{L}_h\mathbf{v}:=\frac{1}{h^2}\begin{bmatrix} (dv)_{1,1}&\cdots&(dv)_{1,m} \\ \vdots&&\vdots \\ (dv)_{m,1}&\cdots&(dv)_{m,m} \end{bmatrix},\quad \mathbf{F}:=\begin{bmatrix} f_{1,1}&\cdots&f_{1,m} \\ \vdots&&\vdots \\ f_{m,1}&\cdots&f_{m,m} \end{bmatrix}.$

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lyche-numerical-linear-algebra- EQ1024	346	■ 0.996	$\lim_{k \rightarrow \infty} \left(\frac{ \lambda_1 }{\lambda_1} \right)^k \mathbf{x}_k = \frac{c_1}{ c_1 } \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }.$		
lyche-numerical-linear-algebra- EQ0031	32	■ 0.996	$\mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \mathbf{b}.$		
lyche-numerical-linear-algebra- EQ0207	86	■ 0.996	$(\mathbf{LU})_{ij} = \sum_{k=1}^n l_{i,k} u_{kj} = \sum_{k=1}^{i-1} a_{pi,k}^{(k)} a_{pk,j}^{(k)} + a_{pi,j}^{(i)} = \sum_{k=1}^{i-1} (a_{pi,j}^{(k)} - a_{pi,j}^{(k+1)}) + a_{pi,j}^{(i)} = a_{pi,j}^{(1)} = a_{pi,j} = (\mathbf{P}^T \mathbf{A})_{ij},$		
lyche-numerical-linear-algebra- EQ0739	261	■ 0.996	$\mathbf{DX} + \mathbf{XD} - \frac{1}{6} \mathbf{DXD} = 4h^4 \mathbf{G}, \text{ where } \mathbf{G} = S\mu \mathbf{FS},$		
lyche-numerical-linear-algebra- EQ0787 (12.27)	277	■ 0.996	$\rho(\mathbf{G}_\omega) = \begin{cases} \frac{1}{4} \left(\omega\beta + \sqrt{(\omega\beta)^2 - 4(\omega - 1)} \right)^2, & \text{for } 0 < \omega \leq \omega^*, \\ \omega - 1, & \text{for } \omega^* < \omega < 2, \end{cases}$		
lyche-numerical-linear-algebra- EQ0819	285	■ 0.996	$d(\omega) := (\omega\mu)^2 - 4(\omega - 1).$		
lyche-numerical-linear-algebra- EQ0850 (13.16)	295	■ 0.996	$\mathbf{r}_j^T \mathbf{A} \mathbf{p}_i \stackrel{(13.13)}{=} \mathbf{r}_j^T \left(\frac{\mathbf{r}_i - \mathbf{r}_{i+1}}{\alpha_i} \right) = 0, \quad i = 0, 1, \dots, j - 2.$		

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lyche-numerical-linear-algebra- E00991	334	0.996	<pre>\begin{aligned} \boldsymbol{A}_{k+1} &= \\ \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k &= \left[\begin{array}{cc} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{array} \right] \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{bmatrix} . \end{aligned}</pre>	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix}$ $= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{bmatrix} .$	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix}$ $= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{bmatrix} .$
lyche-numerical-linear-algebra- E00454 (7.14)	181	0.997	<pre>\boldsymbol{X}_{k+1}=\frac{1}{2}\left(\boldsymbol{X}_k+\boldsymbol{X}_k^{-T}\right), k=0,1,2,\ldots \text{ with } \boldsymbol{X}_0=\boldsymbol{A},</pre>	$\boldsymbol{X}_{k+1} = \frac{1}{2} \left(\boldsymbol{X}_k + \boldsymbol{X}_k^{-T} \right), k = 0, 1, 2, \dots \text{ with } \boldsymbol{X}_0 = \boldsymbol{A},$	$\boldsymbol{X}_{k+1} = \frac{1}{2} (\boldsymbol{X}_k + \boldsymbol{X}_k^{-T}), k = 0, 1, 2, \dots \text{ with } \boldsymbol{X}_0 = \boldsymbol{A},$
lyche-numerical-linear-algebra- E00135	64	0.997	<pre>\left[\begin{array}{l} A_1 \\ A_2 \end{array} \right] B = \left[\begin{array}{l} A_1 B \\ A_2 B \end{array} \right] .</pre>	$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} B = \begin{bmatrix} A_1 B \\ A_2 B \end{bmatrix} .$	$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} B = \begin{bmatrix} A_1 B \\ A_2 B \end{bmatrix} .$
lyche-numerical-linear-algebra- E00269 (4.11)	110	0.997	<pre>\begin{aligned} &\text{if } A(1,1)>0 \\ &A(1,1)=\sqrt{A(1,1)} \\ &A(2:n,1)=A(2:n,1)/A(1,1) \\ &\text{for } j=2:n \\ &A(j:n,j)=A(j:n,j)-A(j,1)*A(j:n,1) \end{aligned}</pre>	if $A(1,1) > 0$ $A(1,1) = \sqrt{A(1,1)}$ $A(2:n,1) = A(2:n,1)/A(1,1)$ for $j = 2:n$ $A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1)$	if $A(1,1) > 0$ $A(1,1) = \sqrt{A(1,1)}$ $A(2:n,1) = A(2:n,1)/A(1,1)$ for $j = 2:n$ $A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1)$
lyche-numerical-linear-algebra- E00431	173	0.997	<pre>\boldsymbol{B}\left[\begin{array}{l} 1 \\ 1 \end{array} \right]=4\left[\begin{array}{l} 1 \\ 1 \end{array} \right], \quad \boldsymbol{B}\left[\begin{array}{l} 1 \\ -1 \end{array} \right]=0\left[\begin{array}{l} 1 \\ -1 \end{array} \right],</pre>	$\boldsymbol{B} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{B} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix},$	$\boldsymbol{B} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{B} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix},$
lyche-numerical-linear-algebra- E00656 (10.1)	238	0.997	<pre>\begin{gathered} -\Delta u:=-\frac{\partial^2}{\partial x^2} u-\frac{\partial^2}{\partial y^2} u=f \\ \text{on } \Omega, \quad u:=0 \text{ on } \partial \Omega . \end{gathered}</pre>	$-\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega,$ $u := 0 \text{ on } \partial \Omega.$	$-\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega,$ $u := 0 \text{ on } \partial \Omega.$
lyche-numerical-linear-algebra- E00666 (10.7)	240	0.997	<pre>\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad \boldsymbol{b} \in \mathbb{R}^n, \quad n=m^2,</pre>	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad \boldsymbol{b} \in \mathbb{R}^n, \quad n = m^2,$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad \boldsymbol{b} \in \mathbb{R}^n, \quad n = m^2,$

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lyche-numerical-linear-algebra- EQ0671 (10.11)	242	■ 0.997	$\boldsymbol{T}_2=\left[\begin{array}{rrr rrr rrr} 2d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2d \end{array}\right].$	$\boldsymbol{T}_2=\left[\begin{array}{c c c} \begin{matrix} 2d & a & 0 \\ a & 2d & a \\ 0 & a & 2d \end{matrix} & \begin{matrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{matrix} & \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix} \\ \hline \begin{matrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{matrix} & \begin{matrix} 2d & a & 0 \\ a & 2d & a \\ 0 & a & 2d \end{matrix} & \begin{matrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{matrix} \\ \hline \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix} & \begin{matrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{matrix} & \begin{matrix} 2d & a & 0 \\ a & 2d & a \\ 0 & a & 2d \end{matrix} \end{array}\right].$	
lyche-numerical-linear-algebra- EQ1010 (14.10)	340	■ 0.997	$a_{kj}=\left\{\begin{array}{l} 1, \quad j=k+1, \\ 0, \quad \text{otherwise} \end{array}\right., \quad e_{kj}=\left\{\begin{array}{l} \epsilon, \quad k=n, j=1, \\ 0, \quad \text{otherwise} \end{array}\right.,$	$a_{kj}=\left\{\begin{array}{l} 1, \quad j=k+1, \\ 0, \quad \text{otherwise} \end{array}\right., \quad e_{kj}=\left\{\begin{array}{l} \epsilon, \quad k=n, j=1, \\ 0, \quad \text{otherwise} \end{array}\right.,$	$a_{kj}=\left\{\begin{array}{l} 1, \quad j=k+1, \\ 0, \quad \text{otherwise,} \end{array}\right. \quad e_{kj}=\left\{\begin{array}{l} \epsilon, \quad k=n, j=1, \\ 0, \quad \text{otherwise,} \end{array}\right.$
lyche-numerical-linear-algebra- EQ0294 (5.10)	122	■ 0.997	$s_0=\sum_{i=1}^k\frac{\left\langle \boldsymbol{v},\boldsymbol{v}_i\right\rangle }{\left\langle \boldsymbol{v}_i,\boldsymbol{v}_i\right\rangle }\boldsymbol{v}_i.$	$s_0=\sum_{i=1}^k\frac{\langle \boldsymbol{v},\boldsymbol{v}_i\rangle }{\langle \boldsymbol{v}_i,\boldsymbol{v}_i\rangle }\boldsymbol{v}_i.$	$s_0=\sum_{i=1}^k\frac{\langle \boldsymbol{v},\boldsymbol{v}_i\rangle }{\langle \boldsymbol{v}_i,\boldsymbol{v}_i\rangle }\boldsymbol{v}_i.$
lyche-numerical-linear-algebra- EQ0416	166	■ 0.997	$B_1:=\left\{ \boldsymbol{x}\in \mathbb{R}^n\mid \left\ \boldsymbol{x}\right\ _2=1\right\} .$	$B_1:=\left\{ \boldsymbol{x}\in \mathbb{R}^n\mid \left\ \boldsymbol{x}\right\ _2=1\right\} .$	$B_1:=\left\{ \boldsymbol{x}\in \mathbb{R}^n\mid \left\ \boldsymbol{x}\right\ _2=1\right\} .$
lyche-numerical-linear-algebra- EQ0694 (10.24)	248	■ 0.997	$\begin{array}{l} -\Delta v(s,t) = f(s,t) \\ -\Delta u(s,t) = v(s,t) \\ u(s,t) = v(s,t) = 0 \end{array} \quad \begin{array}{l} (s,t) \in \Omega \\ (s,t) \in \Omega \\ (s,t) \in \partial \Omega \end{array}$	$\begin{array}{l} -\Delta v(s,t) = f(s,t) \\ -\Delta u(s,t) = v(s,t) \\ u(s,t) = v(s,t) = 0 \end{array} \quad \begin{array}{l} (s,t) \in \Omega \\ (s,t) \in \Omega \\ (s,t) \in \partial \Omega \end{array}$	$\begin{array}{l} -\Delta v(s,t) = f(s,t) \\ -\Delta u(s,t) = v(s,t) \\ u(s,t) = v(s,t) = 0 \end{array} \quad \begin{array}{l} (s,t) \in \Omega \\ (s,t) \in \Omega \\ (s,t) \in \partial \Omega. \end{array}$
lyche-numerical-linear-algebra- EQ0768 (12.16)	272	■ 0.997	$\boldsymbol{x}_k-\boldsymbol{x}=\boldsymbol{G}^k(\boldsymbol{x}_0-\boldsymbol{x}), \quad k=0,1,2,\ldots$	$\boldsymbol{x}_k-\boldsymbol{x}=\boldsymbol{G}^k(\boldsymbol{x}_0-\boldsymbol{x}), \quad k=0,1,2,\ldots$	$\boldsymbol{x}_k-\boldsymbol{x}=\boldsymbol{G}^k(\boldsymbol{x}_0-\boldsymbol{x}), \quad k=0,1,2,\ldots$
lyche-numerical-linear-algebra- EQ0815	284	■ 0.997	$\boldsymbol{G}_J\boldsymbol{v}=(\boldsymbol{L}+\boldsymbol{R})\boldsymbol{v}=\frac{\lambda+\omega-1}{\omega\lambda^{1/2}}\boldsymbol{v}=\mu\boldsymbol{v}.$	$\boldsymbol{G}_J\boldsymbol{v}=(\boldsymbol{L}+\boldsymbol{R})\boldsymbol{v}=\frac{\lambda+\omega-1}{\omega\lambda^{1/2}}\boldsymbol{v}=\mu\boldsymbol{v}.$	$\boldsymbol{G}_J\boldsymbol{v}=(\boldsymbol{L}+\boldsymbol{R})\boldsymbol{v}=\frac{\lambda+\omega-1}{\omega\lambda^{1/2}}\boldsymbol{v}=\mu\boldsymbol{v}.$
lyche-numerical-linear-algebra- EQ0847 (13.11)	294	■ 0.997	$\begin{array}{l} \boldsymbol{t}_{2k-2}=3*4^{1-k}\left[\begin{array}{c} 1 \\ -1/2 \end{array}\right], \quad \boldsymbol{x}_{2k-1}=-4^{-k}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{2k-1}=3*4^{-k}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ \boldsymbol{t}_{2k-1}=3*4^{-k}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \boldsymbol{x}_{2k}=-4^{-k}\left[\begin{array}{c} 1 \\ 1/2 \end{array}\right], \quad \boldsymbol{r}_{2k}=3*4^{-k}\left[\begin{array}{c} 1/2 \\ 0 \end{array}\right]. \end{array}$	$\begin{array}{l} \boldsymbol{t}_{2k-2}=3*4^{1-k}\left[\begin{array}{c} 1 \\ -1/2 \end{array}\right], \quad \boldsymbol{x}_{2k-1}=-4^{-k}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{2k-1}=3*4^{-k}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ \boldsymbol{t}_{2k-1}=3*4^{-k}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \boldsymbol{x}_{2k}=-4^{-k}\left[\begin{array}{c} 1 \\ 1/2 \end{array}\right], \quad \boldsymbol{r}_{2k}=3*4^{-k}\left[\begin{array}{c} 1/2 \\ 0 \end{array}\right]. \end{array}$	$\begin{array}{l} \boldsymbol{t}_{2k-2}=3*4^{1-k}\left[\begin{array}{c} 1 \\ -1/2 \end{array}\right], \quad \boldsymbol{x}_{2k-1}=-4^{-k}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{2k-1}=3*4^{-k}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ \boldsymbol{t}_{2k-1}=3*4^{-k}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \boldsymbol{x}_{2k}=-4^{-k}\left[\begin{array}{c} 1 \\ 1/2 \end{array}\right], \quad \boldsymbol{r}_{2k}=3*4^{-k}\left[\begin{array}{c} 1/2 \\ 0 \end{array}\right]. \end{array}$

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lyche-numerical-linear-algebra-EQ0355 (5.22)	142	■ 0.998	$P_{i_1,n+1} P_{i_2,n+1} \cdots P_{i_n,n+1} \left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right] = \left[\begin{array}{c} \boldsymbol{R}' \\ 0 \end{array} \right],$		
lyche-numerical-linear-algebra-EQ0390 (6.11)	157	■ 0.998	$\lambda_k \leq \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$		
lyche-numerical-linear-algebra-EQ0448	179	■ 0.998	$A := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$		
lyche-numerical-linear-algebra-EQ0765	272	■ 0.998	$x_{k+1}(1) = \frac{1}{2} x_k(2) + \frac{1}{2}, \quad x_{k+1}(2) = \frac{1}{2} x_{k+1}(1) + \frac{1}{2}.$		
lyche-numerical-linear-algebra-EQ0603 (9.8)	222	■ 0.998	$\boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger = \boldsymbol{A}^\dagger, (\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{A}^\dagger \boldsymbol{A}, (\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{A} \boldsymbol{A}^\dagger.$		
lyche-numerical-linear-algebra-EQ0981	331	■ 0.998	$\boldsymbol{X} \boldsymbol{D}_1^{-1} \boldsymbol{X}^{-1} \boldsymbol{r} = (\boldsymbol{X}(\boldsymbol{D} - \mu \boldsymbol{I}) \boldsymbol{X}^{-1})^{-1} \boldsymbol{r} = (\boldsymbol{A} - \mu \boldsymbol{I})^{-1} (\boldsymbol{A} - \mu \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{x}.$		
lyche-numerical-linear-algebra-EQ0228 (3.19)	92	■ 0.998	$\boldsymbol{A} := \begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m1} & \cdots & \boldsymbol{A}_{mm} \end{bmatrix},$		
lyche-numerical-linear-algebra-EQ0691	247	■ 0.998	$\boldsymbol{A} = \begin{bmatrix} 2d & a & a & 0 \\ a & 2d & 0 & a \\ a & 0 & 2d & a \\ 0 & a & a & 2d \end{bmatrix}.$		
lyche-numerical-linear-algebra-EQ1035 (15.8)	351	■ 0.998	$\begin{aligned} & \boldsymbol{A}_1 = \boldsymbol{A} \\ & \text{for } k = 1, 2, \dots \\ & \boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k \quad (\text{QR factorization of } \boldsymbol{A}_k) \\ & \boldsymbol{A}_{k+1} = \boldsymbol{R}_k \boldsymbol{Q}_k. \end{aligned}$		

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lyche-numerical-linear-algebra-EQ0012	24	■ 0.998	$\boldsymbol{A}\left(r_{1}:r_{2},c_{1}:c_{2}\right):=\left[\begin{array}{cccc}a_{r_{1},c_{1}} & a_{r_{1},c_{1}+1} & \cdots & a_{r_{1},c_{2}} \\ a_{r_{1}+1,c_{1}} & a_{r_{1}+1,c_{1}+1} & \cdots & a_{r_{1}+1,c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2},c_{1}} & a_{r_{2},c_{1}+1} & \cdots & a_{r_{2},c_{2}}\end{array}\right]$	$\boldsymbol{A}\left(r_{1}:r_{2},c_{1}:c_{2}\right):=\left[\begin{array}{cccc}a_{r_{1},c_{1}} & a_{r_{1},c_{1}+1} & \cdots & a_{r_{1},c_{2}} \\ a_{r_{1}+1,c_{1}} & a_{r_{1}+1,c_{1}+1} & \cdots & a_{r_{1}+1,c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2},c_{1}} & a_{r_{2},c_{1}+1} & \cdots & a_{r_{2},c_{2}}\end{array}\right]$	$\boldsymbol{A}\left(r_{1}:r_{2},c_{1}:c_{2}\right):=\left[\begin{array}{cccc}a_{r_{1},c_{1}} & a_{r_{1},c_{1}+1} & \cdots & a_{r_{1},c_{2}} \\ a_{r_{1}+1,c_{1}} & a_{r_{1}+1,c_{1}+1} & \cdots & a_{r_{1}+1,c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2},c_{1}} & a_{r_{2},c_{1}+1} & \cdots & a_{r_{2},c_{2}}\end{array}\right]$
lyche-numerical-linear-algebra-EQ0247 (4.2)	101	■ 0.998	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{[k]} & \boldsymbol{B}_k^* \\ \boldsymbol{B}_k & \boldsymbol{F}_k\end{array}\right]=\left[\begin{array}{cc}\boldsymbol{L}_{[k]} & \boldsymbol{0} \\ \boldsymbol{M}_k & \boldsymbol{N}_k\end{array}\right]\left[\begin{array}{cc}\boldsymbol{D}_{[k]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_k\end{array}\right]\left[\begin{array}{cc}\boldsymbol{L}_{[k]}^* & \boldsymbol{M}_k^* \\ \boldsymbol{0} & \boldsymbol{N}_k^*\end{array}\right]=\boldsymbol{L}\boldsymbol{D}\boldsymbol{U},$	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{[k]} & \boldsymbol{B}_k^* \\ \boldsymbol{B}_k & \boldsymbol{F}_k\end{array}\right]=\left[\begin{array}{cc}\boldsymbol{L}_{[k]} & \boldsymbol{0} \\ \boldsymbol{M}_k & \boldsymbol{N}_k\end{array}\right]\left[\begin{array}{cc}\boldsymbol{D}_{[k]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_k\end{array}\right]\left[\begin{array}{cc}\boldsymbol{L}_{[k]}^* & \boldsymbol{M}_k^* \\ \boldsymbol{0} & \boldsymbol{N}_k^*\end{array}\right]=\boldsymbol{L}\boldsymbol{D}\boldsymbol{U},$	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{[k]} & \boldsymbol{B}_k^* \\ \boldsymbol{B}_k & \boldsymbol{F}_k\end{array}\right]=\left[\begin{array}{cc}\boldsymbol{L}_{[k]} & \boldsymbol{0} \\ \boldsymbol{M}_k & \boldsymbol{N}_k\end{array}\right]\left[\begin{array}{cc}\boldsymbol{D}_{[k]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_k\end{array}\right]\left[\begin{array}{cc}\boldsymbol{L}_{[k]}^* & \boldsymbol{M}_k^* \\ \boldsymbol{0} & \boldsymbol{N}_k^*\end{array}\right]=\boldsymbol{L}\boldsymbol{D}\boldsymbol{U},$
lyche-numerical-linear-algebra-EQ0306 (5.13)	125	■ 0.998	$\boldsymbol{v}^*\boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^*(\boldsymbol{x}-\boldsymbol{y})=2\boldsymbol{x}^*\boldsymbol{x}-2\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=2\boldsymbol{v}^*\boldsymbol{x}.$	$\boldsymbol{v}^*\boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^*(\boldsymbol{x}-\boldsymbol{y})=2\boldsymbol{x}^*\boldsymbol{x}-2\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=2\boldsymbol{v}^*\boldsymbol{x}.$	$\boldsymbol{v}^*\boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^*(\boldsymbol{x}-\boldsymbol{y})=2\boldsymbol{x}^*\boldsymbol{x}-2\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=2\boldsymbol{v}^*\boldsymbol{x}.$
lyche-numerical-linear-algebra-EQ0320	129	■ 0.998	$\boldsymbol{A}=\left[\begin{array}{lll}u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33}\end{array}\right]\text{ or }\boldsymbol{A}=\left[\begin{array}{lll}r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33}\end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{lll}u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43}\end{array}\right]\text{ or }\boldsymbol{A}=\left[\begin{array}{lll}r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43}\end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{lll}u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43}\end{array}\right]\text{ or }\boldsymbol{A}=\left[\begin{array}{lll}r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43}\end{array}\right].$
lyche-numerical-linear-algebra-EQ0438	175	■ 0.998	$1=\ z\ _2^2=\left\ \begin{array}{c}\boldsymbol{V}_1 \\ \boldsymbol{\Sigma}_1^{-1}\end{array}\boldsymbol{y}\right\ _2^2=\left\ \begin{array}{c}\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\end{array}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$	$1=\ z\ _2^2=\left\ \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$	$1=\ z\ _2^2=\left\ \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$
lyche-numerical-linear-algebra-EQ0477 (8.10)	189	■ 0.998	$\ \boldsymbol{A}^k\ \leq\ \boldsymbol{A}\ ^k\text{ for }\boldsymbol{A}\in\mathbb{C}^{n\times n}\text{ and }k\in\mathbb{N}.$	$\ \boldsymbol{A}^k\ \leq\ \boldsymbol{A}\ ^k\text{ for }\boldsymbol{A}\in\mathbb{C}^{n\times n}\text{ and }k\in\mathbb{N}.$	$\ \boldsymbol{A}^k\ \leq\ \boldsymbol{A}\ ^k\text{ for }\boldsymbol{A}\in\mathbb{C}^{n\times n}\text{ and }k\in\mathbb{N}.$
lyche-numerical-linear-algebra-EQ0487 (8.17)	191	■ 0.998	$\ \boldsymbol{A}\ _p:=\max_{\boldsymbol{x}\neq\boldsymbol{0}}\frac{\ \boldsymbol{Ax}\ _p}{\ \boldsymbol{x}\ _p}=\max_{\ \boldsymbol{y}\ _p=1}\ \boldsymbol{Ay}\ _p.$	$\ \boldsymbol{A}\ _p:=\max_{\boldsymbol{x}\neq\boldsymbol{0}}\frac{\ \boldsymbol{Ax}\ _p}{\ \boldsymbol{x}\ _p}=\max_{\ \boldsymbol{y}\ _p=1}\ \boldsymbol{Ay}\ _p.$	$\ \boldsymbol{A}\ _p:=\max_{\boldsymbol{x}\neq\boldsymbol{0}}\frac{\ \boldsymbol{Ax}\ _p}{\ \boldsymbol{x}\ _p}=\max_{\ \boldsymbol{y}\ _p=1}\ \boldsymbol{Ay}\ _p.$
lyche-numerical-linear-algebra-EQ0517 (8.32)	201	■ 0.998	$a_1^{\lambda_1}a_2^{\lambda_2}\cdots a_n^{\lambda_n}\leq\sum_{j=1}^n\lambda_ja_j,$	$a_1^{\lambda_1}a_2^{\lambda_2}\cdots a_n^{\lambda_n}\leq\sum_{j=1}^n\lambda_ja_j,$	$a_1^{\lambda_1}a_2^{\lambda_2}\cdots a_n^{\lambda_n}\leq\sum_{j=1}^n\lambda_ja_j,$

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lyche-numerical-linear-algebra- EQ0544	206	■ 0.998	$\backslash\boldsymbol{A}=\left[\backslash\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\backslash\right]$.	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0997	336	■ 0.998	$p_{\{2\}(-M)}>0, \quad \backslash\text{quad } p_{\{2\}\left(d_{\{1\}}\right)}<0, \quad \backslash\text{quad } p_{\{2\}(+M)}>0$.	$p_2(-M) > 0, \quad p_2(d_1) < 0, \quad p_2(+M) > 0.$	$p_2(-M) > 0, \quad p_2(d_1) < 0, \quad p_2(+M) > 0.$
lyche-numerical-linear-algebra- EQ1049	358	■ 0.998	$D_{\{i\}} f(\backslash\boldsymbol{x}):=\frac{\partial f(\backslash\boldsymbol{x})}{\partial x_{\{i\}}}:=\lim_{h\rightarrow\backslash\mathbf{0}}\frac{f(\backslash\boldsymbol{x}+h\backslash\boldsymbol{e}_i)-f(\backslash\boldsymbol{x})}{h}, \quad \backslash\text{quad } \backslash\boldsymbol{x}\in\backslash\mathbb{R}^n,$	$D_i f(\boldsymbol{x}) := \frac{\partial f(\boldsymbol{x})}{\partial x_i} := \lim_{h \rightarrow \mathbf{0}} \frac{f(\boldsymbol{x} + h \boldsymbol{e}_i) - f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$	$D_i f(\boldsymbol{x}) := \frac{\partial f(\boldsymbol{x})}{\partial x_i} := \lim_{h \rightarrow \mathbf{0}} \frac{f(\boldsymbol{x} + h \boldsymbol{e}_i) - f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$
lyche-numerical-linear-algebra- EQ0027 (1.10)	31	■ 0.998	$\backslash\langle\backslash\boldsymbol{x}, \backslash\boldsymbol{y}\backslash\rangle:=\backslash\boldsymbol{y}^*\backslash\boldsymbol{x}=\backslash\boldsymbol{x}^T\backslash\overline{\boldsymbol{y}}=\backslash\overline{\sum_{j=1}^n x_j y_j}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$
lyche-numerical-linear-algebra- EQ0318 (5.15)	128	■ 0.998	$\backslash\begin{aligned} \backslash\boldsymbol{A}_{\{k\}} &= \backslash\left[\backslash\begin{array}{cccc} a_{\{1,1\}}^{\{1\}} & \cdots & a_{\{1,k-1\}}^{\{1\}} & \cdots & a_{\{1,k\}}^{\{1\}} & \cdots & a_{\{1,j\}}^{\{1\}} & \cdots & a_{\{1,n\}}^{\{1\}} \\ & & \ddots & & \vdots & & \vdots & & \vdots \\ & & & a_{\{k-1,k-1\}}^{\{k-1\}} & a_{\{k-1,k\}}^{\{k-1\}} & \cdots & a_{\{k-1,j\}}^{\{k-1\}} & \cdots & a_{\{k-1,n\}}^{\{k-1\}} \\ & & & & a_{\{k,k\}}^{\{k\}} & \cdots & a_{\{k,j\}}^{\{k\}} & \cdots & a_{\{k,n\}}^{\{k\}} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{\{i,k\}}^{\{k\}} & \cdots & a_{\{i,j\}}^{\{k\}} & \cdots & a_{\{i,n\}}^{\{k\}} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{\{m,k\}}^{\{k\}} & \cdots & a_{\{m,j\}}^{\{k\}} & \cdots & a_{\{m,n\}}^{\{k\}} \end{array} \backslash\right] \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix} . \end{aligned}$	$\boldsymbol{A}_k = \left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & & \ddots & & \vdots & & \vdots & & \vdots \\ & & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ & & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{m,k}^{(k)} & \cdots & a_{m,j}^{(k)} & \cdots & a_{m,n}^{(k)} \end{array} \right]$ $= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}.$	$\boldsymbol{A}_k = \left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & & \ddots & & \vdots & & \vdots & & \vdots \\ & & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ & & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{m,k}^{(k)} & \cdots & a_{m,j}^{(k)} & \cdots & a_{m,n}^{(k)} \end{array} \right]$ $= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0362 (6.1)	147	■ 0.998	$\backslash\pi_{\{A\} C}(\backslash\lambda)=\backslash\lambda^{m-n} \backslash\pi_{\{C\} A}(\backslash\lambda), \quad \backslash\text{quad } \backslash\lambda \in \backslash\mathbb{C}.$	$\pi_{AC}(\lambda) = \lambda^{m-n} \pi_{CA}(\lambda), \quad \lambda \in \mathbb{C}.$	$\pi_{AC}(\lambda) = \lambda^{m-n} \pi_{CA}(\lambda), \quad \lambda \in \mathbb{C}.$
lyche-numerical-linear-algebra- EQ0872 (13.28)	303	■ 0.998	$\frac{\backslash\left \backslash\boldsymbol{x}-\backslash\boldsymbol{x}_{\{k\}}\right }{\backslash\left \backslash\boldsymbol{x}-\backslash\boldsymbol{x}_{\{0\}}\right } \leq \min _{\substack{Q \in \Pi_k \\ Q(0)=1}} \max _{a \leq x \leq b} Q(x) ,$	$\frac{\ \boldsymbol{x}-\boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x}-\boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{a \leq x \leq b} Q(x) ,$	$\frac{\ \boldsymbol{x}-\boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x}-\boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{a \leq x \leq b} Q(x) ,$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-EQ0611	224	■ 0.998	$\begin{array}{l} \boldsymbol{A}:=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right], \\ \boldsymbol{A}^{\dagger}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right] \\ \frac{1}{2} \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right]=\frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{b}=\left[\begin{array}{l} 1 / 2 \\ 1 / 2 \end{array}\right] . \end{array}$	$\boldsymbol{A}:=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right], \boldsymbol{A}^{\dagger}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right]=\frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{b}=\left[\begin{array}{l} 1 / 2 \\ 1 / 2 \end{array}\right] .$	$\boldsymbol{A}:=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right], \boldsymbol{A}^{\dagger}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right]=\frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{b}=\left[\begin{array}{l} 1 / 2 \\ 1 / 2 \end{array}\right] .$
lyche-numerical-linear-algebra-EQ0616 (1.4)	28	■ 0.999	$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S} \text { and } \boldsymbol{t} \in \mathcal{T}\}.$	$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S} \text { and } \boldsymbol{t} \in \mathcal{T}\}.$	$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S} \text { and } \boldsymbol{t} \in \mathcal{T}\}.$
lyche-numerical-linear-algebra-EQ0673	44	■ 0.999	$b_{j, k}=\left(\alpha_k+\beta_j\right) A_k\left(-\beta_j\right) B_j\left(-\alpha_k\right),$	$b_{j, k}=\left(\alpha_k+\beta_j\right) A_k\left(-\beta_j\right) B_j\left(-\alpha_k\right),$	$b_{j, k}=\left(\alpha_k+\beta_j\right) A_k\left(-\beta_j\right) B_j\left(-\alpha_k\right),$
lyche-numerical-linear-algebra-EQ0101 (2.18)	55	■ 0.999	$\begin{aligned} \left u_{k+1}\right &=\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right } \\ &\geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right . \end{aligned}$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right } \\ \geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right $	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right } \\ \geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right .$
lyche-numerical-linear-algebra-EQ0156	71	■ 0.999	$\Delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	$\delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	$\delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$
lyche-numerical-linear-algebra-EQ0250	102	■ 0.999	$\begin{array}{l} \left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{ll} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{ll} 1 & 1 / 3 \\ 0 & 1 \end{array}\right] \\ \left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{ll} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{ll} 1 & 1 / 3 \\ 0 & 1 \end{array}\right] \end{array}$	$\left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{ll} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{ll} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$	$\left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{ll} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{ll} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$
lyche-numerical-linear-algebra-EQ0296 (5.11)	123	■ 0.999	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$
lyche-numerical-linear-algebra-EQ0361	147	■ 0.999	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) &=\det\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=\det\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S}\right) \\ &=\det\left(\boldsymbol{S}^{-1}\right) \det(\boldsymbol{A}-\lambda \boldsymbol{I}) \det(\boldsymbol{S})=\det\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \det(\boldsymbol{A}-\lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda), \end{aligned}$	$\pi_{\boldsymbol{B}}(\lambda)=\det\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=\det\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S}\right) \\ =\det\left(\boldsymbol{S}^{-1}\right) \det(\boldsymbol{A}-\lambda \boldsymbol{I}) \det(\boldsymbol{S})=\det\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \det(\boldsymbol{A}-\lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda),$	$\pi_{\boldsymbol{B}}(\lambda)=\det\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=\det\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S}\right) \\ =\det\left(\boldsymbol{S}^{-1}\right) \det(\boldsymbol{A}-\lambda \boldsymbol{I}) \det(\boldsymbol{S})=\det\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \det(\boldsymbol{A}-\lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda),$

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lyche-numerical-linear-algebra- EQ0381	154	■ 0.999	$e_{ii} = \sum_{k=1}^n \bar{b}_{ki} b_{ki} = \sum_{k=1}^i b_{ki} ^2 \text{ and } f_{ii} = \sum_{k=1}^n b_{ik} \bar{b}_{ik} = \sum_{k=i}^n b_{ik} ^2.$		
lyche-numerical-linear-algebra- EQ0406	162	■ 0.999	$\mathbf{A} = \begin{bmatrix} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$		
lyche-numerical-linear-algebra- EQ0446 (7.10)	178	■ 0.999	$\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \mathbf{U} \mathbf{D} \mathbf{U}^T$		
lyche-numerical-linear-algebra- EQ0471 (8.7)	187	■ 0.999	$m\ \mathbf{x}\ ' \leq \ \mathbf{x}\ \leq M\ \mathbf{x}\ '.$		
lyche-numerical-linear-algebra- EQ0481 (8.14)	190	■ 0.999	$\mathcal{S} := \{x \in \mathbb{C}^n : \ \mathbf{x}\ = 1\}$		
lyche-numerical-linear-algebra- EQ0538	204	■ 0.999	$\ \mathbf{x}\ _{\mathbf{A}} := \sqrt{\mathbf{x}^T \mathbf{A} \mathbf{x}}, \quad \mathbf{x} \in \mathbb{R}^n.$		
lyche-numerical-linear-algebra- EQ0634	230	■ 0.999	$\ C - D\ _F^2 = \left\ \begin{bmatrix} \mathbf{0} & A - B \\ A^* - B^* & \mathbf{0} \end{bmatrix} \right\ _F^2 = \ B - A\ _F^2 + \ (B - A)^*\ _F^2 = 2\ B - A\ _F^2.$		
lyche-numerical-linear-algebra- EQ0816	284	■ 0.999	$\frac{1}{4} \left(\lambda^{1/2} w_{i-1,j} + \lambda^{1/2} w_{i,j-1} + \lambda^{-1/2} w_{i+1,j} + \lambda^{-1/2} w_{i,j+1} \right) = \mu w_{i,j}.$		
lyche-numerical-linear-algebra- EQ0861	300	■ 0.999	$\kappa_A = \frac{5 + 4 \cos(\pi h)}{5 - 4 \cos(\pi h)} \leq 9,$		
lyche-numerical-linear-algebra- EQ0954 (13.67)	322	■ 0.999	$\ \mathbf{x}^* - \mathbf{x}\ _2 \leq \ \mathbf{A}\ _2 \min_{\mathbf{w} \in \mathcal{W}} \ \mathbf{x}^* - \mathbf{w}\ _2.$		

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lyche-numerical-linear-algebra-EQ0208	86	■ 0.999	$\begin{aligned} & (\boldsymbol{L} \boldsymbol{U})_{ij} = \sum_{k=1}^n l_{ik}^{(k)} u_{kj} = \sum_{k=1}^{j-1} a_{pi,k}^{(k)} a_{pk,j}^{(k)} + a_{pi,j}^{(k)} a_{pj,j}^{(j)} \\ & = \sum_{k=1}^{j-1} \left(a_{pi,j}^{(k)} - a_{pi,j}^{(k+1)} \right) + a_{pi,j}^{(j)} = a_{pi,j}^{(1)} = a_{pi,j} = (\boldsymbol{P}^T \boldsymbol{A})_{ij}. \end{aligned}$	$(\boldsymbol{LU})_{ij} = \sum_{k=1}^n l_{ik}^{(k)} u_{kj} = \sum_{k=1}^{j-1} a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} + a_{p_i,j}^{(k)} a_{p_j,j}^{(j)}$ $= \sum_{k=1}^{j-1} \left(a_{p_i,j}^{(k)} - a_{p_i,j}^{(k+1)} \right) + a_{p_i,j}^{(j)} = a_{p_i,j}^{(1)} = a_{p_i,j} = (\boldsymbol{P}^T \boldsymbol{A})_{ij}.$	$(\boldsymbol{LU})_{ij} = \sum_{k=1}^n l_{ik}^{(k)} u_{kj} = \sum_{k=1}^{j-1} a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} + a_{p_i,j}^{(k)} a_{p_j,j}^{(j)}$ $= \sum_{k=1}^{j-1} (a_{p_i,j}^{(k)} - a_{p_i,j}^{(k+1)}) + a_{p_i,j}^{(j)} = a_{p_i,j}^{(1)} = a_{p_i,j} = (\boldsymbol{P}^T \boldsymbol{A})_{ij}.$
lyche-numerical-linear-algebra-EQ0265 (4.9)	109	■ 0.999	$\begin{aligned} & 0 \leq \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix} \\ & = [0, -(\boldsymbol{y}^* \boldsymbol{v}) \boldsymbol{v}^* / \alpha + \boldsymbol{y}^* \boldsymbol{B}] \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix} \\ & = -\boldsymbol{y}^* \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{y} / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{C} \boldsymbol{y}. \end{aligned}$	$0 \leq \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= [0, -(\boldsymbol{y}^* \boldsymbol{v}) \boldsymbol{v}^* / \alpha + \boldsymbol{y}^* \boldsymbol{B}] \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= -\boldsymbol{y}^* \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{y} / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{C} \boldsymbol{y}.$	$0 \leq \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= [0, -(\boldsymbol{y}^* \boldsymbol{v}) \boldsymbol{v}^* / \alpha + \boldsymbol{y}^* \boldsymbol{B}] \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= -\boldsymbol{y}^* \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{y} / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{C} \boldsymbol{y}.$
lyche-numerical-linear-algebra-EQ0309	126	■ 0.999	$\begin{aligned} & \boldsymbol{H} := \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ & \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$	$\boldsymbol{H} := \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$ $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$\boldsymbol{H} := \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$ $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

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lyche-numerical-linear-algebra-E08645	232	■ 0.999	$\begin{aligned} B &= (B \ A) \ B = \left(A^* \right) B^* B = \left(A^* C^* \right) A^* B^* B = C A (A^* B^*) B \\ &= C A (B A B) = (C) A B = C (A C) A B = C C^* A^* (A B) \\ &= C C^* (A^* B^* A^*) = C (C^* A^*) = C A C = C. \end{aligned}$	$B = (BA)B = (A^*)B^*B = (A^*C^*)A^*B^*B = CA(A^*B^*)B \\ = CA(BAB) = (C)AB = C(AC)AB = CC^*A^*(AB) \\ = CC^*(A^*B^*A^*) = C(C^*A^*) = CAC = C.$	$B = (BA)B = (A^*)B^*B = (A^*C^*)A^*B^*B = CA(A^*B^*)B \\ = CA(BAB) = (C)AB = C(AC)AB = CC^*A^*(AB) \\ = CC^*(A^*B^*A^*) = C(C^*A^*) = CAC = C.$
lyche-numerical-linear-algebra-E08827	287	■ 0.999	$\begin{aligned} \boldsymbol{x} &:= \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{cc} 0 & a \\ a & 0 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] + \left[\begin{array}{c} 1-a \\ 1-a \end{array} \right] =: \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}. \end{aligned}$	$\boldsymbol{x} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1-a \\ 1-a \end{bmatrix} =: \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}.$	$\boldsymbol{x} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1-a \\ 1-a \end{bmatrix} =: \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}.$
lyche-numerical-linear-algebra-E08942	319	■ 0.999	$\mathbb{W}_k := \operatorname{span} \left(\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \dots, \boldsymbol{A}^{k-1}\boldsymbol{b} \right).$	$\mathbb{W}_k := \operatorname{span} (\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \dots, \boldsymbol{A}^{k-1}\boldsymbol{b}).$	$\mathbb{W}_k := \operatorname{span} (\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \dots, \boldsymbol{A}^{k-1}\boldsymbol{b}).$
lyche-numerical-linear-algebra-E08662 (1.21)	41	■ 0.999	$\overline{\boldsymbol{C}} = \boldsymbol{C} \left(\boldsymbol{I} + \frac{1}{\lambda} \boldsymbol{e}_i (\boldsymbol{e}_i^T - \boldsymbol{a}^T \boldsymbol{C}) \right)$	$\overline{\boldsymbol{C}} = \boldsymbol{C} \left(\boldsymbol{I} + \frac{1}{\lambda} \boldsymbol{e}_i (\boldsymbol{e}_i^T - \boldsymbol{a}^T \boldsymbol{C}) \right)$	$\overline{\boldsymbol{C}} = \boldsymbol{C} \left(\boldsymbol{I} + \frac{1}{\lambda} \boldsymbol{e}_i (\boldsymbol{e}_i^T - \boldsymbol{a}^T \boldsymbol{C}) \right)$
lyche-numerical-linear-algebra-E08282 (3.12)	85	■ 0.999	$\begin{aligned} \boldsymbol{P}^T &= \boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_1 = \boldsymbol{I}(\boldsymbol{p}, :), \quad \boldsymbol{P} = \boldsymbol{P}_1 \boldsymbol{P}_2 \cdots \boldsymbol{P}_{n-1} = \boldsymbol{I}(:, \boldsymbol{p}). \\ \boldsymbol{P}_1 &= \boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_1 = \boldsymbol{I}(\boldsymbol{p}, :), \quad \boldsymbol{P} = \boldsymbol{P}_1 \boldsymbol{P}_2 \cdots \boldsymbol{P}_{n-1} = \boldsymbol{I}(:, \boldsymbol{p}). \end{aligned}$	$\boldsymbol{P}^T = \boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_1 = \boldsymbol{I}(\boldsymbol{p}, :), \quad \boldsymbol{P} = \boldsymbol{P}_1 \boldsymbol{P}_2 \cdots \boldsymbol{P}_{n-1} = \boldsymbol{I}(:, \boldsymbol{p}).$	$\boldsymbol{P}^T = \boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_1 = \boldsymbol{I}(\boldsymbol{p}, :), \quad \boldsymbol{P} = \boldsymbol{P}_1 \boldsymbol{P}_2 \cdots \boldsymbol{P}_{n-1} = \boldsymbol{I}(:, \boldsymbol{p}).$
lyche-numerical-linear-algebra-E08254	104	■ 0.999	$\begin{aligned} \boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} &= 2 \sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i \\ &= \sum_{i=1}^{n-1} x_i^2 - 2 \sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2 \\ &= x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2. \end{aligned}$	$\begin{aligned} \boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} &= 2 \sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i \\ &= \sum_{i=1}^{n-1} x_i^2 - 2 \sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2 \\ &= x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2. \end{aligned}$	$\begin{aligned} \boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} &= 2 \sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i \\ &= \sum_{i=1}^{n-1} x_i^2 - 2 \sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2 \\ &= x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2. \end{aligned}$
lyche-numerical-linear-algebra-E08286	119	■ 0.999	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 \leq \ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{x}\ \ \boldsymbol{y}\ + \ \boldsymbol{y}\ ^2 = (\ \boldsymbol{x}\ + \ \boldsymbol{y}\)^2.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 \leq \ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{x}\ \ \boldsymbol{y}\ + \ \boldsymbol{y}\ ^2 = (\ \boldsymbol{x}\ + \ \boldsymbol{y}\)^2.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 \leq \ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{x}\ \ \boldsymbol{y}\ + \ \boldsymbol{y}\ ^2 = (\ \boldsymbol{x}\ + \ \boldsymbol{y}\)^2.$
lyche-numerical-linear-algebra-E08310	126	■ 0.999	$\mathcal{M} := \left\{ \boldsymbol{w} \in \mathbb{R}^3 : \boldsymbol{w}^T \boldsymbol{v} = 0 \right\} = \left\{ \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} : 2w_1 = 0 \right\}$	$\mathcal{M} := \left\{ \boldsymbol{w} \in \mathbb{R}^3 : \boldsymbol{w}^T \boldsymbol{v} = 0 \right\} = \left\{ \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} : 2w_1 = 0 \right\}$	$\mathcal{M} := \left\{ \boldsymbol{w} \in \mathbb{R}^3 : \boldsymbol{w}^T \boldsymbol{v} = 0 \right\} = \left\{ \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} : 2w_1 = 0 \right\}$

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lyche-numerical-linear-algebra- EQ0348	141	■ 0.999	$\left[\begin{array}{ll} p_{i\ i} & p_{i\ j} \\ p_{j\ i} & p_{j\ j} \end{array}\right] = \left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right],$	$\left[\begin{array}{cc} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{array}\right] = \left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right],$	$\left[\begin{array}{cc} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{array}\right] = \left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right],$
lyche-numerical-linear-algebra- EQ0412 (6.20)	165	■ 0.999	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{u}_j = \sum_{i=1}^n c_{ij} \boldsymbol{u}_i, \quad \text{where } c_{ij} = \frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{Ax} = \sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \quad \text{where } c_j = \frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{Ax} = \sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \quad \text{where } c_j = \frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$
lyche-numerical-linear-algebra- EQ0440	176	■ 0.999	$\frac{\left(\frac{3}{5}x_1 + \frac{4}{5}x_2\right)^2}{9} + \frac{\left(-\frac{4}{5}x_1 + \frac{3}{5}x_2\right)^2}{1} = 1,$	$\frac{\left(\frac{3}{5}x_1 + \frac{4}{5}x_2\right)^2}{9} + \frac{\left(-\frac{4}{5}x_1 + \frac{3}{5}x_2\right)^2}{1} = 1,$	$\frac{\left(\frac{3}{5}x_1 + \frac{4}{5}x_2\right)^2}{9} + \frac{\left(-\frac{4}{5}x_1 + \frac{3}{5}x_2\right)^2}{1} = 1,$
lyche-numerical-linear-algebra- EQ0499	195	■ 0.999	$x_1 + \left(1 + 10^{-16}\right)x_2 = 20 - 10^{-15},$	$x_1 + \left(1 + 10^{-16}\right)x_2 = 20 - 10^{-15},$	$x_1 + (1 + 10^{-16})x_2 = 20 - 10^{-15},$
lyche-numerical-linear-algebra- EQ0510	198	■ 0.999	$\left\ \boldsymbol{B}^{-1}\right\ \leq \left\ \boldsymbol{A}^{-1}\right\ + \left\ \boldsymbol{A}^{-1} \boldsymbol{EB}^{-1}\right\ \leq \left\ \boldsymbol{A}^{-1}\right\ + r\left\ \boldsymbol{B}^{-1}\right\ .$	$\left\ \boldsymbol{B}^{-1}\right\ \leq \left\ \boldsymbol{A}^{-1}\right\ + \left\ \boldsymbol{A}^{-1} \boldsymbol{EB}^{-1}\right\ \leq \left\ \boldsymbol{A}^{-1}\right\ + r\left\ \boldsymbol{B}^{-1}\right\ .$	$\left\ \boldsymbol{B}^{-1}\right\ \leq \left\ \boldsymbol{A}^{-1}\right\ + \left\ \boldsymbol{A}^{-1} \boldsymbol{EB}^{-1}\right\ \leq \left\ \boldsymbol{A}^{-1}\right\ + r\left\ \boldsymbol{B}^{-1}\right\ .$
lyche-numerical-linear-algebra- EQ0559	209	■ 0.999	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }.$	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }.$	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }.$
lyche-numerical-linear-algebra- EQ0617 (9.14)	226	■ 0.999	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}.$	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}.$	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}.$
lyche-numerical-linear-algebra- EQ0661 (10.5)	239	■ 0.999	$\boldsymbol{TV} + \boldsymbol{VT} = h^2 \boldsymbol{F} \quad \text{with } h = 1/(m+1),$	$\boldsymbol{TV} + \boldsymbol{VT} = h^2 \boldsymbol{F} \quad \text{with } h = 1/(m+1),$	$\boldsymbol{TV} + \boldsymbol{VT} = h^2 \boldsymbol{F} \quad \text{with } h = 1/(m+1),$
lyche-numerical-linear-algebra- EQ0723	258	■ 0.999	$\boldsymbol{D}_2 = \text{diag}(1, \omega_4) = \left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right].$	$\boldsymbol{D}_2 = \text{diag}(1, \omega_4) = \left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right].$	$\boldsymbol{D}_2 = \text{diag}(1, \omega_4) = \left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right].$
lyche-numerical-linear-algebra- EQ0846	293	■ 0.999	$\begin{aligned} & \boldsymbol{t}_0 = 3 \left[\begin{array}{c} 1 \\ -1/2 \end{array}\right], \quad \alpha_0 = \frac{1}{2}, \quad \boldsymbol{x}_1 = -4^{-1} \left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_1 = 3 * 4^{-1} \left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \boldsymbol{t}_1 = 3 * 4^{-1} \left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_1 = \frac{1}{2}, \quad \boldsymbol{x}_2 = -4^{-1} \left[\begin{array}{c} 1 \\ 1/2 \end{array}\right], \quad \boldsymbol{r}_2 = 3 * 4^{-1} \left[\begin{array}{c} 1/2 \\ 0 \end{array}\right], \end{aligned}$	$\begin{aligned} & \boldsymbol{t}_0 = 3 \left[\begin{array}{c} 1 \\ -1/2 \end{array}\right], \quad \alpha_0 = \frac{1}{2}, \quad \boldsymbol{x}_1 = -4^{-1} \left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_1 = 3 * 4^{-1} \left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \boldsymbol{t}_1 = 3 * 4^{-1} \left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_1 = \frac{1}{2}, \quad \boldsymbol{x}_2 = -4^{-1} \left[\begin{array}{c} 1 \\ 1/2 \end{array}\right], \quad \boldsymbol{r}_2 = 3 * 4^{-1} \left[\begin{array}{c} 1/2 \\ 0 \end{array}\right], \end{aligned}$	$\begin{aligned} & \boldsymbol{t}_0 = 3 \left[\begin{array}{c} 1 \\ -1/2 \end{array}\right], \quad \alpha_0 = \frac{1}{2}, \quad \boldsymbol{x}_1 = -4^{-1} \left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_1 = 3 * 4^{-1} \left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \boldsymbol{t}_1 = 3 * 4^{-1} \left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_1 = \frac{1}{2}, \quad \boldsymbol{x}_2 = -4^{-1} \left[\begin{array}{c} 1 \\ 1/2 \end{array}\right], \quad \boldsymbol{r}_2 = 3 * 4^{-1} \left[\begin{array}{c} 1/2 \\ 0 \end{array}\right], \end{aligned}$

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lyche-numerical-linear-algebra-EQ08899 (13.40)	308	■ 0.999	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad k=0,1,2,\ldots,$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad k=0,1,2,\ldots,$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad k=0,1,2,\ldots,$
lyche-numerical-linear-algebra-EQ0937	317	■ 0.999	$c_{\{0\}}\left(x^T\boldsymbol{A}_{\{p\}}x\right) \leq x^T\boldsymbol{A}x \leq c_{\{1\}}\left(x^T\boldsymbol{A}_{\{p\}}x\right)$	$c_0\left(x^T\boldsymbol{A}_px\right) \leq x^T\boldsymbol{A}x \leq c_1\left(x^T\boldsymbol{A}_px\right)$	$c_0(x^T\boldsymbol{A}_px) \leq x^T\boldsymbol{A}x \leq c_1(x^T\boldsymbol{A}_px)$
lyche-numerical-linear-algebra-EQ0945	320	■ 0.999	$\boldsymbol{A}\boldsymbol{p}_{k-1} \in \operatorname{span}\left(\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}, \boldsymbol{p}_k\right).$	$\boldsymbol{A}\boldsymbol{p}_{k-1} \in \operatorname{span}\left(\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}, \boldsymbol{p}_k\right).$	$\boldsymbol{A}\boldsymbol{p}_{k-1} \in \operatorname{span}(\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}, \boldsymbol{p}_k).$
lyche-numerical-linear-algebra-EQ0960 (13.69)	323	■ 0.999	$\begin{aligned} &\boldsymbol{x}_{k+1}=\left(1-\omega_k\right) \boldsymbol{x}_{k-1}+\omega_k \boldsymbol{x}_k \\ &\boldsymbol{r}_{k+1}=\left(1-\omega_k\right) \boldsymbol{r}_{k-1}+\omega_k \boldsymbol{B} \boldsymbol{r}_k, \end{aligned}$	$\begin{aligned} \boldsymbol{x}_{k+1} &= (1-\omega_k) \boldsymbol{x}_{k-1} + \omega_k (\boldsymbol{x}_k + \boldsymbol{r}_k), \\ \boldsymbol{r}_{k+1} &= (1-\omega_k) \boldsymbol{r}_{k-1} + \omega_k \boldsymbol{B} \boldsymbol{r}_k, \end{aligned}$	$\begin{aligned} \boldsymbol{x}_{k+1} &= (1-\omega_k) \boldsymbol{x}_{k-1} + \omega_k (\boldsymbol{x}_k + \boldsymbol{r}_k), \\ \boldsymbol{r}_{k+1} &= (1-\omega_k) \boldsymbol{r}_{k-1} + \omega_k \boldsymbol{B} \boldsymbol{r}_k, \end{aligned}$
lyche-numerical-linear-algebra-EQ0986	332	■ 0.999	$\frac{1}{\lambda} \leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\ \boldsymbol{A}\ _p}$	$\frac{1}{\lambda} \leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\ \boldsymbol{A}\ _p}$	$\frac{1}{\lambda} \leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\ \boldsymbol{A}\ _p}$
lyche-numerical-linear-algebra-EQ1025	346	■ 0.999	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1 \lambda_1^k}{\left c_1 \lambda_1^k\right }\left\ \frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n\right\ }, \quad k=0,1,2,\ldots,$	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1 \lambda_1^k}{\left c_1 \lambda_1^k\right }\left\ \frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n\right\ }, \quad k=0,1,2,\ldots,$	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1 \lambda_1^k}{\left c_1 \lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k \boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k \boldsymbol{v}_n\right\ }, \quad k=0,1,2,\ldots,$
lyche-numerical-linear-algebra-EQ1034	349	■ 0.999	$\begin{aligned} &s_k=\frac{\boldsymbol{y}_k^* \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k}=s_{k-1}+\frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k}=s_{k-1}+\frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k}=s_{k-1}+\rho_k, \\ &\boldsymbol{r}_k=\boldsymbol{A} \boldsymbol{x}_k-s_k \boldsymbol{x}_k=\frac{\boldsymbol{A} \boldsymbol{y}_k-\left(s_{k-1}+\rho_k\right) \boldsymbol{y}_k}{\left\ \boldsymbol{y}_k\right\ _2}=\frac{\boldsymbol{x}_{k-1}-\rho_k \boldsymbol{y}_k}{\left\ \boldsymbol{y}_k\right\ _2}=\boldsymbol{w}_k-\rho_k \boldsymbol{x}_k. \end{aligned}$	$\begin{aligned} s_k &= \frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k, \\ \boldsymbol{r}_k &= \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\left\ \boldsymbol{y}_k\right\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\left\ \boldsymbol{y}_k\right\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k. \end{aligned}$	$\begin{aligned} s_k &= \frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k, \\ \boldsymbol{r}_k &= \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\left\ \boldsymbol{y}_k\right\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\left\ \boldsymbol{y}_k\right\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k. \end{aligned}$

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lyche-numerical-linear-algebra-EQ0097	54	■ 0.999	$\left[\begin{array}{lll} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array}\right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right], \quad \left[\begin{array}{ccc} u_1 & c_1 & 0 \\ 0 & 0 & u_2 \\ 0 & 0 & u_3 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right].$		
lyche-numerical-linear-algebra-EQ0380	154	■ 0.999	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$		
lyche-numerical-linear-algebra-EQ0394 (6.13)	158	■ 0.999	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$		
lyche-numerical-linear-algebra-EQ0445	178	■ 0.999	$\ \boldsymbol{A} - \boldsymbol{A}'\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \ \boldsymbol{A} - \boldsymbol{B}\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$		
lyche-numerical-linear-algebra-EQ0621	227	■ 0.999	$\frac{\ \boldsymbol{A}\boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A}\boldsymbol{x})^*(\boldsymbol{A}\boldsymbol{x})}{\boldsymbol{x}^*\boldsymbol{x}} = \frac{\boldsymbol{x}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}} = R_{\boldsymbol{A}^*\boldsymbol{A}}(\boldsymbol{x}),$		
lyche-numerical-linear-algebra-EQ0903	309	■ 0.999	$\ \boldsymbol{\epsilon}_k\ _2^2 - \ \boldsymbol{\epsilon}_{k+1}\ _2^2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} \left(\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2 \right)$		
lyche-numerical-linear-algebra-EQ0003	22	■ 1.000	$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$		

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lyche-numerical-linear-algebra-EQ0008	23	■ 1.000	$\boldsymbol{e}_1 := \left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_2 := \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_3 := \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array} \right], \quad \dots, \quad \boldsymbol{e}_n := \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array} \right],$		$\boldsymbol{e}_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_2 := \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_3 := \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad \boldsymbol{e}_n := \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0025	30	■ 1.000	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$		$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$
lyche-numerical-linear-algebra-EQ0053	39	■ 1.000	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \dots + c_0,$	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \dots + c_0,$	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \dots + c_0,$
lyche-numerical-linear-algebra-EQ0078 (2.2)	46	■ 1.000	$g(x_i) = y_i, \text{ for } i = 1, \dots, n+1.$	$g(x_i) = y_i, \text{ for } i = 1, \dots, n+1.$	$g(x_i) = y_i, \text{ for } i = 1, \dots, n+1.$
lyche-numerical-linear-algebra-EQ0087 (2.9)	51	■ 1.000	$\mu_{i-1} + 4\mu_i + \mu_{i+1} = \frac{6}{h^2} (y_{i+1} - 2y_i + y_{i-1}), \quad i = 2, \dots, n,$	$\mu_{i-1} + 4\mu_i + \mu_{i+1} = \frac{6}{h^2} (y_{i+1} - 2y_i + y_{i-1}), \quad i = 2, \dots, n,$	$\mu_{i-1} + 4\mu_i + \mu_{i+1} = \frac{6}{h^2} (y_{i+1} - 2y_i + y_{i-1}), \quad i = 2, \dots, n,$
lyche-numerical-linear-algebra-EQ0099 (2.16)	54	■ 1.000	$u_1 = d_1, \quad l_k = a_k / u_k, \quad u_{k+1} = d_{k+1} - l_k c_k, \quad k = 1, 2, \dots, n-1,$	$u_1 = d_1, \quad l_k = a_k / u_k, \quad u_{k+1} = d_{k+1} - l_k c_k, \quad k = 1, 2, \dots, n-1,$	$u_1 = d_1, \quad l_k = a_k / u_k, \quad u_{k+1} = d_{k+1} - l_k c_k, \quad k = 1, 2, \dots, n-1,$
lyche-numerical-linear-algebra-EQ0102 (2.19)	56	■ 1.000	$ l_k = \frac{ a_k }{ u_k } \leq \frac{ a_k }{ d_k - a_{k-1} }, u_{k+1} \leq d_{k+1} + \frac{ a_k c_k }{ d_k - a_{k-1} }, k = 1, \dots, n-1.$	$ l_k = \frac{ a_k }{ u_k } \leq \frac{ a_k }{ d_k - a_{k-1} }, u_{k+1} \leq d_{k+1} + \frac{ a_k c_k }{ d_k - a_{k-1} }, k = 1, \dots, n-1.$	$ l_k = \frac{ a_k }{ u_k } \leq \frac{ a_k }{ d_k - a_{k-1} }, u_{k+1} \leq d_{k+1} + \frac{ a_k c_k }{ d_k - a_{k-1} }, k = 1, \dots, n-1.$
lyche-numerical-linear-algebra-EQ0103	56	■ 1.000	$ l_k = \frac{ a_k }{ u_k } \leq \frac{ a_k }{ d_k - a_{k-1} }, u_{k+1} \leq d_{k+1} + l_k c_k \leq d_{k+1} + \frac{ a_k c_k }{ d_k - a_{k-1} }.$	$ l_k = \frac{ a_k }{ u_k } \leq \frac{ a_k }{ d_k - a_{k-1} }, u_{k+1} \leq d_{k+1} + l_k c_k \leq d_{k+1} + \frac{ a_k c_k }{ d_k - a_{k-1} }.$	$ l_k = \frac{ a_k }{ u_k } \leq \frac{ a_k }{ d_k - a_{k-1} }, u_{k+1} \leq d_{k+1} + l_k c_k \leq d_{k+1} + \frac{ a_k c_k }{ d_k - a_{k-1} }.$
lyche-numerical-linear-algebra-EQ0121 (2.31)	61	■ 1.000	$\boldsymbol{S} = [s_1, s_2, s_3] = \begin{bmatrix} \sin \frac{\pi}{4} \sin \frac{2\pi}{4} \sin \frac{3\pi}{4} \\ \sin \frac{2\pi}{4} \sin \frac{4\pi}{4} \sin \frac{6\pi}{4} \\ \sin \frac{3\pi}{4} \sin \frac{6\pi}{4} \sin \frac{9\pi}{4} \end{bmatrix} = \begin{bmatrix} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{bmatrix}, \quad t := \frac{1}{\sqrt{2}}.$	$\boldsymbol{S} = [s_1, s_2, s_3] = \begin{bmatrix} \sin \frac{\pi}{4} \sin \frac{2\pi}{4} \sin \frac{3\pi}{4} \\ \sin \frac{2\pi}{4} \sin \frac{4\pi}{4} \sin \frac{6\pi}{4} \\ \sin \frac{3\pi}{4} \sin \frac{6\pi}{4} \sin \frac{9\pi}{4} \end{bmatrix} = \begin{bmatrix} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{bmatrix}, \quad t := \frac{1}{\sqrt{2}}.$	$\boldsymbol{S} = [s_1, s_2, s_3] = \begin{bmatrix} \sin \frac{\pi}{4} \sin \frac{2\pi}{4} \sin \frac{3\pi}{4} \\ \sin \frac{2\pi}{4} \sin \frac{4\pi}{4} \sin \frac{6\pi}{4} \\ \sin \frac{3\pi}{4} \sin \frac{6\pi}{4} \sin \frac{9\pi}{4} \end{bmatrix} = \begin{bmatrix} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{bmatrix}, \quad t := \frac{1}{\sqrt{2}}.$

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lyche-numerical-linear-algebra- EQ0447 (7.11)	178	■ 1.000	$\boldsymbol{A}:=\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & 0 \end{array}\right]\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]=:\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$	$\boldsymbol{A}:=\left[\begin{array}{cc} 1 & -1 \\ 1 & -1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & 0 \end{array}\right]\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]=:\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$	$\boldsymbol{A}:=\left[\begin{array}{cc} 1 & -1 \\ 1 & -1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & 0 \end{array}\right]\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]=:\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$
lyche-numerical-linear-algebra- EQ0457	181	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 2 \\ -1 & 3 \end{array}\right]$.	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right]$.	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0463	183	■ 1.000	$\boldsymbol{T}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]$.	$\boldsymbol{T}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]$.	$\boldsymbol{T}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0484 (8.16)	190	■ 1.000	$\ \boldsymbol{A}\ =\ \boldsymbol{Ax}^*\ $ for some $\boldsymbol{x}^*\in\mathcal{S}$.	$\ \boldsymbol{A}\ =\ \boldsymbol{Ax}^*\ $ for some $\boldsymbol{x}^*\in\mathcal{S}$.	$\ \boldsymbol{A}\ =\ \boldsymbol{Ax}^*\ $ for some $\boldsymbol{x}^*\in\mathcal{S}$.
lyche-numerical-linear-algebra- EQ0490	192	■ 1.000	$\ \boldsymbol{A}\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^na_{kj}x_j\right \leq\max_{1\leq k\leq m}\sum_{j=1}^n a_{kj} x_j \leq K_{\infty}$.	$\ \boldsymbol{Ax}\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^na_{kj}x_j\right \leq\max_{1\leq k\leq m}\sum_{j=1}^n a_{kj} x_j \leq K_{\infty}$.	$\ \boldsymbol{Ax}\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^na_{kj}x_j\right \leq\max_{1\leq k\leq m}\sum_{j=1}^n a_{kj} x_j \leq K_{\infty}$.
lyche-numerical-linear-algebra- EQ0493 (8.21)	193	■ 1.000	$\ \boldsymbol{A}\ _2\ \boldsymbol{v}\ _1=\sigma^2\ \boldsymbol{v}\ _1=\ \sigma^2\boldsymbol{v}\ _1=\ \boldsymbol{A}^*\boldsymbol{Av}\ _1\leq\ \boldsymbol{A}^*\ _1\ \boldsymbol{A}\ _1\ \boldsymbol{v}\ _1$.	$\ \boldsymbol{A}\ _2^2\ \boldsymbol{v}\ _1=\sigma^2\ \boldsymbol{v}\ _1=\ \sigma^2\boldsymbol{v}\ _1=\ \boldsymbol{A}^*\boldsymbol{Av}\ _1\leq\ \boldsymbol{A}^*\ _1\ \boldsymbol{A}\ _1\ \boldsymbol{v}\ _1$.	$\ \boldsymbol{A}\ _2^2\ \boldsymbol{v}\ _1=\sigma^2\ \boldsymbol{v}\ _1=\ \sigma^2\boldsymbol{v}\ _1=\ \boldsymbol{A}^*\boldsymbol{Av}\ _1\leq\ \boldsymbol{A}^*\ _1\ \boldsymbol{A}\ _1\ \boldsymbol{v}\ _1$.
lyche-numerical-linear-algebra- EQ0505 (8.27)	198	■ 1.000	$\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }\leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }$,	$\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }\leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }$,	$\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }\leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }$,
lyche-numerical-linear-algebra- EQ0509	198	■ 1.000	$\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ \leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \ \boldsymbol{B}^{-1}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }\ \boldsymbol{B}^{-1}\ $.	$\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ \leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \ \boldsymbol{B}^{-1}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }\ \boldsymbol{B}^{-1}\ $.	$\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ \leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \ \boldsymbol{B}^{-1}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }\ \boldsymbol{B}^{-1}\ $.
lyche-numerical-linear-algebra- EQ0521 (8.34)	201	■ 1.000	$a^{\frac{1}{p}}b^{\frac{1}{q}}\leq\frac{1}{p}a+\frac{1}{q}b$, where $\frac{1}{p}+\frac{1}{q}=1$.	$a^{\frac{1}{p}}b^{\frac{1}{q}}\leq\frac{1}{p}a+\frac{1}{q}b$, where $\frac{1}{p}+\frac{1}{q}=1$.	$a^{\frac{1}{p}}b^{\frac{1}{q}}\leq\frac{1}{p}a+\frac{1}{q}b$, where $\frac{1}{p}+\frac{1}{q}=1$.
lyche-numerical-linear-algebra- EQ0528 (8.36)	202	■ 1.000	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}(\ \boldsymbol{x}+\boldsymbol{y}\ ^2-\ \boldsymbol{x}-\boldsymbol{y}\ ^2)$, $\boldsymbol{x},\boldsymbol{y}\in\mathcal{V}$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}(\ \boldsymbol{x}+\boldsymbol{y}\ ^2-\ \boldsymbol{x}-\boldsymbol{y}\ ^2)$, $\boldsymbol{x},\boldsymbol{y}\in\mathcal{V}$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}(\ \boldsymbol{x}+\boldsymbol{y}\ ^2-\ \boldsymbol{x}-\boldsymbol{y}\ ^2)$, $\boldsymbol{x},\boldsymbol{y}\in\mathcal{V}$

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lyche-numerical-linear-algebra- EQ0551	208	■ 1.000	$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{l} b_1 \\ b_2 \end{array}\right]=\left[\begin{array}{l} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{l} e_1 \\ e_2 \end{array}\right], \quad \ e\ _2=0.1.$	$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{l} b_1 \\ b_2 \end{array}\right]=\left[\begin{array}{l} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{l} e_1 \\ e_2 \end{array}\right], \quad \ e\ _2=0.1.$	$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{l} b_1 \\ b_2 \end{array}\right]=\left[\begin{array}{l} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{l} e_1 \\ e_2 \end{array}\right], \quad \ e\ _2=0.1.$
lyche-numerical-linear-algebra- EQ0598	220	■ 1.000	$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{A}^{*} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_1 \boldsymbol{x}=\boldsymbol{c}_1, \quad \boldsymbol{c}_1:=\boldsymbol{Q}_1^{*} \boldsymbol{b}.$	$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{A}^{*} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_1 \boldsymbol{x}=\boldsymbol{c}_1, \quad \boldsymbol{c}_1:=\boldsymbol{Q}_1^{*} \boldsymbol{b}.$	$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{A}^{*} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_1 \boldsymbol{x}=\boldsymbol{c}_1, \quad \boldsymbol{c}_1:=\boldsymbol{Q}_1^{*} \boldsymbol{b}.$
lyche-numerical-linear-algebra- EQ0608 (9.11)	223	■ 1.000	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*}.$	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*}.$	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*}.$
lyche-numerical-linear-algebra- EQ0682 (10.15)	245	■ 1.000	$\lambda_j=d+2 a \cos (j \pi h), \quad h:=\frac{1}{m+1},$	$\lambda_j=d+2 a \cos (j \pi h), \quad h:=\frac{1}{m+1},$	$\lambda_j=d+2 a \cos (j \pi h), \quad h:=\frac{1}{m+1},$
lyche-numerical-linear-algebra- EQ0684 (10.17)	246	■ 1.000	$\boldsymbol{s}_{\boldsymbol{j}}^T \boldsymbol{s}_{\boldsymbol{k}}=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m.$	$\boldsymbol{s}_{\boldsymbol{j}}^T \boldsymbol{s}_{\boldsymbol{k}}=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m.$	$\boldsymbol{s}_{\boldsymbol{j}}^T \boldsymbol{s}_{\boldsymbol{k}}=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m.$
lyche-numerical-linear-algebra- EQ0692 (10.22)	248	■ 1.000	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^2 \mu \boldsymbol{F}$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^2 \mu \boldsymbol{F}$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^2 \mu \boldsymbol{F}.$
lyche-numerical-linear-algebra- EQ0695 (10.21)	248	■ 1.000	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^2 \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U}+\boldsymbol{U} \boldsymbol{T}=h^2 \boldsymbol{V}.$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^2 \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U}+\boldsymbol{U} \boldsymbol{T}=h^2 \boldsymbol{V}.$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^2 \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U}+\boldsymbol{U} \boldsymbol{T}=h^2 \boldsymbol{V}.$
lyche-numerical-linear-algebra- EQ0721	257	■ 1.000	$\boldsymbol{A} \boldsymbol{P}_N=\left[\boldsymbol{a}_1, \boldsymbol{a}_3, \ldots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \ldots, \boldsymbol{a}_N\right],$	$\boldsymbol{A} \boldsymbol{P}_N=\left[\boldsymbol{a}_1, \boldsymbol{a}_3, \ldots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \ldots, \boldsymbol{a}_N\right],$	$\boldsymbol{A} \boldsymbol{P}_N=\left[\boldsymbol{a}_1, \boldsymbol{a}_3, \ldots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \ldots, \boldsymbol{a}_N\right],$
lyche-numerical-linear-algebra- EQ0754 (12.9)	270	■ 1.000	$\boldsymbol{M} \boldsymbol{x}_{k+1}=(\boldsymbol{M}-\boldsymbol{A}) \boldsymbol{x}_k+\boldsymbol{b}$	$\boldsymbol{M} \boldsymbol{x}_{k+1}=(\boldsymbol{M}-\boldsymbol{A}) \boldsymbol{x}_k+\boldsymbol{b}$	$\boldsymbol{M} \boldsymbol{x}_{k+1}=(\boldsymbol{M}-\boldsymbol{A}) \boldsymbol{x}_k+\boldsymbol{b}$
lyche-numerical-linear-algebra- EQ0789 (12.29)	277	■ 1.000	$\rho\left(\boldsymbol{G}_{\omega}\right) > \rho\left(\boldsymbol{G}_{\omega^{*}}\right) \text { for } \omega \in(0,2) \backslash\left\{\omega^{*}\right\}.$	$\rho\left(\boldsymbol{G}_{\omega}\right) > \rho\left(\boldsymbol{G}_{\omega^{*}}\right) \text { for } \omega \in(0,2) \backslash\left\{\omega^{*}\right\}.$	$\rho\left(\boldsymbol{G}_{\omega}\right) > \rho\left(\boldsymbol{G}_{\omega^{*}}\right) \text { for } \omega \in(0,2) \backslash\left\{\omega^{*}\right\}.$
lyche-numerical-linear-algebra- EQ0791	278	■ 1.000	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ \leq\left\ \boldsymbol{G}^k\right\ \left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ \approx \rho(\boldsymbol{G})^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ .$	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ \leq\left\ \boldsymbol{G}^k\right\ \left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ \approx \rho(\boldsymbol{G})^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ .$	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ \leq\left\ \boldsymbol{G}^k\right\ \left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ \approx \rho(\boldsymbol{G})^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ .$
lyche-numerical-linear-algebra- EQ0820	285	■ 1.000	$d^{\{\prime\}}(\omega)=2\left(\omega \mu^2-2\right) < 2(\omega-2) < 0.$	$d^{\prime}(\omega)=2\left(\omega \mu^2-2\right) < 2(\omega-2) < 0.$	$d^{\prime}(\omega)=2\left(\omega \mu^2-2\right) < 2(\omega-2) < 0.$

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lyche-numerical-linear-algebra-2 EQ0840 (13.6)	302	1.000	$\begin{aligned} & \frac{\partial}{\partial y_i} Q(\boldsymbol{y}) &:= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (Q(\boldsymbol{y} + \varepsilon \boldsymbol{e}_i) - Q(\boldsymbol{y})) \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left(-\varepsilon \boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}) + \frac{1}{2} \varepsilon^2 \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i \right) = -\boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}), \quad i = 1, \dots, n, \end{aligned}$	$\frac{\partial}{\partial y_i} Q(\boldsymbol{y}) := \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (Q(\boldsymbol{y} + \varepsilon \boldsymbol{e}_i) - Q(\boldsymbol{y}))$ $= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left(-\varepsilon \boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}) + \frac{1}{2} \varepsilon^2 \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i \right) = -\boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}), \quad i = 1, \dots, n,$	$\frac{\partial}{\partial y_i} Q(\boldsymbol{y}) := \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (Q(\boldsymbol{y} + \varepsilon \boldsymbol{e}_i) - Q(\boldsymbol{y}))$ $= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left(-\varepsilon \boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}) + \frac{1}{2} \varepsilon^2 \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i \right) = -\boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}), \quad i = 1, \dots, n,$
lyche-numerical-linear-algebra-2 EQ0866 (13.23)	301	1.000	$\begin{aligned} & \boldsymbol{r}_k^T \boldsymbol{w} = \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k. \\ & \boldsymbol{p}_k^T \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k, \\ & \boldsymbol{w} \in \mathbb{W}_k. \end{aligned}$	$\boldsymbol{r}_k^T \boldsymbol{w} = \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k.$	$\boldsymbol{r}_k^T \boldsymbol{w} = \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k.$
lyche-numerical-linear-algebra-2 EQ0874 (13.29)	303	1.000	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{\lambda_{\min} \leq x \leq \lambda_{\max}} Q(x) = \frac{2}{a^{-k} + a^k}, \quad a := \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1},$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{\lambda_{\min} \leq x \leq \lambda_{\max}} Q(x) = \frac{2}{a^{-k} + a^k}, \quad a := \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1},$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{\lambda_{\min} \leq x \leq \lambda_{\max}} Q(x) = \frac{2}{a^{-k} + a^k}, \quad a := \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1},$
lyche-numerical-linear-algebra-2 EQ0875 (13.30)	303	1.000	$\frac{x-1}{x+1} < e^{-2/x} \quad \text{for } x > 1$	$\frac{x-1}{x+1} < e^{-2/x} \quad \text{for } x > 1$	$\frac{x-1}{x+1} < e^{-2/x} \quad \text{for } x > 1$
lyche-numerical-linear-algebra-2 EQ0929 (13.58)	315	1.000	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle := h^2 \sum_{j,k=1}^m v_{j,k} w_{j,k},$	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle := h^2 \sum_{j,k=1}^m v_{j,k} w_{j,k},$	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle := h^2 \sum_{j,k=1}^m v_{j,k} w_{j,k},$
lyche-numerical-linear-algebra-2 EQ0930 (13.59)	315	1.000	$\begin{aligned} & \langle L_h v, \boldsymbol{W} \rangle = \sum_{j=1}^m \sum_{k=0}^m c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) (w_{j,k+1} - w_{j,k}) \\ & + \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) (w_{j+1,k} - w_{j,k}). \end{aligned}$	$\langle L_h v, \boldsymbol{W} \rangle = \sum_{j=1}^m \sum_{k=0}^m c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) (w_{j,k+1} - w_{j,k})$ $+ \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) (w_{j+1,k} - w_{j,k}).$	$\langle L_h v, \boldsymbol{W} \rangle = \sum_{j=1}^m \sum_{k=0}^m c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) (w_{j,k+1} - w_{j,k})$ $+ \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) (w_{j+1,k} - w_{j,k}).$
lyche-numerical-linear-algebra-2 EQ0946	320	1.000	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$
lyche-numerical-linear-algebra-2 EQ0957	322	1.000	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \quad \text{and} \quad \rho_k := \ \boldsymbol{r}_k\ _2^2.$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \quad \text{and} \quad \rho_k := \ \boldsymbol{r}_k\ _2^2.$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \quad \text{and} \quad \rho_k := \ \boldsymbol{r}_k\ _2^2.$

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lyche-numerical-linear-algebra- EQ09967	326	■ 1.000	$\begin{aligned} \boldsymbol{A} \boldsymbol{D} &= \operatorname{diag}(\boldsymbol{A}_1, \dots, \boldsymbol{A}_r) \operatorname{diag}(\boldsymbol{X}_1, \dots, \boldsymbol{X}_r) = \operatorname{diag}(\boldsymbol{A}_1 \boldsymbol{X}_1, \dots, \boldsymbol{A}_r \boldsymbol{X}_r) \\ &= \operatorname{diag}(\boldsymbol{X}_1 \boldsymbol{D}_1, \dots, \boldsymbol{X}_r \boldsymbol{D}_r) = \boldsymbol{X} \boldsymbol{D}. \end{aligned}$	$\boldsymbol{AD} = \operatorname{diag}(\boldsymbol{A}_1, \dots, \boldsymbol{A}_r) \operatorname{diag}(\boldsymbol{X}_1, \dots, \boldsymbol{X}_r) = \operatorname{diag}(\boldsymbol{A}_1 \boldsymbol{X}_1, \dots, \boldsymbol{A}_r \boldsymbol{X}_r) \\ = \operatorname{diag}(\boldsymbol{X}_1 \boldsymbol{D}_1, \dots, \boldsymbol{X}_r \boldsymbol{D}_r) = \boldsymbol{XD}.$	$\boldsymbol{AD} = \operatorname{diag}(\boldsymbol{A}_1, \dots, \boldsymbol{A}_r) \operatorname{diag}(\boldsymbol{X}_1, \dots, \boldsymbol{X}_r) = \operatorname{diag}(\boldsymbol{A}_1 \boldsymbol{X}_1, \dots, \boldsymbol{A}_r \boldsymbol{X}_r) \\ = \operatorname{diag}(\boldsymbol{X}_1 \boldsymbol{D}_1, \dots, \boldsymbol{X}_r \boldsymbol{D}_r) = \boldsymbol{XD}.$
lyche-numerical-linear-algebra- EQ1030	348	■ 1.000	$\mu_1(s) = \left(\lambda_1 - s \right)^{-1}, \mu_2(s) = \left(\lambda_2 - s \right)^{-1}, \dots, \mu_n(s) = \left(\lambda_n - s \right)^{-1}.$	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1}.$	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1}.$
lyche-numerical-linear-algebra- EQ1041	352	■ 1.000	$\boldsymbol{A}_{14} = \begin{bmatrix} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{bmatrix}.$	$\boldsymbol{A}_{14} = \begin{bmatrix} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{bmatrix}.$	$\boldsymbol{A}_{14} = \begin{bmatrix} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ1045	353	■ 1.000	$\tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k = \tilde{\boldsymbol{Q}}_{k-1} (\boldsymbol{Q}_k \boldsymbol{R}_k) \tilde{\boldsymbol{R}}_{k-1} = \tilde{\boldsymbol{Q}}_{k-1} \boldsymbol{A}_k \tilde{\boldsymbol{R}}_{k-1} = (\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{Q}}_{k-1}^*) \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^k.$	$\tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k = \tilde{\boldsymbol{Q}}_{k-1} (\boldsymbol{Q}_k \boldsymbol{R}_k) \tilde{\boldsymbol{R}}_{k-1} = \tilde{\boldsymbol{Q}}_{k-1} \boldsymbol{A}_k \tilde{\boldsymbol{R}}_{k-1} = (\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{Q}}_{k-1}^*) \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^k.$	$\tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k = \tilde{\boldsymbol{Q}}_{k-1} (\boldsymbol{Q}_k \boldsymbol{R}_k) \tilde{\boldsymbol{R}}_{k-1} = \tilde{\boldsymbol{Q}}_{k-1} \boldsymbol{A}_k \tilde{\boldsymbol{R}}_{k-1} = (\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{Q}}_{k-1}^*) \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^k.$
lyche-numerical-linear-algebra- EQ1047	354	■ 1.000	$\hat{\boldsymbol{A}}_k = \tilde{\boldsymbol{Q}}_{k-1}^* (\boldsymbol{A} + \boldsymbol{E}) \tilde{\boldsymbol{Q}}_{k-1}, \quad \boldsymbol{E} = \tilde{\boldsymbol{Q}}_{k-1} \left(\boldsymbol{a}_{i+1,i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_i^T \right) \tilde{\boldsymbol{Q}}_{k-1}^*.$	$\hat{\boldsymbol{A}}_k = \tilde{\boldsymbol{Q}}_{k-1}^* (\boldsymbol{A} + \boldsymbol{E}) \tilde{\boldsymbol{Q}}_{k-1}, \quad \boldsymbol{E} = \tilde{\boldsymbol{Q}}_{k-1} \left(\boldsymbol{a}_{i+1,i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_i^T \right) \tilde{\boldsymbol{Q}}_{k-1}^*.$	$\hat{\boldsymbol{A}}_k = \tilde{\boldsymbol{Q}}_{k-1}^* (\boldsymbol{A} + \boldsymbol{E}) \tilde{\boldsymbol{Q}}_{k-1}, \quad \boldsymbol{E} = \tilde{\boldsymbol{Q}}_{k-1} (\boldsymbol{a}_{i+1,i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_i^T) \tilde{\boldsymbol{Q}}_{k-1}^*.$
lyche-numerical-linear-algebra- EQ0002	22	■ 1.000	$\boldsymbol{x} + \boldsymbol{y} := \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \boldsymbol{ax} := \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \quad \boldsymbol{a} \in \mathbb{R}.$	$\boldsymbol{x} + \boldsymbol{y} := \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \boldsymbol{ax} := \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \quad \boldsymbol{a} \in \mathbb{R}.$	$\boldsymbol{x} + \boldsymbol{y} := \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \boldsymbol{ax} := \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \quad \boldsymbol{a} \in \mathbb{R}.$
lyche-numerical-linear-algebra- EQ0004	22	■ 1.000	$\boldsymbol{a}_{:j} := \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \quad \boldsymbol{a}_{i:}^T := [a_{i1}, a_{i2}, \dots, a_{in}], \quad \boldsymbol{A} = [\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \dots, \boldsymbol{a}_{:n}] = \begin{bmatrix} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{bmatrix}$	$\boldsymbol{a}_{:j} := \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \quad \boldsymbol{a}_{i:}^T := [a_{i1}, a_{i2}, \dots, a_{in}], \quad \boldsymbol{A} = [\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \dots, \boldsymbol{a}_{:n}] = \begin{bmatrix} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{bmatrix}$	$\boldsymbol{a}_{:j} := \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \quad \boldsymbol{a}_{i:}^T := [a_{i1}, a_{i2}, \dots, a_{in}], \quad \boldsymbol{A} = [\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \dots, \boldsymbol{a}_{:n}] = \begin{bmatrix} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{bmatrix}$

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lyche-numerical-linear-algebra-EQ0005	22	■ 1.000	$e^{x}=e^{a+i\,b}:=e^{a}(\cos b+i\sin b)$.	$e^x=e^{a+ib}:=e^a(\cos b+i\sin b)$.	$e^x=e^{a+ib}:=e^a(\cos b+i\sin b)$.
lyche-numerical-linear-algebra-EQ0006	22	■ 1.000	$e^{i\,\pi/2}=i$, $\quad e^{i\pi}=-1$, $\quad e^{2i\pi}=1$.	$e^{i\pi/2}=i$, $\quad e^{i\pi}=-1$, $\quad e^{2i\pi}=1$.	$e^{i\pi/2}=i$, $\quad e^{i\pi}=-1$, $\quad e^{2i\pi}=1$.
lyche-numerical-linear-algebra-EQ0007	23	■ 1.000	$x=a+i\,b=r\,e^{i\,\theta}$, $\quad r= x =\sqrt{a^2+b^2}$, $\quad \cos\theta=\frac{a}{r}$, $\quad \sin\theta=\frac{b}{r}$.	$x=a+ib=re^{i\theta}$, $\quad r= x =\sqrt{a^2+b^2}$, $\quad \cos\theta=\frac{a}{r}$, $\quad \sin\theta=\frac{b}{r}$.	$x=a+ib=re^{i\theta}$, $\quad r= x =\sqrt{a^2+b^2}$, $\quad \cos\theta=\frac{a}{r}$, $\quad \sin\theta=\frac{b}{r}$.
lyche-numerical-linear-algebra-EQ0014 (1.2)	25	■ 1.000	$p(t):=a_{\{0\}}+a_{\{1\}}\,t+a_{\{2\}}\,t^2+\cdots+a_{\{n\}}\,t^n$,	$p(t):=a_0+a_1t+a_2t^2+\cdots+a_nt^n$,	$p(t):=a_0+a_1t+a_2t^2+\cdots+a_nt^n$,
lyche-numerical-linear-algebra-EQ0017 (1.5)	28	■ 1.000	$\mathcal{S}\,\cap\,\mathcal{T}:=\{\boldsymbol{x}:\boldsymbol{x}\in\mathcal{S}\,\text{and}\,\boldsymbol{x}\in\mathcal{T}\}$ $\boldsymbol{x}\in\mathcal{S}\,\,\text{and}\,\,\boldsymbol{x}\in\mathcal{T}$.	$\mathcal{S}\cap\mathcal{T}:=\{\boldsymbol{x}:\boldsymbol{x}\in\mathcal{S}\,\text{and}\,\boldsymbol{x}\in\mathcal{T}\}$.	$\mathcal{S}\cap\mathcal{T}:=\{\boldsymbol{x}:\boldsymbol{x}\in\mathcal{S}\,\text{and}\,\boldsymbol{x}\in\mathcal{T}\}$.
lyche-numerical-linear-algebra-EQ0018 (1.6)	28	■ 1.000	$\mathcal{S}\,\cup\,\mathcal{T}:=\{\boldsymbol{x}:\boldsymbol{x}\in\mathcal{S}\,\text{or}\,\boldsymbol{x}\in\mathcal{T}\}$ $\boldsymbol{x}\in\mathcal{S}\,\,\text{or}\,\,\boldsymbol{x}\in\mathcal{T}$.	$\mathcal{S}\cup\mathcal{T}:=\{\boldsymbol{x}:\boldsymbol{x}\in\mathcal{S}\,\text{or}\,\boldsymbol{x}\in\mathcal{T}\}$.	$\mathcal{S}\cup\mathcal{T}:=\{\boldsymbol{x}:\boldsymbol{x}\in\mathcal{S}\,\text{or}\,\boldsymbol{x}\in\mathcal{T}\}$.
lyche-numerical-linear-algebra-EQ0020 (1.8)	29	■ 1.000	$\operatorname{dim}(\mathcal{S}\,\oplus\,\mathcal{T})=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})$.	$\dim(\mathcal{S}\oplus\mathcal{T})=\dim(\mathcal{S})+\dim(\mathcal{T})$.	$\dim(\mathcal{S}\oplus\mathcal{T})=\dim(\mathcal{S})+\dim(\mathcal{T})$.
lyche-numerical-linear-algebra-EQ0021	29	■ 1.000	$\left\{\boldsymbol{u}_{\{1\}},\,\ldots,\,\boldsymbol{u}_{\{p\}},\,\boldsymbol{s}_{\{1\}},\,\ldots,\,\boldsymbol{s}_{\{q\}},\,\boldsymbol{t}_{\{1\}},\,\ldots,\,\boldsymbol{t}_{\{r\}}\right\}$	$\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_p,\boldsymbol{s}_1,\ldots,\boldsymbol{s}_q,\boldsymbol{t}_1,\ldots,\boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_p,\boldsymbol{s}_1,\ldots,\boldsymbol{s}_q,\boldsymbol{t}_1,\ldots,\boldsymbol{t}_r\}$
lyche-numerical-linear-algebra-EQ0022	29	■ 1.000	$\left\{\boldsymbol{u}_{\{1\}},\,\ldots,\,\boldsymbol{u}_{\{p\}},\,\boldsymbol{s}_{\{1\}},\,\ldots,\,\boldsymbol{s}_{\{q\}}\right\}$	$\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_p,\boldsymbol{s}_1,\ldots,\boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_p,\boldsymbol{s}_1,\ldots,\boldsymbol{s}_q\}$
lyche-numerical-linear-algebra-EQ0023	29	■ 1.000	$\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})=p+q+r=(p+q)+(p+r)-p=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S}\cap\mathcal{T})$	$\dim(\mathcal{S}+\mathcal{T})=p+q+r=(p+q)+(p+r)-p=\dim(\mathcal{S})+\dim(\mathcal{T})-\dim(\mathcal{S}\cap\mathcal{T})$	$\dim(\mathcal{S}+\mathcal{T})=p+q+r=(p+q)+(p+r)-p=\dim(\mathcal{S})+\dim(\mathcal{T})-\dim(\mathcal{S}\cap\mathcal{T})$
lyche-numerical-linear-algebra-EQ0026	30	■ 1.000	$\mathcal{R}(\boldsymbol{X}):=\left\{\boldsymbol{X}\boldsymbol{c}:\boldsymbol{c}\in\mathbb{R}^n\right\}=\operatorname{span}(\mathcal{X})$.	$\mathcal{R}(\boldsymbol{X}):=\{\boldsymbol{X}\boldsymbol{c}:\boldsymbol{c}\in\mathbb{R}^n\}=\operatorname{span}(\mathcal{X})$.	$\mathcal{R}(\boldsymbol{X}):=\{\boldsymbol{X}\boldsymbol{c}:\boldsymbol{c}\in\mathbb{R}^n\}=\operatorname{span}(\mathcal{X})$.
lyche-numerical-linear-algebra-EQ0030	31	■ 1.000	$x_{\{1\}}\,\boldsymbol{a}_{\{1\}}+x_{\{2\}}\,\boldsymbol{a}_{\{2\}}+\cdots+x_{\{n\}}\,\boldsymbol{a}_{\{n\}}=\boldsymbol{b}$,	$x_1\boldsymbol{a}_1+x_2\boldsymbol{a}_2+\cdots+x_n\boldsymbol{a}_n=\boldsymbol{b}$,	$x_1\boldsymbol{a}_1+x_2\boldsymbol{a}_2+\cdots+x_n\boldsymbol{a}_n=\boldsymbol{b}$,
lyche-numerical-linear-algebra-EQ0035 (1.12)	35	■ 1.000	$\operatorname{det}(\boldsymbol{A})=\sum_{\sigma\in S_n}\operatorname{sign}(\sigma)\,a_{\sigma(1),1}\,a_{\sigma(2),2}\cdots a_{\sigma(n),n}$	$\det(\boldsymbol{A})=\sum_{\sigma\in S_n}\operatorname{sign}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$	$\det(\boldsymbol{A})=\sum_{\sigma\in S_n}\operatorname{sign}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$.
lyche-numerical-linear-algebra-EQ0036	35	■ 1.000	$\left \begin{array}{cccc}a_{\{11\}} & a_{\{12\}} & \cdots & a_{\{1n\}} \\ a_{\{21\}} & a_{\{22\}} & \cdots & a_{\{2n\}} \\ \vdots & \vdots & & \vdots \\ a_{\{n1\}} & a_{\{n2\}} & \cdots & a_{\{nn\}}\end{array}\right $.	$\left \begin{array}{cccc}a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}\end{array}\right $.	$\left \begin{array}{cccc}a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}\end{array}\right $.
lyche-numerical-linear-algebra-EQ0037	35	■ 1.000	$\left \begin{array}{ll}a_{\{11\}} & a_{\{12\}} \\ a_{\{21\}} & a_{\{22\}}\end{array}\right =a_{\{11\}}\,a_{\{22\}}-a_{\{21\}}\,a_{\{12\}}$.	$\left \begin{array}{ll}a_{11} & a_{12} \\ a_{21} & a_{22}\end{array}\right =a_{11}a_{22}-a_{21}a_{12}$.	$\left \begin{array}{ll}a_{11} & a_{12} \\ a_{21} & a_{22}\end{array}\right =a_{11}a_{22}-a_{21}a_{12}$.
lyche-numerical-linear-algebra-EQ0039 (1.13)	36	■ 1.000	$\operatorname{det}(\boldsymbol{B})=\operatorname{det}(\boldsymbol{A})$.	$\det(\boldsymbol{B})=\det(\boldsymbol{A})$.	$\det(\boldsymbol{B})=\det(\boldsymbol{A})$.

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lyche-numerical-linear-algebra-EQ0041	36	■ 1.000	$\operatorname{det}(\boldsymbol{A}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$	$\det(\boldsymbol{A}) := \begin{vmatrix} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{vmatrix} = 0$	$\det(\boldsymbol{A}) := \begin{vmatrix} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{vmatrix} = 0$
lyche-numerical-linear-algebra-EQ0043	36	■ 1.000	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$
lyche-numerical-linear-algebra-EQ0044	37	■ 1.000	$x_j = \frac{\operatorname{det}(\boldsymbol{A}_j(\boldsymbol{b}))}{\operatorname{det}(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$	$x_j = \frac{\det(\boldsymbol{A}_j(\boldsymbol{b}))}{\det(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$	$x_j = \frac{\det(\boldsymbol{A}_j(\boldsymbol{b}))}{\det(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$
lyche-numerical-linear-algebra-EQ0045	37	■ 1.000	$\boldsymbol{A}^{-1} = \frac{1}{\operatorname{det}(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$	$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$	$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$
lyche-numerical-linear-algebra-EQ0048	38	■ 1.000	$\operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \left \begin{array}{ccc} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{array} \right .$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}.$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}.$
lyche-numerical-linear-algebra-EQ0050 (1.16)	38	■ 1.000	$\operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) + r(\lambda),$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) + r(\lambda),$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) + r(\lambda),$
lyche-numerical-linear-algebra-EQ0051 (1.17)	39	■ 1.000	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \operatorname{det}(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$
lyche-numerical-linear-algebra-EQ0052 (1.18)	39	■ 1.000	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$
lyche-numerical-linear-algebra-EQ0054	39	■ 1.000	$\pi_A(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$	$\pi_A(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$	$\pi_A(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$
lyche-numerical-linear-algebra-EQ0055 (1.19)	39	■ 1.000	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \operatorname{det}(\boldsymbol{A}) = 0.$	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0.$	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0.$
lyche-numerical-linear-algebra-EQ0056	40	■ 1.000	$\left[\begin{array}{cc} W & X \\ Y & Z \end{array} \right] = \left[\begin{array}{cc} A & B \\ C & D \end{array} \right] \left[\begin{array}{cc} E & F \\ G & H \end{array} \right],$	$\begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix},$	$\begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0059	41	■ 1.000	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0063	42	■ 1.000	$\left[\begin{array}{cc} 1 & 2 \\ 2 & 1 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{c} 3 \\ 6 \end{array} \right]$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$
lyche-numerical-linear-algebra-EQ0065	42	■ 1.000	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix}.$	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix}.$	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix}.$

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lyche-numerical-linear-algebra-E00066	42	■ 1.000	$\operatorname{adj}(\boldsymbol{A}) \boldsymbol{A}=\left[\begin{array}{ccc} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{array}\right]=\operatorname{det}(\boldsymbol{A})\boldsymbol{I}.$	$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A}=\left[\begin{array}{ccc} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{array}\right]=\det(\boldsymbol{A})\boldsymbol{I}.$	$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A}=\left[\begin{array}{ccc} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{array}\right]=\det(\boldsymbol{A})\boldsymbol{I}.$
lyche-numerical-linear-algebra-E00067	42	■ 1.000	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right =0$	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right =0$	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right =0.$
lyche-numerical-linear-algebra-E00068	43	■ 1.000	$A(T)=\frac{1}{2}\left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array}\right .$	$A(T)=\frac{1}{2}\left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array}\right .$	$A(T)=\frac{1}{2}\left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array}\right .$
lyche-numerical-linear-algebra-E00069	43	■ 1.000	$\left \begin{array}{cccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{array}\right =\prod_{i>j}(x_i-x_j),$	$\left \begin{array}{cccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{array}\right =\prod_{i>j}(x_i-x_j),$	$\left \begin{array}{cccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{array}\right =\prod_{i>j}(x_i-x_j),$
lyche-numerical-linear-algebra-E00070	43	■ 1.000	$\operatorname{det}(\boldsymbol{A})=P\ g(\boldsymbol{\alpha})\ g(\boldsymbol{\beta})$	$\det(\boldsymbol{A})=Pg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$	$\det(\boldsymbol{A})=Pg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$
lyche-numerical-linear-algebra-E00071	44	■ 1.000	$g(\boldsymbol{\gamma})=\prod_{i=2}^n(\gamma_i-\gamma_1)\left(\gamma_i-\gamma_2\right) \cdots\left(\gamma_i-\gamma_{i-1}\right)$	$g(\boldsymbol{\gamma})=\prod_{i=2}^n(\gamma_i-\gamma_1)\left(\gamma_i-\gamma_2\right) \cdots\left(\gamma_i-\gamma_{i-1}\right)$	$g(\boldsymbol{\gamma})=\prod_{i=2}^n(\gamma_i-\gamma_1)\left(\gamma_i-\gamma_2\right) \cdots\left(\gamma_i-\gamma_{i-1}\right)$
lyche-numerical-linear-algebra-E00072	44	■ 1.000	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})=k\ g(\boldsymbol{\alpha})\ g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})=kg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})=kg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$
lyche-numerical-linear-algebra-E00074	44	■ 1.000	$A_{\{k\}}(x)=\prod_{s \neq k}\left(\frac{\alpha_s-x}{\alpha_s-\alpha_k}\right), \quad B_{\{k\}}(x)=\prod_{s \neq k}\left(\frac{\beta_s-x}{\beta_s-\beta_k}\right).$	$A_k(x)=\prod_{s \neq k}\left(\frac{\alpha_s-x}{\alpha_s-\alpha_k}\right), \quad B_k(x)=\prod_{s \neq k}\left(\frac{\beta_s-x}{\beta_s-\beta_k}\right).$	$A_k(x)=\prod_{s \neq k}\left(\frac{\alpha_s-x}{\alpha_s-\alpha_k}\right), \quad B_k(x)=\prod_{s \neq k}\left(\frac{\beta_s-x}{\beta_s-\beta_k}\right).$
lyche-numerical-linear-algebra-E00075	44	■ 1.000	$t_{\{i, j\}^n}=\frac{f(i) f(j)}{i+j-1},$	$t_{i, j}^n=\frac{f(i) f(j)}{i+j-1},$	$t_{i, j}^n=\frac{f(i) f(j)}{i+j-1},$
lyche-numerical-linear-algebra-E00076	44	■ 1.000	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n.$	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n.$	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n.$
lyche-numerical-linear-algebra-E00077 (2.1)	46	■ 1.000	$x_i=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1$	$x_i=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1$	$x_i=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1$
lyche-numerical-linear-algebra-E00079 (2.3)	47	■ 1.000	$g(x)=y_1+\frac{y_2-y_1}{x_2-x_1}(x-x_1),$	$g(x)=y_1+\frac{y_2-y_1}{x_2-x_1}(x-x_1),$	$g(x)=y_1+\frac{y_2-y_1}{x_2-x_1}(x-x_1),$
lyche-numerical-linear-algebra-E00080	47	■ 1.000	$f(x)=\arctan (10 x)+\pi / 2, \quad x \in[-1,1] .$	$f(x)=\arctan (10 x)+\pi / 2, \quad x \in[-1,1] .$	$f(x)=\arctan (10 x)+\pi / 2, \quad x \in[-1,1] .$

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lyche-numerical-linear-algebra-E00082 (2.5)	49	1.000	$g(x):= \begin{cases} p_1(x)=-\frac{1}{2}x+\frac{3}{2}x^3 & \text{ if } 0 \leq x < 1, \\ p_2(x)=1+4(x-1)+\frac{9}{2}(x-1)^2+\frac{13}{2}(x-1)^3, & \text{ if } 1 \leq x \leq 2, \end{cases}$	$g(x):= \begin{cases} p_1(x)=-\frac{1}{2}x+\frac{3}{2}x^3, & \text{ if } 0 \leq x < 1, \\ p_2(x)=1+4(x-1)+\frac{9}{2}(x-1)^2+\frac{13}{2}(x-1)^3, & \text{ if } 1 \leq x \leq 2, \end{cases}$	$g(x):= \begin{cases} p_1(x)=-\frac{1}{2}x+\frac{3}{2}x^3, & \text{ if } 0 \leq x < 1, \\ p_2(x)=1+4(x-1)+\frac{9}{2}(x-1)^2+\frac{13}{2}(x-1)^3, & \text{ if } 1 \leq x \leq 2, \end{cases}$
lyche-numerical-linear-algebra-E00083	49	1.000	$p_{i-1}^{(j)}(x_i)=p_i^{(j)}(x_i), \quad j=0,1,2, \quad i=2,\dots,n.$	$p_{i-1}^{(j)}(x_i)=p_i^{(j)}(x_i), \quad j=0,1,2, \quad i=2,\dots,n.$	$p_{i-1}^{(j)}(x_i)=p_i^{(j)}(x_i), \quad j=0,1,2, \quad i=2,\dots,n.$
lyche-numerical-linear-algebra-E00085 (2.7)	50	1.000	$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\dots,n,$	$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\dots,n,$	$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\dots,n,$
lyche-numerical-linear-algebra-E00086 (2.8)	50	1.000	$c_{i,1}=y_i, \quad c_{i,2}=\frac{y_{i+1}-y_i}{h}-\frac{\frac{y_{i+1}-y_i}{h}-\frac{y_{i+1}-y_i}{h}}{6}\mu_{i+1}, \quad c_{i,3}=\frac{\frac{y_{i+1}-y_i}{h}-\frac{y_{i+1}-y_i}{h}}{2}, \quad c_{i,4}=\frac{\mu_{i+1}-\mu_i}{6h}.$	$c_{i,1}=y_i, c_{i,2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}, c_{i,3}=\frac{\mu_i}{2}, c_{i,4}=\frac{\mu_{i+1}-\mu_i}{6h}.$	$c_{i,1}=y_i, \quad c_{i,2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}, \quad c_{i,3}=\frac{\mu_i}{2}, \quad c_{i,4}=\frac{\mu_{i+1}-\mu_i}{6h}.$
lyche-numerical-linear-algebra-E00088 (2.10)	51	1.000	$\begin{aligned} & p_{i-1}^{\prime}(x_i) &= c_{i-1,2}+2hc_{i-1,3}+3h^2c_{i-1,4} \\ & &= \frac{y_i-y_{i-1}}{h}-\frac{h}{3}\mu_{i-1}-\frac{h}{6}\mu_i+2h\frac{\mu_{i-1}}{2}+3h^2\frac{\mu_i-\mu_{i-1}}{6h} \\ & &= \frac{y_i-y_{i-1}}{h}+\frac{h}{6}\mu_{i-1}+\frac{h}{3}\mu_i \\ & p_i^{\prime}(x_i) &= c_{i,2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}. \end{aligned}$	$\begin{aligned} p_{i-1}^{\prime}(x_i) &= c_{i-1,2}+2hc_{i-1,3}+3h^2c_{i-1,4} \\ &= \frac{y_i-y_{i-1}}{h}-\frac{h}{3}\mu_{i-1}-\frac{h}{6}\mu_i+2h\frac{\mu_{i-1}}{2}+3h^2\frac{\mu_i-\mu_{i-1}}{6h} \\ &= \frac{y_i-y_{i-1}}{h}+\frac{h}{6}\mu_{i-1}+\frac{h}{3}\mu_i \\ p_i^{\prime}(x_i) &= c_{i,2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}. \end{aligned}$	$\begin{aligned} p_{i-1}^{\prime}(x_i) &= c_{i-1,2}+2hc_{i-1,3}+3h^2c_{i-1,4} \\ &= \frac{y_i-y_{i-1}}{h}-\frac{h}{3}\mu_{i-1}-\frac{h}{6}\mu_i+2h\frac{\mu_{i-1}}{2}+3h^2\frac{\mu_i-\mu_{i-1}}{6h} \\ &= \frac{y_i-y_{i-1}}{h}+\frac{h}{6}\mu_{i-1}+\frac{h}{3}\mu_i \\ p_i^{\prime}(x_i) &= c_{i,2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}. \end{aligned}$
lyche-numerical-linear-algebra-E00090 (2.12)	52	1.000	$ a_{ii} >\sum_{j \neq i} a_{ij} , \quad i=1,\dots,n.$	$ a_{ii} >\sum_{j \neq i} a_{ij} , \quad i=1,\dots,n.$	$ a_{ii} >\sum_{j \neq i} a_{ij} , \quad i=1,\dots,n.$
lyche-numerical-linear-algebra-E00091 (2.13)	52	1.000	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n}\left(\frac{ b_i }{\sigma_i}\right), \quad \text{where } \sigma_i:= a_{ii} -\sum_{j \neq i} a_{ij} .$	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n}\left(\frac{ b_i }{\sigma_i}\right), \quad \text{where } \sigma_i:= a_{ii} -\sum_{j \neq i} a_{ij} .$	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n}\left(\frac{ b_i }{\sigma_i}\right), \quad \text{where } \sigma_i:= a_{ii} -\sum_{j \neq i} a_{ij} .$
lyche-numerical-linear-algebra-E00092	52	1.000	$ b_k = a_{kk}x_k+\sum_{j \neq k}a_{kj}x_j \geq a_{kk} x_k -\sum_{j \neq k} a_{kj} x_j \geq x_k (a_{kk} -\sum_{j \neq k} a_{kj}),$	$ b_k = \left a_{kk}x_k+\sum_{j \neq k}a_{kj}x_j \right \geq a_{kk} x_k -\sum_{j \neq k} a_{kj} x_j \geq x_k \left(a_{kk} -\sum_{j \neq k} a_{kj} \right),$	$ b_k = a_{kk}x_k+\sum_{j \neq k}a_{kj}x_j \geq a_{kk} x_k -\sum_{j \neq k} a_{kj} x_j \geq x_k (a_{kk} -\sum_{j \neq k} a_{kj}),$
lyche-numerical-linear-algebra-E00093	52	1.000	$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 2 \end{bmatrix}.$	$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 2 \end{bmatrix}.$	$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 2 \end{bmatrix}.$

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lyche-numerical-linear-algebra-EQ0094 (2.14)	53	1.000	$g(x) := \begin{cases} p_1(x) = \frac{1}{6}x^3, & \text{if } 0 \leq x < 1, \\ p_2(x) = \frac{1}{6} + \frac{1}{2}(x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{2}(x-1)^3, & \text{if } 1 \leq x < 2, \\ p_3(x) = \frac{2}{3} - (x-2)^2 + \frac{1}{2}(x-2)^3, & \text{if } 2 \leq x < 3, \\ p_4(x) = \frac{1}{6} - \frac{1}{2}(x-3) + \frac{1}{2}(x-3)^2 - \frac{1}{6}(x-3)^3, & \text{if } 3 \leq x \leq 4, \end{cases}$		
lyche-numerical-linear-algebra-EQ0095 (2.15)	54	1.000	$\left[\begin{array}{cccc} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{array} \right] = \left[\begin{array}{cccc} 1 & & & \\ l_1 & 1 & & \\ & \ddots & \ddots & \\ & & l_{n-1} & 1 \end{array} \right] \left[\begin{array}{cccc} u_1 & c_1 & & \\ & \ddots & \ddots & \\ & & u_{n-1} & c_{n-1} \\ & & & u_n \end{array} \right].$		
lyche-numerical-linear-algebra-EQ0100 (2.17)	55	1.000	$ u_k \geq \sigma_k + c_k , \quad \text{where, } \sigma_k := d_k - a_{k-1} - c_k > 0, k = 1, \dots, n,$		
lyche-numerical-linear-algebra-EQ0104 (2.20)	56	1.000	$-u''(x) = f(x), \quad x \in [0, 1], \quad u(0) = 0, u(1) = 0,$		
lyche-numerical-linear-algebra-EQ0105	56	1.000	$g'(x) = \lim_{h \rightarrow 0} \frac{g(x + \frac{h}{2}) - g(x - \frac{h}{2})}{h}$		
lyche-numerical-linear-algebra-EQ0107	57	1.000	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} = f(jh), \quad j = 1, \dots, m, \quad v_0 = v_{m+1} = 0.$		
lyche-numerical-linear-algebra-EQ0109 (2.22)	57	1.000	$ a_{ii} \geq \sum_{j \neq i} a_{ij} , i = 1, \dots, n.$		
lyche-numerical-linear-algebra-EQ0111 (2.23)	58	1.000	$ u_{k+1} = d_{k+1} - l_k c_k = \left d_{k+1} - \frac{a_k c_k}{u_k} \right \geq d_{k+1} - \frac{ a_k c_k }{ u_k } > d_{k+1} - a_k .$		
lyche-numerical-linear-algebra-EQ0112 (2.24)	59	1.000	$Ry''(x) = -Fy(x), \quad y(0) = y(L) = 0,$		
lyche-numerical-linear-algebra-EQ0113 (2.25)	59	1.000	$u''(t) = -Ku(t), \quad u(0) = u(1) = 0, \quad K := \frac{FL^2}{R}.$		
lyche-numerical-linear-algebra-EQ0115	59	1.000	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} = Kv_j, \quad j = 1, \dots, m, h = \frac{1}{m+1}, \quad v_0 = v_{m+1} = 0,$		

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lyche-numerical-linear-algebra-EQ0116 (2.26)	60	■ 1.000	$\boldsymbol{T} \ \boldsymbol{v}=\lambda \boldsymbol{v},$ $\text{ with } \boldsymbol{v}=\left[v_1, \ldots, v_m\right]^T,$ $\right]^T,$	$T \boldsymbol{v}=\lambda \boldsymbol{v}, \text { with } \boldsymbol{v}=\left[v_1, \ldots, v_m\right]^T,$	$T \boldsymbol{v}=\lambda \boldsymbol{v}, \text { with } \boldsymbol{v}=\left[v_1, \ldots, v_m\right]^T,$
lyche-numerical-linear-algebra-EQ0118	60	■ 1.000	$F=\frac{4 \sin ^2(\pi h / 2) R}{h^2 L^2} .$	$F=\frac{4 \sin ^2(\pi h / 2) R}{h^2 L^2} .$	$F=\frac{4 \sin ^2(\pi h / 2) R}{h^2 L^2} .$
lyche-numerical-linear-algebra-EQ0119 (2.28)	60	■ 1.000	$\boldsymbol{T}_1:=\operatorname{tridiag}(a, d, a)$	$T_1:=\operatorname{tridiag}(a, d, a)$	$T_1:=\operatorname{tridiag}(a, d, a)$
lyche-numerical-linear-algebra-EQ0120 (2.29)	60	■ 1.000	$\boldsymbol{S}=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^m,$ $k=1\}^m \in \mathbb{R}^{m \times m} .$	$S=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^m \in \mathbb{R}^{m \times m} .$	$S=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^m \in \mathbb{R}^{m \times m} .$
lyche-numerical-linear-algebra-EQ0126	62	■ 1.000	$\sum_{k=0}^m \cos (2 k j \pi h)+i \sum_{k=0}^m \sin (2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$ $(2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$	$\sum_{k=0}^m \cos (2 k j \pi h)+i \sum_{k=0}^m \sin (2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$	$\sum_{k=0}^m \cos (2 k j \pi h)+i \sum_{k=0}^m \sin (2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$
lyche-numerical-linear-algebra-EQ0130	63	■ 1.000	$\boldsymbol{A} \ \boldsymbol{B}=\left[\boldsymbol{A} \ \boldsymbol{b}_{: 1}, \boldsymbol{A} \ \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \ \boldsymbol{b}_{: n}\right] .$ $\boldsymbol{b}_{: 1}, \boldsymbol{A} \ \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \ \boldsymbol{b}_{: n}\right] .$	$A B=\left[A b_{: 1}, A b_{: 2}, \ldots, A b_{: n}\right] .$	$A B=\left[A b_{: 1}, A b_{: 2}, \ldots, A b_{: n}\right] .$
lyche-numerical-linear-algebra-EQ0131	63	■ 1.000	$\boldsymbol{A}=\boldsymbol{A} \ \boldsymbol{I}=\left[\boldsymbol{A} \ \boldsymbol{e}_{: 1}, \boldsymbol{A} \ \boldsymbol{e}_{: 2}, \ldots, \boldsymbol{A} \ \boldsymbol{e}_{: p}\right] .$ $\boldsymbol{e}_{: 1}, \boldsymbol{A} \ \boldsymbol{e}_{: 2}, \ldots, \boldsymbol{A} \ \boldsymbol{e}_{: p}\right] .$	$A=A I=A\left[e_1, e_2, \ldots, e_p\right]=\left[A e_1, A e_2, \ldots, A e_p\right]$	$A=A I=A\left[e_1, e_2, \ldots, e_p\right]=\left[A e_1, A e_2, \ldots, A e_p\right]$
lyche-numerical-linear-algebra-EQ0133	63	■ 1.000	$\boldsymbol{A} \ \boldsymbol{x}=\boldsymbol{x}_{: 1} \boldsymbol{a}_{: 1}+\boldsymbol{x}_{: 2} \boldsymbol{a}_{: 2}+\cdots+\boldsymbol{x}_p \boldsymbol{a}_{: p} .$ $\boldsymbol{a}_{: 1}+\boldsymbol{x}_{: 2} \boldsymbol{a}_{: 2}+\cdots+\boldsymbol{x}_p \boldsymbol{a}_{: p} .$	$A x=x_1 a_{: 1}+x_2 a_{: 2}+\cdots+x_p a_{: p} .$	$A x=x_1 a_{: 1}+x_2 a_{: 2}+\cdots+x_p a_{: p} .$
lyche-numerical-linear-algebra-EQ0134	64	■ 1.000	$\boldsymbol{A}\left[\boldsymbol{B}_1, \boldsymbol{B}_2\right]=\left[\boldsymbol{A} \boldsymbol{B}_1, \boldsymbol{A} \boldsymbol{B}_2\right] .$	$A\left[B_1, B_2\right]=\left[A B_1, A B_2\right] .$	$A\left[B_1, B_2\right]=\left[A B_1, A B_2\right] .$
lyche-numerical-linear-algebra-EQ0136	64	■ 1.000	$\left[\boldsymbol{A}_1, \boldsymbol{A}_2\right]\left[\begin{array}{l} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{array}\right]=\left[\boldsymbol{A}_1 \boldsymbol{B}_1+\boldsymbol{A}_2 \boldsymbol{B}_2\right] .$	$\left[A_1, A_2\right]\left[\begin{array}{l} B_1 \\ B_2 \end{array}\right]=\left[A_1 B_1+A_2 B_2\right] .$	$\left[A_1, A_2\right]\left[\begin{array}{l} B_1 \\ B_2 \end{array}\right]=\left[A_1 B_1+A_2 B_2\right] .$
lyche-numerical-linear-algebra-EQ0140	65	■ 1.000	$\boldsymbol{C}_{i j}=\sum_{k=1}^s \boldsymbol{A}_{i k} \boldsymbol{B}_{k j}, \quad i=1, \ldots, p, j=1, \ldots, q$	$C_{ij}=\sum_{k=1}^s A_{ik} B_{kj}, \quad i=1, \ldots, p, j=1, \ldots, q$	$C_{ij}=\sum_{k=1}^s A_{ik} B_{kj}, \quad i=1, \ldots, p, j=1, \ldots, q$

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lyche-numerical-linear-algebra-EQ0141	65	1.000	$\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p1} & \cdots & \boldsymbol{A}_{ps} \end{array}\right] \left[\begin{array}{ccc} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s1} & \cdots & \boldsymbol{B}_{sq} \end{array}\right] = \left[\begin{array}{ccc} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p1} & \cdots & \boldsymbol{C}_{pq} \end{array}\right].$	$\begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p1} & \cdots & \boldsymbol{A}_{ps} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s1} & \cdots & \boldsymbol{B}_{sq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p1} & \cdots & \boldsymbol{C}_{pq} \end{bmatrix}.$	
lyche-numerical-linear-algebra-EQ0143 (2.33)	65	1.000	$\boldsymbol{A}^{\{-1\}}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{\{-1\}} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{\{-1\}} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],$
lyche-numerical-linear-algebra-EQ0146	66	1.000	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{\{-1\}}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{\{-1\}}.$	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}.$	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}.$
lyche-numerical-linear-algebra-EQ0148	66	1.000	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & a_{k+1,\,k+1} \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & a_{k+1,\,k+1} \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & a_{k+1,\,k+1} \end{array}\right]$
lyche-numerical-linear-algebra-EQ0149	66	1.000	$\boldsymbol{A}^{\{-1\}}=\left[\begin{array}{cc} \boldsymbol{A}_k^{\{-1\}} & \boldsymbol{c} \\ \boldsymbol{0} & a_{k+1,\,k+1}^{\{-1\}} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_k^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & a_{k+1,\,k+1}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_k^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & a_{k+1,\,k+1}^{-1} \end{array}\right],$
lyche-numerical-linear-algebra-EQ0150	67	1.000	$\max_{2\leq j\leq n} \mu_j \leq\frac{3}{h^2}\max_{2\leq i\leq n} y_{i+1}-2y_i+y_{i-1} .$	$\max_{2\leq j\leq n} \mu_j \leq\frac{3}{h^2}\max_{2\leq i\leq n} y_{i+1}-2y_i+y_{i-1} .$	$\max_{2\leq j\leq n} \mu_j \leq\frac{3}{h^2}\max_{2\leq i\leq n} y_{i+1}-2y_i+y_{i-1} .$
lyche-numerical-linear-algebra-EQ0151 (2.34)	67	1.000	$\left[\begin{array}{cccc} 2&1&&\\&1&4&1\\&&\ddots&\ddots&\ddots\\&&&1&4&1\\&&&&1&2 \end{array}\right]\left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^2}\left[\begin{array}{c} y_2-y_1-hs_1 \\ \delta^2y_2 \\ \delta^2y_3 \\ \vdots \\ \delta^2y_{n-1} \\ \delta^2y_n \\ hs_{n+1}-y_{n+1}+y_n \end{array}\right],$	$\left[\begin{array}{ccc} 2&1&\\1&4&1\\&\ddots&\ddots&\ddots\\&&1&4&1\\&&&1&2 \end{array}\right]\left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^2}\left[\begin{array}{c} y_2-y_1-hs_1 \\ \delta^2y_2 \\ \delta^2y_3 \\ \vdots \\ \delta^2y_{n-1} \\ \delta^2y_n \\ hs_{n+1}-y_{n+1}+y_n \end{array}\right],$	$\left[\begin{array}{ccc} 2&1&\\1&4&1\\&\ddots&\ddots&\ddots\\&&1&4&1\\&&&1&2 \end{array}\right]\left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^2}\left[\begin{array}{c} y_2-y_1-hs_1 \\ \delta^2y_2 \\ \delta^2y_3 \\ \vdots \\ \delta^2y_{n-1} \\ \delta^2y_n \\ hs_{n+1}-y_{n+1}+y_n \end{array}\right],$
lyche-numerical-linear-algebra-EQ0152	68	1.000	$\int_a^b\left(g^{\{\prime\prime\}}(x)\right)^2\,d\,x\leq\int_a^b\left(h^{\{\prime\prime\}}(x)\right)^2\,d\,x$	$\int_a^b\left(g''(x)\right)^2\,dx\leq\int_a^b\left(h''(x)\right)^2\,dx$	$\int_a^b\left(g''(x)\right)^2\,dx\leq\int_a^b\left(h''(x)\right)^2\,dx$
lyche-numerical-linear-algebra-EQ0153 (2.35)	68	1.000	$\int_a^b g^{\{\prime\prime\}}e^{\{\prime\prime\}}=0,\quad e:=h-g.$	$\int_a^b g''e''=0,\quad e:=h-g.$	$\int_a^b g''e''=0,\quad e:=h-g.$

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lyche-numerical-linear-algebra-EQ0173	76	1.000	$\begin{aligned} & \text{\& } x_3=b_3^{\{3\}}/a_{33}^{\{3\}} \text{\& } \\ & \text{\& } x_2=\left(b_2^{\{2\}}-a_{23}^{\{2\}}x_3\right)/a_{22}^{\{2\}} \text{\& } \\ & \text{\& } x_1=\left(b_1^{\{1\}}-a_{12}^{\{1\}}x_2-a_{13}^{\{1\}}x_3\right)/a_{11}^{\{1\}} . \end{aligned}$	$x_3 = b_3^{(3)}/a_{33}^{(3)}$ $x_2 = \left(b_2^{(2)} - a_{23}^{(2)}x_3\right)/a_{22}^{(2)}$ $x_1 = \left(b_1^{(1)} - a_{12}^{(1)}x_2 - a_{13}^{(1)}x_3\right)/a_{11}^{(1)} .$	$x_3 = b_3^{(3)}/a_{33}^{(3)}$ $x_2 = (b_2^{(2)} - a_{23}^{(2)}x_3)/a_{22}^{(2)}$ $x_1 = (b_1^{(1)} - a_{12}^{(1)}x_2 - a_{13}^{(1)}x_3)/a_{11}^{(1)} .$
lyche-numerical-linear-algebra-EQ0177 (3.2)	77	1.000	$\text{\texttt{for }} j=k:n$	$\text{for } j = k : n$	$\text{for } j = k : n$
lyche-numerical-linear-algebra-EQ0178	77	1.000	$a_{ij}^{\{k+1\}}=a_{ij}^{\{k\}}-l_{ik}^{\{k\}}a_{kj}^{\{k\}}$	$a_{ij}^{(k+1)} = a_{ij}^{(k)} - l_{ik}^{(k)}a_{kj}^{(k)}$	$a_{ij}^{(k+1)} = a_{ij}^{(k)} - l_{ik}^{(k)}a_{kj}^{(k)}$
lyche-numerical-linear-algebra-EQ0179	78	1.000	$\boldsymbol{A}_{\{[k]\}}:=\boldsymbol{A}(1:k,1:k)=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{array}\right]$	$\boldsymbol{A}_{[k]} := \boldsymbol{A}(1:k,1:k) = \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$	$\boldsymbol{A}_{[k]} := \boldsymbol{A}(1:k,1:k) = \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$
lyche-numerical-linear-algebra-EQ0180	78	1.000	$b_{\{i,j\}}=a_{r_i}^{\{i\}}, \quad i,j=1,\ldots,k.$	$b_{i,j} = a_{r_i,r_j}, \quad i,j = 1,\ldots,k.$	$b_{i,j} = a_{r_i,r_j}, \quad i,j = 1,\ldots,k.$
lyche-numerical-linear-algebra-EQ0182	78	1.000	$[1],\left[\begin{array}{l} 1 \\ 2 \\ 4 \\ 5 \end{array}\right],\boldsymbol{A}.$	$[1], \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}, \boldsymbol{A}.$	$[1], \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}, \boldsymbol{A}.$
lyche-numerical-linear-algebra-EQ0183 (3.3)	78	1.000	$\operatorname{det}\left(\boldsymbol{A}_{\{[k]\}}\right)=a_{11}^{\{1\}}a_{22}^{\{2\}}\cdots a_{kk}^{\{k\}}, \quad k=1,\ldots,n.$	$\det(\boldsymbol{A}_{[k]}) = a_{11}^{(1)}a_{22}^{(2)}\cdots a_{kk}^{(k)}, \quad k = 1,\ldots,n.$	$\det(\boldsymbol{A}_{[k]}) = a_{11}^{(1)}a_{22}^{(2)}\cdots a_{kk}^{(k)}, \quad k = 1,\ldots,n.$
lyche-numerical-linear-algebra-EQ0187 (3.6)	79	1.000	$(\boldsymbol{L}\boldsymbol{U})_{ij}=\sum_{k=1}^{j-1}l_{ik}^{\{k\}}a_{kj}^{\{k\}}+l_{ij}a_{jj}^{\{j\}}=\sum_{k=1}^{j-1}\left(a_{ij}^{\{k\}}-a_{ij}^{\{k+1\}}\right)+a_{ij}^{\{j\}}=a_{ij}.$	$(\boldsymbol{LU})_{ij} = \sum_{k=1}^{j-1} l_{ik}^{(k)}a_{kj}^{(k)} + l_{ij}a_{jj}^{(j)} = \sum_{k=1}^{j-1} \left(a_{ij}^{(k)} - a_{ij}^{(k+1)}\right) + a_{ij}^{(j)} = a_{ij}.$	$(\boldsymbol{LU})_{ij} = \sum_{k=1}^{j-1} l_{ik}^{(k)}a_{kj}^{(k)} + l_{ij}a_{jj}^{(j)} = \sum_{k=1}^{j-1} \left(a_{ij}^{(k)} - a_{ij}^{(k+1)}\right) + a_{ij}^{(j)} = a_{ij}.$
lyche-numerical-linear-algebra-EQ0188	80	1.000	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array}\right]\left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right]=\left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right].$	$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$	$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0191	80	1.000	$\left[\begin{array}{cccccc} a_{11} & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} & 0 \end{array}\right], \quad \left[\begin{array}{cccccc} a_{11} & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} & 0 \end{array}\right]$	$\begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{bmatrix}, \quad \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{bmatrix}$	$\begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{bmatrix}, \quad \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{bmatrix}$

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lyche-numerical-linear-algebra-EQ0192	81	1.000	$\left[\begin{array}{cccc} a_{k,k} & 0 & \cdots & 0 \\ a_{k+1,k} & a_{k+1,k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n,k} & & \cdots & a_{n \times n} \end{array}\right] \left[\begin{array}{c} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{array}\right] = \left[\begin{array}{c} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{array}\right].$	$\begin{bmatrix} a_{k,k} & 0 & \cdots & 0 \\ a_{k+1,k} & a_{k+1,k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n,k} & & \cdots & a_{n \times n} \end{bmatrix} \begin{bmatrix} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{bmatrix}.$	$\begin{bmatrix} a_{k,k} & 0 & \cdots & 0 \\ a_{k+1,k} & a_{k+1,k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n,k} & & \cdots & a_{n \times n} \end{bmatrix} \begin{bmatrix} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0194	82	1.000	$b((k+1):n)=b((k+1):n)-x(k) * A((k+1):n, k) \text{ .}$	$b((k+1):n)=b((k+1):n)-x(k)*A((k+1):n,k).$	$b((k+1):n)=b((k+1):n)-x(k)*A((k+1):n,k).$
lyche-numerical-linear-algebra-EQ0195 (3.9)	82	1.000	$\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.3 cforwardsolve} \[N_{LU}:=\frac{2}{3} n^3-\frac{1}{2} n^2-\frac{1}{6} n \]$	(not rendered)	$N_{LU}:=\frac{2}{3}n^3-\frac{1}{2}n^2-\frac{1}{6}n$
lyche-numerical-linear-algebra-EQ0196	83	1.000	$M+D+A+S=\frac{2}{3} n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2} n(n-1)=N_{LU}$	$M+D+A+S=\frac{2}{3}n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2}n(n-1)=N_{LU}$	$M+D+A+S=\frac{2}{3}n(n-1)(n-\frac{1}{2})+\frac{1}{2}n(n-1)=N_{LU}$
lyche-numerical-linear-algebra-EQ0197	83	1.000	$M+S=2 \sum_{k=1}^{n-1}(n-k)^2 \approx 2 \int_1^{n-1}(n-k)^2 d k \approx 2 \int_0^n(n-k)^2 d k=\frac{2}{3} n^3$	$M+S=2 \sum_{k=1}^{n-1}(n-k)^2 \approx 2 \int_1^{n-1}(n-k)^2 d k \approx 2 \int_0^n(n-k)^2 d k=\frac{2}{3} n^3$	$M+S=2 \sum_{k=1}^{n-1}(n-k)^2 \approx 2 \int_1^{n-1}(n-k)^2 d k \approx 2 \int_0^n(n-k)^2 d k=\frac{2}{3} n^3$
lyche-numerical-linear-algebra-EQ0198	83	1.000	$N_S=2 \sum_{k=1}^n(2 k-1) \approx 2 \int_1^n(2 k-1) d k \approx 4 \int_0^n k d k=2 n^2$	$N_S=2 \sum_{k=1}^n(2 k-1) \approx 2 \int_1^n(2 k-1) d k \approx 4 \int_0^n k d k=2 n^2$	$N_S=2 \sum_{k=1}^n(2 k-1) \approx 2 \int_1^n(2 k-1) d k \approx 4 \int_0^n k d k=2 n^2.$
lyche-numerical-linear-algebra-EQ0199	84	1.000	$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \ldots, \boldsymbol{e}_{i_n}\right] \in \mathbb{R}^{n \times n},$	$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \ldots, \boldsymbol{e}_{i_n}\right] \in \mathbb{R}^{n \times n},$	$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \ldots, \boldsymbol{e}_{i_n}\right] \in \mathbb{R}^{n \times n},$
lyche-numerical-linear-algebra-EQ0200 (3.10)	84	1.000	$\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^T \boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$	$\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^T \boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$	$\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^T \boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$
lyche-numerical-linear-algebra-EQ0206	86	1.000	$a_{p_i, k}^{(k)} a_{p_k, j}^{(k)}=a_{p_i, j}^{(k)}, j \wedge\{(k+1)\} \text { for } k \leq \min (i, j), \text { and } a_{p_i, j}^{(k)} a_{p_j, j}^{(j)}=a_{p_i, j}^{(j)} \text { for } i > j \text{ .}$	$a_{p_i, k}^{(k)} a_{p_k, j}^{(k)}=a_{p_i, j}^{(k)}-a_{p_i, j}^{(k+1)} \text { for } k<\min (i, j), \text { and } a_{p_i, j}^{(k)} a_{p_j, j}^{(j)}=a_{p_i, j}^{(j)} \text { for } i>j.$	$a_{p_i, k}^{(k)} a_{p_k, j}^{(k)}=a_{p_i, j}^{(k)}-a_{p_i, j}^{(k+1)} \text { for } k<\min (i, j), \text { and } a_{p_i, j}^{(k)} a_{p_j, j}^{(j)}=a_{p_i, j}^{(j)} \text { for } i>j.$
lyche-numerical-linear-algebra-EQ0209	87	1.000	$\left a_{r_k, k}^{(k)}\right :=\max \left\{\left a_{i, k}^{(k)}\right : k \leq i \leq n\right\}$	$\left a_{r_k, k}^{(k)}\right :=\max \left\{\left a_{i, k}^{(k)}\right : k \leq i \leq n\right\}$	$\left a_{r_k, k}^{(k)}\right :=\max \left\{\left a_{i, k}^{(k)}\right : k \leq i \leq n\right\}$
lyche-numerical-linear-algebra-EQ0211	87	1.000	$\begin{aligned} 10^{-4} x_1+2 x_2 &=4 \\ \left(1-2 \times 10^4\right) x_2 &=3-4 \times 10^4 \end{aligned}$	$\begin{aligned} 10^{-4} x_1+2 x_2 &=4 \\ \left(1-2 \times 10^4\right) x_2 &=3-4 \times 10^4 \end{aligned}$	$\begin{aligned} 10^{-4} x_1+2 x_2 &=4 \\ \left(1-2 \times 10^4\right) x_2 &=3-4 \times 10^4 \end{aligned}$
lyche-numerical-linear-algebra-EQ0212	87	1.000	$x_2=\frac{-39997}{-19999} \approx 2, \quad x_1=\frac{4-2 x_2}{10^{-4}}=\frac{20000}{19999} \approx 1 \text{ .}$	$x_2=\frac{-39997}{-19999} \approx 2, \quad x_1=\frac{4-2 x_2}{10^{-4}}=\frac{20000}{19999} \approx 1.$	$x_2=\frac{-39997}{-19999} \approx 2, \quad x_1=\frac{4-2 x_2}{10^{-4}}=\frac{20000}{19999} \approx 1.$
lyche-numerical-linear-algebra-EQ0214	87	1.000	$\begin{aligned} x_1+x_2 &=3 \\ 10^{-4} x_1+2 x_2 &=4 \end{aligned}$	$\begin{aligned} x_1+x_2 &=3 \\ 10^{-4} x_1+2 x_2 &=4 \end{aligned}$	$\begin{aligned} x_1+x_2 &=3 \\ 10^{-4} x_1+2 x_2 &=4 \end{aligned}$

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lyche-numerical-linear-algebra-EQ0215	87	■ 1.000	$\begin{aligned} x_1+x_2 &\&=3 \\ \left(2-10^{-4}\right) x_2 &\&=4-3 \times 10^{-4} \end{aligned}$	$\begin{aligned} x_1+x_2 &=3 \\ \left(2-10^{-4}\right) x_2 &=4-3 \times 10^{-4} \end{aligned}$	$\begin{aligned} x_1+x_2 &=3 \\ \left(2-10^{-4}\right) x_2 &=4-3 \times 10^{-4} \end{aligned}$
lyche-numerical-linear-algebra-EQ0217	88	■ 1.000	$\frac{\left \left a_{r_k,k}\right \right }{s_k}:=\max \left\{\frac{\left a_{i,k}\right }{s_k}: k \leq i \leq n\right\}, \quad s_k:=\max _{1 \leq j \leq n}\left a_{k j}\right .$	$\frac{\left a_{r_k,k}\right }{s_k}:=\max \left\{\frac{\left a_{i,k}\right }{s_k}: k \leq i \leq n\right\}, \quad s_k:=\max _{1 \leq j \leq n}\left a_{k j}\right .$	$\frac{\left a_{r_k,k}\right }{s_k}:=\max \left\{\frac{\left a_{i,k}\right }{s_k}: k \leq i \leq n\right\}, \quad s_k:=\max _{1 \leq j \leq n}\left a_{k j}\right .$
lyche-numerical-linear-algebra-EQ0218	88	■ 1.000	$a_{\{r_k\},\{s_k\}}^{(k)}:=\max \left\{\left \left a_{i, j}\right \right : k \leq i, j \leq n\right\}$	$a_{r_k,s_k}^{(k)}:=\max \left\{\left a_{i, j}\right : k \leq i, j \leq n\right\}$	$a_{r_k,s_k}^{(k)}:=\max \left\{\left a_{i, j}\right : k \leq i, j \leq n\right\}$
lyche-numerical-linear-algebra-EQ0219	88	■ 1.000	$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{array}\right] .$	$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{array}\right] .$	$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{array}\right] .$
lyche-numerical-linear-algebra-EQ0221 (3.15)	89	■ 1.000	$u_1=a, \quad u_2=b, \quad a l_1=c, \quad b l_1+u_3=d .$	$u_1=a, \quad u_2=b, \quad a l_1=c, \quad b l_1+u_3=d .$	$u_1=a, \quad u_2=b, \quad a l_1=c, \quad b l_1+u_3=d .$
lyche-numerical-linear-algebra-EQ0222	89	■ 1.000	$\boldsymbol{A}_{\{1\}}:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_{\{2\}}:=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{\{3\}}:=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{\{4\}}:=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right] .$	$\boldsymbol{A}_1:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_2:=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_3:=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_4:=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right] .$	$\boldsymbol{A}_1:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_2:=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_3:=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_4:=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right] .$
lyche-numerical-linear-algebra-EQ0223	89	■ 1.000	$\boldsymbol{A}_{\{[k]\}}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$
lyche-numerical-linear-algebra-EQ0225 (3.17)	90	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^T & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_n \\ 0 & u_{nn} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1} \boldsymbol{u}_n \\ \boldsymbol{l}_n^T \boldsymbol{U}_{n-1} & \boldsymbol{l}_n^T \boldsymbol{u}_n+u_{nn} \end{array}\right],$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^T & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_n \\ 0 & u_{nn} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1} \boldsymbol{u}_n \\ \boldsymbol{l}_n^T \boldsymbol{U}_{n-1} & \boldsymbol{l}_n^T \boldsymbol{u}_n+u_{nn} \end{array}\right],$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^T & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_n \\ 0 & u_{nn} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1} \boldsymbol{u}_n \\ \boldsymbol{l}_n^T \boldsymbol{U}_{n-1} & \boldsymbol{l}_n^T \boldsymbol{u}_n+u_{nn} \end{array}\right],$
lyche-numerical-linear-algebra-EQ0226 (3.18)	90	■ 1.000	$\boldsymbol{U}_{n-1}^T \boldsymbol{l}_n=\boldsymbol{r}_n, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_n=\boldsymbol{c}_n, \quad u_{nn}=a_{nn}-\boldsymbol{l}_n^T \boldsymbol{u}_n .$	$\boldsymbol{U}_{n-1}^T \boldsymbol{l}_n=\boldsymbol{r}_n, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_n=\boldsymbol{c}_n, \quad u_{nn}=a_{nn}-\boldsymbol{l}_n^T \boldsymbol{u}_n .$	$\boldsymbol{U}_{n-1}^T \boldsymbol{l}_n=\boldsymbol{r}_n, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_n=\boldsymbol{c}_n, \quad u_{nn}=a_{nn}-\boldsymbol{l}_n^T \boldsymbol{u}_n .$

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lyche-numerical-linear-algebra-EQ0227	91	1.000	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k, \quad \left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \mathbf{0} \\ \boldsymbol{l}_k^T & 1 \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_k \\ u_{kk} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{c}_k \\ a_{kk} \end{array} \right].$		$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k, \quad \left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \mathbf{0} \\ \boldsymbol{l}_k^T & 1 \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_k \\ u_{kk} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{c}_k \\ a_{kk} \end{array} \right].$
lyche-numerical-linear-algebra-EQ0230	92	1.000	$\boldsymbol{A}_{\{k\}} := \left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k1} & \cdots & \boldsymbol{A}_{kk} \end{array} \right]$		$\boldsymbol{A}_{\{k\}} := \left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k1} & \cdots & \boldsymbol{A}_{kk} \end{array} \right]$
lyche-numerical-linear-algebra-EQ0232 (3.22)	93	1.000	$\begin{array}{l} \begin{array}{c} \text{\captionsetup{labelformat=empty}} \\ \text{\caption{Listing 3.5 cbacksolve} \[\boldsymbol{A}((k+1):} \\ \text{n, (k+1): n) \boldsymbol{b}_{\{k\}((k+1): n)} = -} \\ \text{\boldsymbol{A}((k+1): n, k) \boldsymbol{b}_{\{k\}(k), \quad k=1, \ldots,} \\ \text{n-1 . \]} \end{array} \end{array}$	(not rendered)	$\boldsymbol{A}((k+1):n, (k+1):n) \boldsymbol{b}_k((k+1):n) = -\boldsymbol{A}((k+1):n, k) \boldsymbol{b}_k(k), \quad k = 1, \dots, n-1.$
lyche-numerical-linear-algebra-EQ0233 (3.23)	94	1.000	$\boldsymbol{A}(1:k, 1:k) \boldsymbol{b}_k(1:k) = \boldsymbol{I}(1:k, k), \quad k = n, n-1, \dots, 1,$	$\boldsymbol{A}(1:k, 1:k) \boldsymbol{b}_k(1:k) = \boldsymbol{I}(1:k, k), \quad k = n, n-1, \dots, 1,$	$\boldsymbol{A}(1:k, 1:k) \boldsymbol{b}_k(1:k) = \boldsymbol{I}(1:k, k), \quad k = n, n-1, \dots, 1,$
lyche-numerical-linear-algebra-EQ0235 (3.28)	95	1.000	$\boldsymbol{C} := \boldsymbol{U} + \boldsymbol{v} \boldsymbol{e}_1^T,$	$\boldsymbol{C} := \boldsymbol{U} + \boldsymbol{v} \boldsymbol{e}_1^T,$	$\boldsymbol{C} := \boldsymbol{U} + \boldsymbol{v} \boldsymbol{e}_1^T,$
lyche-numerical-linear-algebra-EQ0236 (3.29)	95	1.000	$\boldsymbol{P} := \boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1,n},$	$\boldsymbol{P} := \boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1,n},$	$\boldsymbol{P} := \boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1,n},$
lyche-numerical-linear-algebra-EQ0237 (3.30)	96	1.000	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{w} \boldsymbol{e}_1^T.$	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{w} \boldsymbol{e}_1^T.$	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{w} \boldsymbol{e}_1^T.$
lyche-numerical-linear-algebra-EQ0238	96	1.000	$\operatorname{det}(\boldsymbol{A}_{\{k\}}) = u_{11} u_{22} \cdots u_{kk} \text{ for } k=1, \dots, n,$	$\det(\boldsymbol{A}_{[k]}) = u_{11} u_{22} \cdots u_{kk} \text{ for } k = 1, \dots, n.$	$\det(\boldsymbol{A}_{[k]}) = u_{11} u_{22} \cdots u_{kk} \text{ for } k = 1, \dots, n.$
lyche-numerical-linear-algebra-EQ0239 (3.31)	97	1.000	$u_{11} = a_{11}, \quad u_{kk} = \frac{\operatorname{det}(\boldsymbol{A}_{\{k\}})}{\operatorname{det}(\boldsymbol{A}_{\{k-1\}})}, \text{ for } k = 2, \dots, n.$	$u_{11} = a_{11}, \quad u_{kk} = \frac{\det(\boldsymbol{A}_{[k]})}{\det(\boldsymbol{A}_{[k-1]})}, \text{ for } k = 2, \dots, n.$	$u_{11} = a_{11}, \quad u_{kk} = \frac{\det(\boldsymbol{A}_{[k]})}{\det(\boldsymbol{A}_{[k-1]})}, \text{ for } k = 2, \dots, n.$
lyche-numerical-linear-algebra-EQ0242	98	1.000	$\boldsymbol{B} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \dots, \boldsymbol{a}_n, \boldsymbol{b}] \in \mathbb{R}^{n \times n}.$	$\boldsymbol{B} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \dots, \boldsymbol{a}_n, \boldsymbol{b}] \in \mathbb{R}^{n \times n}.$	$\boldsymbol{B} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \dots, \boldsymbol{a}_n, \boldsymbol{b}] \in \mathbb{R}^{n \times n}.$
lyche-numerical-linear-algebra-EQ0243	98	1.000	$\boldsymbol{A} := \left[\begin{array}{cc} -3 & -2 \\ 4 & 2 \end{array} \right].$	$\boldsymbol{A} := \left[\begin{array}{cc} -3 & -2 \\ 4 & 2 \end{array} \right].$	$\boldsymbol{A} := \left[\begin{array}{cc} -3 & -2 \\ 4 & 2 \end{array} \right].$
lyche-numerical-linear-algebra-EQ0244	98	1.000	$\boldsymbol{P} := [\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \dots, \boldsymbol{e}_1].$	$\boldsymbol{P} := [\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \dots, \boldsymbol{e}_1].$	$\boldsymbol{P} := [\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \dots, \boldsymbol{e}_1].$
lyche-numerical-linear-algebra-EQ0246 (4.1)	101	1.000	$d_1 = a. \quad a_1 = b, \quad d_2 = d - a l_1 ^2.$	$d_1 = a. \quad a_1 = b, \quad d_2 = d - a l_1 ^2.$	$d_1 = a. \quad a_1 = b, \quad d_2 = d - a l_1 ^2.$
lyche-numerical-linear-algebra-EQ0251	102	1.000	$f(\boldsymbol{x}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \bar{x}_i x_j$	$f(\boldsymbol{x}) = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \bar{x}_i x_j$	$f(\boldsymbol{x}) = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \bar{x}_i x_j$

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lyche-numerical-linear-algebra-EQ0255 (4.5)	104	■ 1.000	$\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) = \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X} := [\boldsymbol{e}_{r_1}, \dots, \boldsymbol{e}_{r_k}] \in \mathbb{C}^{n \times k}.$	$\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) = \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X} := [\boldsymbol{e}_{r_1}, \dots, \boldsymbol{e}_{r_k}] \in \mathbb{C}^{n \times k}.$	
lyche-numerical-linear-algebra-EQ0256 (4.6)	104	■ 1.000	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y} = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^k, \quad \boldsymbol{x} := \boldsymbol{X} \boldsymbol{y}.$	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y} = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^k, \quad \boldsymbol{x} := \boldsymbol{X} \boldsymbol{y}.$	
lyche-numerical-linear-algebra-EQ0257 (0)	105	■ 1.000	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array} \right] \left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array} \right] \left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} \sqrt{2} & 0 \\ -1/\sqrt{2} & \sqrt{3/2} \end{array} \right] \left[\begin{array}{cc} \sqrt{2} & -1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{array} \right].$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array} \right] \left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array} \right] \left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} \sqrt{2} & 0 \\ -1/\sqrt{2} & \sqrt{3/2} \end{array} \right] \left[\begin{array}{cc} \sqrt{2} & -1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{array} \right].$	
lyche-numerical-linear-algebra-EQ0258 (4.7)	106	■ 1.000	$\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 4.2 bandcholesky} \[\theta < (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A} (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2\text{Re}(\bar{\alpha}\beta a_{ij}). \end{figure}$	$(not\ rendered)$	$0 < (\leq)(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A} (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2\text{Re}(\bar{\alpha}\beta a_{ij}).$
lyche-numerical-linear-algebra-EQ0259	106	■ 1.000	$\theta < \left a_{ij} \right ^2 a_{ii} + a_{ii}^2 a_{jj} - 2 \left a_{ij} \right ^2 a_{ii} = a_{ii} \left(a_{ii} a_{jj} - \left a_{ij} \right ^2 \right).$	$0 < \left a_{ij} \right ^2 a_{ii} + a_{ii}^2 a_{jj} - 2 \left a_{ij} \right ^2 a_{ii} = a_{ii} \left(a_{ii} a_{jj} - \left a_{ij} \right ^2 \right).$	$0 < \left a_{ij} \right ^2 a_{ii} + a_{ii}^2 a_{jj} - 2 \left a_{ij} \right ^2 a_{ii} = a_{ii} (a_{ii} a_{jj} - \left a_{ij} \right ^2).$
lyche-numerical-linear-algebra-EQ0260	106	■ 1.000	$\left a_{ij} \right = \left b_{ij} \right < \sqrt{b_{ii} b_{jj}} = \sqrt{(a_{ii} + \varepsilon)(a_{jj} + \varepsilon)}, \quad i \neq j.$	$\left a_{ij} \right = \left b_{ij} \right < \sqrt{b_{ii} b_{jj}} = \sqrt{(a_{ii} + \varepsilon)(a_{jj} + \varepsilon)}, \quad i \neq j.$	$\left a_{ij} \right = \left b_{ij} \right < \sqrt{b_{ii} b_{jj}} = \sqrt{(a_{ii} + \varepsilon)(a_{jj} + \varepsilon)}, \quad i \neq j.$
lyche-numerical-linear-algebra-EQ0261	107	■ 1.000	$\boldsymbol{A}_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array} \right], \quad \boldsymbol{A}_2 = \left[\begin{array}{cc} 1 & 2 \\ 2 & 2 \end{array} \right], \quad \boldsymbol{A}_3 = \left[\begin{array}{cc} -2 & 1 \\ 1 & 2 \end{array} \right].$	$\boldsymbol{A}_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array} \right], \quad \boldsymbol{A}_2 = \left[\begin{array}{cc} 1 & 2 \\ 2 & 2 \end{array} \right], \quad \boldsymbol{A}_3 = \left[\begin{array}{cc} -2 & 1 \\ 1 & 2 \end{array} \right].$	$\boldsymbol{A}_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array} \right], \quad \boldsymbol{A}_2 = \left[\begin{array}{cc} 1 & 2 \\ 2 & 2 \end{array} \right], \quad \boldsymbol{A}_3 = \left[\begin{array}{cc} -2 & 1 \\ 1 & 2 \end{array} \right].$
lyche-numerical-linear-algebra-EQ0266 (4.10)	109	■ 1.000	$\boldsymbol{L}^* := \left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right], \quad \beta := \sqrt{\alpha},$	$\boldsymbol{L}^* := \left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right], \quad \beta := \sqrt{\alpha},$	$\boldsymbol{L}^* := \left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right], \quad \beta := \sqrt{\alpha},$
lyche-numerical-linear-algebra-EQ0271	113	■ 1.000	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + (2-a)x_1x_2 + ax_2x_1 + x_2^2 = x_1^2 + (x_1 + x_2)^2 > 0.$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + (2-a)x_1x_2 + ax_2x_1 + x_2^2 = x_1^2 + (x_1 + x_2)^2 > 0.$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + (2-a)x_1x_2 + ax_2x_1 + x_2^2 = x_1^2 + (x_1 + x_2)^2 > 0.$
lyche-numerical-linear-algebra-EQ0273	114	■ 1.000	$\boldsymbol{A} = \boldsymbol{L} (\boldsymbol{I} + \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{L}^T,$	$\boldsymbol{A} = \boldsymbol{L} (\boldsymbol{I} + \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{L}^T,$	$\boldsymbol{A} = \boldsymbol{L} (\boldsymbol{I} + \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{L}^T,$

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lyche-numerical-linear-algebra- EQ0275	114 114	■ 1.000	$\boldsymbol{L}_{+}=\left[\begin{array}{cc}\boldsymbol{L} & 0 \\ \boldsymbol{y}^T & \lambda\end{array}\right],$	$L_{+}=\left[\begin{array}{cc}L & 0 \\ \boldsymbol{y}^T & \lambda\end{array}\right],$	$L_{+}=\left[\begin{array}{cc}L & 0 \\ \boldsymbol{y}^T & \lambda\end{array}\right],$
lyche-numerical-linear-algebra- EQ0276	117 117	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \bar{a} \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \bar{b} \langle \boldsymbol{z}, \boldsymbol{x} \rangle$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \bar{a} \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \bar{b} \langle \boldsymbol{z}, \boldsymbol{x} \rangle$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \bar{a} \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \bar{b} \langle \boldsymbol{z}, \boldsymbol{x} \rangle$
lyche-numerical-linear-algebra- EQ0277 (5.1)	117 117	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z} \rangle = \bar{a} \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \bar{b} \langle \boldsymbol{x}, \boldsymbol{z} \rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} \rangle = a ^2 \langle \boldsymbol{x}, \boldsymbol{y} \rangle.$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z} \rangle = \bar{a} \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \bar{b} \langle \boldsymbol{x}, \boldsymbol{z} \rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} \rangle = a ^2 \langle \boldsymbol{x}, \boldsymbol{y} \rangle.$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} + \boldsymbol{b} \boldsymbol{z} \rangle = \bar{a} \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \bar{b} \langle \boldsymbol{x}, \boldsymbol{z} \rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y} \rangle = a ^2 \langle \boldsymbol{x}, \boldsymbol{y} \rangle.$
lyche-numerical-linear-algebra- EQ0280 (5.2)	117 117	■ 1.000	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \mapsto \ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \mapsto \ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \mapsto \ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra- EQ0281 (5.3)	118 118	■ 1.000	$ \langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \ \boldsymbol{x}\ \ \boldsymbol{y}\ ,$	$ \langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \ \boldsymbol{x}\ \ \boldsymbol{y}\ ,$	$ \langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \ \boldsymbol{x}\ \ \boldsymbol{y}\ ,$
lyche-numerical-linear-algebra- EQ0285 (5.5)	119 119	■ 1.000	$\ \boldsymbol{x} + \boldsymbol{a} \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + a \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \bar{a} \langle \boldsymbol{x}, \boldsymbol{y} \rangle + a ^2 \ \boldsymbol{y}\ ^2, \quad a \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$	$\ \boldsymbol{x} + \boldsymbol{a} \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + a \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \bar{a} \langle \boldsymbol{x}, \boldsymbol{y} \rangle + a ^2 \ \boldsymbol{y}\ ^2, \quad a \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$	$\ \boldsymbol{x} + \boldsymbol{a} \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + a \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \bar{a} \langle \boldsymbol{x}, \boldsymbol{y} \rangle + a ^2 \ \boldsymbol{y}\ ^2, \quad a \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$
lyche-numerical-linear-algebra- EQ0287 (5.6)	119 119	■ 1.000	$\cos \theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$
lyche-numerical-linear-algebra- EQ0288 (5.7)	119 119	■ 1.000	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \ \boldsymbol{y}\ ^2, \quad \text{if } \boldsymbol{x} \perp \boldsymbol{y}.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \ \boldsymbol{y}\ ^2, \quad \text{if } \boldsymbol{x} \perp \boldsymbol{y}.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \ \boldsymbol{y}\ ^2, \quad \text{if } \boldsymbol{x} \perp \boldsymbol{y}.$
lyche-numerical-linear-algebra- EQ0290	120 120	■ 1.000	$\left\{ \boldsymbol{u}_1, \dots, \boldsymbol{u}_k \right\} := \left\{ \frac{\boldsymbol{v}_1}{\ \boldsymbol{v}_1\ }, \dots, \frac{\boldsymbol{v}_k}{\ \boldsymbol{v}_k\ } \right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\} := \left\{ \frac{\boldsymbol{v}_1}{\ \boldsymbol{v}_1\ }, \dots, \frac{\boldsymbol{v}_k}{\ \boldsymbol{v}_k\ } \right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\} := \left\{ \frac{\boldsymbol{v}_1}{\ \boldsymbol{v}_1\ }, \dots, \frac{\boldsymbol{v}_k}{\ \boldsymbol{v}_k\ } \right\}$
lyche-numerical-linear-algebra- EQ0291	120 120	■ 1.000	$\langle \boldsymbol{v}_j, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle}{\langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle} \langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle = 0$	$\langle \boldsymbol{v}_j, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle}{\langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle} \langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle = 0$	$\langle \boldsymbol{v}_j, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle}{\langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle} \langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle = 0$

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lyche-numerical-linear-algebra-121 EQ0292	121	■ 1.000	$\operatorname{dim}(\mathcal{S} \oplus \mathcal{T}) = \operatorname{dim}(\mathcal{S}) + \operatorname{dim}(\mathcal{T})$.	$\dim(S \oplus \mathcal{T}) = \dim(S) + \dim(\mathcal{T})$.	$\dim(S \oplus \mathcal{T}) = \dim(S) + \dim(\mathcal{T})$.
lyche-numerical-linear-algebra-122 EQ0295	122	■ 1.000	$\left\langle \boldsymbol{s}_0, \boldsymbol{v}_j \right\rangle = \left\langle \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \langle \boldsymbol{v}, \boldsymbol{v}_j \rangle$.	$\langle s_0, v_j \rangle = \left\langle \sum_{i=1}^k \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} v_i, v_j \right\rangle = \sum_{i=1}^k \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} \langle v_i, v_j \rangle = \langle v, v_j \rangle$.	$\langle s_0, v_j \rangle = \langle \sum_{i=1}^k \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} v_i, v_j \rangle = \sum_{i=1}^k \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} \langle v_i, v_j \rangle = \langle v, v_j \rangle$.
lyche-numerical-linear-algebra-123 EQ0297	123	■ 1.000	$\ \boldsymbol{v} - \boldsymbol{s}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2 + \ \boldsymbol{u}\ ^2 > \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2$.	$\ \boldsymbol{v} - \boldsymbol{s}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2 + \ \boldsymbol{u}\ ^2 > \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2$.	$\ \boldsymbol{v} - \boldsymbol{s}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2 + \ \boldsymbol{u}\ ^2 > \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2$.
lyche-numerical-linear-algebra-123 EQ0299	123	■ 1.000	$\langle \boldsymbol{e}_i, \boldsymbol{A} \boldsymbol{e}_j \rangle = \langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j$.	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j$.	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j$.
lyche-numerical-linear-algebra-124 EQ0302	124	■ 1.000	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u} \in \mathbb{C}^n$ and $\boldsymbol{u}^* \boldsymbol{u} = 2$	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u} \in \mathbb{C}^n$ and $\boldsymbol{u}^* \boldsymbol{u} = 2$	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u} \in \mathbb{C}^n$ and $\boldsymbol{u}^* \boldsymbol{u} = 2$
lyche-numerical-linear-algebra-124 EQ0304	124	■ 1.000	$\boldsymbol{H}^* \boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^* + \boldsymbol{u} (\boldsymbol{u}^* \boldsymbol{u}) \boldsymbol{u}^* = \boldsymbol{I}$.	$\boldsymbol{H}^* \boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^* + \boldsymbol{u} (\boldsymbol{u}^* \boldsymbol{u}) \boldsymbol{u}^* = \boldsymbol{I}$.	$\boldsymbol{H}^* \boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^* + \boldsymbol{u} (\boldsymbol{u}^* \boldsymbol{u}) \boldsymbol{u}^* = \boldsymbol{I}$.
lyche-numerical-linear-algebra-124 EQ0305 (5.12)	124	■ 1.000	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$
lyche-numerical-linear-algebra-125 EQ0307	125	■ 1.000	$\boldsymbol{H} = \boldsymbol{I} - \frac{2 \boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$, where $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$,	$\boldsymbol{H} = \boldsymbol{I} - \frac{2 \boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$, where $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$,	$\boldsymbol{H} = \boldsymbol{I} - \frac{2 \boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$, where $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$,
lyche-numerical-linear-algebra-128 EQ0315	128	■ 1.000	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$, $\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$, $\left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array} \right]$.	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$, $\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$, $\left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array} \right]$.	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$, $\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$, $\left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array} \right]$.

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lyche-numerical-linear-algebra- EQ0339	137	■ 1.000	$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right]$.	$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right]$.	$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0340	137	■ 1.000	$\boldsymbol{H} \ \boldsymbol{x}=\boldsymbol{y}$, $\quad \text{where} \quad \boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}$.	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}$, $\quad \text{where} \quad \boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}$.	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}$, $\quad \text{where} \quad \boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}$.
lyche-numerical-linear-algebra- EQ0343	137	■ 1.000	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right]$.	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right]$.	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0344 (5.21)	138	■ 1.000	$ \operatorname{det}(\boldsymbol{A}) \leq \prod_{j=1}^n\left\ \boldsymbol{a}_{\{j\}}\right\ _2$.	$ \det(\boldsymbol{A}) \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2$.	$ \det(\boldsymbol{A}) \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2$.
lyche-numerical-linear-algebra- EQ0346	139	■ 1.000	$\boldsymbol{A}:=\left[\begin{array}{rr} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{cc} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{cc} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$
lyche-numerical-linear-algebra- EQ0347	140	■ 1.000	$\boldsymbol{G}_3 \boldsymbol{G}_2 \boldsymbol{G}_1 \boldsymbol{H}=\boldsymbol{R}$	$\boldsymbol{G}_3 \boldsymbol{G}_2 \boldsymbol{G}_1 \boldsymbol{H}=\boldsymbol{R}$	$\boldsymbol{G}_3 \boldsymbol{G}_2 \boldsymbol{G}_1 \boldsymbol{H}=\boldsymbol{R}$
lyche-numerical-linear-algebra- EQ0349	141	■ 1.000	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$
lyche-numerical-linear-algebra- EQ0351	141	■ 1.000	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n, n}\right)$	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n, n}\right)$	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n, n}\right)$
lyche-numerical-linear-algebra- EQ0352	141	■ 1.000	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n, n-1}\right)$.	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n, n-1}\right)$.	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n, n-1}\right)$.
lyche-numerical-linear-algebra- EQ0353	142	■ 1.000	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$
lyche-numerical-linear-algebra- EQ0356	145	■ 1.000	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$.	$\pi_{\boldsymbol{A}}(\lambda)=\det(\boldsymbol{A}-\lambda \boldsymbol{I})$.	$\pi_{\boldsymbol{A}}(\lambda)=\det(\boldsymbol{A}-\lambda \boldsymbol{I})$.
lyche-numerical-linear-algebra- EQ0357	146	■ 1.000	$\sum_{j=1}^m c_{\{j\}} \boldsymbol{x}_{\{j\}}=\mathbf{0}$ $\Rightarrow \sum_{j=1}^m c_{\{j\}} \boldsymbol{x}_{\{j\}}=\sum_{j=1}^m c_{\{j\}} \lambda_{\{j\}} \boldsymbol{x}_{\{j\}}=\mathbf{0}$.	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\mathbf{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j=\sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j=\mathbf{0}$.	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\mathbf{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j=\sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j=\mathbf{0}$.
lyche-numerical-linear-algebra- EQ0358	146	■ 1.000	$\boldsymbol{I}:=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]$, $\quad \boldsymbol{J}:=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right]$.	$\boldsymbol{I}:=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$, $\quad \boldsymbol{J}:=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$.	$\boldsymbol{I}:=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$, $\quad \boldsymbol{J}:=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0359	146	■ 1.000	$\boldsymbol{x}=\sum_{j=1}^n c_{\{j\}} \boldsymbol{x}_{\{j\}}$ $\text{for some scalars } c_{\{1\}}, \ldots, c_{\{n\}}$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{x}_j$ for some scalars $c_1, \ldots, c_n$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{x}_j$ for some scalars $c_1, \ldots, c_n$.
lyche-numerical-linear-algebra- EQ0360	146	■ 1.000	$\boldsymbol{x}=\frac{x_{\{1\}}+x_{\{2\}}}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]+\frac{x_{\{1\}}-x_{\{2\}}}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right]$.	$\boldsymbol{x}=\frac{x_1+x_2}{2}\left[\begin{array}{c} 1 \\ 1 \end{array}\right]+\frac{x_1-x_2}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right]$.	$\boldsymbol{x}=\frac{x_1+x_2}{2}\left[\begin{array}{c} 1 \\ 1 \end{array}\right]+\frac{x_1-x_2}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right]$.

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lyche-numerical-linear-algebra- EQ0365 (6.3)	148	■ 1.000	$\mathcal{N}(\boldsymbol{A}-\lambda\boldsymbol{I}) := \left\{\boldsymbol{x} \in \mathbb{C}^n : (\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}\right\}$	$\mathcal{N}(\boldsymbol{A}-\lambda\boldsymbol{I}) := \{\boldsymbol{x} \in \mathbb{C}^n : (\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}\}$	$\mathcal{N}(\boldsymbol{A}-\lambda\boldsymbol{I}) := \{\boldsymbol{x} \in \mathbb{C}^n : (\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}\}$
lyche-numerical-linear-algebra- EQ0368 (6.6)	150	■ 1.000	$\boldsymbol{U}_i = \operatorname{diag}\left(\boldsymbol{J}_{m_i,1}(\lambda_i), \ldots, \boldsymbol{J}_{m_i,g_i}(\lambda_i)\right).$	$\boldsymbol{U}_i = \operatorname{diag}\left(\boldsymbol{J}_{m_{i,1}}(\lambda_i), \ldots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)\right).$	$\boldsymbol{U}_i = \operatorname{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i), \ldots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)).$
lyche-numerical-linear-algebra- EQ0372	152	■ 1.000	$\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{V}^*\boldsymbol{A}\boldsymbol{v}_1 = \lambda_1\boldsymbol{V}^*\boldsymbol{v}_1 = \lambda_1\boldsymbol{e}_1.$	$\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{V}^*\boldsymbol{A}\boldsymbol{v}_1 = \lambda_1\boldsymbol{V}^*\boldsymbol{v}_1 = \lambda_1\boldsymbol{e}_1.$	$\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{V}^*\boldsymbol{A}\boldsymbol{v}_1 = \lambda_1\boldsymbol{V}^*\boldsymbol{v}_1 = \lambda_1\boldsymbol{e}_1.$
lyche-numerical-linear-algebra- EQ0376	153	■ 1.000	$\boldsymbol{V} = \left[\begin{array}{ccc} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{array}\right].$	$\boldsymbol{V} = \left[\begin{array}{ccc} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{array}\right].$	$\boldsymbol{V} = \left[\begin{array}{ccc} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{array}\right].$
lyche-numerical-linear-algebra- EQ0377	153	■ 1.000	$\boldsymbol{M} = \left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{array}\right].$	$\boldsymbol{M} = \left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{array}\right].$	$\boldsymbol{M} = \left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{array}\right].$
lyche-numerical-linear-algebra- EQ0378	153	■ 1.000	$\boldsymbol{A}^*\boldsymbol{A} = \operatorname{diag}\left(\overline{d_1}d_1, \ldots, \overline{d_n}d_n\right) = \operatorname{diag}\left(d_1 ^2, \ldots, d_n ^2\right) = \boldsymbol{A}\boldsymbol{A}^*,$	$\boldsymbol{A}^*\boldsymbol{A} = \operatorname{diag}\left(\overline{d_1}d_1, \ldots, \overline{d_n}d_n\right) = \operatorname{diag}\left(d_1 ^2, \ldots, d_n ^2\right) = \boldsymbol{A}\boldsymbol{A}^*,$	$\boldsymbol{A}^*\boldsymbol{A} = \operatorname{diag}(\overline{d_1}d_1, \ldots, \overline{d_n}d_n) = \operatorname{diag}(d_1 ^2, \ldots, d_n ^2) = \boldsymbol{A}\boldsymbol{A}^*,$
lyche-numerical-linear-algebra- EQ0382	154	■ 1.000	$e_{ii} = \sum_{k=1}^i b_{ki} ^2 = b_{ii} ^2 = \sum_{k=i}^n b_{ik} ^2 = f_{ii}$	$e_{ii} = \sum_{k=1}^i b_{ki} ^2 = b_{ii} ^2 = \sum_{k=i}^n b_{ik} ^2 = f_{ii}$	$e_{ii} = \sum_{k=1}^i b_{ki} ^2 = b_{ii} ^2 = \sum_{k=i}^n b_{ik} ^2 = f_{ii}$
lyche-numerical-linear-algebra- EQ0385 (6.10)	155	■ 1.000	$\boldsymbol{M} = \left[\begin{array}{cc} \mu & v \\ -v & \mu \end{array}\right]$	$\boldsymbol{M} = \left[\begin{array}{cc} \mu & v \\ -v & \mu \end{array}\right]$	$\boldsymbol{M} = \left[\begin{array}{cc} \mu & v \\ -v & \mu \end{array}\right]$
lyche-numerical-linear-algebra- EQ0388	156	■ 1.000	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$
lyche-numerical-linear-algebra- EQ0389	156	■ 1.000	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$
lyche-numerical-linear-algebra- EQ0391	157	■ 1.000	$\operatorname{dim}\left(\mathcal{S} \cap \mathcal{S}'\right) = \operatorname{dim}(\mathcal{S}) + \operatorname{dim}(\mathcal{S}') - \operatorname{dim}(\mathcal{S} + \mathcal{S}') \geq (n - k + 1) + k - n = 1.$	$\dim(\mathcal{S} \cap \mathcal{S}') = \dim(\mathcal{S}) + \dim(\mathcal{S}') - \dim(\mathcal{S} + \mathcal{S}') \geq (n - k + 1) + k - n = 1.$	$\dim(\mathcal{S} \cap \mathcal{S}') = \dim(\mathcal{S}) + \dim(\mathcal{S}') - \dim(\mathcal{S} + \mathcal{S}') \geq (n - k + 1) + k - n = 1.$
lyche-numerical-linear-algebra- EQ0395 (6.14)	158	■ 1.000	$\alpha_k + \varepsilon_n \leq \beta_k \leq \alpha_k + \varepsilon_1, \text{ for } k = 1, \ldots, n,$	$\alpha_k + \varepsilon_n \leq \beta_k \leq \alpha_k + \varepsilon_1, \text{ for } k = 1, \ldots, n,$	$\alpha_k + \varepsilon_n \leq \beta_k \leq \alpha_k + \varepsilon_1, \text{ for } k = 1, \ldots, n,$
lyche-numerical-linear-algebra- EQ0398 (6.16)	159	■ 1.000	$\boldsymbol{v} = \sum_{j=1}^n \frac{\boldsymbol{y}_j^* \boldsymbol{v}}{\boldsymbol{y}_j^* \boldsymbol{x}_j} \boldsymbol{x}_j = \sum_{k=1}^n \frac{\boldsymbol{x}_k^* \boldsymbol{v}}{\boldsymbol{y}_k^* \boldsymbol{x}_k} \boldsymbol{y}_k, \quad \boldsymbol{v} \in \mathbb{C}^n.$	$\boldsymbol{v} = \sum_{j=1}^n \frac{\boldsymbol{y}_j^* \boldsymbol{v}}{\boldsymbol{y}_j^* \boldsymbol{x}_j} \boldsymbol{x}_j = \sum_{k=1}^n \frac{\boldsymbol{x}_k^* \boldsymbol{v}}{\boldsymbol{y}_k^* \boldsymbol{x}_k} \boldsymbol{y}_k, \quad \boldsymbol{v} \in \mathbb{C}^n.$	$\boldsymbol{v} = \sum_{j=1}^n \frac{\boldsymbol{y}_j^* \boldsymbol{v}}{\boldsymbol{y}_j^* \boldsymbol{x}_j} \boldsymbol{x}_j = \sum_{k=1}^n \frac{\boldsymbol{x}_k^* \boldsymbol{v}}{\boldsymbol{y}_k^* \boldsymbol{x}_k} \boldsymbol{y}_k, \quad \boldsymbol{v} \in \mathbb{C}^n.$

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lyche-numerical-linear-algebra- EQ0399	160	■ 1.000	$\backslash\boldsymbol{X}:=\backslashleft[\backslash\boldsymbol{x}_{_1}, \backslashdots, \backslash\boldsymbol{x}_{_n}\backslashright], \backslashquad \backslash\boldsymbol{Y}:=\backslashleft[\backslash\boldsymbol{y}_{_1}, \backslashdots, \backslash\boldsymbol{y}_{_n}\backslashright], \backslashquad \backslash\boldsymbol{D}:=\backslashoperatorname{diag}\backslashleft(\backslash\lambda_{_1}, \backslashdots, \backslash\lambda_{_n}\backslashright) \text{ .}$	$\boldsymbol{X} := [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n], \quad \boldsymbol{Y} := [\boldsymbol{y}_1, \dots, \boldsymbol{y}_n], \quad \boldsymbol{D} := \text{diag}(\lambda_1, \dots, \lambda_n).$	$\boldsymbol{X} := [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n], \quad \boldsymbol{Y} := [\boldsymbol{y}_1, \dots, \boldsymbol{y}_n], \quad \boldsymbol{D} := \text{diag}(\lambda_1, \dots, \lambda_n).$
lyche-numerical-linear-algebra- EQ0400 (6.17)	160	■ 1.000	$\backslash\boldsymbol{v}=\backslashsum_{j=1}^n \backslashfrac{\backslash\boldsymbol{x}_{_j}^*\backslash\boldsymbol{v}}{\backslash\boldsymbol{x}_{_j}^*\backslash\boldsymbol{x}_{_j}}\backslash\boldsymbol{x}_{_j} \text{ .}$	$\boldsymbol{v} = \sum_{j=1}^n \frac{\boldsymbol{x}_j^* \boldsymbol{v}}{\boldsymbol{x}_j^* \boldsymbol{x}_j} \boldsymbol{x}_j.$	$\boldsymbol{v} = \sum_{j=1}^n \frac{\boldsymbol{x}_j^* \boldsymbol{v}}{\boldsymbol{x}_j^* \boldsymbol{x}_j} \boldsymbol{x}_j.$
lyche-numerical-linear-algebra- EQ0402	161	■ 1.000	$\backslashleft(\backslash\boldsymbol{V}^*\backslash\boldsymbol{A}^*\backslash\boldsymbol{V}\backslashright) \backslash\boldsymbol{u}=\backslash\boldsymbol{V}^*\backslash\boldsymbol{A}^*\backslash\boldsymbol{y}=\overline{\backslash\lambda} \backslash\boldsymbol{V}^*\backslash\boldsymbol{y}=\overline{\backslash\lambda} \backslash\boldsymbol{u},$	$(\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}) \boldsymbol{u} = \boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{V}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{u},$	$(\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}) \boldsymbol{u} = \boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{V}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{u},$
lyche-numerical-linear-algebra- EQ0405 (6.18)	162	■ 1.000	$\backslash\boldsymbol{A}^k \backslash\boldsymbol{x}=\backslashsum_{j=1}^n \text{ c}_{_j} \backslash\lambda_{_j}^k \backslash\boldsymbol{x}_{_j} \text{ for some scalars } \text{ c}_{_1}, \backslashdots, \text{ c}_{_n}$	$\boldsymbol{A}^k \boldsymbol{x} = \sum_{j=1}^n c_j \lambda_j^k \boldsymbol{x}_j \text{ for some scalars } c_1, \dots, c_n$	$\boldsymbol{A}^k \boldsymbol{x} = \sum_{j=1}^n c_j \lambda_j^k \boldsymbol{x}_j \text{ for some scalars } c_1, \dots, c_n.$
lyche-numerical-linear-algebra- EQ0411	164	■ 1.000	$p(\backslash\boldsymbol{A}):=\backslashsum_{j=0}^r \text{ b}_{_j} \backslash\boldsymbol{A}^j,$	$p(\boldsymbol{A}) := \sum_{j=0}^r b_j \boldsymbol{A}^j,$	$p(\boldsymbol{A}) := \sum_{j=0}^r b_j \boldsymbol{A}^j,$
lyche-numerical-linear-algebra- EQ0417 (7.1)	168	■ 1.000	$\backslash\boldsymbol{A}=\backslash\boldsymbol{U} \backslash\boldsymbol{D} \backslash\boldsymbol{U}^*, \text{ where } \backslash\boldsymbol{U}^* \backslash\boldsymbol{U}=\backslash\boldsymbol{I} \text{ .}$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^*, \text{ where } \boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}.$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^*, \text{ where } \boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}.$
lyche-numerical-linear-algebra- EQ0419	169	■ 1.000	$\backslash\lambda_{_1} \backslashgeq \backslashcdots \backslashgeq \backslash\lambda_{_r}>0=\backslash\lambda_{_r+1}=\backslashcdots=\backslash\lambda_{_n} \text{ .}$	$\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n.$	$\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n.$
lyche-numerical-linear-algebra- EQ0421 (7.3)	170	■ 1.000	$\backslash\lambda=\backslashfrac{\backslash\boldsymbol{v}^*\backslash\boldsymbol{A}^*\backslash\boldsymbol{v}}{\backslash\boldsymbol{v}^*\backslash\boldsymbol{v}}=\backslashfrac{\backslash\ \backslash\boldsymbol{A} \backslash\boldsymbol{v}\ _2^2}{\backslash\ \backslash\boldsymbol{v}\ _2^2} \backslashgeq 0 \text{ .}$	$\lambda = \frac{\boldsymbol{v}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}}{\boldsymbol{v}^* \boldsymbol{v}} = \frac{\ \boldsymbol{A} \boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2} \geq 0.$	$\lambda = \frac{\boldsymbol{v}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}}{\boldsymbol{v}^* \boldsymbol{v}} = \frac{\ \boldsymbol{A} \boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2} \geq 0.$
lyche-numerical-linear-algebra- EQ0423	171	■ 1.000	$\backslash\boldsymbol{B}:=\backslash\boldsymbol{A}^T \backslash\boldsymbol{A}=\backslashfrac{1}{25}\backslashleft[\backslashbegin{array}{ll} 52 \& 36 \backslash\backslash 36 \& 73 \backslashend{array}\backslashright]$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A} = \frac{1}{25} \begin{bmatrix} 52 & 36 \\ 36 & 73 \end{bmatrix}$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A} = \frac{1}{25} \begin{bmatrix} 52 & 36 \\ 36 & 73 \end{bmatrix}$
lyche-numerical-linear-algebra- EQ0424	171	■ 1.000	$\backslash\boldsymbol{B}\backslashleft[\backslashbegin{array}{l} 3 \backslash\backslash 4 \backslashend{array}\backslashright]=4\backslashleft[\backslashbegin{array}{l} 3 \backslash\backslash 4 \backslashend{array}\backslashright], \backslashquad \backslash\boldsymbol{B}\backslashleft[\backslashbegin{array}{r} 4 \backslash\backslash -3 \backslashend{array}\backslashright]=1\backslashleft[\backslashbegin{array}{r} 4 \backslash\backslash -3 \backslashend{array}\backslashright] \text{ .}$	$\boldsymbol{B} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \boldsymbol{B} \begin{bmatrix} 4 \\ -3 \end{bmatrix} = 1 \begin{bmatrix} 4 \\ -3 \end{bmatrix}.$	$\boldsymbol{B} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \boldsymbol{B} \begin{bmatrix} 4 \\ -3 \end{bmatrix} = 1 \begin{bmatrix} 4 \\ -3 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0428 (7.6)	172	■ 1.000	$\backslash\boldsymbol{A}=\backslashsum_{j=1}^r \backslashsigma_{_j} \backslash\boldsymbol{u}_{_j} \backslash\boldsymbol{v}_{_j}^* \text{ .}$	$\boldsymbol{A} = \sum_{j=1}^r \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^*.$	$\boldsymbol{A} = \sum_{j=1}^r \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^*.$
lyche-numerical-linear-algebra- EQ0430	172	■ 1.000	$\backslash\boldsymbol{B}:=\backslash\boldsymbol{A}^T \backslash\boldsymbol{A}=\backslashleft[\backslashbegin{array}{ll} 2 \& 2 \backslash\backslash 2 \& 2 \backslashend{array}\backslashright]$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$
lyche-numerical-linear-algebra- EQ0433	173	■ 1.000	$\backslash\boldsymbol{A}=\backslashfrac{1}{\sqrt{2}}\backslashleft[\backslashbegin{array}{l} 1 \backslash\backslash 1 \backslash\backslash 0 \backslashend{array}\backslashright][2] \backslashfrac{1}{\sqrt{2}}\backslashleft[\backslashbegin{array}{ll} 1 \& 1 \backslashend{array}\backslashright] \text{ .}$	$\boldsymbol{A} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} [2] \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}.$	$\boldsymbol{A} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} [2] \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}.$

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lyche-numerical-linear-algebra- EQ0434	173	■ 1.000	$\boldsymbol{s}_1=\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_2=\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_3=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],$	$\boldsymbol{s}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{s}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{s}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$	$\boldsymbol{s}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{s}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{s}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0437	174	■ 1.000	$\mathcal{E}=\boldsymbol{U}_1\tilde{\mathcal{E}} \text{ where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}=\left[y_1,\ldots,y_n\right]^T\in\mathbb{R}^n:\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}=1\right\}.$	$\mathcal{E} = \boldsymbol{U}_1\tilde{\mathcal{E}} \text{ where } \tilde{\mathcal{E}} := \left\{\boldsymbol{y} = [y_1, \dots, y_n]^T \in \mathbb{R}^n : \frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2} = 1\right\}.$	$\mathcal{E} = \boldsymbol{U}_1\tilde{\mathcal{E}} \text{ where } \tilde{\mathcal{E}} := \{\boldsymbol{y} = [y_1, \dots, y_n]^T \in \mathbb{R}^n : \frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2} = 1\}.$
lyche-numerical-linear-algebra- EQ0441 (7.9)	176	■ 1.000	$\ \boldsymbol{A}\ _F:=\left(\sum_{i=1}^m\sum_{j=1}^n\left a_{ij}\right ^2\right)^{1/2}.$	$\ \boldsymbol{A}\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2\right)^{1/2}.$	$\ \boldsymbol{A}\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2\right)^{1/2}.$
lyche-numerical-linear-algebra- EQ0444	177	■ 1.000	$\left\ \boldsymbol{A}-\boldsymbol{A}^{\prime}\right\ _F=\left\ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \boldsymbol{0} \end{array}\right] \boldsymbol{V}^*\right\ _F=\left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \boldsymbol{0} \end{array}\right]\right\ _F=\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}<\epsilon.$	$\ \boldsymbol{A}-\boldsymbol{A}'\ _F = \left\ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{array}\right]\boldsymbol{V}^*\right\ _F = \left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{array}\right]\right\ _F = \sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2} < \epsilon.$	$\ \boldsymbol{A}-\boldsymbol{A}'\ _F = \ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{array}\right]\boldsymbol{V}^*\ _F = \left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{array}\right]\right\ _F = \sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2} < \epsilon.$
lyche-numerical-linear-algebra- EQ0449	180	■ 1.000	$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array}\right] \in \mathbb{R}^{(m+n) \times(m+n)}$	$\boldsymbol{C} := \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}$	$\boldsymbol{C} := \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}$
lyche-numerical-linear-algebra- EQ0450	180	■ 1.000	$\left\{\left(\sigma_1, \boldsymbol{p}_1\right), \ldots,\left(\sigma_n, \boldsymbol{p}_n\right),\left(-\sigma_1, \boldsymbol{q}_1\right), \ldots,\left(-\sigma_n, \boldsymbol{q}_n\right),\left(0, \boldsymbol{r}_{n+1}\right), \ldots,\left(0, \boldsymbol{r}_m\right)\right\},$	$\{(\sigma_1, \boldsymbol{p}_1), \dots, (\sigma_n, \boldsymbol{p}_n), (-\sigma_1, \boldsymbol{q}_1), \dots, (-\sigma_n, \boldsymbol{q}_n), (0, \boldsymbol{r}_{n+1}), \dots, (0, \boldsymbol{r}_m)\},$	$\{(\sigma_1, \boldsymbol{p}_1), \dots, (\sigma_n, \boldsymbol{p}_n), (-\sigma_1, \boldsymbol{q}_1), \dots, (-\sigma_n, \boldsymbol{q}_n), (0, \boldsymbol{r}_{n+1}), \dots, (0, \boldsymbol{r}_m)\},$
lyche-numerical-linear-algebra- EQ0452 (7.12)	180	■ 1.000	$\boldsymbol{Q}:=\boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P}:=\boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$\boldsymbol{Q} := \boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P} := \boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$\boldsymbol{Q} := \boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P} := \boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$
lyche-numerical-linear-algebra- EQ0453 (7.13)	180	■ 1.000	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$
lyche-numerical-linear-algebra- EQ0455 (7.15)	181	■ 1.000	$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2} \boldsymbol{X}_k^{-T}\left(\boldsymbol{X}_k^T-\boldsymbol{Q}^T\right)\left(\boldsymbol{X}_k-\boldsymbol{Q}\right)$	$\boldsymbol{X}_{k+1} - \boldsymbol{Q} = \frac{1}{2} \boldsymbol{X}_k^{-T} (\boldsymbol{X}_k^T - \boldsymbol{Q}^T) (\boldsymbol{X}_k - \boldsymbol{Q})$	$\boldsymbol{X}_{k+1} - \boldsymbol{Q} = \frac{1}{2} \boldsymbol{X}_k^{-T} (\boldsymbol{X}_k^T - \boldsymbol{Q}^T) (\boldsymbol{X}_k - \boldsymbol{Q})$
lyche-numerical-linear-algebra- EQ0456 (7.16)	181	■ 1.000	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\ _F \leq \frac{1}{2}\left\ \boldsymbol{X}_k^{-1}\right\ _F\left\ \boldsymbol{X}_k-\boldsymbol{Q}\right\ _F^2.$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{Q}\ _F \leq \frac{1}{2} \ \boldsymbol{X}_k^{-1}\ _F \ \boldsymbol{X}_k - \boldsymbol{Q}\ _F^2.$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{Q}\ _F \leq \frac{1}{2} \ \boldsymbol{X}_k^{-1}\ _F \ \boldsymbol{X}_k - \boldsymbol{Q}\ _F^2.$
lyche-numerical-linear-algebra- EQ0458	181	■ 1.000	$\boldsymbol{B}=\left[\begin{array}{ccc} 1 & 0 & -1 \\ 1 & 1 & 1 \end{array}\right].$	$\boldsymbol{B} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$	$\boldsymbol{B} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$

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lyche-numerical-linear-algebra- EQ0461	182	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{rrrr} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right]$.	$\boldsymbol{A}=\begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.	$\boldsymbol{A}=\begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra- EQ0464	183	■ 1.000	$\left \boldsymbol{A}-\boldsymbol{A}'\right _F^2=\sigma_{r+1}^2+\cdots+\sigma_n^2$.	$\ \boldsymbol{A}-\boldsymbol{A}'\ _F^2=\sigma_{r+1}^2+\cdots+\sigma_n^2$.	$\ \boldsymbol{A}-\boldsymbol{A}'\ _F^2=\sigma_{r+1}^2+\cdots+\sigma_n^2$.
lyche-numerical-linear-algebra- EQ0472	187	■ 1.000	$\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}: \boldsymbol{y}'=1\right\}$.	$\mathcal{S}:=\{\boldsymbol{y} \in \mathcal{V}: \ \boldsymbol{y}\}'=1\}$.	$\mathcal{S}:=\{\boldsymbol{y} \in \mathcal{V}: \ \boldsymbol{y}\}'=1\}$.
lyche-numerical-linear-algebra- EQ0473 (8.8)	187	■ 1.000	$m \leq \boldsymbol{y} \leq M, \quad \boldsymbol{y} \in \mathcal{S}.$	$m \leq \ \boldsymbol{y}\ \leq M, \quad \boldsymbol{y} \in \mathcal{S}.$	$m \leq \ \boldsymbol{y}\ \leq M, \quad \boldsymbol{y} \in \mathcal{S}.$
lyche-numerical-linear-algebra- EQ0474	188	■ 1.000	$\left \boldsymbol{A}\right _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}$	$\ \boldsymbol{A}\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}$	$\ \boldsymbol{A}\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}$
lyche-numerical-linear-algebra- EQ0475	188	■ 1.000	$\mu\ \boldsymbol{A}\ \leq\left\ \boldsymbol{A}'\right\ \leq M\ \boldsymbol{A}\ $	$\mu\ \boldsymbol{A}\ \leq\ \boldsymbol{A}\ ' \leq M\ \boldsymbol{A}\ $	$\mu\ \boldsymbol{A}\ \leq\ \boldsymbol{A}\ ' \leq M\ \boldsymbol{A}\ $
lyche-numerical-linear-algebra- EQ0476 (8.9)	188	■ 1.000	$\left \boldsymbol{A}\right _S:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right , \quad\left \boldsymbol{A}\right _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}, \quad\left \boldsymbol{A}\right _M:=\max _{i, j}\left a_{i j}\right .$	$\ \boldsymbol{A}\ _S:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right , \quad\ \boldsymbol{A}\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}, \quad\ \boldsymbol{A}\ _M:=\max _{i, j}\left a_{i j}\right .$	$\ \boldsymbol{A}\ _S:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right , \quad\ \boldsymbol{A}\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}, \quad\ \boldsymbol{A}\ _M:=\max _{i, j}\left a_{i j}\right .$
lyche-numerical-linear-algebra- EQ0478 (8.11)	189	■ 1.000	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ ,$	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ ,$	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ ,$
lyche-numerical-linear-algebra- EQ0479 (8.12)	190	■ 1.000	$\left\ \boldsymbol{A}\right\ :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{x}\right\ }{\ \boldsymbol{x}\ }.$	$\ \boldsymbol{A}\ :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ }.$	$\ \boldsymbol{A}\ :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ }.$
lyche-numerical-linear-algebra- EQ0480 (8.13)	190	■ 1.000	$\left\ \boldsymbol{A}\right\ =\max _{\ \boldsymbol{x}\ =1}\left\ \boldsymbol{A} \boldsymbol{x}\right\ .$	$\ \boldsymbol{A}\ =\max _{\ \boldsymbol{x}\ =1}\ \boldsymbol{A} \boldsymbol{x}\ .$	$\ \boldsymbol{A}\ =\max _{\ \boldsymbol{x}\ =1}\ \boldsymbol{A} \boldsymbol{x}\ .$
lyche-numerical-linear-algebra- EQ0483 (8.15)	190	■ 1.000	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra- EQ0489	192	■ 1.000	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _1=\sum_{k=1}^m\left \sum_{j=1}^n a_{k j} x_j\right \leq \sum_{k=1}^m \sum_{j=1}^n\left a_{k j}\right \left x_j\right =\sum_{j=1}^n\left(\sum_{k=1}^m\left a_{k j}\right \right)\left x_j\right \leq K_1.$	$\ \boldsymbol{A} \boldsymbol{x}\ _1=\sum_{k=1}^m\left \sum_{j=1}^n a_{k j} x_j\right \leq \sum_{k=1}^m \sum_{j=1}^n\left a_{k j}\right \left x_j\right =\sum_{j=1}^n\left(\sum_{k=1}^m\left a_{k j}\right \right)\left x_j\right \leq K_1.$	$\ \boldsymbol{A} \boldsymbol{x}\ _1=\sum_{k=1}^m\left \sum_{j=1}^n a_{k j} x_j\right \leq \sum_{k=1}^m \sum_{j=1}^n\left a_{k j}\right \left x_j\right =\sum_{j=1}^n\left(\sum_{k=1}^m\left a_{k j}\right \right)\left x_j\right \leq K_1.$
lyche-numerical-linear-algebra- EQ0491	192	■ 1.000	$\left\ \boldsymbol{A}\right\ _1=\frac{29}{15}, \quad\left\ \boldsymbol{A}\right\ _2=2, \quad\left\ \boldsymbol{A}\right\ _\infty=\frac{37}{15}, \quad\left\ \boldsymbol{A}\right\ _F=\sqrt{5}.$	$\ \boldsymbol{A}\ _1=\frac{29}{15}, \quad\ \boldsymbol{A}\ _2=2, \quad\ \boldsymbol{A}\ _\infty=\frac{37}{15}, \quad\ \boldsymbol{A}\ _F=\sqrt{5}.$	$\ \boldsymbol{A}\ _1=\frac{29}{15}, \quad\ \boldsymbol{A}\ _2=2, \quad\ \boldsymbol{A}\ _\infty=\frac{37}{15}, \quad\ \boldsymbol{A}\ _F=\sqrt{5}.$
lyche-numerical-linear-algebra- EQ0495	194	■ 1.000	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _2=\max _{\ \boldsymbol{x}\ _2=1}\left\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{x}\right\ _2=\max _{\ \boldsymbol{x}\ _2=1}\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2=\ \boldsymbol{A}\ _2.$	$\ \boldsymbol{U} \boldsymbol{A}\ _2=\max _{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{x}\ _2=\max _{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{A} \boldsymbol{x}\ _2=\ \boldsymbol{A}\ _2.$	$\ \boldsymbol{U} \boldsymbol{A}\ _2=\max _{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{x}\ _2=\max _{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{A} \boldsymbol{x}\ _2=\ \boldsymbol{A}\ _2.$
lyche-numerical-linear-algebra- EQ0496	194	■ 1.000	$\left\ \boldsymbol{A} \boldsymbol{V}\right\ _2=\left\ (\boldsymbol{A} \boldsymbol{V})^*\right\ _2=\left\ \boldsymbol{V}^* \boldsymbol{A}^*\right\ _2=\left\ \boldsymbol{A}^*\right\ _2=\ \boldsymbol{A}\ _2.$	$\ \boldsymbol{A} \boldsymbol{V}\ _2=\ (\boldsymbol{A} \boldsymbol{V})^*\ _2=\ \boldsymbol{V}^* \boldsymbol{A}^*\ _2=\ \boldsymbol{A}^*\ _2=\ \boldsymbol{A}\ _2.$	$\ \boldsymbol{A} \boldsymbol{V}\ _2=\ (\boldsymbol{A} \boldsymbol{V})^*\ _2=\ \boldsymbol{V}^* \boldsymbol{A}^*\ _2=\ \boldsymbol{A}^*\ _2=\ \boldsymbol{A}\ _2.$

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lyche-numerical-linear-algebra-E09501 (8.22)	195	■ 1.000	$\frac{1}{K(\mathbf{A})} \frac{\ \mathbf{e}\ }{\ \mathbf{b}\ } \leq \frac{\ \mathbf{y} - \mathbf{x}\ }{\ \mathbf{x}\ } \leq K(\mathbf{A}) \frac{\ \mathbf{e}\ }{\ \mathbf{b}\ },$		
lyche-numerical-linear-algebra-E09503 (8.24)	196	■ 1.000	$\begin{aligned} & \frac{\ \mathbf{y} - \mathbf{x}\ }{\ \mathbf{y}\ } \leq r \leq K(\mathbf{A}) \frac{\ \mathbf{E}\ }{\ \mathbf{A}\ }, \\ & \frac{\ \mathbf{y} - \mathbf{x}\ }{\ \mathbf{x}\ } \leq \frac{r}{1-r} \leq \frac{K(\mathbf{A}) \ \mathbf{E}\ }{1-r \ \mathbf{A}\ }. \end{aligned}$		
lyche-numerical-linear-algebra-E09506 (8.28)	198	■ 1.000	$\frac{1}{1+r} \leq \frac{\ \mathbf{B}^{-1}\ }{\ \mathbf{A}^{-1}\ } \leq \frac{1}{1-r}.$		
lyche-numerical-linear-algebra-E09507 (8.29)	198	■ 1.000	$\frac{\ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ }{\ \mathbf{A}^{-1}\ } \leq \frac{r}{1-r} \leq \frac{K(\mathbf{A}) \ \mathbf{E}\ }{1-r \ \mathbf{A}\ }.$		
lyche-numerical-linear-algebra-E09508 (8.30)	198	■ 1.000	$\mathbf{B}^{-1} - \mathbf{A}^{-1} = -\mathbf{A}^{-1} \mathbf{E} \mathbf{B}^{-1} = -\mathbf{B}^{-1} \mathbf{E} \mathbf{A}^{-1}.$		
lyche-numerical-linear-algebra-E09512 (8.31)	199	■ 1.000	$f((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda)f(x_1) + \lambda f(x_2)$		
lyche-numerical-linear-algebra-E09514	200	■ 1.000	$x = \frac{x_2 - x}{x_2 - x_1} x_1 + \frac{x - x_1}{x_2 - x_1} x_2 = (1-\lambda)x_1 + \lambda x_2$		
lyche-numerical-linear-algebra-E09515	200	■ 1.000	$f\left(\sum_{j=1}^n \lambda_j z_j\right) \leq \sum_{j=1}^n \lambda_j f(z_j).$		
lyche-numerical-linear-algebra-E09516	200	■ 1.000	$f\left(\sum_{j=1}^n \lambda_j z_j\right) = f(\lambda_1 z_1 + (1-\lambda_1)u) \leq \lambda_1 f(z_1) + (1-\lambda_1)f(u) \leq \sum_{j=1}^n \lambda_j f(z_j)$		
lyche-numerical-linear-algebra-E09518	201	■ 1.000	$-\log\left(\sum_{j=1}^n \lambda_j a_j\right) \leq -\sum_{j=1}^n \lambda_j \log(a_j) = -\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n})$		
lyche-numerical-linear-algebra-E09519 (8.33)	201	■ 1.000	$(a_1 a_2 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^n a_j.$		

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-201-E09529	201	1.000	$\sum_{j=1}^n \left x_j y_j \right \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \text{ where } \frac{1}{p} + \frac{1}{q} = 1$	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \text{ where } \frac{1}{p} + \frac{1}{q} = 1$	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \text{ where } \frac{1}{p} + \frac{1}{q} = 1.$
lyche-numerical-linear-algebra-201-E09522	201	1.000	$\frac{1}{\ \mathbf{x}\ _p \ \mathbf{y}\ _q} \sum_{j=1}^n x_j y_j = \sum_{j=1}^n \left(\frac{ x_j ^p}{\ \mathbf{x}\ _p^p} \right)^{\frac{1}{p}} \left(\frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right)^{\frac{1}{q}} \leq \sum_{j=1}^n \left(\frac{1}{p} \frac{ x_j ^p}{\ \mathbf{x}\ _p^p} + \frac{1}{q} \frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right) = 1$	$\frac{1}{\ \mathbf{x}\ _p \ \mathbf{y}\ _q} \sum_{j=1}^n x_j y_j = \sum_{j=1}^n \left(\frac{ x_j ^p}{\ \mathbf{x}\ _p^p} \right)^{\frac{1}{p}} \left(\frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right)^{\frac{1}{q}} \leq \sum_{j=1}^n \left(\frac{1}{p} \frac{ x_j ^p}{\ \mathbf{x}\ _p^p} + \frac{1}{q} \frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right) = 1$	$\frac{1}{\ \mathbf{x}\ _p \ \mathbf{y}\ _q} \sum_{j=1}^n x_j y_j = \sum_{j=1}^n \left(\frac{ x_j ^p}{\ \mathbf{x}\ _p^p} \right)^{\frac{1}{p}} \left(\frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right)^{\frac{1}{q}} \leq \sum_{j=1}^n \left(\frac{1}{p} \frac{ x_j ^p}{\ \mathbf{x}\ _p^p} + \frac{1}{q} \frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right) = 1$
lyche-numerical-linear-algebra-202-E09523	202	1.000	$\ \mathbf{x} + \mathbf{y}\ _p \leq \ \mathbf{x}\ _p + \ \mathbf{y}\ _p.$	$\ \mathbf{x} + \mathbf{y}\ _p \leq \ \mathbf{x}\ _p + \ \mathbf{y}\ _p.$	$\ \mathbf{x} + \mathbf{y}\ _p \leq \ \mathbf{x}\ _p + \ \mathbf{y}\ _p.$
lyche-numerical-linear-algebra-202-E09524	202	1.000	$\ \mathbf{x} + \mathbf{y}\ _p^p = \sum_{j=1}^n x_j + y_j ^p \leq \sum_{j=1}^n x_j x_j + y_j ^{p-1} + \sum_{j=1}^n y_j x_j + y_j ^{p-1}.$	$\ \mathbf{x} + \mathbf{y}\ _p^p = \sum_{j=1}^n x_j + y_j ^p \leq \sum_{j=1}^n x_j x_j + y_j ^{p-1} + \sum_{j=1}^n y_j x_j + y_j ^{p-1}.$	$\ \mathbf{x} + \mathbf{y}\ _p^p = \sum_{j=1}^n x_j + y_j ^p \leq \sum_{j=1}^n x_j x_j + y_j ^{p-1} + \sum_{j=1}^n y_j x_j + y_j ^{p-1}.$
lyche-numerical-linear-algebra-202-E09525	202	1.000	$\ \mathbf{x} + \mathbf{y}\ _p^p \leq \ \mathbf{x}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} + \ \mathbf{y}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} = (\ \mathbf{x}\ _p + \ \mathbf{y}\ _p) \ \mathbf{x} + \mathbf{y}\ _p^{p-1},$	$\ \mathbf{x} + \mathbf{y}\ _p^p \leq \ \mathbf{x}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} + \ \mathbf{y}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} = (\ \mathbf{x}\ _p + \ \mathbf{y}\ _p) \ \mathbf{x} + \mathbf{y}\ _p^{p-1},$	$\ \mathbf{x} + \mathbf{y}\ _p^p \leq \ \mathbf{x}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} + \ \mathbf{y}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} = (\ \mathbf{x}\ _p + \ \mathbf{y}\ _p) \ \mathbf{x} + \mathbf{y}\ _p^{p-1},$
lyche-numerical-linear-algebra-202-E09526 (8.35)	202	1.000	$\ \mathbf{x} + \mathbf{y}\ ^2 + \ \mathbf{x} - \mathbf{y}\ ^2 = 2\ \mathbf{x}\ ^2 + 2\ \mathbf{y}\ ^2,$	$\ \mathbf{x} + \mathbf{y}\ ^2 + \ \mathbf{x} - \mathbf{y}\ ^2 = 2\ \mathbf{x}\ ^2 + 2\ \mathbf{y}\ ^2,$	$\ \mathbf{x} + \mathbf{y}\ ^2 + \ \mathbf{x} - \mathbf{y}\ ^2 = 2\ \mathbf{x}\ ^2 + 2\ \mathbf{y}\ ^2,$
lyche-numerical-linear-algebra-203-E09534	203	1.000	$\left\langle \frac{n}{m} \mathbf{x}, \mathbf{y} \right\rangle = \frac{n}{m} \langle \mathbf{x}, \mathbf{y} \rangle.$	$\left\langle \frac{n}{m} \mathbf{x}, \mathbf{y} \right\rangle = \frac{n}{m} \langle \mathbf{x}, \mathbf{y} \rangle.$	$\left\langle \frac{n}{m} \mathbf{x}, \mathbf{y} \right\rangle = \frac{n}{m} \langle \mathbf{x}, \mathbf{y} \rangle.$
lyche-numerical-linear-algebra-204-E09539	204	1.000	$\tilde{\mathbf{b}}_k := \mathbf{b}_k - \mathbf{B}_{k-1} (\mathbf{B}_{k-1}^T \mathbf{A} \mathbf{B}_{k-1})^{-1} \mathbf{B}_{k-1}^T \mathbf{A} \mathbf{b}_k, \quad k = 2, \dots, n.$	$\tilde{\mathbf{b}}_k := \mathbf{b}_k - \mathbf{B}_{k-1} (\mathbf{B}_{k-1}^T \mathbf{A} \mathbf{B}_{k-1})^{-1} \mathbf{B}_{k-1}^T \mathbf{A} \mathbf{b}_k, \quad k = 2, \dots, n.$	$\tilde{\mathbf{b}}_k := \mathbf{b}_k - \mathbf{B}_{k-1} (\mathbf{B}_{k-1}^T \mathbf{A} \mathbf{B}_{k-1})^{-1} \mathbf{B}_{k-1}^T \mathbf{A} \mathbf{b}_k, \quad k = 2, \dots, n.$
lyche-numerical-linear-algebra-205-E09540	205	1.000	$\ \mathbf{b}_1\ _A \ \mathbf{b}_2\ _A \cdots \ \mathbf{b}_n\ _A \leq 2^{n(n-1)/4} \sqrt{\det(\mathbf{A})}.$	$\ \mathbf{b}_1\ _A \ \mathbf{b}_2\ _A \cdots \ \mathbf{b}_n\ _A \leq 2^{n(n-1)/4} \sqrt{\det(\mathbf{A})}.$	$\ \mathbf{b}_1\ _A \ \mathbf{b}_2\ _A \cdots \ \mathbf{b}_n\ _A \leq 2^{n(n-1)/4} \sqrt{\det(\mathbf{A})}.$
lyche-numerical-linear-algebra-205-E09541	205	1.000	$\ \mathbf{A}\ := \sqrt{mn} \ \mathbf{A}\ _M, \quad \mathbf{A} \in \mathbb{C}^{m \times n}$	$\ \mathbf{A}\ := \sqrt{mn} \ \mathbf{A}\ _M, \quad \mathbf{A} \in \mathbb{C}^{m \times n}$	$\ \mathbf{A}\ := \sqrt{mn} \ \mathbf{A}\ _M, \quad \mathbf{A} \in \mathbb{C}^{m \times n}$
lyche-numerical-linear-algebra-205-E09542	205	1.000	$\ \mathbf{A}\ _2 = \max_{\ \mathbf{x}\ _2 = \ \mathbf{y}\ _2 = 1} \mathbf{y}^* \mathbf{A} \mathbf{x} .$	$\ \mathbf{A}\ _2 = \max_{\ \mathbf{x}\ _2 = \ \mathbf{y}\ _2 = 1} \mathbf{y}^* \mathbf{A} \mathbf{x} .$	$\ \mathbf{A}\ _2 = \max_{\ \mathbf{x}\ _2 = \ \mathbf{y}\ _2 = 1} \mathbf{y}^* \mathbf{A} \mathbf{x} .$

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lyche-numerical-linear-algebra-207 EQ0545	207	1.000	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\frac{1}{\lambda_{k+1}}\boldsymbol{r}_k$, where $\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$.	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\frac{1}{\lambda_{k+1}}\boldsymbol{r}_k$, where $\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$.	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\frac{1}{\lambda_{k+1}}\boldsymbol{r}_k$, where $\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$.
lyche-numerical-linear-algebra-207 EQ0546	207	1.000	$\boldsymbol{r}_k=\sum_{i=1}^n c_{ik}\boldsymbol{u}_i$,	$\boldsymbol{r}_k=\sum_{i=1}^n c_{ik}\boldsymbol{u}_i$,	$\boldsymbol{r}_k=\sum_{i=1}^n c_{ik}\boldsymbol{u}_i$,
lyche-numerical-linear-algebra-207 EQ0548	207	1.000	$\lambda_j=d+2c\cos\left(\frac{j\pi}{n+1}\right)$, $j=1,\ldots,n$.	$\lambda_j=d+2c\cos\left(\frac{j\pi}{n+1}\right)$, $j=1,\ldots,n$.	$\lambda_j=d+2c\cos\left(\frac{j\pi}{n+1}\right)$, $j=1,\ldots,n$.
lyche-numerical-linear-algebra-207 EQ0549 (8.40)	207	1.000	$\boldsymbol{T}\boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B}\boldsymbol{x}_k$.	$\boldsymbol{T}\boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B}\boldsymbol{x}_k$.	$\boldsymbol{T}\boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B}\boldsymbol{x}_k$.
lyche-numerical-linear-algebra-207 EQ0550	207	1.000	$\min\left\{\max_i\sum_{j=1}^n b_{ij} ,\max_j\sum_{i=1}^n b_{ij} \right\}<d-2c$.	$\min\left\{\max_i\sum_{j=1}^n b_{ij} ,\max_j\sum_{i=1}^n b_{ij} \right\}<d-2c$.	$\min\left\{\max_i\sum_{j=1}^n b_{ij} ,\max_j\sum_{i=1}^n b_{ij} \right\}<d-2c$.
lyche-numerical-linear-algebra-208 EQ0552 (8.41)	208	1.000	$\frac{4}{\pi^2}(m+1)^2-2/3<\operatorname{cond}_p(\boldsymbol{T})\leq\frac{1}{2}(m+1)^2$, $p=1,2,\infty$,	$\frac{4}{\pi^2}(m+1)^2-2/3<\operatorname{cond}_p(\boldsymbol{T})\leq\frac{1}{2}(m+1)^2$, $p=1,2,\infty$,	$\frac{4}{\pi^2}(m+1)^2-2/3<\operatorname{cond}_p(\boldsymbol{T})\leq\frac{1}{2}(m+1)^2$, $p=1,2,\infty$,
lyche-numerical-linear-algebra-208 EQ0554	208	1.000	$\operatorname{cond}_2(\boldsymbol{T})=\cot^2\left(\frac{\pi h}{2}\right)=1/\tan^2\left(\frac{\pi h}{2}\right)$.	$\operatorname{cond}_2(\boldsymbol{T})=\cot^2\left(\frac{\pi h}{2}\right)=1/\tan^2\left(\frac{\pi h}{2}\right)$.	$\operatorname{cond}_2(\boldsymbol{T})=\cot^2\left(\frac{\pi h}{2}\right)=1/\tan^2\left(\frac{\pi h}{2}\right)$.
lyche-numerical-linear-algebra-209 EQ0555 (8.43)	209	1.000	$\frac{4}{\pi^2}h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2}h^{-2}$.	$\frac{4}{\pi^2}h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2}h^{-2}$.	$\frac{4}{\pi^2}h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2}h^{-2}$.
lyche-numerical-linear-algebra-209 EQ0556	209	1.000	$\frac{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ }{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ }\leq\ \boldsymbol{E}\ $	$\frac{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ }{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ }\leq\ \boldsymbol{E}\ $	$\frac{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ }{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ }\leq\ \boldsymbol{E}\ $
lyche-numerical-linear-algebra-209 EQ0557	209	1.000	$\frac{1}{1+\ \boldsymbol{E}\ }\leq\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ \leq\frac{1}{1-\ \boldsymbol{E}\ }$	$\frac{1}{1+\ \boldsymbol{E}\ }\leq\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ \leq\frac{1}{1-\ \boldsymbol{E}\ }$	$\frac{1}{1+\ \boldsymbol{E}\ }\leq\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ \leq\frac{1}{1-\ \boldsymbol{E}\ }$
lyche-numerical-linear-algebra-209 EQ0558	209	1.000	$\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ \leq\frac{\ \boldsymbol{E}\ }{1-\ \boldsymbol{E}\ }$.	$\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ \leq\frac{\ \boldsymbol{E}\ }{1-\ \boldsymbol{E}\ }$.	$\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ \leq\frac{\ \boldsymbol{E}\ }{1-\ \boldsymbol{E}\ }$.
lyche-numerical-linear-algebra-209 EQ0560	209	1.000	$\frac{K(\boldsymbol{B})^{-1}\ \boldsymbol{E}\ }{1+r}\leq\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{A}^{-1}\ }$.	$\frac{K(\boldsymbol{B})^{-1}\ \boldsymbol{E}\ }{1+r}\leq\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{A}^{-1}\ }$.	$\frac{K(\boldsymbol{B})^{-1}\ \boldsymbol{E}\ }{1+r}\leq\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{A}^{-1}\ }$.
lyche-numerical-linear-algebra-209 EQ0561	209	1.000	$a=x_0<x_1<\cdots<x_n=b$,	$a=x_0<x_1<\cdots<x_n=b$,	$a=x_0<x_1<\cdots<x_n=b$,
lyche-numerical-linear-algebra-210 EQ0562	210	1.000	$\begin{array}{l} g(x_i)=y_i, \quad i=0,1,\ldots,n, \\ g'(a)=g'(b), \quad g''(a)=g''(b). \end{array}$	$\begin{array}{l} g(x_i)=y_i, \quad i=0,1,\ldots,n, \\ g'(a)=g'(b), \quad g''(a)=g''(b). \end{array}$	$\begin{array}{l} g(x_i)=y_i, \quad i=0,1,\ldots,n, \\ g'(a)=g'(b), \quad g''(a)=g''(b). \end{array}$

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lyche-numerical-linear-algebra-2019-09-05E09565	210	1.000	$\lambda_{i}:=\frac{h_{i}}{h_{i-1}+h_{i}}, \quad \mu_{i}:=\frac{h_{i-1}}{h_{i-1}+h_{i}}, \quad i=1, \ldots, n,$	$\lambda_i := \frac{h_i}{h_{i-1} + h_i}, \quad \mu_i := \frac{h_{i-1}}{h_{i-1} + h_i}, \quad , i = 1, \dots, n,$	$\lambda_i := \frac{h_i}{h_{i-1} + h_i}, \quad \mu_i := \frac{h_{i-1}}{h_{i-1} + h_i}, \quad , i = 1, \dots, n,$
lyche-numerical-linear-algebra-2019-09-05E09566	210	1.000	$h_{i}=x_{i+1}-x_{i}, \quad i=0, \ldots, n-1, \text{ and } h_n=h_0.$	$h_i = x_{i+1} - x_i, \quad i = 0, \dots, n - 1, \text{ and } h_n = h_0.$	$h_i = x_{i+1} - x_i, \quad i = 0, \dots, n - 1, \text{ and } h_n = h_0.$
lyche-numerical-linear-algebra-2019-09-05E09567	210	1.000	$\frac{1}{2} \leq \frac{h_i}{h_{i-1}} \leq 2, \quad i=1, \ldots, n.$	$\frac{1}{2} \leq \frac{h_i}{h_{i-1}} \leq 2, \quad i = 1, \dots, n.$	$\frac{1}{2} \leq \frac{h_i}{h_{i-1}} \leq 2, \quad i = 1, \dots, n.$
lyche-numerical-linear-algebra-2019-09-05E09568	211	1.000	$\ \boldsymbol{A}\ _{\infty}=3 \quad \text{and that} \quad \ \boldsymbol{A}\ _1 \leq \frac{10}{3}.$	$\ \boldsymbol{A}\ _{\infty} = 3 \quad \text{and that} \quad \ \boldsymbol{A}\ _1 \leq \frac{10}{3}.$	$\ \boldsymbol{A}\ _{\infty} = 3 \quad \text{and that} \quad \ \boldsymbol{A}\ _1 \leq \frac{10}{3}.$
lyche-numerical-linear-algebra-2019-09-05E09571 (8.45)	211	1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 + i \ \boldsymbol{x} + i \boldsymbol{y}\ ^2 - i \ \boldsymbol{x} - i \boldsymbol{y}\ ^2 \right),$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 + i \ \boldsymbol{x} + i \boldsymbol{y}\ ^2 - i \ \boldsymbol{x} - i \boldsymbol{y}\ ^2 \right),$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 + i \ \boldsymbol{x} + i \boldsymbol{y}\ ^2 - i \ \boldsymbol{x} - i \boldsymbol{y}\ ^2 \right),$
lyche-numerical-linear-algebra-2019-09-05E09574	213	1.000	$E(\boldsymbol{x}) := \ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2,$	$E(\boldsymbol{x}) := \ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2,$	$E(\boldsymbol{x}) := \ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2,$
lyche-numerical-linear-algebra-2019-09-05E09576	214	1.000	$\ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2 = (x_1 - 1)^2 + (x_1 - 1)^2 + (x_1 - 2)^2 = 3x_1^2 - 8x_1 + 6.$	$\ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2 = (x_1 - 1)^2 + (x_1 - 1)^2 + (x_1 - 2)^2 = 3x_1^2 - 8x_1 + 6.$	$\ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2 = (x_1 - 1)^2 + (x_1 - 1)^2 + (x_1 - 2)^2 = 3x_1^2 - 8x_1 + 6.$
lyche-numerical-linear-algebra-2019-09-05E09581	215	1.000	$y=\boldsymbol{u}^* \boldsymbol{x}=\sum_{i=1}^n u_i x_i,$	$y = \boldsymbol{u}^* \boldsymbol{x} = \sum_{i=1}^n u_i x_i,$	$y = \boldsymbol{u}^* \boldsymbol{x} = \sum_{i=1}^n u_i x_i,$
lyche-numerical-linear-algebra-2019-09-05E09585 (9.2)	216	1.000	$E(\boldsymbol{x}) := \ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2 = \sum_{k=1}^m \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$	$E(\boldsymbol{x}) := \ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2 = \sum_{k=1}^m \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$	$E(\boldsymbol{x}) := \ \boldsymbol{Ax} - \boldsymbol{b}\ _2^2 = \sum_{k=1}^m \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$
lyche-numerical-linear-algebra-2019-09-05E09586	217	1.000	$E(\boldsymbol{x}) := \sum_{k=1}^m w_k \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$	$E(\boldsymbol{x}) := \sum_{k=1}^m w_k \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$	$E(\boldsymbol{x}) := \sum_{k=1}^m w_k \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$
lyche-numerical-linear-algebra-2019-09-05E09587 (9.3)	217	1.000	$p(t_k) := \sum_{j=1}^n x_j \phi_j(t_k) = 0, \quad k = 1, \dots, m \Rightarrow x_1 = \dots = x_n = 0.$	$p(t_k) := \sum_{j=1}^n x_j \phi_j(t_k) = 0, \quad k = 1, \dots, m \Rightarrow x_1 = \dots = x_n = 0.$	$p(t_k) := \sum_{j=1}^n x_j \phi_j(t_k) = 0, \quad k = 1, \dots, m \Rightarrow x_1 = \dots = x_n = 0.$
lyche-numerical-linear-algebra-2019-09-05E09595	219	1.000	$(\boldsymbol{A}^* \boldsymbol{b})_i = \sum_{k=1}^m \bar{a}_{k,i} b_k, i = 1, \dots, n,$	$(\boldsymbol{A}^* \boldsymbol{b})_i = \sum_{k=1}^m \bar{a}_{k,i} b_k, i = 1, \dots, n,$	$(\boldsymbol{A}^* \boldsymbol{b})_i = \sum_{k=1}^m \bar{a}_{k,i} b_k, i = 1, \dots, n,$
lyche-numerical-linear-algebra-2019-09-05E09596 (9.6)	220	1.000	$K_2(\boldsymbol{A}) := \ \boldsymbol{A}^* \boldsymbol{A}\ _2 \left\ (\boldsymbol{A}^* \boldsymbol{A})^{-1} \right\ _2 = \frac{\lambda_1}{\lambda_n} = \frac{\sigma_1^2}{\sigma_n^2} = K_2(\boldsymbol{A})^2,$	$K_2(\boldsymbol{A}^* \boldsymbol{A}) := \ \boldsymbol{A}^* \boldsymbol{A}\ _2 \left\ (\boldsymbol{A}^* \boldsymbol{A})^{-1} \right\ _2 = \frac{\lambda_1}{\lambda_n} = \frac{\sigma_1^2}{\sigma_n^2} = K_2(\boldsymbol{A})^2,$	$K_2(\boldsymbol{A}^* \boldsymbol{A}) := \ \boldsymbol{A}^* \boldsymbol{A}\ _2 \left\ (\boldsymbol{A}^* \boldsymbol{A})^{-1} \right\ _2 = \frac{\lambda_1}{\lambda_n} = \frac{\sigma_1^2}{\sigma_n^2} = K_2(\boldsymbol{A})^2,$

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lyche-numerical-linear-algebra-230 EQ0633	230	■ 1.000	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 = \sum_{i=1}^t \alpha_i - \beta_i ^2 + \sum_{i=1}^t -\alpha_i + \beta_i ^2 = 2 \sum_{i=1}^t \alpha_i - \beta_i ^2$		
lyche-numerical-linear-algebra-230 EQ0635 (9.22)	230	■ 1.000	$\left(t_i - c_1\right)^2 + \left(y_i - c_2\right)^2 = r^2, i = 1, \ldots, m,$		
lyche-numerical-linear-algebra-230 EQ0636 (9.23)	230	■ 1.000	$t_i x_1 + y_i x_2 + x_3 = t_i^2 + y_i^2, i = 1, \ldots, m,$		
lyche-numerical-linear-algebra-231 EQ0637	231	■ 1.000	$\sum_{i=1}^3 \left\ p\left(x_i\right)-y_i\right\ ^2,$		
lyche-numerical-linear-algebra-231 EQ0638	231	■ 1.000	$F(\boldsymbol{x}):=\ \boldsymbol{Ax}-\boldsymbol{b}\ _2^2+\ \boldsymbol{x}\ _2^2.$		
lyche-numerical-linear-algebra-231 EQ0639	231	■ 1.000	$F(\boldsymbol{x})=\boldsymbol{x}^T\boldsymbol{Bx}-2\boldsymbol{c}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{b}, \quad \text{where} \quad \boldsymbol{c}=\boldsymbol{A}^T\boldsymbol{b}.$		
lyche-numerical-linear-algebra-231 EQ0640 (9.24)	231	■ 1.000	$\ \boldsymbol{r}(\boldsymbol{x})\ _D^2:=\sum_{i=1}^m r_i(x)^2 d_i, \quad \boldsymbol{x} \in \mathbb{R}^n$		
lyche-numerical-linear-algebra-232 EQ0642	232	■ 1.000	$\boldsymbol{A}^T\boldsymbol{DAx}=\boldsymbol{A}^T\boldsymbol{Db}.$		
lyche-numerical-linear-algebra-232 EQ0643	232	■ 1.000	$K_2\left(\boldsymbol{A}^T\boldsymbol{DA}\right)\leq K_2\left(\boldsymbol{A}^T\boldsymbol{A}\right)K_2(\boldsymbol{D}),$		
lyche-numerical-linear-algebra-233 EQ0646 (9.25)	233	■ 1.000	$\begin{aligned} & \boldsymbol{P}_{\mathcal{R}(A^*)}=\boldsymbol{A}^\dagger\boldsymbol{A}=\boldsymbol{V}_1\boldsymbol{V}_1^*=\sum_{j=1}^r\boldsymbol{v}_j\boldsymbol{v}_j^*\in\mathbb{C}^{n\times n} \\ & \boldsymbol{P}_{\mathcal{N}(A)}=\boldsymbol{I}-\boldsymbol{A}^\dagger\boldsymbol{A}=\boldsymbol{V}_2\boldsymbol{V}_2^*=\sum_{j=r+1}^n\boldsymbol{v}_j\boldsymbol{v}_j^*\in\mathbb{C}^{n\times n}. \end{aligned}$		
lyche-numerical-linear-algebra-234 EQ0652	234	■ 1.000	$A=\begin{bmatrix} B \\ C \end{bmatrix}$		
lyche-numerical-linear-algebra-235 EQ0653	235	■ 1.000	$\boldsymbol{A}=\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \boldsymbol{b}=\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$		

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lyche-numerical-linear-algebra-235 EQ0655	235	■ 1.000	$\left \sigma_i(\epsilon)-\sigma_i(\epsilon)\right \leq\left \epsilon\right ,\quad i=1,\ldots,n,$	$ \sigma_i(\epsilon)-\sigma_i(0) \leq \epsilon ,\quad i=1,\ldots,n,$	$ \sigma_i(\epsilon)-\sigma_i(0) \leq \epsilon ,\quad i=1,\ldots,n,$
lyche-numerical-linear-algebra-238 EQ0657	238	■ 1.000	$\bar{\Omega}_h:=\{(j\,h,\,k\,h):j,\,k=0,1,\ldots,m+1\},$ $\quad\quad\text{where}\quad h=1/(m+1)\,.$	$\bar{\Omega}_h:=\{(jh,kh):j,k=0,1,\ldots,m+1\},\quad\text{where}\quad h=1/(m+1).$	$\bar{\Omega}_h:=\{(jh,kh):j,k=0,1,\ldots,m+1\},\quad\text{where}\quad h=1/(m+1).$
lyche-numerical-linear-algebra-239 EQ0658 (10.2)	239	■ 1.000	$\frac{\partial^2 u(j\,h,\,k\,h)}{\partial x^2}\approx\frac{v_{j-1,\,k}-2\,v_{j,\,k}+v_{j+1,\,k}}{h^2},$ $\frac{\partial^2 u(j\,h,\,k\,h)}{\partial y^2}\approx\frac{v_{j,\,k-1}-2\,v_{j,\,k}+v_{j,\,k+1}}{h^2}.$	$\frac{\partial^2 u(jh,kh)}{\partial x^2}\approx\frac{v_{j-1,k}-2v_{j,k}+v_{j+1,k}}{h^2},\frac{\partial^2 u(jh,kh)}{\partial y^2}\approx\frac{v_{j,k-1}-2v_{j,k}+v_{j,k+1}}{h^2}.$	$\frac{\partial^2 u(jh,kh)}{\partial x^2}\approx\frac{v_{j-1,k}-2v_{j,k}+v_{j+1,k}}{h^2},\frac{\partial^2 u(jh,kh)}{\partial y^2}\approx\frac{v_{j,k-1}-2v_{j,k}+v_{j,k+1}}{h^2}.$
lyche-numerical-linear-algebra-239 EQ0660 (10.4)	239	■ 1.000	$-\Delta_h v_{j,\,k}:=\frac{-v_{j-1,\,k}+2\,v_{j,\,k}-v_{j+1,\,k}}{h^2}+\frac{-v_{j,\,k-1}+2\,v_{j,\,k}-v_{j,\,k+1}}{h^2}.$	$-\Delta_h v_{j,k}:=\frac{-v_{j-1,k}+2v_{j,k}-v_{j+1,k}}{h^2}+\frac{-v_{j,k-1}+2v_{j,k}-v_{j,k+1}}{h^2}.$	$-\Delta_h v_{j,k}:=\frac{-v_{j-1,k}+2v_{j,k}-v_{j+1,k}}{h^2}+\frac{-v_{j,k-1}+2v_{j,k}-v_{j,k+1}}{h^2}.$
lyche-numerical-linear-algebra-239 EQ0662 (10.6)	239	■ 1.000	$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{array}\right]\in\mathbb{R}^{m\times m},\quad\boldsymbol{F}:=\left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{array}\right]\in\mathbb{R}^{m\times m}.$	$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{array}\right]\in\mathbb{R}^{m\times m},\quad\boldsymbol{F}:=\left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{array}\right]\in\mathbb{R}^{m\times m}.$	$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{array}\right]\in\mathbb{R}^{m\times m},\quad\boldsymbol{F}:=\left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{array}\right]\in\mathbb{R}^{m\times m}.$
lyche-numerical-linear-algebra-239 EQ0663	239	■ 1.000	$\sum_{i=1}^m\boldsymbol{T}_{j,i}v_{i,\,k}+\sum_{i=1}^mv_{j,i}\boldsymbol{T}_{i,\,k}=-v_{j-1,\,k}+2\,v_{j,\,k}-v_{j+1,\,k}-v_{j,\,k-1}+2\,v_{j,\,k}-v_{j,\,k+1},$	$\sum_{i=1}^m\boldsymbol{T}_{j,i}v_{i,k}+\sum_{i=1}^mv_{j,i}\boldsymbol{T}_{i,k}=-v_{j-1,k}+2v_{j,k}-v_{j+1,k}-v_{j,k-1}+2v_{j,k}-v_{j,k+1},$	$\sum_{i=1}^m\boldsymbol{T}_{j,i}v_{i,k}+\sum_{i=1}^mv_{j,i}\boldsymbol{T}_{i,k}=-v_{j-1,k}+2v_{j,k}-v_{j+1,k}-v_{j,k-1}+2v_{j,k}-v_{j,k+1},$
lyche-numerical-linear-algebra-239 EQ0664	239	■ 1.000	$\operatorname{vec}(\boldsymbol{B}):=\left[b_{11},\ldots,b_{m1},b_{12},\ldots,b_{m2},\ldots,b_{1n},\ldots,b_{mn}\right]^T\in\mathbb{R}^{mn}$	$\operatorname{vec}(\boldsymbol{B}):=[b_{11},\ldots,b_{m1},b_{12},\ldots,b_{m2},\ldots,b_{1n},\ldots,b_{mn}]^T\in\mathbb{R}^{mn}$	$\operatorname{vec}(\boldsymbol{B}):=[b_{11},\ldots,b_{m1},b_{12},\ldots,b_{m2},\ldots,b_{1n},\ldots,b_{mn}]^T\in\mathbb{R}^{mn}$
lyche-numerical-linear-algebra-240 EQ0665	240	■ 1.000	$4\,x_{i}-x_{i-1}-x_{i+1}-x_{i-m}-x_{i+m}=b_i,$	$4x_i-x_{i-1}-x_{i+1}-x_{i-m}-x_{i+m}=b_i,$	$4x_i-x_{i-1}-x_{i+1}-x_{i-m}-x_{i+m}=b_i,$
lyche-numerical-linear-algebra-241 EQ0669 (10.9)	241	■ 1.000	$\boldsymbol{T}_1:=\operatorname{tridiag}(a,\,d,\,a)\,.$	$\boldsymbol{T}_1:=\operatorname{tridiag}(a,d,a).$	$\boldsymbol{T}_1:=\operatorname{tridiag}(a,d,a).$
lyche-numerical-linear-algebra-242 EQ0672	242	■ 1.000	$\boldsymbol{C}:=\left[\begin{array}{cccc} \boldsymbol{b}_{1,1} & \boldsymbol{b}_{1,2} & \cdots & \boldsymbol{b}_{1,s} \\ \boldsymbol{b}_{2,1} & \boldsymbol{b}_{2,2} & \cdots & \boldsymbol{b}_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{b}_{r,1} & \boldsymbol{b}_{r,2} & \cdots & \boldsymbol{b}_{r,s} \end{array}\right].$	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A}b_{1,1} & \boldsymbol{A}b_{1,2} & \cdots & \boldsymbol{A}b_{1,s} \\ \boldsymbol{A}b_{2,1} & \boldsymbol{A}b_{2,2} & \cdots & \boldsymbol{A}b_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}b_{r,1} & \boldsymbol{A}b_{r,2} & \cdots & \boldsymbol{A}b_{r,s} \end{array}\right].$	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A}b_{1,1} & \boldsymbol{A}b_{1,2} & \cdots & \boldsymbol{A}b_{1,s} \\ \boldsymbol{A}b_{2,1} & \boldsymbol{A}b_{2,2} & \cdots & \boldsymbol{A}b_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}b_{r,1} & \boldsymbol{A}b_{r,2} & \cdots & \boldsymbol{A}b_{r,s} \end{array}\right].$
lyche-numerical-linear-algebra-242 EQ0673	242	■ 1.000	$\boldsymbol{T}_1=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right],\quad\boldsymbol{I}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\boldsymbol{T}_1=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right],\quad\boldsymbol{I}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\boldsymbol{T}_1=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right],\quad\boldsymbol{I}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$

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lyche-numerical-linear-algebra-E00675 (10.12)	243	■ 1.000	$\boldsymbol{T}_2=\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1, \quad \boldsymbol{T}_1, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 = \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1, \quad \boldsymbol{T}_1, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 = \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1, \quad \boldsymbol{T}_1, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2) \times (m^2)}.$
lyche-numerical-linear-algebra-E00678	243	■ 1.000	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r,1} & \cdots & \boldsymbol{A} b_{r,t} \end{array} \right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t,1} & \cdots & \boldsymbol{C} d_{t,s} \end{array} \right].$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r,1} & \cdots & \boldsymbol{A} b_{r,t} \end{array} \right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t,1} & \cdots & \boldsymbol{C} d_{t,s} \end{array} \right].$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r,1} & \cdots & \boldsymbol{A} b_{r,t} \end{array} \right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t,1} & \cdots & \boldsymbol{C} d_{t,s} \end{array} \right].$
lyche-numerical-linear-algebra-E00681	245	■ 1.000	$\begin{aligned} &\left(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ &\Leftrightarrow (\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T + \boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow (\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T + \boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow (\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T + \boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$
lyche-numerical-linear-algebra-E00683 (10.16)	246	■ 1.000	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$
lyche-numerical-linear-algebra-E00685 (10.18)	246	■ 1.000	$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j, k = 1, \dots, m,$	$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j, k = 1, \dots, m,$	$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j, k = 1, \dots, m,$
lyche-numerical-linear-algebra-E00688 (10.20)	246	■ 1.000	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}$	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}$	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}.$
lyche-numerical-linear-algebra-E00689	246	■ 1.000	$\lambda_{j,k} = 4 - 2 \cos(2jw) - 2 \cos(2kw) = 4 \sin^2(jw) + 4 \sin^2(kw), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 4 - 2 \cos(2jw) - 2 \cos(2kw) = 4 \sin^2(jw) + 4 \sin^2(kw), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 4 - 2 \cos(2jw) - 2 \cos(2kw) = 4 \sin^2(jw) + 4 \sin^2(kw), \quad j, k = 1, \dots, m.$
lyche-numerical-linear-algebra-E00690	246	■ 1.000	$\lambda_{\max} = 8 \cos^2 w, \quad \lambda_{\min} = 8 \sin^2 w.$	$\lambda_{\max} = 8 \cos^2 w, \quad \lambda_{\min} = 8 \sin^2 w.$	$\lambda_{\max} = 8 \cos^2 w, \quad \lambda_{\min} = 8 \sin^2 w.$
lyche-numerical-linear-algebra-E00696	249	■ 1.000	$(\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V}) = h^2 \operatorname{vec}(\boldsymbol{F}), \quad (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U}) = h^2 \operatorname{vec}(\boldsymbol{V}).$	$(\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V}) = h^2 \operatorname{vec}(\boldsymbol{F}), \quad (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U}) = h^2 \operatorname{vec}(\boldsymbol{V}).$	$(\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V}) = h^2 \operatorname{vec}(\boldsymbol{F}), \quad (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U}) = h^2 \operatorname{vec}(\boldsymbol{V}).$
lyche-numerical-linear-algebra-E00697 (10.25)	249	■ 1.000	$\boldsymbol{T}^2 \boldsymbol{U} + 2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T} + \boldsymbol{U} \boldsymbol{T}^2 = h^4 \boldsymbol{F} \quad \text{and} \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b},$	$\boldsymbol{T}^2 \boldsymbol{U} + 2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T} + \boldsymbol{U} \boldsymbol{T}^2 = h^4 \boldsymbol{F} \quad \text{and} \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b},$	$\boldsymbol{T}^2 \boldsymbol{U} + 2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T} + \boldsymbol{U} \boldsymbol{T}^2 = h^4 \boldsymbol{F} \quad \text{and} \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b},$

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lyche-numerical-linear-algebra-252 EQ0700 (11.2)	252	■ 1.000	$\boldsymbol{U}_{\{1\}}=\boldsymbol{D}_{\{1\}},$ $\quad \boldsymbol{U}_{\{k\}}=\boldsymbol{A}_{\{k\}}$ $\boldsymbol{U}_{\{k\}}^{-1}, \quad \boldsymbol{U}_{\{k+1\}}=$ $\boldsymbol{D}_{\{k+1\}}-\boldsymbol{L}_{\{k\}}$ $\boldsymbol{C}_{\{k\}}, \quad k=1,2, \ldots, m-1 .$	$\boldsymbol{U}_1 = \boldsymbol{D}_1, \quad \boldsymbol{L}_k = \boldsymbol{A}_k \boldsymbol{U}_k^{-1}, \quad \boldsymbol{U}_{k+1} = \boldsymbol{D}_{k+1} - \boldsymbol{L}_k \boldsymbol{C}_k, \quad k = 1, 2, \dots, m-1.$	$\boldsymbol{U}_1 = \boldsymbol{D}_1, \quad \boldsymbol{L}_k = \boldsymbol{A}_k \boldsymbol{U}_k^{-1}, \quad \boldsymbol{U}_{k+1} = \boldsymbol{D}_{k+1} - \boldsymbol{L}_k \boldsymbol{C}_k, \quad k = 1, 2, \dots, m-1.$
lyche-numerical-linear-algebra-253 EQ0704 (11.4)	253	■ 1.000	$\boldsymbol{S}:=\left[\boldsymbol{s}_{\{1\}}, \ldots, \right.$ $\left.\boldsymbol{s}_{\{m\}}\right]=\left[\sin \left(j k \pi h\right)\right]_{j, k=1}^m$ $\in \mathbb{R}^{m \times m}, \quad \boldsymbol{D}=\operatorname{diag}\left(\lambda_1, \ldots, \lambda_m\right) .$	$\boldsymbol{S} := [\boldsymbol{s}_1, \dots, \boldsymbol{s}_m] = [\sin(jk\pi h)]_{j,k=1}^m \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_m).$	$\boldsymbol{S} := [\boldsymbol{s}_1, \dots, \boldsymbol{s}_m] = [\sin(jk\pi h)]_{j,k=1}^m \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_m).$
lyche-numerical-linear-algebra-253 EQ0705 (2h)	253	■ 1.000	$\boldsymbol{T} \boldsymbol{S}=\left[\boldsymbol{T} \boldsymbol{s}_{\{1\}}, \ldots, \boldsymbol{T} \boldsymbol{s}_{\{m\}}\right]=$ $\left[\lambda_1 \boldsymbol{s}_{\{1\}}, \ldots, \lambda_m \boldsymbol{s}_{\{m\}}\right]=$ $\boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2=\boldsymbol{S}^T \boldsymbol{S}=\frac{1}{2 h} \boldsymbol{I} .$	$\boldsymbol{T} \boldsymbol{S} = [\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$	$\boldsymbol{T} \boldsymbol{S} = [\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$
lyche-numerical-linear-algebra-253 EQ0708	253	■ 1.000	$x_{\{j, k\}}=4 h^4 g_{\{j, k\}} / \left(\lambda_{\{j\}}+\lambda_{\{k\}}\right)=$ $h^4 g_{\{j, k\}} / \left(\sigma_{\{j\}}+\sigma_{\{k\}}\right), \quad \sigma_{\{j\}}:=$ $\lambda_{\{j\}} / 4=\sin ^2\left(j \pi h / 2\right) .$	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$
lyche-numerical-linear-algebra-255 EQ0710	255	■ 1.000	$w_{\{j\}}=\sum_{k=1}^m \sin \left(\frac{j k \pi}{m+1}\right) v_{\{k\}},$ $\quad j=1, \ldots, m$	$w_j = \sum_{k=1}^m \sin \left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$	$w_j = \sum_{k=1}^m \sin \left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$
lyche-numerical-linear-algebra-255 EQ0711 (11.5)	255	■ 1.000	$\omega_{\{N\}}=\exp ^{-2 \pi i / N}=\cos (2 \pi / N)-i \sin$ $(2 \pi / N),$	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$
lyche-numerical-linear-algebra-256 EQ0712	256	■ 1.000	$z_{\{j+1\}}=\sum_{k=0}^{N-1} \omega_{\{N\}}^{j k} y_{\{k+1\}}, \quad$ $j=0, \ldots, N-1,$	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$
lyche-numerical-linear-algebra-256 EQ0713	256	■ 1.000	$\omega_{\{4\}}=\exp ^{-2 \pi i / 4}=\cos (\pi / 2)-i \sin (\pi / 2)=-i$	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$
lyche-numerical-linear-algebra-256 EQ0715	256	■ 1.000	$\sin w=\frac{e^{i w}-e^{-i w}}{2 i} .$	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$
lyche-numerical-linear-algebra-256 EQ0716	256	■ 1.000	$\boldsymbol{z}^T=\left[0, \boldsymbol{x}^T, 0,-\boldsymbol{x}_B^T\right] \in \mathbb{R}^{2 m+2}, \quad$ $\boldsymbol{x}_B^T:=\left[x_m, \ldots, x_2, x_1\right] .$	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$
lyche-numerical-linear-algebra-256 EQ0718	256	■ 1.000	$\omega^{\{j k\}}=e^{-\pi i j k /(m+1)}, \quad \omega^{\{(2 m+2-j) k\}}=$ $e^{-2 \pi i j} e^{\pi i j k /(m+1)}=e^{\pi i j k /(m+1)} .$	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$
lyche-numerical-linear-algebra-257 EQ0720	257	■ 1.000	$\boldsymbol{P}_{\{N\}}=\left[\boldsymbol{e}_{\{1\}}, \boldsymbol{e}_{\{3\}}, \ldots, \boldsymbol{e}_{\{N-1\}}, \boldsymbol{e}_{\{2\}},\right.$ $\left.\boldsymbol{e}_{\{4\}}, \ldots, \boldsymbol{e}_{\{N\}}\right],$	$\boldsymbol{P}_N = [\boldsymbol{e}_1, \boldsymbol{e}_3, \dots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_2, \boldsymbol{e}_4, \dots, \boldsymbol{e}_N],$	$\boldsymbol{P}_N = [\boldsymbol{e}_1, \boldsymbol{e}_3, \dots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_2, \boldsymbol{e}_4, \dots, \boldsymbol{e}_N],$
lyche-numerical-linear-algebra-258 EQ0724	258	■ 1.000	$\boldsymbol{F}_{\{2\}}=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right], \quad \boldsymbol{D}_{\{2\}}$ $\boldsymbol{F}_{\{2\}}=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],$	$\boldsymbol{F}_2 = \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right], \quad \boldsymbol{D}_2 \boldsymbol{F}_2 = \left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right],$	$\boldsymbol{F}_2 = \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right], \quad \boldsymbol{D}_2 \boldsymbol{F}_2 = \left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right],$

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lyche-numerical-linear-algebra-EQ0727 (11.8)	258	■ 1.000	$\boldsymbol{D}_{\mathfrak{m}}=\operatorname{diag}\!\left(1,\right. \\ \left.\omega_{\mathfrak{N}},\,\omega_{\mathfrak{N}}^{\mathfrak{2}},\,\ldots,\,\omega_{\mathfrak{N}}^{\mathfrak{m}-1}\right)\,.$	$\boldsymbol{D}_m = \operatorname{diag}\left(1, \omega_N, \omega_N^2, \ldots, \omega_N^{m-1}\right).$	$\boldsymbol{D}_m = \operatorname{diag}(1, \omega_N, \omega_N^2, \ldots, \omega_N^{m-1}).$
lyche-numerical-linear-algebra-EQ0728	258	■ 1.000	$\omega_{\mathfrak{m}}^{\mathfrak{m}}=1,\,\omega_{\mathfrak{2}\,\mathfrak{m}}^{\mathfrak{2}\,\mathfrak{k}}=\omega_{\mathfrak{m}}^{\mathfrak{2}\,\mathfrak{k}}=\omega_{\mathfrak{m}}^{\mathfrak{k}}, \\ \omega_{\mathfrak{2}\,\mathfrak{m}}^{\mathfrak{m}}=-1,\!\!\left(\boldsymbol{F}_{\mathfrak{m}}\right)_{\mathfrak{p},\mathfrak{q}}=\omega_{\mathfrak{m}}^{\mathfrak{j}\mathfrak{k}},\!\!\left(\boldsymbol{D}_{\mathfrak{m}}\boldsymbol{F}_{\mathfrak{m}}\right)_{\mathfrak{p},\mathfrak{q}}=\omega_{\mathfrak{2}\mathfrak{m}}^{\mathfrak{j}}\omega_{\mathfrak{m}}^{\mathfrak{j}\mathfrak{k}}, \\ \omega_{\mathfrak{m}}^{\mathfrak{m}}=\omega_{\mathfrak{2}\,\mathfrak{m}}^{\mathfrak{2}\,\mathfrak{k}},\!\!\left(\boldsymbol{F}_{\mathfrak{m}}\right)_{\mathfrak{p},\mathfrak{q}}=\omega_{\mathfrak{m}}^{\mathfrak{j}\mathfrak{k}},\!\!\left(\boldsymbol{D}_{\mathfrak{m}}\boldsymbol{F}_{\mathfrak{m}}\right)_{\mathfrak{p},\mathfrak{q}}=\omega_{\mathfrak{2}\mathfrak{m}}^{\mathfrak{j}}\omega_{\mathfrak{m}}^{\mathfrak{j}\mathfrak{k}},$	$\omega_m^m = 1, \omega_{2m}^{2k} = \omega_m^k, \omega_{2m}^m = -1, (\boldsymbol{F}_m)_{p,q} = \omega_m^{jk}, (\boldsymbol{D}_m \boldsymbol{F}_m)_{p,q} = \omega_{2m}^j \omega_m^{jk},$	$\omega_m^m = 1, \omega_{2m}^{2k} = \omega_m^k, \omega_{2m}^m = -1, (\boldsymbol{F}_m)_{p,q} = \omega_m^{jk}, (\boldsymbol{D}_m \boldsymbol{F}_m)_{p,q} = \omega_{2m}^j \omega_m^{jk},$
lyche-numerical-linear-algebra-EQ0730	258	■ 1.000	$\boldsymbol{w}_{\mathfrak{1}}^{\mathfrak{T}}=\left[y_{\mathfrak{1}},\,y_{\mathfrak{3}},\,\ldots,\,y_{\mathfrak{2}\,\mathfrak{m}-1}\right],\quad \boldsymbol{w}_{\mathfrak{2}}^{\mathfrak{T}}=\left[y_{\mathfrak{2}},\,y_{\mathfrak{4}},\,\ldots,\,y_{\mathfrak{2}\,\mathfrak{m}}\right]\,.$	$\boldsymbol{w}_1^T = [y_1, y_3, \ldots, y_{2m-1}], \quad \boldsymbol{w}_2^T = [y_2, y_4, \ldots, y_{2m}].$	$\boldsymbol{w}_1^T = [y_1, y_3, \ldots, y_{2m-1}], \quad \boldsymbol{w}_2^T = [y_2, y_4, \ldots, y_{2m}].$
lyche-numerical-linear-algebra-EQ0732	259	■ 1.000	$\boldsymbol{q}_{\mathfrak{1}}=\boldsymbol{F}_{\mathfrak{m}}\boldsymbol{w}_{\mathfrak{1}},\quad \text{and}\quad \boldsymbol{q}_{\mathfrak{2}}=\boldsymbol{D}_{\mathfrak{m}}\left(\boldsymbol{F}_{\mathfrak{m}}\boldsymbol{w}_{\mathfrak{2}}\right).$	$\boldsymbol{q}_1 = \boldsymbol{F}_m \boldsymbol{w}_1, \quad \text{and} \quad \boldsymbol{q}_2 = \boldsymbol{D}_m (\boldsymbol{F}_m \boldsymbol{w}_2).$	$\boldsymbol{q}_1 = \boldsymbol{F}_m \boldsymbol{w}_1, \quad \text{and} \quad \boldsymbol{q}_2 = \boldsymbol{D}_m (\boldsymbol{F}_m \boldsymbol{w}_2).$
lyche-numerical-linear-algebra-EQ0733	260	■ 1.000	$\frac{8\,N^{\mathfrak{2}}}{5\,N\log_2N}\approx 84000\,\,.$	$\frac{8N^2}{5N\log_2N} \approx 84000.$	$\frac{8N^2}{5N\log_2N} \approx 84000.$
lyche-numerical-linear-algebra-EQ0734	260	■ 1.000	$8\,\gamma_{\mathfrak{m}}\mathfrak{m}(\mathfrak{m}+1)\,\log_{\mathfrak{2}}(\mathfrak{2}\,\mathfrak{m}+2)\,\approx 8\,\gamma_{\mathfrak{m}}\mathfrak{m}^{\mathfrak{2}} \\ \log_{\mathfrak{2}}\mathfrak{m}=4\,\gamma_{\mathfrak{n}}\mathfrak{n}\,\log_{\mathfrak{2}}\mathfrak{n},$	$8\gamma m(m+1)\log_2(2m+2) \approx 8\gamma m^2\log_2m = 4\gamma n\log_2n,$	$8\gamma m(m+1)\log_2(2m+2) \approx 8\gamma m^2\log_2m = 4\gamma n\log_2n,$
lyche-numerical-linear-algebra-EQ0735	261	■ 1.000	$\mathfrak{v}_{\mathfrak{i}\,\mathfrak{j}}=\frac{1}{(\mathfrak{m}+1)^{\mathfrak{4}}}\sum_{\mathfrak{p}=1}^{\mathfrak{m}}\sum_{\mathfrak{r}=1}^{\mathfrak{m}}\frac{\sin\left(\frac{\mathfrak{i}\,\mathfrak{p}\,\pi}{\mathfrak{m}+1}\right)\sin\left(\frac{\mathfrak{j}\,\mathfrak{r}\,\pi}{\mathfrak{m}+1}\right)\sin\left(\frac{\mathfrak{k}\,\mathfrak{p}\,\pi}{\mathfrak{m}+1}\right)\sin\left(\frac{\mathfrak{l}\,\mathfrak{r}\,\pi}{\mathfrak{m}+1}\right)}{\left[\sin\left(\frac{\mathfrak{p}\,\pi}{2(\mathfrak{m}+1)}\right)\right]^2+\left[\sin\left(\frac{\mathfrak{r}\,\pi}{2(\mathfrak{m}+1)}\right)\right]^2}f_{\mathfrak{k},\mathfrak{l}}.$	$v_{ij} = \frac{1}{(m+1)^4} \sum_{p=1}^m \sum_{r=1}^m \sum_{k=1}^m \sum_{l=1}^m \frac{\sin\left(\frac{ip\pi}{m+1}\right) \sin\left(\frac{jr\pi}{m+1}\right) \sin\left(\frac{kp\pi}{m+1}\right) \sin\left(\frac{lr\pi}{m+1}\right)}{\left[\sin\left(\frac{p\pi}{2(m+1)}\right)\right]^2 + \left[\sin\left(\frac{r\pi}{2(m+1)}\right)\right]^2} f_{k,l}.$	$v_{ij} = \frac{1}{(m+1)^4} \sum_{p=1}^m \sum_{r=1}^m \sum_{k=1}^m \sum_{l=1}^m \frac{\sin\left(\frac{ip\pi}{m+1}\right) \sin\left(\frac{jr\pi}{m+1}\right) \sin\left(\frac{kp\pi}{m+1}\right) \sin\left(\frac{lr\pi}{m+1}\right)}{\left[\sin\left(\frac{p\pi}{2(m+1)}\right)\right]^2 + \left[\sin\left(\frac{r\pi}{2(m+1)}\right)\right]^2} f_{k,l}.$
lyche-numerical-linear-algebra-EQ0737 (11.9)	261	■ 1.000	$\left(\boldsymbol{T}+\lambda_{\mathfrak{j}}\boldsymbol{I}\right)\boldsymbol{x}_{\mathfrak{j}}=\boldsymbol{c}_{\mathfrak{j}}\quad \mathfrak{j}=1,\ldots,\mathfrak{m},$	$(\boldsymbol{T} + \lambda_j \boldsymbol{I}) \boldsymbol{x}_j = \boldsymbol{c}_j \quad j = 1, \ldots, m,$	$(\boldsymbol{T} + \lambda_j \boldsymbol{I}) \boldsymbol{x}_j = \boldsymbol{c}_j \quad j = 1, \ldots, m,$
lyche-numerical-linear-algebra-EQ0738	261	■ 1.000	$\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}=\frac{1}{6}\boldsymbol{T}\boldsymbol{V}\boldsymbol{T}=h^2\mu\boldsymbol{F},$	$\boldsymbol{TV} + \boldsymbol{VT} - \frac{1}{6} \boldsymbol{TVT} = h^2 \mu \boldsymbol{F},$	$\boldsymbol{TV} + \boldsymbol{VT} - \frac{1}{6} \boldsymbol{TVT} = h^2 \mu \boldsymbol{F},$
lyche-numerical-linear-algebra-EQ0741	262	■ 1.000	$\boldsymbol{T}^{\mathfrak{2}}\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^{\mathfrak{2}}=h^{\mathfrak{4}}\boldsymbol{F}.$	$\boldsymbol{T}^2 \boldsymbol{U} + 2 \boldsymbol{TUT} + \boldsymbol{UT}^2 = h^4 \boldsymbol{F}.$	$\boldsymbol{T}^2 \boldsymbol{U} + 2 \boldsymbol{TUT} + \boldsymbol{UT}^2 = h^4 \boldsymbol{F}.$
lyche-numerical-linear-algebra-EQ0742	262	■ 1.000	$\boldsymbol{D}^{\mathfrak{2}}\boldsymbol{X}+2\boldsymbol{D}\boldsymbol{X}\boldsymbol{D}+\boldsymbol{X}\boldsymbol{D}^{\mathfrak{2}}=4h^{\mathfrak{6}}\boldsymbol{G},\,\text{where}\,\boldsymbol{G}=\boldsymbol{S}\boldsymbol{F}\boldsymbol{S},$	$\boldsymbol{D}^2 \boldsymbol{X} + 2 \boldsymbol{DXD} + \boldsymbol{XD}^2 = 4 h^6 \boldsymbol{G}, \text{ where } \boldsymbol{G} = \boldsymbol{SFS},$	$\boldsymbol{D}^2 \boldsymbol{X} + 2 \boldsymbol{DXD} + \boldsymbol{XD}^2 = 4 h^6 \boldsymbol{G}, \text{ where } \boldsymbol{G} = \boldsymbol{SFS},$
lyche-numerical-linear-algebra-EQ0743	262	■ 1.000	$\mathfrak{x}_{\mathfrak{j},\mathfrak{k}}=\frac{h^{\mathfrak{6}}\mathfrak{g}_{\mathfrak{j},\mathfrak{k}}}{4(\sigma_{\mathfrak{j}}+\sigma_{\mathfrak{k}})^{\mathfrak{2}}},\,\text{where}\,\sigma_{\mathfrak{j}}=\sin^2((\mathfrak{j}\pi\mathfrak{h})/2)\,\text{for}\,\mathfrak{j},\mathfrak{k}=1,2,\ldots,\mathfrak{m}.$	$x_{j,k} = \frac{h^6 g_{j,k}}{4(\sigma_j + \sigma_k)^2}, \text{ where } \sigma_j = \sin^2((j\pi h)/2) \text{ for } j,k = 1, 2, \ldots, m.$	$x_{j,k} = \frac{h^6 g_{j,k}}{4(\sigma_j + \sigma_k)^2}, \text{ where } \sigma_j = \sin^2((j\pi h)/2) \text{ for } j,k = 1, 2, \ldots, m.$
lyche-numerical-linear-algebra-EQ0744 (12.1)	266	■ 1.000	$\boldsymbol{x}(\mathfrak{i})=\left(-\sum_{\mathfrak{j}=1}^{\mathfrak{i}-1}\boldsymbol{a}_{\mathfrak{i}\,\mathfrak{j}}\boldsymbol{x}(\mathfrak{j})-\sum_{\mathfrak{j}=\mathfrak{i}+1}^{\mathfrak{n}}\boldsymbol{a}_{\mathfrak{i}\,\mathfrak{j}}\boldsymbol{x}(\mathfrak{j})+\boldsymbol{b}_{\mathfrak{i}}\right)/\boldsymbol{a}_{\mathfrak{i}\,\mathfrak{i}},\quad \mathfrak{i}=1,2,\ldots,\mathfrak{n}\,.$	$\boldsymbol{x}(i) = \left(-\sum_{j=1}^{i-1} a_{ij} \boldsymbol{x}(j) - \sum_{j=i+1}^n a_{ij} \boldsymbol{x}(j) + b_i\right) / a_{ii}, \quad i = 1, 2, \ldots, n.$	$\boldsymbol{x}(i) = \left(-\sum_{j=1}^{i-1} a_{ij} \boldsymbol{x}(j) - \sum_{j=i+1}^n a_{ij} \boldsymbol{x}(j) + b_i\right) / a_{ii}, \quad i = 1, 2, \ldots, n.$

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lyche-numerical-linear-algebra-EQ0745 (12.2)	266	1.000	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$
lyche-numerical-linear-algebra-EQ0746 (12.3)	266	1.000	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$
lyche-numerical-linear-algebra-EQ0747 (12.4)	266	1.000	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$
lyche-numerical-linear-algebra-EQ0748 (12.5)	267	1.000	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$
lyche-numerical-linear-algebra-EQ0749	267	1.000	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\ldots,m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\ldots,m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\ldots,m,$
lyche-numerical-linear-algebra-EQ0750 (12.6)	267	1.000	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$
lyche-numerical-linear-algebra-EQ0755 (12.10)	270	1.000	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$
lyche-numerical-linear-algebra-EQ0756	270	1.000	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_k=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_{k+1}=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_{k+1}=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$
lyche-numerical-linear-algebra-EQ0758 (12.12)	271	1.000	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$
lyche-numerical-linear-algebra-EQ0759 (12.13)	271	1.000	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$
lyche-numerical-linear-algebra-EQ0760	271	1.000	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

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lyche-numerical-linear-algebra-EQ0761	271	■ 1.000	$\boldsymbol{A}_{\boldsymbol{L}}=\left[\begin{array}{ll} 0 & 0 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{\boldsymbol{R}}=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right],$	$\boldsymbol{A}_L = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{A}_R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$	$\boldsymbol{A}_L = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{A}_R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0762	271	■ 1.000	$\boldsymbol{M}_{\boldsymbol{J}}=\boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{M}_{\omega}=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}=\left[\begin{array}{cc} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{array}\right].$	$\boldsymbol{M}_J = \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{M}_{\omega} = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L = \begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$	$\boldsymbol{M}_J = \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{M}_{\omega} = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L = \begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0764 (12.15)	271	■ 1.000	$\boldsymbol{G}_{\boldsymbol{J}}=\boldsymbol{I}-\boldsymbol{D}^{-1}\boldsymbol{A}=\left[\begin{array}{cc} 0 & 1/2 \\ 1/2 & 0 \end{array}\right], \quad \boldsymbol{G}_1=\left[\begin{array}{cc} 0 & 1/2 \\ 0 & 1/4 \end{array}\right].$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A} = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_1 = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A} = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_1 = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0771 (12.18)	273	■ 1.000	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\alpha(\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k).$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha(\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k).$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha(\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k).$
lyche-numerical-linear-algebra-EQ0773	274	■ 1.000	$\rho_{\alpha}:=\rho(\boldsymbol{I}-\alpha\boldsymbol{A}):=\max_j 1-\alpha\lambda_j =\begin{cases} 1-\alpha\lambda_1, & \text{if } \alpha \leq 0 \\ 1-\alpha\lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha\lambda_1-1, & \text{if } \alpha > \alpha_o, \end{cases}$	$\rho_{\alpha} := \rho(\boldsymbol{I} - \alpha\boldsymbol{A}) := \max_j 1 - \alpha\lambda_j = \begin{cases} 1 - \alpha\lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha\lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha\lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$	$\rho_{\alpha} := \rho(\boldsymbol{I} - \alpha\boldsymbol{A}) := \max_j 1 - \alpha\lambda_j = \begin{cases} 1 - \alpha\lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha\lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha\lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$
lyche-numerical-linear-algebra-EQ0774	274	■ 1.000	$\rho_{\alpha_o}=\alpha_o\lambda_1-1=\frac{\lambda_1-\lambda_n}{\lambda_1+\lambda_n}=\frac{\kappa-1}{\kappa+1}<1.$	$\rho_{\alpha_o} = \alpha_o\lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$	$\rho_{\alpha_o} = \alpha_o\lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$
lyche-numerical-linear-algebra-EQ0776 (12.21)	275	■ 1.000	$\boldsymbol{G}_{\omega}=(\boldsymbol{I}-\omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R}+(1-\omega)\boldsymbol{I}).$	$\boldsymbol{G}_{\omega} = (\boldsymbol{I} - \omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R} + (1 - \omega)\boldsymbol{I}).$	$\boldsymbol{G}_{\omega} = (\boldsymbol{I} - \omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R} + (1 - \omega)\boldsymbol{I}).$
lyche-numerical-linear-algebra-EQ0777 (12.22)	275	■ 1.000	$\omega^{-1}(2-\omega) 1-\lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x}=(1- \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$	$\omega^{-1}(2 - \omega) 1 - \lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x} = (1 - \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$	$\omega^{-1}(2 - \omega) 1 - \lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x} = (1 - \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$
lyche-numerical-linear-algebra-EQ0778	275	■ 1.000	$\boldsymbol{G}_{\omega}\boldsymbol{x}=(\boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1}\boldsymbol{A})\boldsymbol{x}=\boldsymbol{x}-(\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}})^{-1}\boldsymbol{A}\boldsymbol{x}$	$\boldsymbol{G}_{\omega}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{M}_{\omega}^{-1}\boldsymbol{A})\boldsymbol{x} = \boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)^{-1}\boldsymbol{A}\boldsymbol{x}$	$\boldsymbol{G}_{\omega}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{M}_{\omega}^{-1}\boldsymbol{A})\boldsymbol{x} = \boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)^{-1}\boldsymbol{A}\boldsymbol{x}$
lyche-numerical-linear-algebra-EQ0779 (12.23)	276	■ 1.000	$\boldsymbol{A}\boldsymbol{x}=(\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}})\boldsymbol{y}, \quad \boldsymbol{y}:=(1-\lambda)\boldsymbol{x}.$	$\boldsymbol{A}\boldsymbol{x} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{y}, \quad \boldsymbol{y} := (1 - \lambda)\boldsymbol{x}.$	$\boldsymbol{A}\boldsymbol{x} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{y}, \quad \boldsymbol{y} := (1 - \lambda)\boldsymbol{x}.$
lyche-numerical-linear-algebra-EQ0780 (12.24)	276	■ 1.000	$\boldsymbol{E}\boldsymbol{y}=\lambda\boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{E}:=\omega^{-1}\boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}.$	$\boldsymbol{E}\boldsymbol{y} = \lambda\boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{E} := \omega^{-1}\boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D} = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}.$	$\boldsymbol{E}\boldsymbol{y} = \lambda\boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{E} := \omega^{-1}\boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D} = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}.$

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lyche-numerical-linear-algebra- EQ0781	276	■ 1.000	$\boldsymbol{E} \boldsymbol{y} = \left(\boldsymbol{\omega}^{-1} \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A} \right) \boldsymbol{y} = \boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{y} = \boldsymbol{A} \boldsymbol{x} - (1 - \lambda) \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A} \boldsymbol{x}.$		
lyche-numerical-linear-algebra- EQ0784	276	■ 1.000	$\lambda_{j,k} = 4 - 2 \cos(j \pi h) - 2 \cos(k \pi h), \quad j, k = 1, \dots, m, h = 1/(m+1).$		
lyche-numerical-linear-algebra- EQ0785 (12.25)	276	■ 1.000	$\mu_{j,k} = 1 - \frac{1}{4} \lambda_{j,k} = \frac{1}{2} \cos(j \pi h) + \frac{1}{2} \cos(k \pi h), \quad j, k = 1, \dots, m.$		
lyche-numerical-linear-algebra- EQ0786 (12.26)	277	■ 1.000	$\left\ \boldsymbol{x}_k - \boldsymbol{x} \right\ _2 \leq \left\ \boldsymbol{G}_J \right\ _2^k \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ _2 = \cos^k(\pi h) \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ _2, \quad k = 0, 1, 2, \dots$		
lyche-numerical-linear-algebra- EQ0788 (12.28)	277	■ 1.000	$\omega^* := \frac{2}{1 + \sqrt{1 - \beta^2}} > 1.$		
lyche-numerical-linear-algebra- EQ0790 (12.30)	277	■ 1.000	$\omega^* = \frac{2}{1 + \sin(\pi h)}, \quad \rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1 = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)}, \quad h = \frac{1}{m+1}.$		
lyche-numerical-linear-algebra- EQ0792 (12.31)	278	■ 1.000	$\tilde{k} := \frac{s \log(10)}{\eta}$		
lyche-numerical-linear-algebra- EQ0794	279	■ 1.000	$\tilde{k}_n = \frac{2 \log(10)s}{\pi^2} n + O(n^{-1}) = O(n).$		
lyche-numerical-linear-algebra- EQ0795	279	■ 1.000	$\tilde{k}_n = \frac{\log(10)s}{\pi^2} n + O(n^{-1}) = O(n).$		
lyche-numerical-linear-algebra- EQ0796	279	■ 1.000	$\tilde{k}_n = \frac{\log(10)s}{2\pi} \sqrt{n} + O(n^{-1/2}) = O(\sqrt{n}).$		
lyche-numerical-linear-algebra- EQ0797 (12.32)	280	■ 1.000	$\left\ \boldsymbol{x}_k - \boldsymbol{x}_{k-1} \right\ \geq \frac{1 - \ \boldsymbol{G}\ }{\ \boldsymbol{G}\ } \left\ \boldsymbol{x}_k - \boldsymbol{x} \right\ , \quad k \geq 1.$		
lyche-numerical-linear-algebra- EQ0799 (12.33)	280	■ 1.000	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{M}^{-1} \boldsymbol{r}_k, \quad \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k.$		
lyche-numerical-linear-algebra- EQ0800	281	■ 1.000	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0} \iff \rho(\boldsymbol{A}) < 1,$		
lyche-numerical-linear-algebra- EQ0801	281	■ 1.000	$\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} = \begin{bmatrix} \lambda_1 & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_2 & t^{-1} r_{23} \\ 0 & 0 & \lambda_3 \end{bmatrix}.$		

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lyche-numerical-linear-algebra-282- EQ8883 (12.34)	282	■ 1.000	$\lim_{k \rightarrow \infty} \left\ \boldsymbol{A}^k \right\ ^{1/k} = \rho(\boldsymbol{A})$.	$\lim_{k \rightarrow \infty} \left\ \boldsymbol{A}^k \right\ ^{1/k} = \rho(\boldsymbol{A})$.	$\lim_{k \rightarrow \infty} \left\ \boldsymbol{A}^k \right\ ^{1/k} = \rho(\boldsymbol{A})$.
lyche-numerical-linear-algebra-282- EQ8884	282	■ 1.000	$\left\ \boldsymbol{A}^k \right\ = \left\ (\rho(\boldsymbol{A}) + \epsilon)^k \boldsymbol{B}^k \right\ = (\rho(\boldsymbol{A}) + \epsilon)^k \left\ \boldsymbol{B}^k \right\ < (\rho(\boldsymbol{A}) + \epsilon)^k$.	$\left\ \boldsymbol{A}^k \right\ = \left\ (\rho(\boldsymbol{A}) + \epsilon)^k \boldsymbol{B}^k \right\ = (\rho(\boldsymbol{A}) + \epsilon)^k \left\ \boldsymbol{B}^k \right\ < (\rho(\boldsymbol{A}) + \epsilon)^k$.	$\left\ \boldsymbol{A}^k \right\ = \left\ (\rho(\boldsymbol{A}) + \epsilon)^k \boldsymbol{B}^k \right\ = (\rho(\boldsymbol{A}) + \epsilon)^k \left\ \boldsymbol{B}^k \right\ < (\rho(\boldsymbol{A}) + \epsilon)^k$.
lyche-numerical-linear-algebra-282- EQ8885	282	■ 1.000	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$
lyche-numerical-linear-algebra-283- EQ8887	283	■ 1.000	$\left\ \sum_{k=m+1}^l \boldsymbol{B}^k \right\ \leq \sum_{k=m+1}^l \left\ \boldsymbol{B}^k \right\ \leq \left\ \boldsymbol{B} \right\ ^{m+1} \sum_{k=0}^{\infty} \left\ \boldsymbol{B} \right\ ^k = \frac{\left\ \boldsymbol{B} \right\ ^{m+1}}{1 - \left\ \boldsymbol{B} \right\ }$.	$\left\ \boldsymbol{S}_l - \boldsymbol{S}_m \right\ = \left\ \sum_{k=m+1}^l \boldsymbol{B}^k \right\ \leq \sum_{k=m+1}^l \left\ \boldsymbol{B} \right\ ^k \leq \left\ \boldsymbol{B} \right\ ^{m+1} \sum_{k=0}^{\infty} \left\ \boldsymbol{B} \right\ ^k = \frac{\left\ \boldsymbol{B} \right\ ^{m+1}}{1 - \left\ \boldsymbol{B} \right\ }$.	$\left\ \boldsymbol{S}_l - \boldsymbol{S}_m \right\ = \left\ \sum_{k=m+1}^l \boldsymbol{B}^k \right\ \leq \sum_{k=m+1}^l \left\ \boldsymbol{B} \right\ ^k \leq \left\ \boldsymbol{B} \right\ ^{m+1} \sum_{k=0}^{\infty} \left\ \boldsymbol{B} \right\ ^k = \frac{\left\ \boldsymbol{B} \right\ ^{m+1}}{1 - \left\ \boldsymbol{B} \right\ }$.
lyche-numerical-linear-algebra-283- EQ8888 (12.36)	283	■ 1.000	$\left(\sum_{k=0}^m \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I} + \boldsymbol{B} + \cdots + \boldsymbol{B}^m - (\boldsymbol{B} + \cdots + \boldsymbol{B}^{m+1}) = \boldsymbol{I} - \boldsymbol{B}^{m+1}$.	$\left(\sum_{k=0}^m \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I} + \boldsymbol{B} + \cdots + \boldsymbol{B}^m - (\boldsymbol{B} + \cdots + \boldsymbol{B}^{m+1}) = \boldsymbol{I} - \boldsymbol{B}^{m+1}$.	$\left(\sum_{k=0}^m \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I} + \boldsymbol{B} + \cdots + \boldsymbol{B}^m - (\boldsymbol{B} + \cdots + \boldsymbol{B}^{m+1}) = \boldsymbol{I} - \boldsymbol{B}^{m+1}$.
lyche-numerical-linear-algebra-283- EQ8889 (12.37)	283	■ 1.000	$\frac{1}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \dots, m,$	$\frac{1}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \dots, m,$	$\frac{1}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \dots, m,$
lyche-numerical-linear-algebra-283- EQ8810 (12.38)	283	■ 1.000	$(\omega \boldsymbol{R} + \lambda \omega \boldsymbol{L}) \boldsymbol{w} = (\lambda + \omega - 1) \boldsymbol{w}$.	$(\omega \boldsymbol{R} + \lambda \omega \boldsymbol{L}) \boldsymbol{w} = (\lambda + \omega - 1) \boldsymbol{w}$.	$(\omega \boldsymbol{R} + \lambda \omega \boldsymbol{L}) \boldsymbol{w} = (\lambda + \omega - 1) \boldsymbol{w}$.
lyche-numerical-linear-algebra-284- EQ8811 (12.39)	284	■ 1.000	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = (\lambda + \omega - 1) w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = (\lambda + \omega - 1) w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = (\lambda + \omega - 1) w_{i,j}$
lyche-numerical-linear-algebra-284- EQ8812 (12.40)	284	■ 1.000	$\mu := \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}}$	$\mu := \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}}$	$\mu := \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}}$
lyche-numerical-linear-algebra-284- EQ8813 (12.41)	284	■ 1.000	$\mu \omega \lambda^{1/2} = \lambda + \omega - 1$	$\mu \omega \lambda^{1/2} = \lambda + \omega - 1$	$\mu \omega \lambda^{1/2} = \lambda + \omega - 1$
lyche-numerical-linear-algebra-284- EQ8814	284	■ 1.000	$\frac{\omega}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \lambda^{-1/2} (\lambda + \omega - 1) v_{i,j}$.	$\frac{\omega}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \lambda^{-1/2} (\lambda + \omega - 1) v_{i,j}$.	$\frac{\omega}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \lambda^{-1/2} (\lambda + \omega - 1) v_{i,j}$.
lyche-numerical-linear-algebra-284- EQ8817	284	■ 1.000	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = \omega \mu \lambda^{1/2} w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = \omega \mu \lambda^{1/2} w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = \omega \mu \lambda^{1/2} w_{i,j}$
lyche-numerical-linear-algebra-285- EQ8818 (12.42)	285	■ 1.000	$\lambda(\mu) := \frac{1}{4} \left(\omega \mu \pm \sqrt{(\omega \mu)^2 - 4(\omega - 1)} \right)^2$.	$\lambda(\mu) := \frac{1}{4} \left(\omega \mu \pm \sqrt{(\omega \mu)^2 - 4(\omega - 1)} \right)^2$.	$\lambda(\mu) := \frac{1}{4} \left(\omega \mu \pm \sqrt{(\omega \mu)^2 - 4(\omega - 1)} \right)^2$.
lyche-numerical-linear-algebra-285- EQ8821 (12.43)	285	■ 1.000	$\omega = \tilde{\omega}(\mu) := 2 \frac{1 - \sqrt{1 - \mu^2}}{\mu^2} = \frac{2}{1 + \sqrt{1 - \mu^2}}$.	$\omega = \tilde{\omega}(\mu) := 2 \frac{1 - \sqrt{1 - \mu^2}}{\mu^2} = \frac{2}{1 + \sqrt{1 - \mu^2}}$.	$\omega = \tilde{\omega}(\mu) := 2 \frac{1 - \sqrt{1 - \mu^2}}{\mu^2} = \frac{2}{1 + \sqrt{1 - \mu^2}}$.

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lyche-numerical-linear-algebra-293 EQ0844 (13.9)	293	■ 1.000	$\begin{aligned} \boldsymbol{p}_k &= \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= (\boldsymbol{p}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k. \end{aligned}$	$\boldsymbol{p}_k = \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k,$ $\alpha_k = (\boldsymbol{p}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k),$ $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k,$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k.$	
lyche-numerical-linear-algebra-293 EQ0845 (13.10)	293	■ 1.000	$\begin{aligned} \boldsymbol{r}_{k+1} &= \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{A} (\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k. \end{aligned}$	$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{A} (\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k.$	$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{A} (\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k.$
lyche-numerical-linear-algebra-296 EQ0853 (13.17)	296	■ 1.000	$\begin{aligned} \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}. \end{aligned}$	$\boldsymbol{p}_{k+1} := \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$	$\boldsymbol{p}_{k+1} := \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$
lyche-numerical-linear-algebra-298 EQ0856	298	■ 1.000	$\boldsymbol{A} = \text{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}$	$\boldsymbol{A} = \text{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}$	$\boldsymbol{A} = \text{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}.$
lyche-numerical-linear-algebra-300 EQ0859	300	■ 1.000	$\boldsymbol{T}_2 := \text{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 := \text{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 := \text{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}.$
lyche-numerical-linear-algebra-300 EQ0860 (13.21)	300	■ 1.000	$\lambda_{j,k} = 2d + 2a \cos(j\pi h) + 2a \cos(k\pi h), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 2d + 2a \cos(j\pi h) + 2a \cos(k\pi h), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 2d + 2a \cos(j\pi h) + 2a \cos(k\pi h), \quad j, k = 1, \dots, m.$
lyche-numerical-linear-algebra-301 EQ0864	301	■ 1.000	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$
lyche-numerical-linear-algebra-301 EQ0865 (13.22)	301	■ 1.000	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$
lyche-numerical-linear-algebra-301 EQ0867 (13.24)	301	■ 1.000	$\left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _{\boldsymbol{A}} = \min_{\boldsymbol{w} \in \mathbb{W}_k} \left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}}.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _{\boldsymbol{A}} = \min_{\boldsymbol{w} \in \mathbb{W}_k} \left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}}.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _{\boldsymbol{A}} = \min_{\boldsymbol{w} \in \mathbb{W}_k} \left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}}.$
lyche-numerical-linear-algebra-302 EQ0869 (13.25)	302	■ 1.000	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q(\lambda_j)^2, \quad Q(t) := 1 - tP(t).$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q(\lambda_j)^2, \quad Q(t) := 1 - tP(t).$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q(\lambda_j)^2, \quad Q(t) := 1 - tP(t).$
lyche-numerical-linear-algebra-302 EQ0870 (13.26)	302	■ 1.000	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 \right\ _{\boldsymbol{A}}^2 = \boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c} \text{ where } \boldsymbol{c} = (\boldsymbol{I} - \boldsymbol{A}P(\boldsymbol{A})) \boldsymbol{r}_0 = Q(\boldsymbol{A}) \boldsymbol{r}_0.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 \right\ _{\boldsymbol{A}}^2 = \boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c} \text{ where } \boldsymbol{c} = (\boldsymbol{I} - \boldsymbol{A}P(\boldsymbol{A})) \boldsymbol{r}_0 = Q(\boldsymbol{A}) \boldsymbol{r}_0.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 \right\ _{\boldsymbol{A}}^2 = \boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c} \text{ where } \boldsymbol{c} = (\boldsymbol{I} - \boldsymbol{A}P(\boldsymbol{A})) \boldsymbol{r}_0 = Q(\boldsymbol{A}) \boldsymbol{r}_0.$

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lyche-numerical-linear-algebra-302 EQ0871 (13.27)	302	■ 1.000	$\boldsymbol{c}=\sum_{j=1}^n\sigma_jQ\left(\lambda_j\right)\boldsymbol{u}_j$, $\boldsymbol{A}^{-1}\boldsymbol{c}=\sum_{i=1}^n\sigma_i\frac{Q\left(\lambda_i\right)}{\lambda_i}\boldsymbol{u}_i$.	$\boldsymbol{c}=\sum_{j=1}^n\sigma_jQ\left(\lambda_j\right)\boldsymbol{u}_j,\quad\boldsymbol{A}^{-1}\boldsymbol{c}=\sum_{i=1}^n\sigma_i\frac{Q\left(\lambda_i\right)}{\lambda_i}\boldsymbol{u}_i.$	$\boldsymbol{c}=\sum_{j=1}^n\sigma_jQ\left(\lambda_j\right)\boldsymbol{u}_j,\quad\boldsymbol{A}^{-1}\boldsymbol{c}=\sum_{i=1}^n\sigma_i\frac{Q\left(\lambda_i\right)}{\lambda_i}\boldsymbol{u}_i.$
lyche-numerical-linear-algebra-303 EQ0876	303	■ 1.000	$e^{2/x}=e^{2y}=\sum_{k=0}^{\infty}\frac{(2y)^k}{k!}<-1+2\sum_{k=0}^{\infty}y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$	$e^{2/x}=e^{2y}=\sum_{k=0}^{\infty}\frac{(2y)^k}{k!}<-1+2\sum_{k=0}^{\infty}y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$	$e^{2/x}=e^{2y}=\sum_{k=0}^{\infty}\frac{(2y)^k}{k!}<-1+2\sum_{k=0}^{\infty}y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$
lyche-numerical-linear-algebra-304 EQ0877	304	■ 1.000	$ f _{\infty}=\max_{a\leq x\leq b} f(x) $.	$\ f\ _{\infty}=\max_{a\leq x\leq b} f(x) .$	$\ f\ _{\infty}=\max_{a\leq x\leq b} f(x) .$
lyche-numerical-linear-algebra-304 EQ0878	304	■ 1.000	$\left Q^*\right _{\infty}=\min_{Q\in\mathcal{S}_k}\ Q\ _{\infty}$.	$\ Q^*\ _{\infty}=\min_{Q\in\mathcal{S}_k}\ Q\ _{\infty}.$	$\ Q^*\ _{\infty}=\min_{Q\in\mathcal{S}_k}\ Q\ _{\infty}.$
lyche-numerical-linear-algebra-304 EQ0879	304	■ 1.000	$T_{n+1}(t)=2tT_n(t)-T_{n-1}(t)$, $n\geq 1$, $t\in\mathbb{R}$,	$T_{n+1}(t)=2tT_n(t)-T_{n-1}(t),\quad n\geq 1,\quad t\in\mathbb{R},$	$T_{n+1}(t)=2tT_n(t)-T_{n-1}(t),\quad n\geq 1,\quad t\in\mathbb{R},$
lyche-numerical-linear-algebra-304 EQ0880	304	■ 1.000	$P_{n+1}(t)+P_{n-1}(t)=\cos(n+1)\phi+\cos(n-1)\phi=2\cos\phi\cos n\phi=2tP_n(t)$,	$P_{n+1}(t)+P_{n-1}(t)=\cos(n+1)\phi+\cos(n-1)\phi=2\cos\phi\cos n\phi=2tP_n(t),$	$P_{n+1}(t)+P_{n-1}(t)=\cos(n+1)\phi+\cos(n-1)\phi=2\cos\phi\cos n\phi=2tP_n(t),$
lyche-numerical-linear-algebra-305 EQ0882	305	■ 1.000	$z_1=t+\sqrt{t^2-1}$, $z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1}$.	$z_1=t+\sqrt{t^2-1},\quad z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1}.$	$z_1=t+\sqrt{t^2-1},\quad z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1}.$
lyche-numerical-linear-algebra-305 EQ0883 (13.32)	305	■ 1.000	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))}$, $u(x)=\frac{b+a-2x}{b-a}$.	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))},\quad u(x)=\frac{b+a-2x}{b-a}.$	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))},\quad u(x)=\frac{b+a-2x}{b-a}.$
lyche-numerical-linear-algebra-305 EQ0884	305	■ 1.000	$\sigma_j=f\left(\mu_{j-1}\right)f\left(\mu_j\right)$.	$\sigma_j=f\left(\mu_{j-1}\right)f\left(\mu_j\right).$	$\sigma_j=f\left(\mu_{j-1}\right)f\left(\mu_j\right).$
lyche-numerical-linear-algebra-305 EQ0885	305	■ 1.000	$Q\left(\mu_j\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=Q^*\left(\mu_j\right)$	$Q\left(\mu_j\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=Q^*\left(\mu_j\right)$	$Q\left(\mu_j\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=Q^*\left(\mu_j\right)$
lyche-numerical-linear-algebra-305 EQ0886	305	■ 1.000	$-Q\left(\mu_{j-1}\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=-Q^*\left(\mu_{j-1}\right)$.	$-Q\left(\mu_{j-1}\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=-Q^*\left(\mu_{j-1}\right).$	$-Q\left(\mu_{j-1}\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=-Q^*\left(\mu_{j-1}\right).$
lyche-numerical-linear-algebra-306 EQ0887 (13.33)	306	■ 1.000	$Q^*(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right)$.	$Q^*(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$	$Q^*(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$
lyche-numerical-linear-algebra-306 EQ0888 (13.34)	306	■ 1.000	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1} T_k(t) =1$.	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1} T_k(t) =1.$	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1} T_k(t) =1.$
lyche-numerical-linear-algebra-306 EQ0889	306	■ 1.000	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}$, $\kappa=b/a$.	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad\kappa=b/a.$	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad\kappa=b/a.$
lyche-numerical-linear-algebra-306 EQ0890 (13.35)	306	■ 1.000	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$
lyche-numerical-linear-algebra-307 EQ0893 (13.38)	307	■ 1.000	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0$, $i=1,\dots,n$ and $\sum_{i=1}^nt_i=1$.	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$

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lyche-numerical-linear-algebra-E08894	307	1.000	$\sqrt{a\,b}=\sqrt{(a\,c)(b\,/\,c)}\,\leq(a\,c+b\,/\,c)\,/\,2=\frac{1}{2}\sum_{i=1}^n\,t_i\left(\lambda_{\mathrm{bda}_{\mathrm{i}}}c+1\,/\,\lambda_{\mathrm{bda}_{\mathrm{i}}}\right)\right)=\frac{1}{2}\sum_{i=1}^n\,t_i\,f\left(\lambda_{\mathrm{bda}_{\mathrm{i}}}c\right),$	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq(ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i\left(\lambda_ic+1/(\lambda_ic)\right)=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq(ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i\left(\lambda_ic+1/(\lambda_ic)\right)=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$
lyche-numerical-linear-algebra-E08895	307	1.000	$\sqrt{a\,b}\,\leq\frac{1}{2}\sum_{m\,c\,\leq\,x\,\leq\,M\,c}\,f(x)\,.$	$\sqrt{ab}\leq\frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$	$\sqrt{ab}\leq\frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$
lyche-numerical-linear-algebra-E08896 (13.39)	307	1.000	$\sqrt{a\,b}\,\leq\frac{1}{2}\max\left\{f(m\,c),\,f(M\,c)\right\}\,.$	$\sqrt{ab}\leq\frac{1}{2}\max\{f(mc),f(Mc)\}.$	$\sqrt{ab}\leq\frac{1}{2}\max\{f(mc),f(Mc)\}.$
lyche-numerical-linear-algebra-E08898	308	1.000	$1=\left(\sum_{i=1}^nt_i\right)^2=\left(\sum_{i=1}^n\left(t_i\lambda_i\right)^{1/2}\left(t_i/\lambda_i\right)^{1/2}\right)^2\leq\left(\sum_{i=1}^nt_i\lambda_i\right)\left(\sum_{i=1}^nt_i/\lambda_i\right)=ab.$	$1=\left(\sum_{i=1}^nt_i\right)^2=\left(\sum_{i=1}^n(t_i\lambda_i)^{1/2}(t_i/\lambda_i)^{1/2}\right)^2\leq\left(\sum_{i=1}^nt_i\lambda_i\right)\left(\sum_{i=1}^nt_i/\lambda_i\right)=ab.$	$1=\left(\sum_{i=1}^nt_i\right)^2=\left(\sum_{i=1}^n(t_i\lambda_i)^{1/2}(t_i/\lambda_i)^{1/2}\right)^2\leq\left(\sum_{i=1}^nt_i\lambda_i\right)\left(\sum_{i=1}^nt_i/\lambda_i\right)=ab.$
lyche-numerical-linear-algebra-E08900	308	1.000	$\boldsymbol{\epsilon}_{k+1}=\boldsymbol{\epsilon}_k-\alpha_k\boldsymbol{r}_k,\quad\alpha_k:=\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}.$	$\boldsymbol{\epsilon}_{k+1}=\boldsymbol{\epsilon}_k-\alpha_k\boldsymbol{r}_k,\quad\alpha_k:=\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}.$	$\boldsymbol{\epsilon}_{k+1}=\boldsymbol{\epsilon}_k-\alpha_k\boldsymbol{r}_k,\quad\alpha_k:=\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}.$
lyche-numerical-linear-algebra-E08902	308	1.000	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2}=1-\frac{\left(\boldsymbol{r}_k^T\boldsymbol{r}_k\right)^2}{\left(\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k\right)\left(\boldsymbol{r}_k^T\boldsymbol{A}^{-1}\boldsymbol{r}_k\right)}\leq1-\frac{4\lambda_{\min}\lambda_{\max}}{\left(\lambda_{\min}+\lambda_{\max}\right)^2}=\left(\frac{\kappa-1}{\kappa+1}\right)^2$	$\frac{\ \boldsymbol{\epsilon}_{k+1}\ _{\boldsymbol{A}}^2}{\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2}=1-\frac{(\boldsymbol{r}_k^T\boldsymbol{r}_k)^2}{(\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k)(\boldsymbol{r}_k^T\boldsymbol{A}^{-1}\boldsymbol{r}_k)}\leq1-\frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min}+\lambda_{\max})^2}=\left(\frac{\kappa-1}{\kappa+1}\right)^2$	$\frac{\ \boldsymbol{\epsilon}_{k+1}\ _{\boldsymbol{A}}^2}{\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2}=1-\frac{(\boldsymbol{r}_k^T\boldsymbol{r}_k)^2}{(\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k)(\boldsymbol{r}_k^T\boldsymbol{A}^{-1}\boldsymbol{r}_k)}\leq1-\frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min}+\lambda_{\max})^2}=\left(\frac{\kappa-1}{\kappa+1}\right)^2$
lyche-numerical-linear-algebra-E08905 (13.41)	309	1.000	$\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i\boldsymbol{p}_i,\quad\alpha_i=\frac{\boldsymbol{r}_i^T\boldsymbol{r}_i}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$	$\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i\boldsymbol{p}_i,\quad\alpha_i=\frac{\boldsymbol{r}_i^T\boldsymbol{r}_i}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$	$\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i\boldsymbol{p}_i,\quad\alpha_i=\frac{\boldsymbol{r}_i^T\boldsymbol{r}_i}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$
lyche-numerical-linear-algebra-E08906 (13.42)	309	1.000	$\left\ \boldsymbol{\epsilon}_j\right\ _{\boldsymbol{A}}^2=\boldsymbol{\epsilon}_j\boldsymbol{A}\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i^2\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i=\sum_{i=j}^m\frac{\left(\boldsymbol{r}_i^T\boldsymbol{r}_i\right)^2}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$	$\ \boldsymbol{\epsilon}_j\ _{\boldsymbol{A}}^2=\boldsymbol{\epsilon}_j\boldsymbol{A}\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i^2\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i=\sum_{i=j}^m\frac{(\boldsymbol{r}_i^T\boldsymbol{r}_i)^2}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$	$\ \boldsymbol{\epsilon}_j\ _{\boldsymbol{A}}^2=\boldsymbol{\epsilon}_j\boldsymbol{A}\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i^2\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i=\sum_{i=j}^m\frac{(\boldsymbol{r}_i^T\boldsymbol{r}_i)^2}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$

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lyche-numerical-linear-algebra-EQ09907	309	1.000	$\begin{aligned} \boldsymbol{p}_{i+1} &= \left(\boldsymbol{r}_i + \beta_{i-1} \boldsymbol{p}_{i-1} \right) \\ \boldsymbol{p}_k &= \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k (\boldsymbol{p}_k^T \boldsymbol{p}_k), \end{aligned}$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = (\boldsymbol{r}_i + \beta_{i-1} \boldsymbol{p}_{i-1})^T \boldsymbol{p}_k = \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k (\boldsymbol{p}_k^T \boldsymbol{p}_k),$	
lyche-numerical-linear-algebra-EQ09908 (13.43)	309	1.000	$\begin{aligned} \boldsymbol{p}_{i+1} &= \frac{\boldsymbol{r}_i \boldsymbol{r}_i^T}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k, \quad i \geq k. \end{aligned}$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k, \quad i \geq k.$	
lyche-numerical-linear-algebra-EQ09909	309	1.000	$\begin{aligned} \left\ \boldsymbol{\epsilon}_{k+1} \right\ _2^2 &= \left\ \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k \right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k \right\ _2^2, \end{aligned}$	$\left\ \boldsymbol{\epsilon}_k \right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k \right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k \right\ _2^2,$	
lyche-numerical-linear-algebra-EQ09913 (13.49)	311	1.000	$\begin{aligned} \boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} &= \boldsymbol{B} \boldsymbol{A}. \end{aligned}$	$\boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$	
lyche-numerical-linear-algebra-EQ09916 (13.50)	311	1.000	$\begin{aligned} \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \\ \boldsymbol{s}_{k+1} &= \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k, \\ \boldsymbol{p}_{k+1} &= \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}. \end{aligned}$	$\begin{aligned} \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \\ \boldsymbol{s}_{k+1} &= \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k, \\ \boldsymbol{p}_{k+1} &= \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}. \end{aligned}$	
lyche-numerical-linear-algebra-EQ09921	313	1.000	$\begin{aligned} I_m &:= \{(j, k) : 1 \leq j, k \leq m\}, \\ \bar{I}_m &:= \{(j, k) : 0 \leq j, k \leq m+1\}, \\ \partial I_m &:= \bar{I}_m \setminus I_m. \end{aligned}$	$\begin{aligned} I_m &:= \{(j, k) : 1 \leq j, k \leq m\}, \\ \bar{I}_m &:= \{(j, k) : 0 \leq j, k \leq m+1\}, \\ \partial I_m &:= \bar{I}_m \setminus I_m. \end{aligned}$	$\begin{aligned} I_m &:= \{(j, k) : 1 \leq j, k \leq m\}, \\ \bar{I}_m &:= \{(j, k) : 0 \leq j, k \leq m+1\}, \\ \partial I_m &:= \bar{I}_m \setminus I_m. \end{aligned}$
lyche-numerical-linear-algebra-EQ09922	313	1.000	$\left\{ (x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m \right\}$	$\{(x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m\}$	

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lyche-numerical-linear-algebra- EQ0923 (13.54)	314	1.000	$\begin{aligned} & \frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f(t + \frac{h}{2}) \frac{d}{dt} g(t + h/2) - f(t - \frac{h}{2}) \frac{d}{dt} g(t - \frac{h}{2}) \right) / h \\ & \approx \left(f(t + \frac{h}{2}) (g(t + h) - g(t)) - f(t - \frac{h}{2}) (g(t) - g(t - h)) \right) / h^2 \end{aligned}$	$\frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f \left(t + \frac{h}{2} \right) \frac{d}{dt} g(t + h/2) - f \left(t - \frac{h}{2} \right) \frac{d}{dt} g \left(t - \frac{h}{2} \right) \right) / h$ $\approx \left(f \left(t + \frac{h}{2} \right) (g(t + h) - g(t)) - f \left(t - \frac{h}{2} \right) (g(t) - g(t - h)) \right) / h^2$	$\frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f(t + \frac{h}{2}) \frac{d}{dt} g(t + h/2) - f(t - \frac{h}{2}) \frac{d}{dt} g(t - \frac{h}{2}) \right) / h$ $\approx \left(f(t + \frac{h}{2}) (g(t + h) - g(t)) - f(t - \frac{h}{2}) (g(t) - g(t - h)) \right) / h^2$
lyche-numerical-linear-algebra- EQ0927	314	1.000	$\mathbf{V} := \left[\begin{array}{ccc} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{array} \right]$	$\mathbf{V} := \left[\begin{array}{ccc} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{array} \right]$	$\mathbf{V} := \left[\begin{array}{ccc} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{array} \right]$
lyche-numerical-linear-algebra- EQ0928 (13.57)	314	1.000	$\mathbf{A} \mathbf{x} = \mathbf{A} \operatorname{vec}(\mathbf{V}) := h^2 \operatorname{vec}(L_h \mathbf{v}).$	$\mathbf{A} \mathbf{x} = \mathbf{A} \operatorname{vec}(\mathbf{V}) := h^2 \operatorname{vec}(L_h \mathbf{v}).$	$\mathbf{A} \mathbf{x} = \mathbf{A} \operatorname{vec}(\mathbf{V}) := h^2 \operatorname{vec}(L_h \mathbf{v}).$
lyche-numerical-linear-algebra- EQ0931 (13.60)	315	1.000	$\sum_{i=1}^m (a_{i-1} (b_i - b_{i-1}) - a_i (b_{i+1} - b_i)) c_i = \sum_{i=0}^m a_i (b_{i+1} - b_i) (c_{i+1} - c_i).$	$\sum_{i=1}^m (a_{i-1} (b_i - b_{i-1}) - a_i (b_{i+1} - b_i)) c_i = \sum_{i=0}^m a_i (b_{i+1} - b_i) (c_{i+1} - c_i).$	$\sum_{i=1}^m (a_{i-1} (b_i - b_{i-1}) - a_i (b_{i+1} - b_i)) c_i = \sum_{i=0}^m a_i (b_{i+1} - b_i) (c_{i+1} - c_i).$
lyche-numerical-linear-algebra- EQ0932	315	1.000	$\sum_{i=0}^m a_i (b_{i+1} - b_i) c_{i+1} - \sum_{i=0}^m a_i (b_{i+1} - b_i) c_i,$	$\sum_{i=0}^m a_i (b_{i+1} - b_i) c_{i+1} - \sum_{i=0}^m a_i (b_{i+1} - b_i) c_i,$	$\sum_{i=0}^m a_i (b_{i+1} - b_i) c_{i+1} - \sum_{i=0}^m a_i (b_{i+1} - b_i) c_i,$
lyche-numerical-linear-algebra- EQ0933	316	1.000	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$
lyche-numerical-linear-algebra- EQ0934	317	1.000	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$
lyche-numerical-linear-algebra- EQ0936	317	1.000	$\mathbf{x}^T \mathbf{A}_p \mathbf{x} = \sum_{i=1}^m \sum_{j=0}^m (v_{i,j+1} - v_{i,j})^2 + \sum_{j=1}^m \sum_{i=0}^m (v_{i+1,j} - v_{i,j})^2.$	$\mathbf{x}^T \mathbf{A}_p \mathbf{x} = \sum_{i=1}^m \sum_{j=0}^m (v_{i,j+1} - v_{i,j})^2 + \sum_{j=1}^m \sum_{i=0}^m (v_{i+1,j} - v_{i,j})^2.$	$\mathbf{x}^T \mathbf{A}_p \mathbf{x} = \sum_{i=1}^m \sum_{j=0}^m (v_{i,j+1} - v_{i,j})^2 + \sum_{j=1}^m \sum_{i=0}^m (v_{i+1,j} - v_{i,j})^2.$
lyche-numerical-linear-algebra- EQ0938	318	1.000	$Q(\mathbf{y}) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j.$	$Q(\mathbf{y}) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j.$	$Q(\mathbf{y}) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j.$
lyche-numerical-linear-algebra- EQ0939	318	1.000	$\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{l} 0 \\ 3 \end{array} \right]$	$\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{l} 0 \\ 3 \end{array} \right]$	$\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{l} 0 \\ 3 \end{array} \right]$
lyche-numerical-linear-algebra- EQ0940	319	1.000	$\mathbf{w}^T \mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{w}^T \mathbf{A}^T \mathbf{b}, \quad \text{for all } \mathbf{w} \in \mathbb{W},$	$\mathbf{w}^T \mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{w}^T \mathbf{A}^T \mathbf{b}, \quad \text{for all } \mathbf{w} \in \mathbb{W},$	$\mathbf{w}^T \mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{w}^T \mathbf{A}^T \mathbf{b}, \quad \text{for all } \mathbf{w} \in \mathbb{W},$
lyche-numerical-linear-algebra- EQ0941	319	1.000	$\ \mathbf{b} - \mathbf{A} \hat{\mathbf{x}}\ _2 \leq \ \mathbf{b} - \mathbf{A} \mathbf{w}\ _2, \quad \text{for all } \mathbf{w} \in \mathbb{W}.$	$\ \mathbf{b} - \mathbf{A} \hat{\mathbf{x}}\ _2 \leq \ \mathbf{b} - \mathbf{A} \mathbf{w}\ _2, \quad \text{for all } \mathbf{w} \in \mathbb{W}.$	$\ \mathbf{b} - \mathbf{A} \hat{\mathbf{x}}\ _2 \leq \ \mathbf{b} - \mathbf{A} \mathbf{w}\ _2, \quad \text{for all } \mathbf{w} \in \mathbb{W}.$

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lyche-numerical-linear-algebra- EQ0943	319	■ 1.000	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$
lyche-numerical-linear-algebra- EQ0944	320	■ 1.000	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}_k.$	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}_k.$	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}_k.$
lyche-numerical-linear-algebra- EQ0952 (13.65)	321	■ 1.000	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$
lyche-numerical-linear-algebra- EQ0953 (13.66)	322	■ 1.000	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k$	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k$	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k.$
lyche-numerical-linear-algebra- EQ0955	322	■ 1.000	$\mathcal{W}_k := \text{span}(\boldsymbol{b}, \boldsymbol{B} \boldsymbol{b}, \dots, \boldsymbol{B}^{k-1} \boldsymbol{b}).$	$\mathcal{W}_k := \text{span}(\boldsymbol{b}, \boldsymbol{B} \boldsymbol{b}, \dots, \boldsymbol{B}^{k-1} \boldsymbol{b}).$	$\mathcal{W}_k := \text{span}(\boldsymbol{b}, \boldsymbol{B} \boldsymbol{b}, \dots, \boldsymbol{B}^{k-1} \boldsymbol{b}).$
lyche-numerical-linear-algebra- EQ0956	322	■ 1.000	$\langle \boldsymbol{A} \boldsymbol{x}_k, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle, \text{ for all } \boldsymbol{w} \in \mathcal{W}_k.$	$\langle \boldsymbol{A} \boldsymbol{x}_k, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle, \text{ for all } \boldsymbol{w} \in \mathcal{W}_k.$	$\langle \boldsymbol{A} \boldsymbol{x}_k, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle, \text{ for all } \boldsymbol{w} \in \mathcal{W}_k.$
lyche-numerical-linear-algebra- EQ0958	322	■ 1.000	$\rho_k \neq 0, \quad k = 0, \dots, m.$	$\rho_k \neq 0, \quad k = 0, \dots, m.$	$\rho_k \neq 0, \quad k = 0, \dots, m.$
lyche-numerical-linear-algebra- EQ0962 (13.72)	323	■ 1.000	$\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1} \rangle = \langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_j \rangle, \quad j = 0, 1, \dots, k.$	$\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1} \rangle = \langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_j \rangle, \quad j = 0, 1, \dots, k.$	$\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1} \rangle = \langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_j \rangle, \quad j = 0, 1, \dots, k.$
lyche-numerical-linear-algebra- EQ0963 (13.70)	323	■ 1.000	$T_n(t) = \cosh(n \operatorname{arccosh} t) \text{ for } t \geq 1,$	$T_n(t) = \cosh(n \operatorname{arccosh} t) \text{ for } t \geq 1,$	$T_n(t) = \cosh(n \operatorname{arccosh} t) \text{ for } t \geq 1,$
lyche-numerical-linear-algebra- EQ0964	324	■ 1.000	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{bmatrix} \begin{bmatrix} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{bmatrix} = \begin{bmatrix} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{bmatrix},$	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{bmatrix} \begin{bmatrix} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{bmatrix} = \begin{bmatrix} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{bmatrix},$	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{bmatrix} \begin{bmatrix} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{bmatrix} = \begin{bmatrix} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{bmatrix},$

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lyche-numerical-linear-algebra- EQ0965	324	1.000	$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array}\right] = \left[\begin{array}{l} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{array}\right].$	$\begin{bmatrix} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{bmatrix} = \begin{bmatrix} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{bmatrix}.$	$\begin{bmatrix} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{bmatrix} = \begin{bmatrix} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0966	326	1.000	$\boldsymbol{A} = \operatorname{diag}(\boldsymbol{A}_1, \boldsymbol{A}_2, \dots, \boldsymbol{A}_r), \quad \boldsymbol{A}_i \in \mathbb{C}^{m_i \times m_i}.$	$\boldsymbol{A} = \operatorname{diag}(\boldsymbol{A}_1, \boldsymbol{A}_2, \dots, \boldsymbol{A}_r), \quad \boldsymbol{A}_i \in \mathbb{C}^{m_i \times m_i}.$	$\boldsymbol{A} = \operatorname{diag}(\boldsymbol{A}_1, \boldsymbol{A}_2, \dots, \boldsymbol{A}_r), \quad \boldsymbol{A}_i \in \mathbb{C}^{m_i \times m_i}.$
lyche-numerical-linear-algebra- EQ0968	327	1.000	$\operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^r \operatorname{det}(\boldsymbol{A}_{ii} - \lambda \boldsymbol{I})$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^r \det(\boldsymbol{A}_{ii} - \lambda \boldsymbol{I})$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^r \det(\boldsymbol{A}_{ii} - \lambda \boldsymbol{I})$
lyche-numerical-linear-algebra- EQ0970	328	1.000	$ \lambda - a_{ii} = \left \sum_{j \neq i} a_{ij} x_j / x_i \right \leq \sum_{j \neq i} a_{ij} x_j / x_i \leq r_i$	$ \lambda - a_{ii} = \left \sum_{j \neq i} a_{ij} x_j / x_i \right \leq \sum_{j \neq i} a_{ij} x_j / x_i \leq r_i$	$ \lambda - a_{ii} = \left \sum_{j \neq i} a_{ij} x_j / x_i \right \leq \sum_{j \neq i} a_{ij} x_j / x_i \leq r_i$
lyche-numerical-linear-algebra- EQ0971	328	1.000	$R_i = \{z \in \mathbb{R} : z - 2 \leq 2\}, \quad i = 2, 3, \dots, m - 1.$	$R_i = \{z \in \mathbb{R} : z - 2 \leq 2\}, \quad i = 2, 3, \dots, m - 1.$	$R_i = \{z \in \mathbb{R} : z - 2 \leq 2\}, \quad i = 2, 3, \dots, m - 1.$
lyche-numerical-linear-algebra- EQ0972	328	1.000	$\lambda_j = 4 \left[\sin \frac{j\pi}{2(m+1)} \right]^2, \quad j = 1, 2, \dots, m.$	$\lambda_j = 4 \left[\sin \frac{j\pi}{2(m+1)} \right]^2, \quad j = 1, 2, \dots, m.$	$\lambda_j = 4 \left[\sin \frac{j\pi}{2(m+1)} \right]^2, \quad j = 1, 2, \dots, m.$
lyche-numerical-linear-algebra- EQ0973	329	1.000	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn}), \quad t \in [0, 1].$	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn}), \quad t \in [0, 1].$	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn}), \quad t \in [0, 1].$
lyche-numerical-linear-algebra- EQ0974	329	1.000	$R_i(t) = \{z \in \mathbb{C} : z - a_{ii} \leq t r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} .$	$R_i(t) = \{z \in \mathbb{C} : z - a_{ii} \leq t r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} .$	$R_i(t) = \{z \in \mathbb{C} : z - a_{ii} \leq t r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} .$
lyche-numerical-linear-algebra- EQ0978	330	1.000	$\begin{aligned} & \left \mu - \lambda_1 \right ^n \leq \prod_{j=1}^n \left \mu - \lambda_j \right = \left \det(\mu \boldsymbol{I} - \boldsymbol{A}) \right = \left \det((\mu \boldsymbol{I} - \boldsymbol{A})[\boldsymbol{u}_1, \dots, \boldsymbol{u}_n]) \right \\ & \leq \ (\mu \boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_1\ _2 \prod_{j=2}^n \ (\mu \boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_j\ _2 \leq \ \boldsymbol{E}\ _2 (\ \boldsymbol{A} + \boldsymbol{E}\ _2 + \ \boldsymbol{A}\ _2)^{n-1}. \end{aligned}$	$\begin{aligned} & \mu - \lambda_1 ^n \leq \prod_{j=1}^n \mu - \lambda_j = \det(\mu \boldsymbol{I} - \boldsymbol{A}) = \det((\mu \boldsymbol{I} - \boldsymbol{A})[\boldsymbol{u}_1, \dots, \boldsymbol{u}_n]) \\ & \leq \ (\mu \boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_1\ _2 \prod_{j=2}^n \ (\mu \boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_j\ _2 \leq \ \boldsymbol{E}\ _2 (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A}\ _2)^{n-1}. \end{aligned}$	$\begin{aligned} & \mu - \lambda_1 ^n \leq \prod_{j=1}^n \mu - \lambda_j = \det(\mu \boldsymbol{I} - \boldsymbol{A}) = \det((\mu \boldsymbol{I} - \boldsymbol{A})[\boldsymbol{u}_1, \dots, \boldsymbol{u}_n]) \\ & \leq \ (\mu \boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_1\ _2 \prod_{j=2}^n \ (\mu \boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_j\ _2 \leq \ \boldsymbol{E}\ _2 (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A}\ _2)^{n-1}. \end{aligned}$

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lyche-numerical-linear-algebra- EQ0979 (14.2)	331	■ 1.000	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$
lyche-numerical-linear-algebra- EQ0980 (14.3)	331	■ 1.000	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$
lyche-numerical-linear-algebra- EQ0983	332	■ 1.000	$\ \boldsymbol{A} \boldsymbol{x}\ _p = \left\ [\lambda_1 x_1, \dots, \lambda_n x_n]^T \right\ _p = \left(\sum_{j=1}^n \lambda_j ^p x_j ^p \right)^{1/p} \leq \rho(\boldsymbol{A}) \ \boldsymbol{x}\ _p.$	$\ \boldsymbol{A} \boldsymbol{x}\ _p = \left\ [\lambda_1 x_1, \dots, \lambda_n x_n]^T \right\ _p = \left(\sum_{j=1}^n \lambda_j ^p x_j ^p \right)^{1/p} \leq \rho(\boldsymbol{A}) \ \boldsymbol{x}\ _p.$	$\ \boldsymbol{A} \boldsymbol{x}\ _p = \left\ [\lambda_1 x_1, \dots, \lambda_n x_n]^T \right\ _p = \left(\sum_{j=1}^n \lambda_j ^p x_j ^p \right)^{1/p} \leq \rho(\boldsymbol{A}) \ \boldsymbol{x}\ _p.$
lyche-numerical-linear-algebra- EQ0984 (14.4)	332	■ 1.000	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$
lyche-numerical-linear-algebra- EQ0985 (14.5)	332	■ 1.000	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$
lyche-numerical-linear-algebra- EQ0987	333	■ 1.000	$(\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I}) \boldsymbol{x} = (\mu \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{E} - \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{A}^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0}.$	$(\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I}) \boldsymbol{x} = (\mu \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{E} - \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{A}^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0}.$	$(\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I}) \boldsymbol{x} = (\mu \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{E} - \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{A}^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0}.$
lyche-numerical-linear-algebra- EQ0988	333	■ 1.000	$\boldsymbol{A}_{k+1}^* = (\boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k)^* = \boldsymbol{H}_k \boldsymbol{A}_k^* \boldsymbol{H}_k = \boldsymbol{A}_{k+1}.$	$\boldsymbol{A}_{k+1}^* = (\boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k)^* = \boldsymbol{H}_k \boldsymbol{A}_k^* \boldsymbol{H}_k = \boldsymbol{A}_{k+1}.$	$\boldsymbol{A}_{k+1}^* = (\boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k)^* = \boldsymbol{H}_k \boldsymbol{A}_k^* \boldsymbol{H}_k = \boldsymbol{A}_{k+1}.$
lyche-numerical-linear-algebra- EQ0990	334	■ 1.000	$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \in \mathbb{C}^{n \times n}.$	$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \in \mathbb{C}^{n \times n}.$	$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \in \mathbb{C}^{n \times n}.$
lyche-numerical-linear-algebra- EQ0993	335	■ 1.000	$\boldsymbol{Q}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix} * \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k - \boldsymbol{v}_k (\boldsymbol{v}_k^T \boldsymbol{U}_k) \end{bmatrix}.$	$\boldsymbol{Q}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix} * \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k - \boldsymbol{v}_k (\boldsymbol{v}_k^T \boldsymbol{U}_k) \end{bmatrix}.$	$\boldsymbol{Q}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix} * \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k - \boldsymbol{v}_k (\boldsymbol{v}_k^T \boldsymbol{U}_k) \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0996 (14.7)	336	■ 1.000	$p_{k+1}(x) = (x - d_{k+1}) p_k(x) - c_k^2 p_{k-1}(x).$	$p_{k+1}(x) = (x - d_{k+1}) p_k(x) - c_k^2 p_{k-1}(x).$	$p_{k+1}(x) = (x - d_{k+1}) p_k(x) - c_k^2 p_{k-1}(x).$
lyche-numerical-linear-algebra- EQ1001	337	■ 1.000	$p_{k+1}(y_j) = (-1)^{k+1-j} p_{k+1}(y_j) \neq 0, \text{ for } j = 0, 1, \dots, k+1.$	$p_{k+1}(y_j) = (-1)^{k+1-j} p_{k+1}(y_j) \neq 0, \text{ for } j = 0, 1, \dots, k+1.$	$p_{k+1}(y_j) = (-1)^{k+1-j} p_{k+1}(y_j) \neq 0, \text{ for } j = 0, 1, \dots, k+1.$

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lyche-numerical-linear-algebra- EQ1002	337 337	■ 1.000	$p_{k+1}\left(y_j\right)=-c_{k}^2\,p_{k-1}\left(y_j\right)-c_{k}^2\left(y_j-z_1\right)\cdots\left(y_j-z_{k-1}\right)$	$p_{k+1}\left(y_j\right)=-c_k^2p_{k-1}\left(y_j\right)=-c_k^2\left(y_j-z_1\right)\cdots\left(y_j-z_{k-1}\right).$	$p_{k+1}(y_j)=-c_k^2p_{k-1}(y_j)=-c_k^2(y_j-z_1)\cdots(y_j-z_{k-1}).$
lyche-numerical-linear-algebra- EQ1003	337 337	■ 1.000	$\sum_{i=1}^n\lambda_i\left y_i\right ^2=\sum_{i=1}^k\lambda_i\left x_i\right ^2>0$	$\boldsymbol{y}^*\boldsymbol{A}\boldsymbol{y}=\sum_{i=1}^n\lambda_i\left y_i\right ^2=\sum_{i=1}^k\lambda_i\left x_i\right ^2>0.$	$\boldsymbol{y}^*\boldsymbol{A}\boldsymbol{y}=\sum_{i=1}^n\lambda_i\left y_i\right ^2=\sum_{i=1}^k\lambda_i\left x_i\right ^2>0.$
lyche-numerical-linear-algebra- EQ1005 (14.8)	338 338	■ 1.000	$d_1(\alpha)=d_1-\alpha,\,d_k(\alpha)=d_k-\alpha-c_{k-1}^2/d_{k-1}(\alpha),\,k=2,3,\ldots,n$	$d_1(\alpha)=d_1-\alpha,\,d_k(\alpha)=d_k-\alpha-c_{k-1}^2/d_{k-1}(\alpha),\,k=2,3,\ldots,n.$	$d_1(\alpha)=d_1-\alpha,\,d_k(\alpha)=d_k-\alpha-c_{k-1}^2/d_{k-1}(\alpha),\,k=2,3,\ldots,n.$
lyche-numerical-linear-algebra- EQ1006	338 338	■ 1.000	$\rho(x):=\#\left\{k:d_k(x)>0\right\}\text{ for }k=1,\ldots,n$	$\rho(x):=\#\left\{k:d_k(x)>0\text{ for }k=1,\ldots,n\right\}$	$\rho(x):=\#\{k:d_k(x)>0\text{ for }k=1,\ldots,n\}$
lyche-numerical-linear-algebra- EQ1007	340 340	■ 1.000	$\boldsymbol{A}=\left(\begin{array}{cccc}4&1&0&0\\1&4&1&0\\0&1&4&1\\0&0&1&4\end{array}\right)$	$\boldsymbol{A}=\left(\begin{array}{cccc}4&1&0&0\\1&4&1&0\\0&1&4&1\\0&0&1&4\end{array}\right).$	$\boldsymbol{A}=\left(\begin{array}{cccc}4&1&0&0\\1&4&1&0\\0&1&4&1\\0&0&1&4\end{array}\right).$
lyche-numerical-linear-algebra- EQ1008	340 340	■ 1.000	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}),\quad\boldsymbol{D}:=\text{diag}(a_{11},\ldots,a_{nn}),\quad t\in\mathbb{R}.$	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}),\quad\boldsymbol{D}:=\text{diag}(a_{11},\ldots,a_{nn}),\quad t\in\mathbb{R}.$	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}),\quad\boldsymbol{D}:=\text{diag}(a_{11},\ldots,a_{nn}),\quad t\in\mathbb{R}.$
lyche-numerical-linear-algebra- EQ1009	340 340	■ 1.000	$\left \lambda-\mu\right \leq C\left(t_2-t_1\right)^{1/n},\text{ where }C\leq 2\left(\left\ \boldsymbol{D}\right\ _2+\left\ \boldsymbol{A}-\boldsymbol{D}\right\ _2\right)$	$\left \lambda-\mu\right \leq C\left(t_2-t_1\right)^{1/n},\text{ where }C\leq 2\left(\left\ \boldsymbol{D}\right\ _2+\left\ \boldsymbol{A}-\boldsymbol{D}\right\ _2\right).$	$\left \lambda-\mu\right \leq C\left(t_2-t_1\right)^{1/n},\text{ where }C\leq 2\left(\left\ \boldsymbol{D}\right\ _2+\left\ \boldsymbol{A}-\boldsymbol{D}\right\ _2\right).$
lyche-numerical-linear-algebra- EQ1013	342 342	■ 1.000	$\boldsymbol{D}^{-1/2}:=\text{diag}\left(d_1^{-1/2},\ldots,d_n^{-1/2}\right)$	$\boldsymbol{D}^{-1/2}:=\text{diag}\left(d_1^{-1/2},\ldots,d_n^{-1/2}\right).$	$\boldsymbol{D}^{-1/2}:=\text{diag}\left(d_1^{-1/2},\ldots,d_n^{-1/2}\right).$
lyche-numerical-linear-algebra- EQ1027	347 347	■ 1.000	$\rho(\lambda)^2=\left \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\right ^2+\left\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}\right\ _2^2,$	$\rho(\lambda)^2=\left \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\right ^2+\left\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}\right\ _2^2,$	$\rho(\lambda)^2=\left \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\right ^2+\left\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}\right\ _2^2,$
lyche-numerical-linear-algebra- EQ1028	347 347	■ 1.000	$\frac{\left \lambda-\mu\right }{\left \lambda\right }\leq K_2(\boldsymbol{X})K_2(\boldsymbol{A})\frac{\left\ \boldsymbol{A}\boldsymbol{u}-\mu\boldsymbol{u}\right\ _2}{\left\ \boldsymbol{A}\right\ _2}.$	$\frac{\left \lambda-\mu\right }{\left \lambda\right }\leq K_2(\boldsymbol{X})K_2(\boldsymbol{A})\frac{\left\ \boldsymbol{A}\boldsymbol{u}-\mu\boldsymbol{u}\right\ _2}{\left\ \boldsymbol{A}\right\ _2}.$	$\frac{\left \lambda-\mu\right }{\left \lambda\right }\leq K_2(\boldsymbol{X})K_2(\boldsymbol{A})\frac{\left\ \boldsymbol{A}\boldsymbol{u}-\mu\boldsymbol{u}\right\ _2}{\left\ \boldsymbol{A}\right\ _2}.$
lyche-numerical-linear-algebra- EQ1033	349 349	■ 1.000	$\rho_k:=\frac{\boldsymbol{y}_k^*\boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^*\boldsymbol{y}_k},\quad\boldsymbol{w}_k:=\frac{\boldsymbol{x}_{k-1}}{\left\ \boldsymbol{y}_k\right\ _2}.$	$\rho_k:=\frac{\boldsymbol{y}_k^*\boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^*\boldsymbol{y}_k},\quad\boldsymbol{w}_k:=\frac{\boldsymbol{x}_{k-1}}{\left\ \boldsymbol{y}_k\right\ _2}.$	$\rho_k:=\frac{\boldsymbol{y}_k^*\boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^*\boldsymbol{y}_k},\quad\boldsymbol{w}_k:=\frac{\boldsymbol{x}_{k-1}}{\left\ \boldsymbol{y}_k\right\ _2}.$
lyche-numerical-linear-algebra- EQ1036 (15.9)	351 351	■ 1.000	$\boldsymbol{A}_{k+1}=\boldsymbol{R}_k\boldsymbol{Q}_k=\boldsymbol{Q}_k^*\boldsymbol{A}_k\boldsymbol{Q}_k,$	$\boldsymbol{A}_{k+1}=\boldsymbol{R}_k\boldsymbol{Q}_k=\boldsymbol{Q}_k^*\boldsymbol{A}_k\boldsymbol{Q}_k,$	$\boldsymbol{A}_{k+1}=\boldsymbol{R}_k\boldsymbol{Q}_k=\boldsymbol{Q}_k^*\boldsymbol{A}_k\boldsymbol{Q}_k,$
lyche-numerical-linear-algebra- EQ1037	351 351	■ 1.000	$\boldsymbol{A}_1=\boldsymbol{A}=\left(\begin{array}{cc}2&1\\1&2\end{array}\right)=\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc}-2&-1\\-1&2\end{array}\right]\right)*\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc}-5&-4\\0&3\end{array}\right]\right)=\boldsymbol{Q}_1\boldsymbol{R}_1$	$\boldsymbol{A}_1=\boldsymbol{A}=\left[\begin{array}{cc}2&1\\1&2\end{array}\right]=\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc}-2&-1\\-1&2\end{array}\right]\right)*\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc}-5&-4\\0&3\end{array}\right]\right)=\boldsymbol{Q}_1\boldsymbol{R}_1$	$\boldsymbol{A}_1=\boldsymbol{A}=\left[\begin{array}{cc}2&1\\1&2\end{array}\right]=\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc}-2&-1\\-1&2\end{array}\right]\right)*\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc}-5&-4\\0&3\end{array}\right]\right)=\boldsymbol{Q}_1\boldsymbol{R}_1$

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lyche-numerical-linear-algebra- EQ1038	351	■ 1.000	$\boldsymbol{A}_{\{2\}}=\boldsymbol{R}_{\{1\}}\boldsymbol{Q}_{\{1\}}=\frac{1}{5}\left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array}\right]*\left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array}\right]=\frac{1}{5}\left[\begin{array}{cc} 14 & -3 \\ -3 & 6 \end{array}\right]=\left[\begin{array}{cc} 2.8 & -0.6 \\ -0.6 & 1.2 \end{array}\right].$	$\boldsymbol{A}_2 = \boldsymbol{R}_1 \boldsymbol{Q}_1 = \frac{1}{5} \begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix} * \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 14 & -3 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 2.8 & -0.6 \\ -0.6 & 1.2 \end{bmatrix}.$	$\boldsymbol{A}_2 = \boldsymbol{R}_1 \boldsymbol{Q}_1 = \frac{1}{5} \begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix} * \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 14 & -3 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 2.8 & -0.6 \\ -0.6 & 1.2 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ1039	351	■ 1.000	$\boldsymbol{A}_{\{4\}}\approx\left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array}\right], \quad \boldsymbol{A}_{\{10\}}\approx\left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array}\right]$	$\boldsymbol{A}_4 \approx \begin{bmatrix} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{bmatrix}, \quad \boldsymbol{A}_{10} \approx \begin{bmatrix} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{bmatrix}$	$\boldsymbol{A}_4 \approx \begin{bmatrix} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{bmatrix}, \quad \boldsymbol{A}_{10} \approx \begin{bmatrix} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{bmatrix}$
lyche-numerical-linear-algebra- EQ1040	352	■ 1.000	$\boldsymbol{A}_{\{1\}}=\boldsymbol{A}=\left[\begin{array}{cccc} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.8214 & 0.9218 & 0.2311 & 0.7621 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \end{array}\right]$	$\boldsymbol{A}_1 = \boldsymbol{A} = \begin{bmatrix} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{bmatrix}$	$\boldsymbol{A}_1 = \boldsymbol{A} = \begin{bmatrix} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{bmatrix}$
lyche-numerical-linear-algebra- EQ1042 (15.10)	352	■ 1.000	$\boldsymbol{\tilde{Q}}_{\{k\}}:=\boldsymbol{Q}_{\{1\}}\cdots\boldsymbol{Q}_{\{k\}} \text{ and } \boldsymbol{\tilde{R}}_{\{k\}}:=\boldsymbol{R}_{\{k\}}\cdots\boldsymbol{R}_{\{1\}},$	$\tilde{\boldsymbol{Q}}_k := \boldsymbol{Q}_1 \cdots \boldsymbol{Q}_k \text{ and } \tilde{\boldsymbol{R}}_k := \boldsymbol{R}_k \cdots \boldsymbol{R}_1,$	$\tilde{\boldsymbol{Q}}_k := \boldsymbol{Q}_1 \cdots \boldsymbol{Q}_k \text{ and } \tilde{\boldsymbol{R}}_k := \boldsymbol{R}_k \cdots \boldsymbol{R}_1,$
lyche-numerical-linear-algebra- EQ1044 (15.11)	353	■ 1.000	$\boldsymbol{A}_{\{k\}}=\boldsymbol{Q}_{\{k-1\}}^*\boldsymbol{Q}_{\{k-1\}}\boldsymbol{A}_{\{k-1\}}=\boldsymbol{Q}_{\{k-1\}}^*\boldsymbol{Q}_{\{k-2\}}^*\boldsymbol{A}_{\{k-2\}}\boldsymbol{Q}_{\{k-2\}}\boldsymbol{Q}_{\{k-1\}}=\cdots=\boldsymbol{Q}_{\{k-1\}}^*\boldsymbol{A}\boldsymbol{Q}_{\{k-1\}}.$	$\boldsymbol{A}_k = \boldsymbol{Q}_{k-1}^* \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}.$	$\boldsymbol{A}_k = \boldsymbol{Q}_{k-1}^* \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}.$
lyche-numerical-linear-algebra- EQ1046	353	■ 1.000	$\boldsymbol{A}^k\boldsymbol{e}_1=\tilde{\boldsymbol{Q}}_k\tilde{\boldsymbol{R}}_k\boldsymbol{e}_1=\tilde{r}_{11}^{(k)}\tilde{\boldsymbol{Q}}_k\boldsymbol{e}_1 \text{ or } \tilde{q}_1^{(k)}:=\tilde{\boldsymbol{Q}}_k\boldsymbol{e}_1=\frac{1}{\tilde{r}_{11}^{(k)}}\boldsymbol{A}^k\boldsymbol{e}_1.$	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{q}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{q}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$
lyche-numerical-linear-algebra- EQ1048	355	■ 1.000	$\boldsymbol{A}_{\{k+1\}}=\boldsymbol{Q}_{\{k\}}^*(\boldsymbol{A}_{\{k\}}-s_k\boldsymbol{I})\boldsymbol{Q}_{\{k\}}+s_k\boldsymbol{I}=\boldsymbol{Q}_{\{k\}}^*\boldsymbol{A}_{\{k\}}\boldsymbol{Q}_{\{k\}}$	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$
lyche-numerical-linear-algebra- EQ1050 (16.1)	358	■ 1.000	$\nabla f:=\left[\begin{array}{c} D_1 f \\ \vdots \\ D_n f \end{array}\right], \quad \boldsymbol{H} f:=\nabla \nabla^T f:=\left[\begin{array}{ccc} D_1 D_1 f & \cdots & D_1 D_n f \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n f \end{array}\right],$	$\nabla f := \begin{bmatrix} D_1 f \\ \vdots \\ D_n f \end{bmatrix}, \quad \boldsymbol{H} f := \nabla \nabla^T f := \begin{bmatrix} D_1 D_1 f & \cdots & D_1 D_n f \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n f \end{bmatrix},$	$\nabla f := \begin{bmatrix} D_1 f \\ \vdots \\ D_n f \end{bmatrix}, \quad \boldsymbol{H} f := \nabla \nabla^T f := \begin{bmatrix} D_1 D_1 f & \cdots & D_1 D_n f \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n f \end{bmatrix},$

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lyche-numerical-linear-algebra-EQ1051	359	■ 1.000	$\nabla^{\mathrm{T}} \boldsymbol{f} := \left[\begin{array}{ccc} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_m & \cdots & D_n f_m \end{array} \right] .$	$\nabla^T \boldsymbol{f} := \begin{bmatrix} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_m & \cdots & D_n f_m \end{bmatrix} .$	$\nabla^T \boldsymbol{f} := \begin{bmatrix} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_m & \cdots & D_n f_m \end{bmatrix} .$
lyche-numerical-linear-algebra-EQ1053 (16.2)	359	■ 1.000	$f(\boldsymbol{x} + \boldsymbol{h}) = f(\boldsymbol{x}) + \boldsymbol{h}^{\mathrm{T}} \nabla f(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^{\mathrm{T}} \nabla \nabla^{\mathrm{T}} f(\boldsymbol{c}) \boldsymbol{h}, \text{ for some } \boldsymbol{c} \in L .$	$f(\boldsymbol{x} + \boldsymbol{h}) = f(\boldsymbol{x}) + \boldsymbol{h}^T \nabla f(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{c}) \boldsymbol{h}, \text{ for some } \boldsymbol{c} \in L.$	$f(\boldsymbol{x} + \boldsymbol{h}) = f(\boldsymbol{x}) + \boldsymbol{h}^T \nabla f(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{c}) \boldsymbol{h}, \text{ for some } \boldsymbol{c} \in L.$
lyche-numerical-linear-algebra-EQ1055	359	■ 1.000	$g(1) = g(0) + g'(\theta) + \frac{1}{2} g''(u), \text{ for some } u \in (0, 1),$	$g(1) = g(0) + g'(0) + \frac{1}{2} g''(u), \text{ for some } u \in (0, 1),$	$g(1) = g(0) + g'(0) + \frac{1}{2} g''(u), \text{ for some } u \in (0, 1),$
lyche-numerical-linear-algebra-EQ1056	360	■ 1.000	$\begin{aligned} D_i \left(\boldsymbol{x}^{\mathrm{T}} \boldsymbol{B} \boldsymbol{x} \right) &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\left(\boldsymbol{x} + h \boldsymbol{e}_i \right)^{\mathrm{T}} \boldsymbol{B} \left(\boldsymbol{x} + h \boldsymbol{e}_i \right) - \boldsymbol{x}^{\mathrm{T}} \boldsymbol{B} \boldsymbol{x} \right) \\ &= \lim_{h \rightarrow 0} \left(\boldsymbol{e}_i^{\mathrm{T}} \boldsymbol{B} \boldsymbol{x} + \boldsymbol{x}^{\mathrm{T}} \boldsymbol{B} \boldsymbol{e}_i + h \boldsymbol{e}_i^{\mathrm{T}} \boldsymbol{e}_i \right) = \boldsymbol{e}_i^{\mathrm{T}} \left(\boldsymbol{B} + \boldsymbol{B}^{\mathrm{T}} \right) \boldsymbol{x}, \end{aligned}$	$D_i \left(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x} \right) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\left(\boldsymbol{x} + h \boldsymbol{e}_i \right)^T \boldsymbol{B} \left(\boldsymbol{x} + h \boldsymbol{e}_i \right) - \boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x} \right) \\ = \lim_{h \rightarrow 0} \left(\boldsymbol{e}_i^T \boldsymbol{B} \boldsymbol{x} + \boldsymbol{x}^T \boldsymbol{B} \boldsymbol{e}_i + h \boldsymbol{e}_i^T \boldsymbol{e}_i \right) = \boldsymbol{e}_i^T \left(\boldsymbol{B} + \boldsymbol{B}^T \right) \boldsymbol{x},$	$D_i(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \lim_{h \rightarrow 0} \frac{1}{h} \left((\boldsymbol{x} + h \boldsymbol{e}_i)^T \boldsymbol{B} (\boldsymbol{x} + h \boldsymbol{e}_i) - \boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x} \right) \\ = \lim_{h \rightarrow 0} \left(\boldsymbol{e}_i^T \boldsymbol{B} \boldsymbol{x} + \boldsymbol{x}^T \boldsymbol{B} \boldsymbol{e}_i + h \boldsymbol{e}_i^T \boldsymbol{e}_i \right) = \boldsymbol{e}_i^T (\boldsymbol{B} + \boldsymbol{B}^T) \boldsymbol{x},$